

PREDICTIVE ANALYTICS MARKET WITH COVID-19 IMPACT ANALYSIS

GLOBAL FORECASTS TO 2026

BY SOLUTION (FINANCIAL ANALYTICS, RISK ANALYTICS, MARKETING ANALYTICS,
WEB & SOCIAL MEDIA ANALYTICS), SERVICE, DEPLOYMENT MODE,
ORGANIZATION SIZE, VERTICAL, AND REGION

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LIST OF ABBREVIATIONS

FULL FORM	ABBREVIATION
AMI	Advanced Metering Infrastructure
APAC	Asia Pacific
ARP	Address Resolution Protocol
B2B	Business-to-Business
B2C	Business-to-Consumer
BCBS	Basel Committee on Banking Supervision
BFSI	Banking, Financial Services, and Insurance
BI	Business Intelligence
BIS	Bank of International Settlement
BMGF	Bill and Melinda Gates Foundation
BPO	Business Process Outsourcing
BU	Business Unit
CAGR	Compound Annual Growth Rate
CAPEX	Capital Expenditure
CIO	Chief Information Officer
CLV	Customer Lifetime Value
CMS	Content Management System
CRM	Customer Relationship Management
CSO	Chief Strategy Officer
CSP	Cloud Service Provider
CTO	Chief Technology Officer
CX	Customer Experience
CXO	Chief Experience Officer
DDoS	Distributed Denial of Service
DNS	Domain Name System
ERP	Enterprise Resource Planning
ETA	Estimated Time of Arrival
EU	European Union

FY	Financial Year
GDP	Gross Domestic Product
GDPR	General Data Protection Regulation
GRC	Governance, Risk Management, and Compliance
HDF	Hortonworks Data Flow
IBP	Integrated Business Planning
ICT	Information and Communications Technology
IDS	Intrusion Detection System
IE	Intelligent Enterprise
IPS	Intrusion Prevention System
ISMS	Information Security Management System
ISV	Independent Software Vendor
IVR	Interactive Voice Response
KSA	Kingdom of Saudi Arabia
LoB	Line of Business
M&A	Merger and Acquisition
MEA	Middle East and Africa
MZR	Multizone Region
NDP	National Development Plan
NLP	Natural Language Processing
OPEX	Operating Expenditure
PAT	Predictive Analytics Technologies
R&D	Research and Development
RED	Real-time Early Detection
ROI	Return On Investment
RPA	Robotic Process Automation
SCADA	Supervisory Control and Data Acquisition
SEC	Securities and Exchange Commission
SI	System Integrator
SME	Small and Medium-sized Enterprise
SPSS	Statistical Package for Social Sciences
TMS	Transportation Management System

TPL	Third-Party Logistics
TRA	Telecommunications Regulatory Authority
UAE	United Arab Emirates
USD	United States Dollar
VAR	Value-Added Reseller
VLAN	Virtual Local Area Network
VP	Vice President
WHO	World Health Organization
WMS	Warehouse Management System
Y-o-Y	Year-on-Year

1 INTRODUCTION

1.1 INTRODUCTION TO COVID-19

COVID-19 is an infectious disease caused by the novel coronavirus. Largely unknown before its outbreak in Wuhan, China, in December 2019, COVID-19 moved from a regional crisis to a global pandemic in just a few weeks. The World Health Organization (WHO) declared COVID-19 as a pandemic on March 11, 2020.

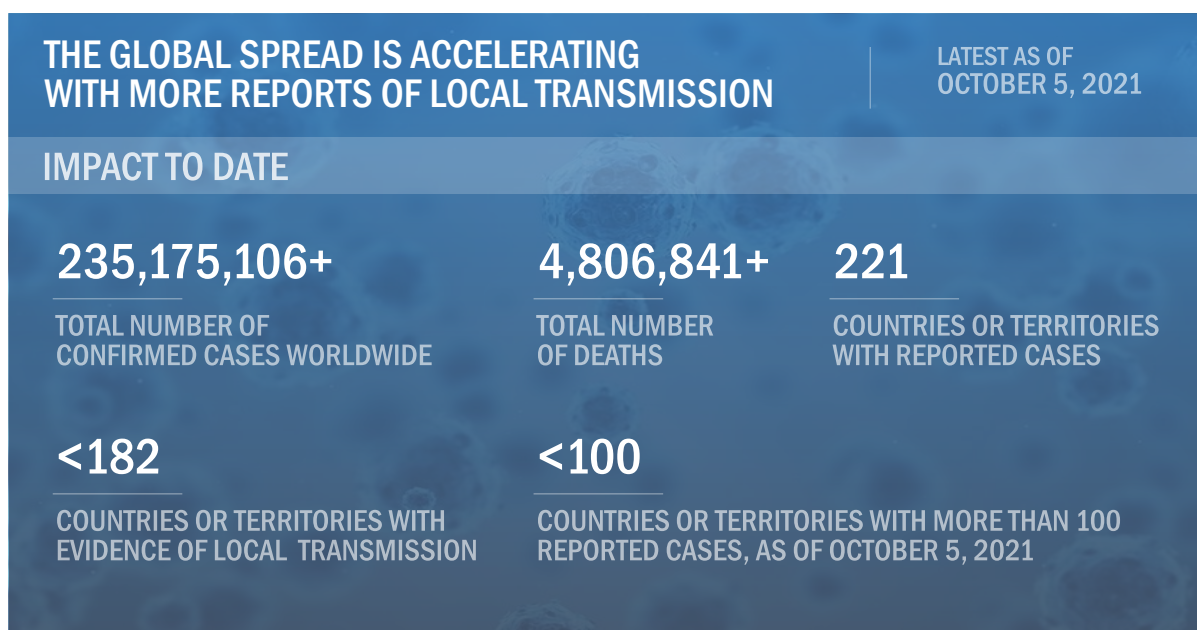
As the virus is spreading rapidly at a global level, countries globally are trying to prevent further contagion by adopting measures such as social distancing, contact tracing, self-quarantining, surveillance, communication, and testing. So far, China, South Korea, Japan, and Singapore have managed to flatten the curve, while the US, Italy, Spain, Germany, France, and Iran have imposed drastic measures to slow down the spread and control fatalities.

Government agencies have announced special financial aid packages to develop preventive and curative drugs, the purchase of critical care medical devices, and the fast-track approval of diagnostic tests. Various organizations across the globe, such as the Bill and Melinda Gates Foundation (BMGF) and Wellcome Trust, are also coming together to accelerate and strengthen the efforts to fight the COVID-19 pandemic.

1.2 COVID-19 HEALTH ASSESSMENT

It is an unprecedented crisis creating an enormous strain on healthcare providers across the globe. According to the WHO, within 67 days, the first 100,000 confirmed cases were reported worldwide. It took the first 11 days to reach 200,000 confirmed cases. In the next four days, the number reached 300,000 confirmed cases, while in another 12 days, the count reached one million confirmed cases.

FIGURE 1 COVID-19: GLOBAL PROPAGATION

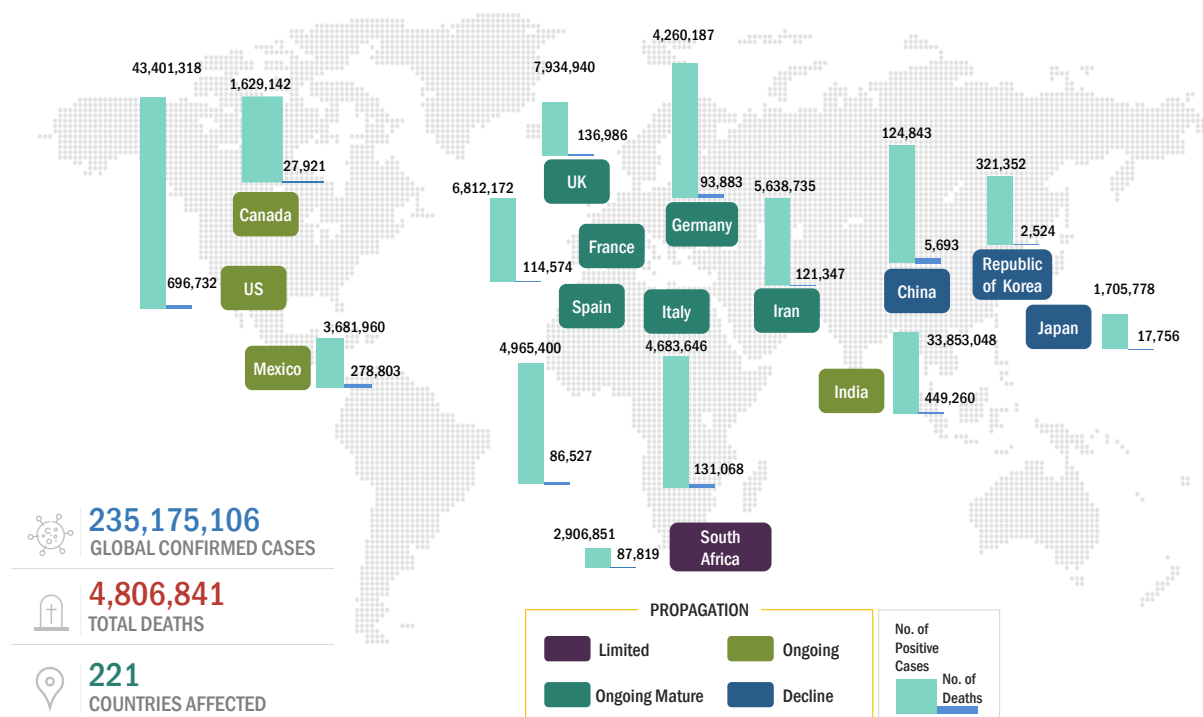


Source: WHO and ECDC

As of October 5, 2021, the virus has spread to more than 221 countries, with over 235,175,106 reported cases and close to 4,806,841 deaths.

The actual number of infected people is probably higher than the official total, as some mild and asymptomatic cases are not tested and counted.

FIGURE 2 COVID-19 PROPAGATION: SELECT COUNTRIES



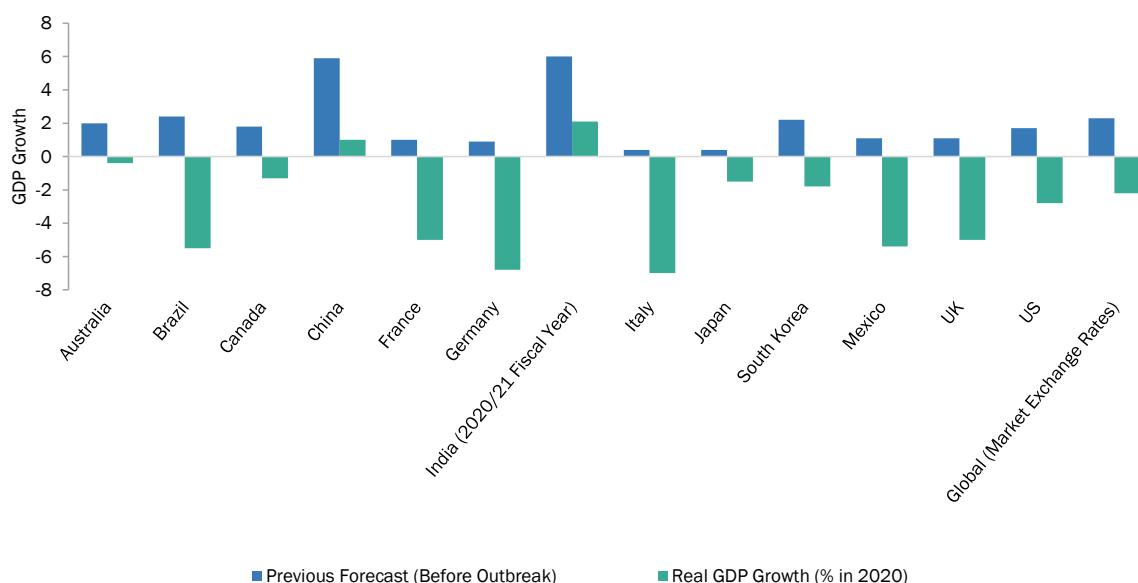
Source: WHO Coronavirus Disease 2019; data as of October 5th, 2021

Disease progression patterns are similar in most countries; however, flattening of the curve and disease containment largely depend on various measures taken by each country. China and South Korea, for example, have managed to control the spread of the disease by implementing severe countermeasures, such as restricting travel; imposing city-wide quarantines, curfews, and lockdowns; building specialty hospitals; increasing medical aid; and ramping up testing. Most other countries are trying to emulate these measures to contain the spread of the disease.

1.3 COVID-19 ECONOMIC ASSESSMENT

The COVID-19 outbreak has not only affected the healthcare sector but also impacted the global economy. It has had a significant economic impact on various financial and industrial sectors, such as energy, oil and gas, transportation and logistics, manufacturing, and aviation. It is predicted that the global economy will go into recession due to the loss of trillions of dollars. With the increasing number of countries imposing and extending lockdowns, economic activities are declining, which would impact the global economy.

FIGURE 3 REVISED GROSS DOMESTIC PRODUCT FORECASTS FOR SELECT G20 COUNTRIES IN 2020



Source: The Economist Intelligent Unit

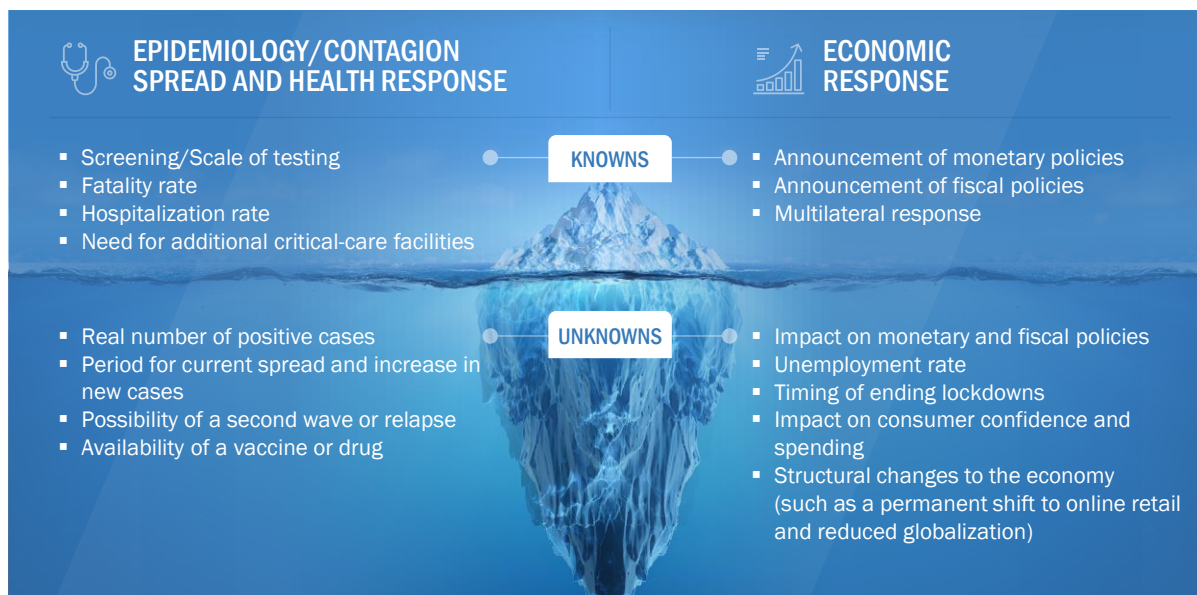
With the combined effects of the pandemic and the slump in global oil prices, investments are expected to contract sharply this year, especially in the energy sector. Export growth is also expected to reduce drastically. A recovery is expected in the second half of the year, provided there is no emergence of another pandemic wave, and countries manage to control the spread quickly and resume routine.

China has managed to contain the spread and is currently on track to resume routine. It is expected to recover in the second half of the year. However, the Eurozone, one of the hardest-hit regions, is expected to undergo a recession throughout 2021. Within the Eurozone, Germany, which is a manufacturing hub, would recover at a much slower pace than the other countries. The US economy is also expected to see a slump, with unemployment set to rise; recovery will be slow and only seen in the second half of the year.

1.3.1 COVID-19 ECONOMIC IMPACT—SCENARIO ASSESSMENT

Recovery from this crisis largely depends on various knowns and unknowns, which most countries will have to deal with and respond to. The speed and strength of the economic recovery will depend on the health and economic responses and the agility and speed of implementing various measures.

FIGURE 4 CRITERIA IMPACTING GLOBAL ECONOMY



In the recent past, the global economy has become substantially more interconnected. The adverse consequences of various steps related to containment are evident from global supply chain disruptions, weaker demand for imported products and services, and the increase in the unemployment rate. Risk aversion has increased in the financial market, with all-time-low interest rates and sharp declines in equity and commodity prices. Consumer and business confidence have also reduced significantly.

This has magnified over time, and the spillover is seen in other countries as well. The latest forecasts by multiple sources predict a four percent decrease in the global Gross Domestic Product (GDP) in 2020 and a slow recovery, making the recession caused by this pandemic worse than the 2009 global recession. Based on these forecasts, the global economy is expected to recover by 2021; however, for some countries, including the UK, Germany, India, and Italy, recovery is predicted to take longer.

In this fast-changing environment, the full impact of the pandemic on the global economy may not entirely be known. MarketsandMarkets uses three scenario-based approaches (epidemiology, health response, and economic response) to assess the economic impact and recovery period at the global level. Countries and regions are expected to have different impacts and recovery periods.

FIGURE 5 SCENARIOS IN TERMS OF RECOVERY OF GLOBAL ECONOMY

SCENARIO		
 OPTIMISTIC	 REALISTIC	 PESSIMISTIC
EPIDEMIOLOGY/CONTAGION SPREAD AND HEALTH RESPONSE		
FATALITY RATE 1.5%–2% HOSPITALIZATION RATE LESS THAN 15% ADDITIONAL ICU BEDS 2.5%–5% ADDITIONAL VENTILATORS 15%	FATALITY RATE 2%–3.5% HOSPITALIZATION RATE 15%–20% ADDITIONAL ICU BEDS 5% ADDITIONAL VENTILATORS 15%–20%	FATALITY RATE MORE THAN 3.5% HOSPITALIZATION RATE MORE THAN 20% ADDITIONAL ICU BEDS 15% ADDITIONAL VENTILATORS 25%
ECONOMIC RESPONSE		
FISCAL STIMULUS USD 7.5–8 TRILLION GLOBALLY EMERGENCY FUNDING TO SUPPORT EFFORTS TO FIGHT COVID-19 USD 100 BILLION GLOBALLY MULTILATERAL AGENCIES FINANCING USD 100 BILLION GLOBALLY	FISCAL STIMULUS USD 5–7.5 TRILLION GLOBALLY EMERGENCY FUNDING TO SUPPORT EFFORTS TO FIGHT COVID-19 USD 75–100 BILLION GLOBALLY MULTILATERAL AGENCIES FINANCING USD 50–75 BILLION GLOBALLY	FISCAL STIMULUS LESS THAN USD 5 TRILLION GLOBALLY EMERGENCY FUNDING TO SUPPORT EFFORTS TO FIGHT COVID-19 LESS THAN USD 75 BILLION GLOBALLY MULTILATERAL AGENCIES FINANCING LESS THAN USD 50 BILLION GLOBALLY
RECOVERY SCENARIO		
▪ Milder recession in 2021 Q1 and Q2; stronger recovery in 2022 and after ▪ US: Recovery by 2021 Q4 ▪ EU5: Recovery by 2021 Q3 ▪ China: Recovery by 2021 Q1 ▪ India: Recovery by 2022 Q4	▪ Recession in 2021 Q1 and Q2; partial bounce back in 2021 Q3, then slow growth ▪ US: Recovery by 2022 Q3 ▪ EU5: Recovery by 2022 Q2 ▪ China: Recovery by 2021 Q4 ▪ India: Recovery by 2022 Q4	▪ Deeper recession in 2021 Q1, Q2, and Q3; modest rebound in 2022 ▪ US: Recovery by 2022 Q4 ▪ EU5: Recovery by 2022 Q3 ▪ China: Recovery by 2021 Q4 ▪ India: Recovery by 2023 Q2

Source: UNCTAD, Moody's, FAS, OECD, and MarketsandMarkets Analysis

With more than four million confirmed cases of COVID-19, businesses are coping with lost revenues and disrupted supply chains as factory shutdowns and quarantine measures are spreading across the globe, restricting movement and business activities. The economic impact is mounting, and the top economies of the world are expected to enter a global recession.

To provide a perspective on various scenarios, we assume the factors that would impact the economic recovery from the global recession.

In a realistic scenario, we assume that possible health and economic responses would help economies bounce back partially in 2021 Q4. In Q1, Q2, and Q3 of 2021, countries will ramp up testing capacities and critical healthcare services and increase financial aid availability. This will help contain the spread of the virus. The economic recession will last until 2022 Q2; a slow recovery will be seen from 2022 Q3. This is expected to be a 'U-shaped' economic recovery, with a gradual improvement in the economy.

1.4 OBJECTIVES OF THE STUDY

- To define, describe, and forecast the predictive analytics market by component, deployment mode, organization size, vertical, and region
- To provide detailed information about major factors (drivers, restraints, opportunities, and industry-specific challenges) influencing the market growth
- To analyze opportunities in the market and provide details of the competitive landscape for stakeholders and market leaders
- To forecast the market size of segments with respect to five main regions: North America, Europe, Asia Pacific (APAC), Middle East and Africa (MEA), and Latin America
- To profile key players and comprehensively analyze their market rankings and core competencies
- To analyze competitive developments, such as mergers and acquisitions, new product developments, and Research and Development (R&D) activities, in the market
- To analyze the impact of the COVID-19 pandemic on the predictive analytics market.

1.5 MARKET DEFINITION

According to SAS Institute, predictive analytics is the use of data, statistical algorithms, and machine learning techniques to identify the likelihood of future outcomes based on historical data. The goal is to analyze current data to make predictions about future events.

Predictive analytics is a statistical and data mining solution that consists of numerous algorithms and methodologies. These algorithms and methodologies are used for both structured as well as unstructured data to extract business insights. Predictive analytics offers flexible, scalable, and advanced solutions to help users make better-informed business decisions. It helps industries understand customer perception by providing a competitive market edge and the ability to rapidly orchestrate business decisions.

1.5.1 INCLUSIONS AND EXCLUSIONS

CATEGORY	INCLUSION	EXCLUSION
Predictive Analytics Market	The Business-to-Business (B2B) approach is considered for revenue estimation.	The Business-to-Consumer (B2C) approach is not considered for calculating revenue.
Solutions	The report includes the following solutions: ■ Financial Analytics (fraud detection, profitability management, governance and compliance management, and others [asset and liability management]) ■ Risk Analytics (cyber risk management, operational risk management, credit and market risk management, and others [reputational risk, environmental risk, third-party risk, and economic risk]) ■ Marketing Analytics (predictive modeling, yield management, product and service, development strategies, and others [demand forecasting and contact optimization]) ■ Sales Analytics (Sales Life Cycle Management, Sales Rep Efficiency Management, and others [territory management and incentive compensation management])	The report does not include hardware and open-source platforms; however, open-source platforms that are integrated with supporting technologies and imply cost toward developing solutions are considered in the report.

- Customer Analytics (customer segmentation and clustering, customer behavior analysis, monitoring customer loyalty and satisfaction, and others [operational efficiency management and campaign performance management])
- Web and Social Media Analytics (social media management, search engine optimization, performance monitoring, and competitor benchmarking)
- Supply Chain Analytics (distribution and logistics optimization, inventory management, manufacturing analysis, and others [vendor performance management and spend analysis])
- Network Analytics (intelligent network optimization, traffic management, and others [managed network security and configuration change management])

Services	The report includes managed and professional services. Professional services include consulting, deployment and integration, and support and maintenance.	Consulting services mentioned in the report are pre-deployment services. Post-deployment services are not included in the current scope of the report. Services such as IT outsourcing, application development, and Business Process Outsourcing (BPO) are not included in the study.
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The solutions segment comprises financial analytics, risk analytics, marketing analytics, sales analytics, customer analytics, web and social media analytics, supply chain analytics, and network analytics. The vertical segment includes banking, financial services, and insurance (BFSI); healthcare and life sciences; retail and eCommerce; media and entertainment; manufacturing; government and defense; energy and utilities; telecom and IT; travel and hospitality; transportation and logistics, others (education and research, and real estate). Based on deployment modes, the predictive analytics market is segmented into cloud and on-premises. Segmentation, by organization size, covers small and medium-sized enterprises (SMEs) and large enterprises. This market study is also segmented into five regions: North America, Europe, APAC, MEA, and Latin America.

1.6 MARKET SCOPE

The study provides an in-depth analysis of the global predictive analytics market based on contemporary market trends and developments and its potential growth from 2021 to 2026. It provides detailed market trends, competitive landscape, market size, forecasts, and analysis of key predictive analytics solution providers.

1.6.1 MARKET SEGMENTATION



*Other financial analytics applications include asset and liability management.

**Other risk analytics applications include reputational risk, environmental risk, third-party risk, and economic risk.

***Other marketing analytics applications include demand forecasting and contact optimization.

****Other sales analytics applications include territory management and incentive compensation management.

*****Other customer analytics applications include operational efficiency management and campaign performance management.

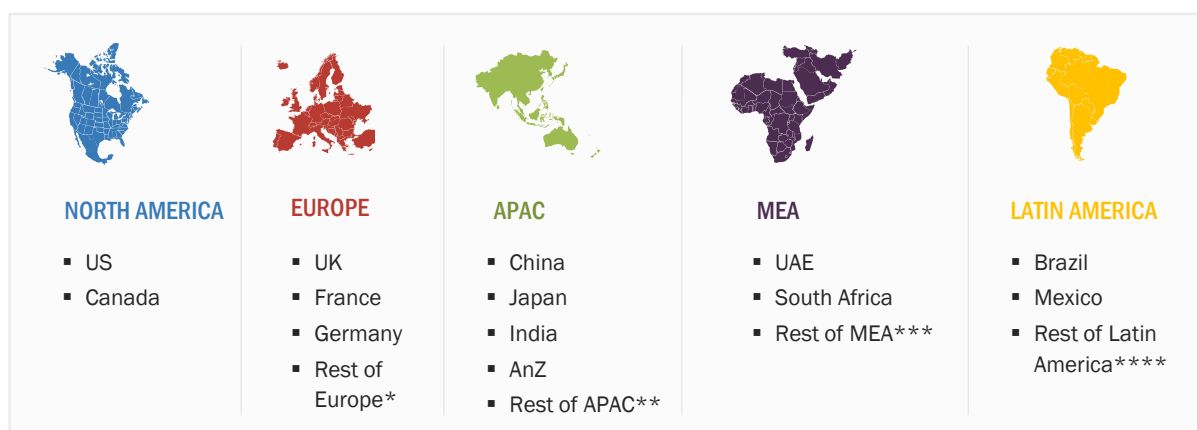
*****Other supply chain analytics applications include vendor performance management and spend analysis.

*****Other network analytics applications include managed network security and configuration change management.

*****Other verticals include education and research and real estate.

Source: MarketsandMarkets Analysis

1.6.2 REGIONS COVERED



*Rest of Europe includes Spain, Sweden, Switzerland, and Denmark.

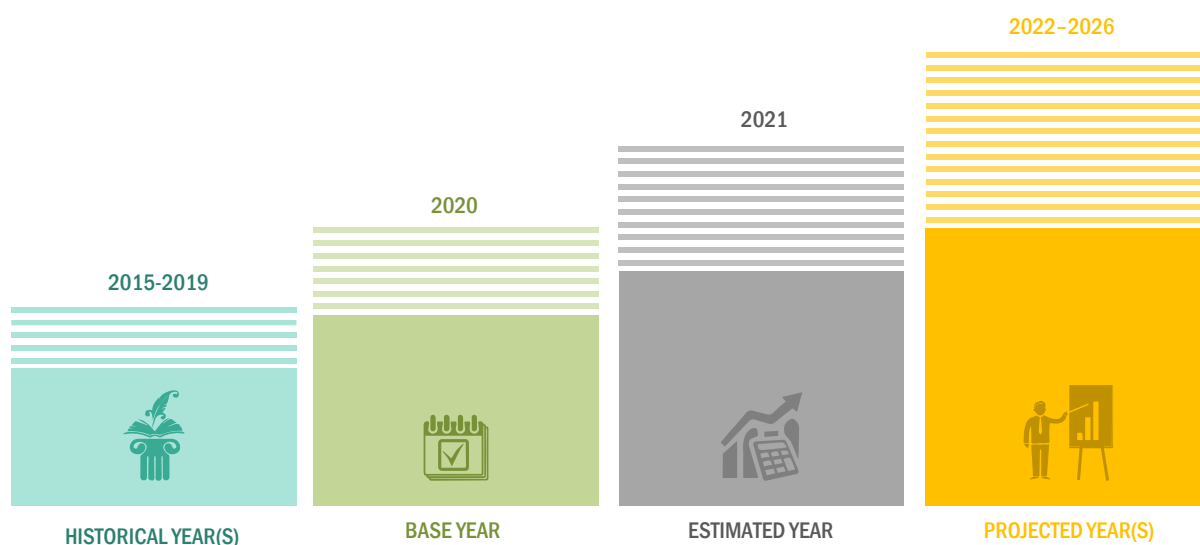
**Rest of APAC includes Singapore, South Korea, and Malaysia.

***Rest of MEA includes Qatar, Saudi Arabia, Israel, and Kuwait.

****Rest of Latin America includes Argentina, Colombia, Peru, and Chile.

Source: MarketsandMarkets Analysis

1.6.3 YEARS CONSIDERED FOR THE STUDY



Note: The forecast period is 2021–2026.

Source: MarketsandMarkets Analysis

1.7 CURRENCY CONSIDERED

The base currency used in the report is the United States Dollar (USD), with the market size indicated in USD million/billion.

- Companies' revenue figures were referred from their annual reports.
- The average annual currency conversion rate was used for companies that reported their revenue in currencies other than the USD.

TABLE 1 UNITED STATES DOLLAR EXCHANGE RATE, 2018–2020

Currency	2018	2019	2020
EUR	0.848	0.893	0.877

Source: The US Internal Revenue Service (IRS)

1.8 STAKEHOLDERS

- Predictive analytics solution providers
- Independent software vendors (ISVs)
- Investors and venture capitalists (VCs)
- Managed service providers
- Support and maintenance service providers
- System integrators (SIs)/migration service providers
- Value-added resellers (VARs) and distributors

1.9 SUMMARY OF CHANGES

- The new research study covers the value chain analysis of the predictive analytics market.
- This version of the report also includes a qualitative assessment of various modeling techniques deployed in predictive analytics.
- The new market study consists of the COVID-19 market drivers for every segment in the predictive analytics market.
- The new market study consists of the historical revenue analysis of major public players in the predictive analytics market.
- The research study consists of the market share analysis of the major public players in the predictive analytics market.
- The new edition of the report provides updated financial information in the context of the predictive analytics market from 2019 to 2020 for each listed company in graphical representation.
- The current report includes the market developments from 2018 to 2021.

2 RESEARCH METHODOLOGY

2.1 RESEARCH DATA

The research study involved the use of extensive secondary sources; directories; as well as several journals and magazines, such as the Institute of Electrical and Electronics Engineers IoT Journal, International Journal of Data Science and Analytics, and IoT Magazine, to identify and collect the information useful for a comprehensive market research study on the predictive analytics market. The primary sources were mainly industry experts from the core and related industries, preferred predictive analytics solution providers, third-party service providers, consulting service providers, end users, and other commercial enterprises. In-depth interviews were conducted with various primary respondents, including key industry participants and subject matter experts, to obtain and verify critical qualitative and quantitative information, as well as assess the market's prospects.

The following figure highlights the market research methodology applied to make the report on the global predictive analytics market:

FIGURE 6 PREDICTIVE ANALYTICS MARKET: RESEARCH DESIGN



Source: MarketsandMarkets Analysis

2.1.1 SECONDARY DATA

The market size of companies offering predictive analytics solutions and services was arrived at based on secondary data available through paid and unpaid sources. It was also arrived at by analyzing the product portfolios of major companies and rating the companies based on their performance and quality.

In the secondary research process, various sources were referred to for identifying and collecting information for the study. The secondary sources included annual reports, press releases, and investor presentations of companies; white papers, journals, and certified publications; and articles from recognized authors, directories, and databases. The data was also collected from other secondary sources, such as IoT conferences and related magazines. Additionally, predictive analytics spending of various countries was extracted from respective sources. Secondary research was mainly used to obtain key information about the industry's value chain and supply chain to identify key players by solution, service, market classification, and segmentation, according to the offerings of major players and industry trends related to solutions, applications, verticals, and regions, and key developments from both market- and technology-oriented perspectives.

2.1.2 PRIMARY DATA

In the primary research process, various primary sources from both supply and demand sides were interviewed to obtain qualitative and quantitative information on the market. The primary sources from the supply-side included various industry experts, including Chief Experience Officers (CXOs), Vice Presidents (VPs), directors from business development, marketing, product development/innovation teams, related key executives from predictive analytics solution vendors, SIs, professional service providers, industry associations, and key opinion leaders.

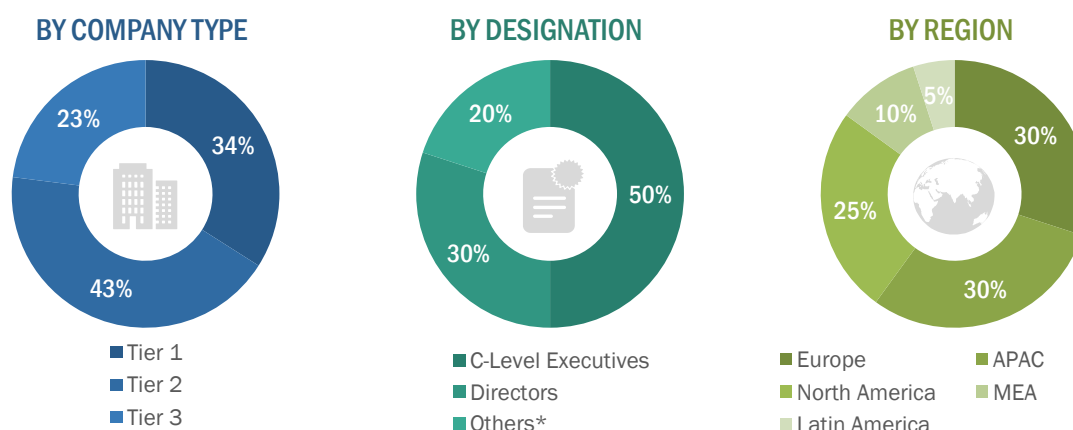
Primary interviews were conducted to gather insights, such as market statistics, data of revenue collected from solutions and services, market breakups, market size estimations, market forecasts, and data triangulation. Primary research also helped in understanding various trends related to technology, application, deployment, and region. Stakeholders, such as Chief Information Officers (CIOs), Chief Technology Officers (CTOs), Chief Strategy Officers (CSOs), and the installation teams of governments/end users using predictive analytics solutions, from the demand-side, were interviewed to understand the buyer's perspective on suppliers, products, service providers, and their current usage of predictive analytics solutions and services that would affect the overall predictive analytics market.

TABLE 2 PRIMARY INTERVIEWS

PREDICTIVE ANALYTICS SOLUTIONS AND SERVICE PROVIDERS	
Qubole	Qlik
Tiger Analytics	IBM
Rapidminer	

Source: Primary Interviews

2.1.2.1 Breakup of primary profiles



*Others include sales managers, marketing managers, and product managers.

Note: Tier 1 companies' overall annual revenue is more than USD 10 billion; tier 2 companies' revenue is between USD 1 and 10 billion; and tier 3 companies' revenue is between USD 500 million–USD 1 billion

Source: Industry Experts

2.1.2.2 Key industry insights

As the eCommerce industry is recovering post COVID-19, predictive analytics is playing a vital role in the recovery process. We see daily changes in the serviceability of areas for logistics. Understanding demands and optimizing supply chain and logistics will give a competitive advantage to eCommerce businesses.

– Puneet Tripathi, Data Science Head
Wakefit

Bringing the right insights into the right hands at the right time is going to be a key driver for analytics.

– Hardik Patel, Product Development Manager
Qlik

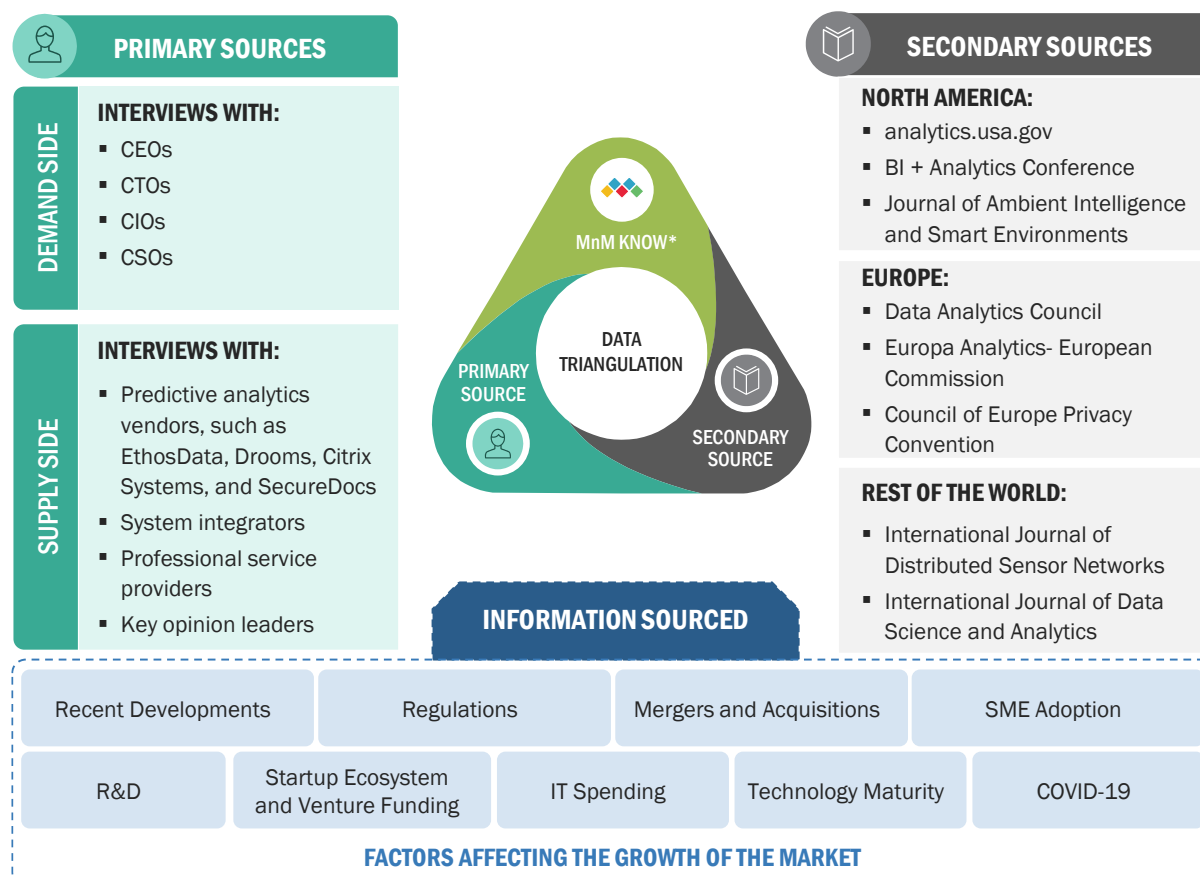
For data science, we need an environment that is more responsive to human interactivity. Current events may influence what happens tomorrow. A lot of continuous iteration will make the system smarter. For the machine to learn, we explore data, visualize it, build a model, ask a question, an answer comes back, bring in other data, and repeat the process. The more real-time and granular we can get, the more responsive and competitive we can be.

– Himanshu Swamy, Data Science Manager
Publicis Sapient

Source: Industry Experts

2.2 MARKET BREAKUP AND DATA TRIANGULATION

FIGURE 7 DATA TRIANGULATION



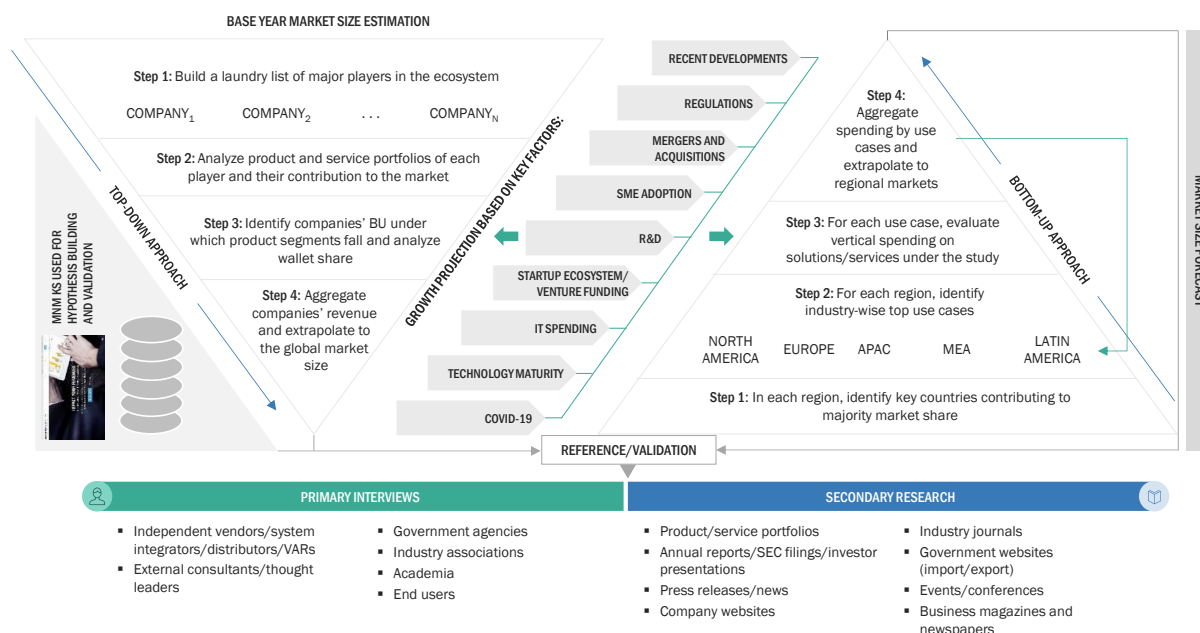
*MnM KNOW stands for MarketsandMarkets' 'Knowledge Asset Management' framework. In this context, it stands for the existing market research knowledge repository of over 5,000 granular markets, our flagship competitive intelligence and market research platform 'Knowledge Store,' subject matter experts, and independent consultants. MnM KNOW acts as an independent source that helps us validate information gathered from primary and secondary sources. Research studies referred from MarketsandMarkets' repository included the Big data Market.

Source: MarketsandMarkets Analysis

2.3 MARKET SIZE ESTIMATION

Multiple approaches were adopted for estimating and forecasting the predictive analytics market. The first approach involves the estimation of the market size by summation of companies' revenue generated through the sale of solutions and services.

FIGURE 8 PREDICTIVE ANALYTICS MARKET: TOP-DOWN AND BOTTOM-UP APPROACHES



Source: MarketsandMarkets Analysis

2.3.1 TOP-DOWN APPROACH

In the top-down approach, an exhaustive list of all the vendors offering solutions and services in the global predictive analytics market was prepared. The revenue contribution of all the vendors in the market was estimated through annual reports, press releases, funding, investor presentations, paid databases, and primary interviews. Each vendor was evaluated on the basis of its products/solutions by technology, service, and application. The aggregate of all the companies' revenue was extrapolated to reach the overall market size. Each subsegment was studied and analyzed for its global market size and regional penetration. The markets were triangulated through both primary and secondary research. The primary procedure included extensive interviews for key insights from the industry leaders, such as Chief Information Officers (CIOs), Chief Executive Officers (CEOs), Vice Presidents (VPs), directors, and marketing executives. The market numbers were further triangulated with the existing MarketsandMarkets repository for validation.

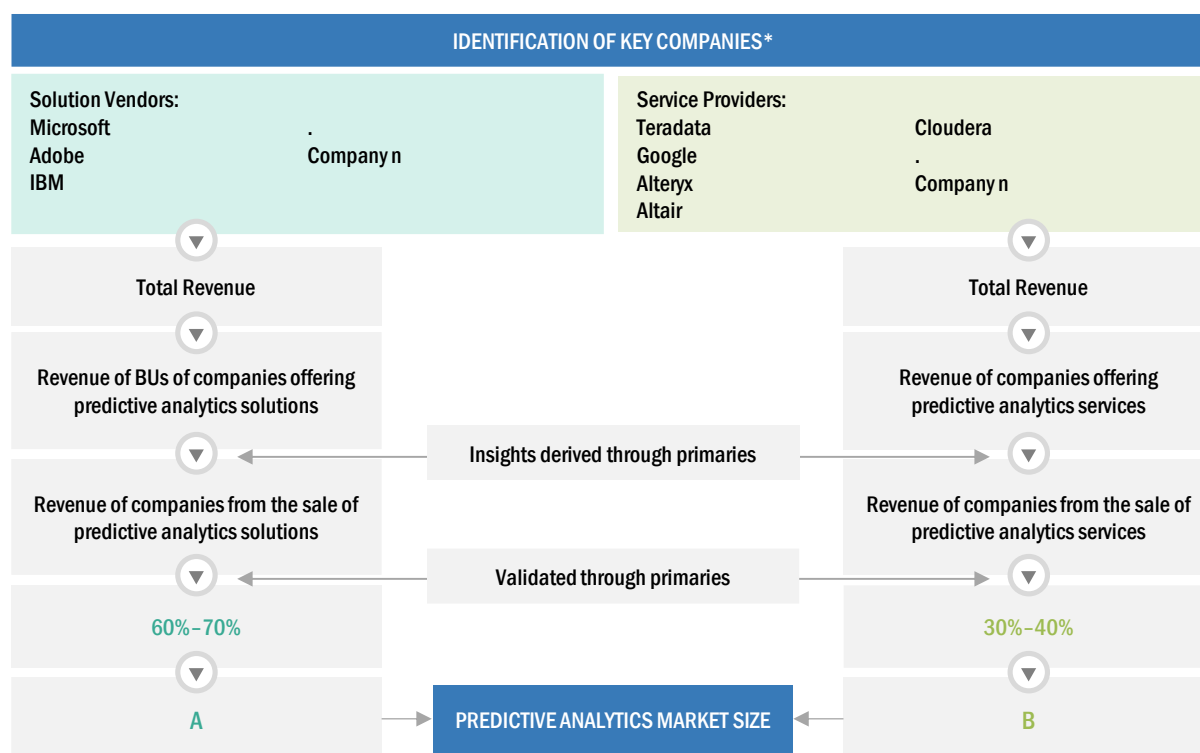
The list of vendors considered for estimating the market is not limited to the vendors profiled in the report. However, MarketsandMarkets has prepared a laundry list of vendors operating in the predictive analytics market (which offers solutions and services) and their product mapping (related to the predictive analytics market) to identify the major vendors operating in the market.

2.3.2 BOTTOM-UP APPROACH

In the bottom-up approach, the adoption of predictive analytics solutions among different end users in key countries, with respect to their regions that contribute the most to the market share, was identified. For cross-validation, the adoption of predictive analytics solutions and services among different industries, along with different use cases with respect to their regions, was identified and extrapolated. In addition, weightages were given to the use cases identified in different regions for the market size calculation.

Based on these numbers, the regional split was determined by primary and secondary sources. The procedure included the analysis of the predictive analytics market's regional penetration. Based on secondary research, the regional spending on Information and Communications Technology (ICT), socioeconomic analysis of each country, strategic vendor analysis of major predictive analytics providers, and organic and inorganic business development activities of regional and global players were estimated. With the data triangulation procedure and validation of data through primaries, the exact values of the overall predictive analytics market size and its segments' market size were determined and confirmed using the study.

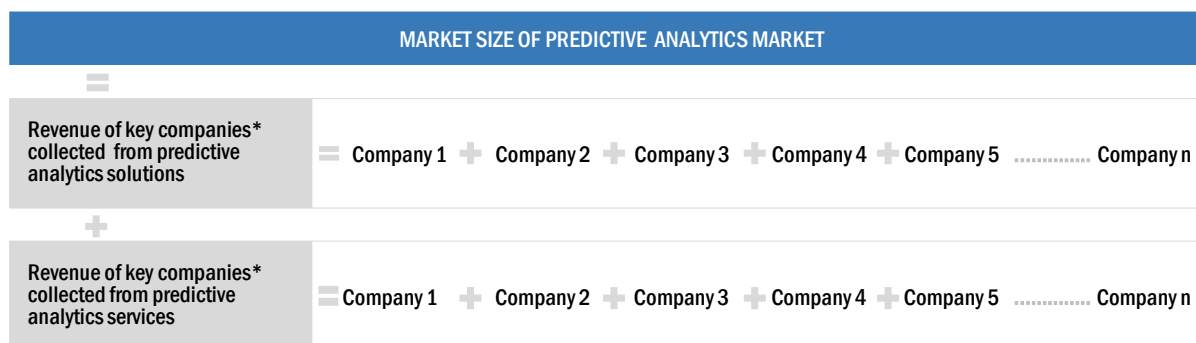
FIGURE 9 MARKET SIZE ESTIMATION METHODOLOGY - APPROACH 1 (SUPPLY SIDE):
REVENUE OF SOLUTIONS/SERVICES OF PREDICTIVE ANALYTICS MARKET



*The key companies should contribute to more than 50% of the total market.

Source: MarketsandMarkets Analysis

FIGURE 10 MARKET SIZE ESTIMATION METHODOLOGY -APPROACH 2 - BOTTOM-UP
(SUPPLY SIDE): COLLECTIVE REVENUE OF ALL SOLUTIONS/SERVICES OF THE
PREDICTIVE ANALYTICS MARKET



*Key companies should contribute to more than 50% of the total market.

Source: MarketsandMarkets Analysis

In this approach for market estimation, key predictive analytics solution and service providers, such as IBM (US), Microsoft (US), Oracle (US), SAP (Germany), SAS Institute (US), Google (US), Salesforce (US), AWS (US), HPE (US), Teradata (US), Alteryx (US), FICO (US), Altair (US), Domo (US), Cloudera (US), Board International (Switzerland), TIBCO Software (US), Hitachi Vantara (US), Qlik (US), Happiest Minds (India), Dataiku (US), RapidMiner (US), Biofourmis (US), In-med Prognostics (India), Aito.ai (Finland), Symend (US), Onward Health (India), Unioncrate (US), CyberLabs (Brazil), Actify Data Labs (India), Amlgo Labs (India), and Verimos (US), contribute almost 60-65% to the global predictive analytics market. After confirming these companies through primary interviews with industry experts, their total revenue through annual reports, Securities and Exchange Commission (SEC) filings, and paid databases was estimated.

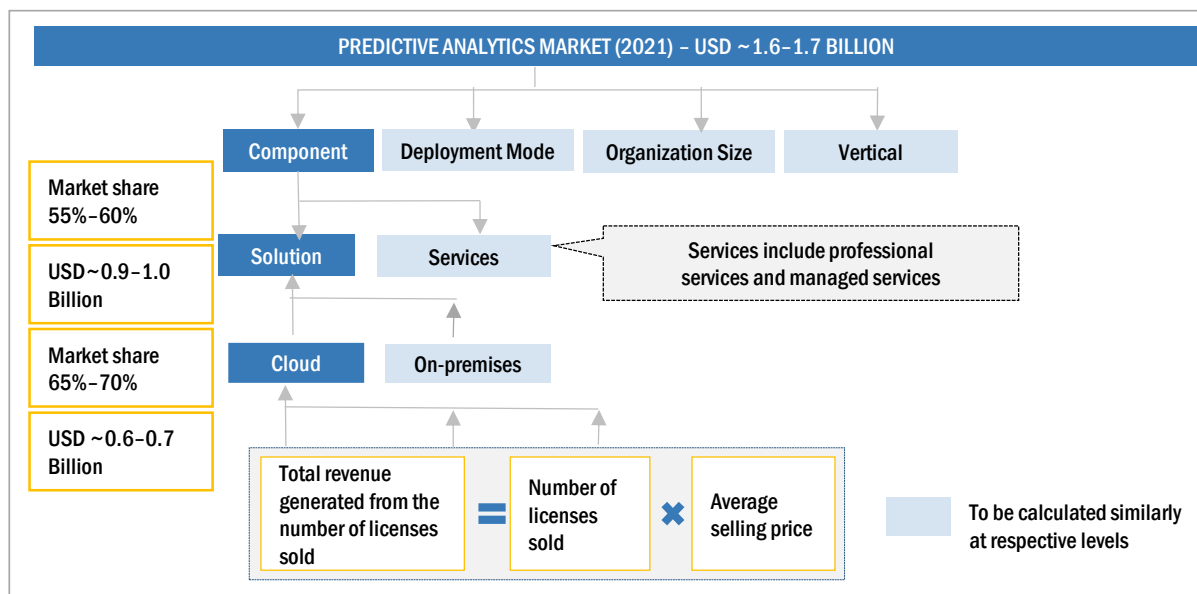
These companies' revenue, pertaining to the Business Units (BUs) that offer predictive analytics solutions, was identified through similar sources. Then, through primaries, the data of revenue generated through specific predictive analytics solutions was collected. The collective revenue of key companies that offer predictive analytics solutions comprises 60%-65% of the market, which was again confirmed through primary interviews with industry experts. With the assumption that the rest of the market is contributed by smaller players (part of the unorganized market), the market size of organized players (60%-65%) and unorganized players (35%-40%) collectively was assumed to be the market size of the global predictive analytics market for Financial Year (FY) 2020.

FIGURE 11 MARKET SIZE ESTIMATION METHODOLOGY-APPROACH 3, BOTTOM-UP (SUPPLY SIDE): COLLECTIVE REVENUE FROM ALL SOFTWARE/SERVICES OF THE PREDICTIVE ANALYTICS MARKET

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5
Identification of 30-35 major predictive analytics companies, such as IBM, Google, Alteryx, Oracle, SAP, Salesforce, HPE, AWS, and Teradata	Estimation of top players' revenue from BUs offering predictive analytics solutions and services	Identification of the revenue contribution of predictive analytics solutions and services to the relevant BU/BUs	Calculation of the total market size of predictive analytics solutions and services using the combined share of top companies and their revenues	Revalidation of the total market size and market share estimations with primary respondents
KEY SOURCE OF INFORMATION:				
Secondary sources such as company financials, press releases, paid databases (Factiva), and primary research	Secondary sources such as company financials, press releases, paid databases (Factiva), and primary research	Primary research (CEOs, VPs, marketing directors, and technology and innovation directors, among other stakeholders)	MarketsandMarkets calculations, company websites, and annual reports	Primary research (directors - product development, VPs - operations, indoor location managers, COOs, and CSOs)

Source: MarketsandMarkets Analysis

FIGURE 12 MARKET SIZE ESTIMATION METHODOLOGY-APPROACH 4, BOTTOM-UP (DEMAND SIDE): SHARE OF PREDICTIVE ANALYTICS THROUGH OVERALL PREDICTIVE ANALYTICS SPENDING



Source: MarketsandMarkets Analysis

Top-down and bottom-up approaches were used to estimate and validate the size of the predictive analytics market and various other dependent subsegments. The research methodology used to estimate the market size included the following details:

- Key market players were not limited to IBM (US), Microsoft (US), Oracle (US), SAP (Germany), SAS Institute (US), Google (US), Salesforce (US), AWS (US), HPE (US), Teradata (US), Alteryx (US), FICO (US), Altair (US), Domo (US), Cloudera (US), Board International (Switzerland), TIBCO Software (US), Hitachi Vantara (US), Qlik (US), Happiest Minds (India), Dataiku (US), RapidMiner (US), Biofourmis (US), In-med Prognostics (India), Aito.ai (Finland), Symend (US), Onward Health (India), Unioncrate (US), CyberLabs (Brazil), Actify Data Labs (India), Amlgo Labs (India), and Verimos (US).
- The market players were identified through extensive secondary research, and their revenue contribution in respective regions was determined through primary and secondary research.
- The entire procedure included studying the annual and financial reports of top market players and extensive interviews for key insights from industry leaders, such as CEOs, VPs, directors, and marketing executives.
- All percentage splits and breakups were determined using secondary sources and verified through primary sources.

All the possible parameters affecting the market covered in the research study have been accounted for, viewed in extensive detail, verified through primary research, and analyzed to get the final quantitative and qualitative data. The data is consolidated and added with detailed inputs and analysis from MarketsandMarkets.

2.4 MARKET FORECAST

Factors that have a direct or indirect impact on the market were analyzed to estimate the overall trend of the market during the forecast period. Each factor was analyzed on a scale of 1–3, 1 being the lowest and 3 being the highest. After analyzing each factor and its impact on the market, various forecasting techniques were used to arrive at the market size.

TABLE 3 FACTOR ANALYSIS

FACTOR	INFERENCE	IMPACT
Recent Developments	Major vendors account for 60%–65% of the developments in the market.	Medium
Regulations	There are various existing regulations related to data analytics, and these regulations significantly affect the predictive analytics market. Hence, the factor of regulations is considered high for the predictive analytics market.	High
SME Adoption	Predictive analytics solutions and services are witnessing increased adoption among SMEs, followed by large enterprises across the globe.	Medium
R&D Spending	The players operating in the predictive analytics market are allocating a significant part of their revenue for R&D activities to innovate their solutions and meet the emerging demand from customers.	Medium
Start-up Ecosystem/ Venture Funding	The number of start-ups in the predictive analytics market is more than 40% compared to the existing players; hence, the growth of start-ups in the predictive analytics market is high. Start-ups offering predictive analytics solutions are receiving significant investments and are coming up with innovative predictive analytics solutions.	High

IT Spending	The top three industry verticals in the predictive analytics market are expected to contribute between 40% and 60% to the overall market share. Various stakeholders across the industry verticals in different regions are expected to adopt predictive analytics solutions for gaining a competitive advantage in the global market.	Medium
Technology Maturity	Analytics and IoT technologies are nearing the maturity phase.	Medium
Merger and Acquisition (M&A)	In the last year, there were more than five acquisition-related developments in the predictive analytics market. Therefore, the M&A factor is considered to have a medium impact on the predictive analytics market.	Medium
COVID-19 Impact	The outbreak of COVID-19 has impacted verticals, such as manufacturing, and transportation and logistics; hence, the pandemic's impact on the predictive analytics market is moderate.	Medium

Source: MarketsandMarkets Analysis

TABLE 4 **IMPACT OF COVID-19**

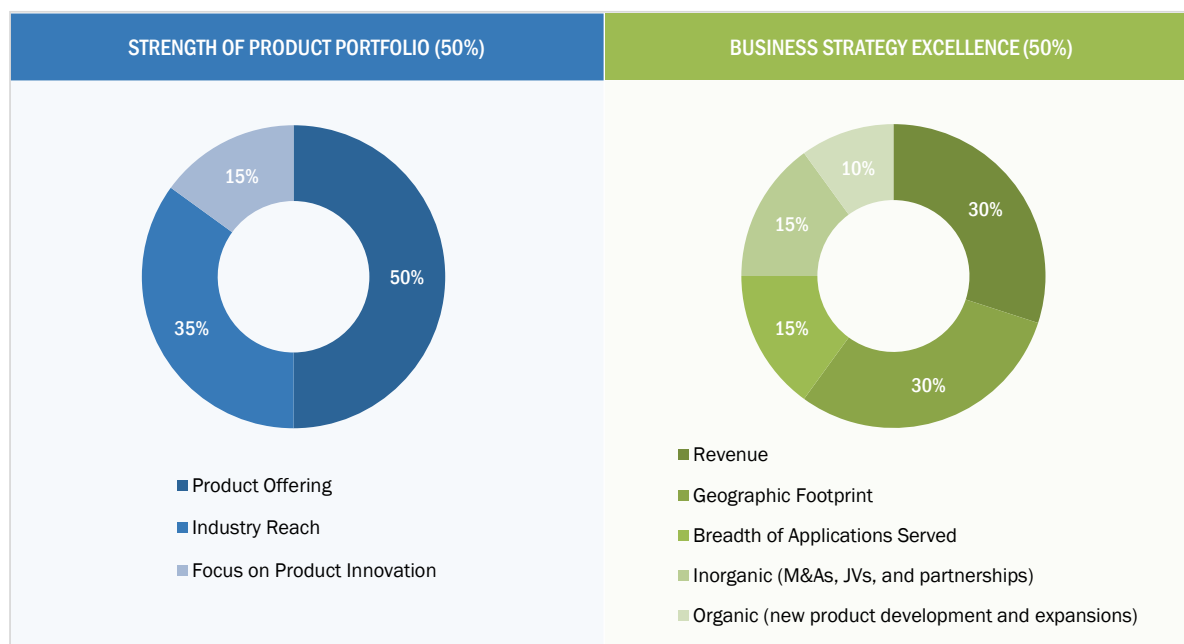
SUPPLY-SIDE FACTORS	IMPACT	DEMAND-SIDE FACTORS	IMPACT
Supply chain operations disruption	Low	Lack of workforce across organizations	Medium
Impact of stringent lockdown on on-premises implementation	High	Monitoring omnichannel advertisements	Medium
Impact of stringent lockdown on cloud implementation	Low	Use of virtual environment	High
Strict government policies	Medium	Penetration of internet	Medium
Impact on ongoing new product launches	Medium	Focus on core operations	High

Source: MarketsandMarkets Analysis

2.5 COMPANY EVALUATION MATRIX

The competitive evaluation matrix reviews the major players offering predictive analytics solutions and outlines the findings and analysis based on their core competencies. Vendor evaluations were based on two broad categories: strength of product portfolio and business strategy excellence. Each category carries various criteria based on which the vendors were evaluated. The figure below depicts the criteria and weightage used for scoring the predictive analytics vendors:

FIGURE 13 COMPANY EVALUATION MATRIX: CRITERIA WEIGHTAGE



Source: MarketsandMarkets Analysis

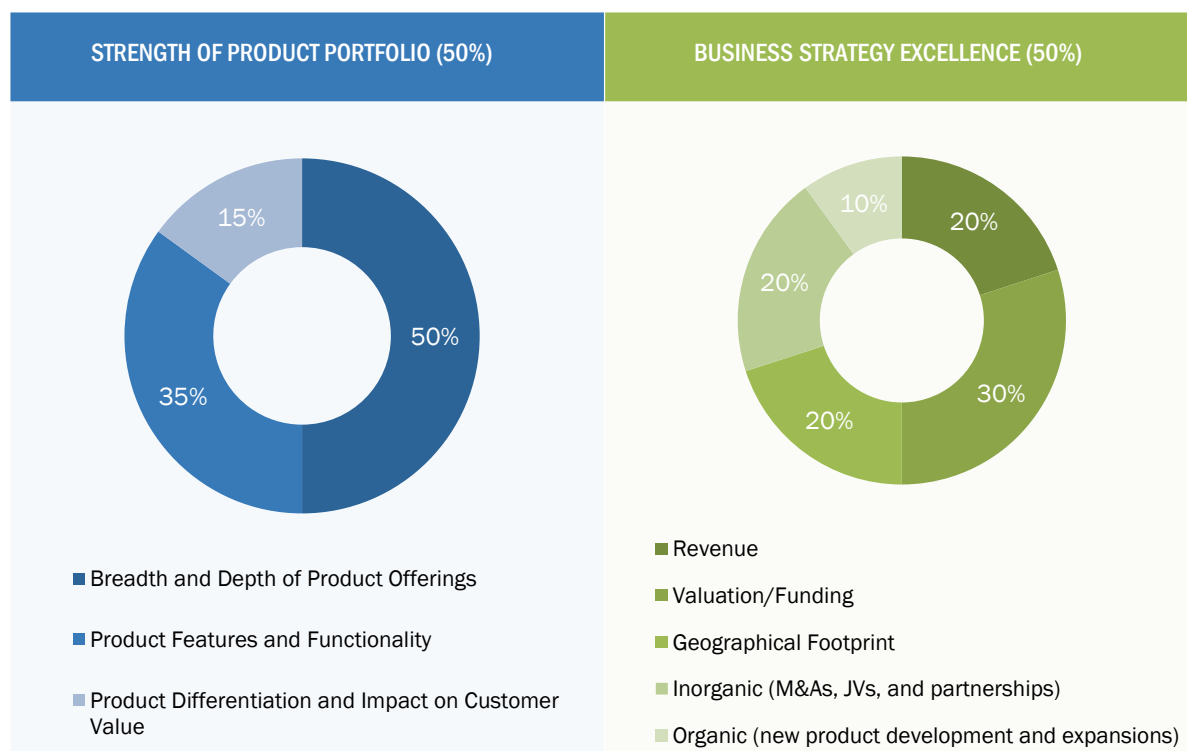
Extensive secondary and primary research was carried out to gather key information about the vendors' strength of product portfolio and business strategy excellence, based on a questionnaire. A scale of 0–10 was formulated for each question, after which each criterion for every vendor was scored on the basis of the collected information. After the completion of the data gathering and verification process, the scores and weightage for shortlisted vendors against each parameter were finalized. A comparison scorecard was prepared after evaluating all the vendors, and each vendor was placed in the competitive evaluation matrix based on its strength of product portfolio and business strategy excellence scores.

2.6 START-UP/SME EVALUATION MATRIX METHODOLOGY

The start-up/small and medium-sized enterprises (SMEs) evaluation matrix reviews the start-ups/SMEs offering predictive analytics solutions and services. Ten start-ups/SMEs founded after 2015 were considered for evaluation in the startup/SME evaluation matrix.

The figure below depicts the criteria and weightage used for scoring start-ups/SMEs in the predictive analytics market.

FIGURE 14 START-UP/SME EVALUATION MATRIX: CRITERIA WEIGHTAGE



Source: MarketsandMarkets Analysis

2.7 ASSUMPTIONS FOR THE STUDY

The following assumptions were taken into consideration to complete the overall market engineering process of the predictive analytics market:

FACTOR	ASSUMPTION	IMPACT
Economy	According to an update released by the International Monetary Fund (IMF) in January 2021, the global economy grew 5.5% this year, a 0.3% increase from October's forecasts.	- A positive global economy is expected to continue until the end of 2026. - In developed countries, a modest and uneven recovery is expected to continue with the gradual narrowing of output gaps.
Exchange Rate	MarketsandMarkets assumes that dollar fluctuations would not be severe enough to affect the forecast period to a significant extent.	A fall in the dollar value is expected to create revenue growth opportunities for US multinationals, whereas stability in the dollar value is expected to stabilize the price of imports and exports.

Financial Statistics	MarketsandMarkets extracts all the revenue and financial insights from company websites or annual reports.	It ensures the authenticity of the financials mentioned in the report.
Solution	Solution spending has increased over the last two years, and organizations are investing in adopting solutions to achieve digital transformation.	Organizations are expected to increase their spending on IT and business services.
Service	- The growth of the services segment has been steady. Government austerity programs have included direct reductions in the level of IT services spending. Cannibalization from the cloud is still expected to be a drag on the overall growth. - On the upside, it is expected that some downstream spending from solutions might happen, but lesser than previous cycles.	Solution spending would increase the adoption of services.
COVID-19	- MarketsandMarkets assumes a short-term effect on the technology sector, affecting the electronics value chain and initiating an inflationary risk on products. - MarketsandMarkets assumes delays in scheduled new projects and reduced discretionary spending by enterprises.	The impact of COVID-19 is believed to be short-term and may have a minimal effect on the forecast period.
Market Estimation	- The pricing trend is assumed to vary over time. - All forecasts are made with the standard assumption that the accepted currency is USD. - For the conversion of various currencies to USD, average historical exchange rates are used according to the year specified. For all the historical and current exchange rates required for calculations and currency conversions, the US Internal Revenue Service's website is used. - All the forecasts are made under the standard assumption that the globally accepted currency, USD, remains constant during the next five years. - Vendor-side analysis: The market size estimates of associated solutions and services are factored in from the vendor side by assuming an average of licensing and subscription-based models of leading and innovative vendors in the market. - Demand/end-user analysis: End users operating in verticals across regions are analyzed in terms of market spending on predictive analytics solutions based on some of the key use cases. These factors for the predictive analytics industry per region are separately analyzed, and the average spending was extrapolated with an approximation based on assumed weightage. This factor is derived by averaging various market influencers, including recent developments, regulations, mergers and acquisitions, enterprise/SME adoption, start-up ecosystem, IT spending, technology propensity and maturity, use cases, and the estimated number of organizations per region.	NA

Source: MarketsandMarkets Analysis

2.8 LIMITATIONS OF THE STUDY

The following limitations were taken into consideration to complete the overall market engineering process of the predictive analytics market:

METHODOLOGY-RELATED LIMITATIONS



- The report focuses on the offerings of major vendors in the predictive analytics market. The market size estimation is done based on the approximation of the market ranking/contribution of these vendors, along with corresponding investments by various regulatory bodies in the market.
- The overall market size is arrived at using a holistic and systematic approach.

SCOPE-RELATED LIMITATIONS



- The report is focused on the B2B business model; the B2C model is not considered in the report.
- Predictive analytics solutions offered by MSPs are not considered for forecasting the market size.

Source: MarketsandMarkets Analysis

3 EXECUTIVE SUMMARY

Traditional analytics platforms use static and stored data to analyze simple or complex patterns and react to any business situation. These platforms take days to analyze and weeks to act on the stored data. However, predictive analytics has the advanced analytics capability of extracting meaningful information from the present and historical data sets. It assists companies in determining data patterns, predicting future outcomes and trends, anticipating events, spotting anomalies, forecasting using what-if simulations, and understanding customer behavior changes. Predictive models and analysis are used to estimate future probabilities with acceptable levels of consistency. Predictive analytics technology is the core enabler of big data, allowing businesses to use historical data and combine it with customer insights to predict future events. Big data is an ever-increasing technology that assists businesses in optimizing processes and minimizing operational costs. The combination of real-time data streams, AI, machine learning (ML), and predictive analytics can deliver competitive business advantages. Traditional analytics and BI systems use the deductive approach for analyzing data. This approach works well with structured data. Predictive analytics, on the other hand, applies the inductive reasoning analysis approach, which deals with large datasets derived from ML, robotics, sensors, and AI. It uses algorithms that carry out complex calculations on large data sets and discover interrelationships and patterns between them.

The predictive analytics technology is finally witnessing large-scale adoption. The history of predictive analytics dates back to the early 1900s. Around 1940, organizations first started using basic computational models for neural networks and linear programming. In the 1960s, corporations and research institutions started using non-linear programming tools, which has led to the commercialization of analytics. Predictive modeling was then applied to credit risk decisions, marketing techniques, and travel and logistics. By the 1990s, predictive analytics models were implemented by corporate organizations and tech start-ups. Today, many forward-looking companies have started using predictive analytics across a range of disparate data sets to achieve greater efficiency. The technology is operationalized by companies for their business processes. Globally, the predictive analytics market is witnessing significant growth because of the transformation from traditional BI techniques to advanced analytics techniques. Many studies related to predictive analytics have revealed that organizations that strategically use predictive analytics outperform other organizations that use traditional data analytics approaches. From predicting market trends to gaining insights into customer needs, predictive analytics can help businesses tap the power of their data to exploit new opportunities and gain an advantage over their competitors.

Predictive analytics uses mathematical, regression modeling, statistical, neural nets, ML, genetic algorithms, text mining, decision trees, clustering, and data exploration techniques to extract insights from historical and current data. ML, AI, Natural Language Processing (NLP), and advanced condition prediction are the most crucial features of an effective predictive analytics solution. NLP is considered a cognitive capability for IoT that helps users interact with systems and devices using simple, human language. It evaluates structured and unstructured data, such as emails, chats, and social media interactions. Predictive analytics estimates which type of involvement will be required, where, and when.

The key benefit of predictive analytics is that it enables organizations to identify untapped opportunities and exposes the hidden risks associated with large data volumes. Predictive insights enable organizations to make the right choices at the right time for shaping the course of their business. The major growth drivers for the predictive analytics market include the increasing volume of data generated across verticals, growing focus on market and competitive intelligence, and rising need for advanced analytics for future predictions. The predictive analytics market also faces many challenges, including changing regulations leading to the restructuring of the analytics model, ownership and privacy of collected data, and the lack of appropriate analytical skills. Factors such as complex analytical workflows and the diversity of data models are expected to limit market growth. The market has many new opportunities, such as the demand for real-time streaming analytics and the high adoption rate in small and medium-sized

enterprises (SMEs). These factors are expected to shape the future of the predictive analytics market. Predictive analytics is very effective at the point of decision, be it for Customer Relationship Management (CRM), Enterprise Resource Planning (ERP), or any industry-specific software. Companies can use this technology for solving business concerns and identifying new opportunities, some of which are mentioned below:

- Identifying new business/market opportunities
- Increasing overall business efficiency and lowering associated operational costs
- Enhancing customer experiences and creating deeper engagements with customers
- Reducing risks and mitigating problems by anticipating future trends and events

The spread of COVID-19 has been disrupting the world, businesses, and economies and has impacted the way of living of the masses and approaches adopted by enterprises for their business management. The ability of enterprises to sustain this pandemic has become new normal for them as they shift their focus from growth opportunities and concentrate on implementing measures to mitigate the impact of COVID-19. The upcoming analytics projects are kept on hold owing to the pandemic. Several companies are competing with each other to gain a single project. Businesses have already started their efforts to return to normal and are facing multiple challenges in terms of the impact of the pandemic on their customer base and operations. Meeting customer expectations in terms of optimization of processes and increased security concerns due to the presence of various connected networks, rise in connectivity issues, and decline in industrial and manufacturing operations are the key challenges faced by businesses. New practices such as work from home and social distancing are creating the requirement of the remote health monitoring of infected patients, smart payment technologies, and the development of digital infrastructures for large-scale technology deployments. With an increased focus on health, there has been a rise in the demand for health-related wearable devices. For instance, in August 2020, Fitbit announced ~34% growth in its smartwatch sales in the second quarter of 2020.

Based on components, the predictive analytics market is segmented into solutions and services. The solutions considered for the predictive analytics market include financial analytics, risk analytics, marketing analytics, sales analytics, customer analytics, web and social media analytics, supply chain analytics, network analytics, and others (HR analytics and legal analytics). The services segment includes managed services and professional services. Professional services are further classified into consulting, and deployment and integration. The risk analytics segment is expected to account for the largest market share and contributed a share of 17.1% in the predictive analytics market. The customer analytics segment is expected to witness the highest growth rate during the forecast period, growing at a Compound Annual Growth Rate (CAGR) of 28.2% during the forecast period.

Among deployment modes, the cloud segment is expected to grow at a higher CAGR due to improved scalability and operational efficiency. By organization size, the SMEs segment is expected to witness a higher CAGR due to the increased adoption of cloud-based predictive analytics solutions and services.

On the basis of verticals, the market is segmented into BFSI, manufacturing, retail and eCommerce, government and defense, healthcare and life sciences, energy and utilities, telecommunications and IT, transportation and logistics, media and entertainment, travel and hospitality, and others (research and education and real estate). The BFSI vertical is expected to dominate the market because of the prominent use of financial analytics, risk analytics, and customer analytics solutions across BFSI operations.

On the basis of regions, the market is divided into North America, Europe, APAC, Latin America, and MEA. North America is expected to dominate the market due to early technology adoption and the presence of a large number of solution providers. The growing technology investments and improving penetration of the retail and manufacturing vertical are expected to drive the market growth of APAC during the forecast period.

Major vendors offering predictive analytics solutions and services across the globe are SAS Institute, Inc. (SAS Institute), IBM Corporation (IBM), Oracle Corporation (Oracle), SAP SE (SAP), Google LLC (Google), Microsoft Corporation (Microsoft), Salesforce.com, Inc. (Salesforce), Amazon Web Services (AWS), Hewlett Packard Enterprise (HPE), Teradata Corporation (Teradata), Alteryx, Altair Engineering Inc. (Altair), Fair, Isaac and Company (FICO), Domo, Inc. (Domo), Cloudera, Inc. (Cloudera), BOARD International (Board), TIBCO Software Inc. (TIBCO Software), Hitachi Vantara Corporation (Hitachi Vantara), Happiest Minds Technologies (Happiest Minds), Dataiku (US), QlikTech International AB (Qlik), RapidMiner, Inc. (RapidMiner), Biofourmis Inc. (Biofourmis), Prognostics In-Med Pvt. Ltd (In-Med Prognostics), Aito Intelligence Oy (Aito.ai), Symend, Onward Health, Unioncrate, CyberLabs AI (CyberLabs), Actify Data Labs Inc (Actify Data Labs), Amlgo Labs Pvt Ltd (Amlgo Labs), and Verimos, Inc. (Verimos). These vendors have adopted different organic and inorganic growth strategies, such as new product launches, partnerships and collaborations, and mergers and acquisitions, to expand their offerings in the predictive analytics market.

The predictive analytics market is projected to grow from USD 7,216 million in 2020 to USD 28,144 million by 2026, at a CAGR of 21.7% during the forecast period.

TABLE 5 PREDICTIVE ANALYTICS MARKET SIZE AND GROWTH RATE, 2015–2020 (USD MILLION, Y-O-Y%)

Predictive Analytics	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Market Size	3,285	3,851	4,566	5,460	6,637	7,216	17.0%
Y-o-Y		17%	19%	20%	22%	9%	

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

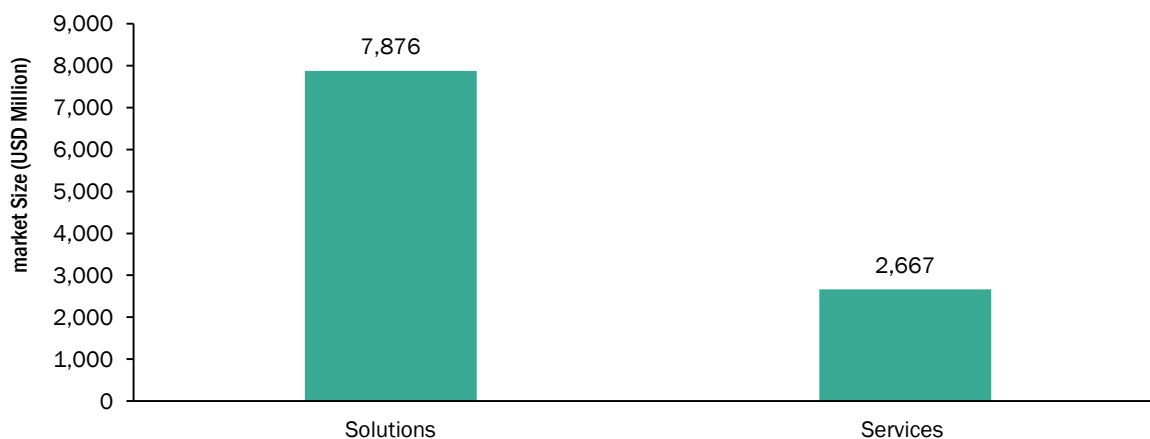
TABLE 6 PREDICTIVE ANALYTICS MARKET SIZE AND GROWTH RATE, 2021–2026 (USD MILLION, Y-O-Y%)

Predictive Analytics	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Market Size	10,543	13,781	16,954	20,675	24,554	28,144	21.7%
Y-o-Y	46%	31%	23%	22%	19%	15%	

e: estimated; p: projected

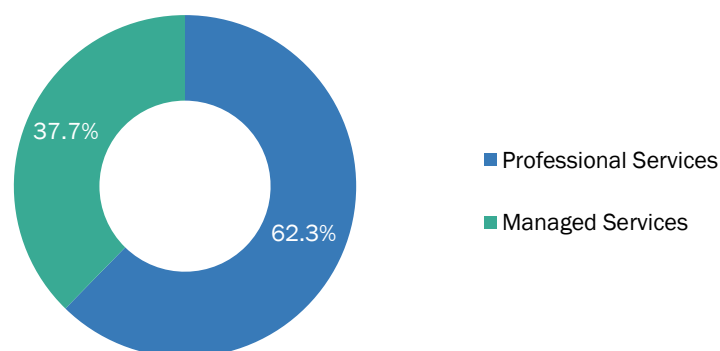
Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 15 SOLUTIONS SEGMENT, BY COMPONENT, EXPECTED TO ACHIEVE A LARGER MARKET SIZE IN 2021

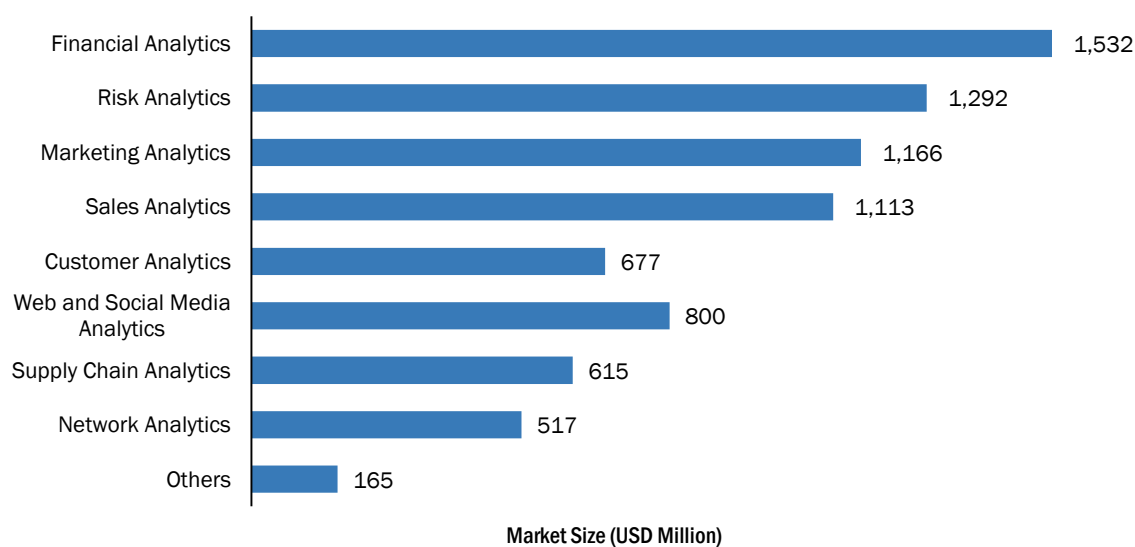


Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 16 PROFESSIONAL SERVICES SEGMENT TO ACCOUNT FOR A LARGER MARKET SHARE IN 2021

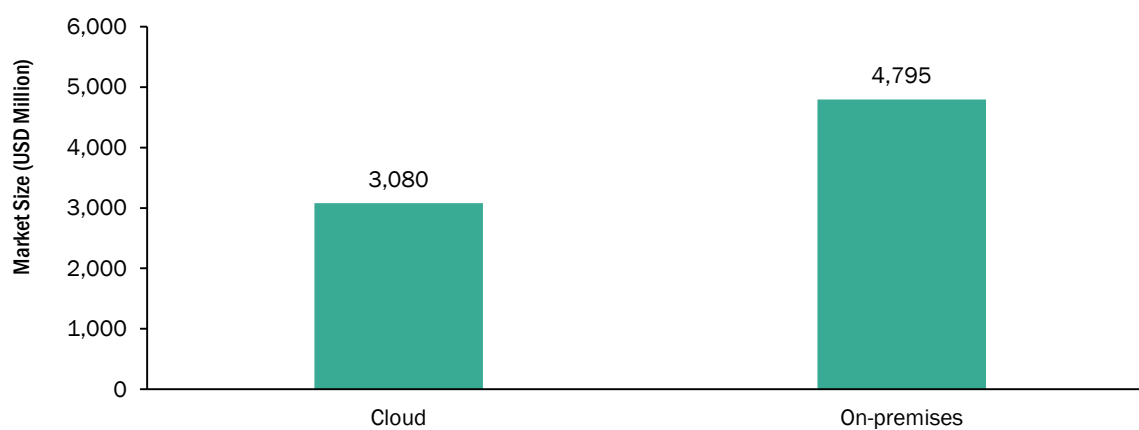


Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 17 FINANCIAL ANALYTICS SOLUTIONS TO DOMINATE THE MARKET IN 2021

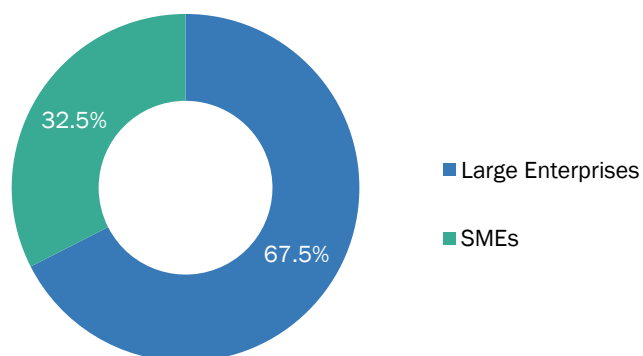
*Other solutions include HR analytics and legal analytics.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 18 ON-PREMISE DEPLOYMENT TO LEAD THE MARKET IN 2021

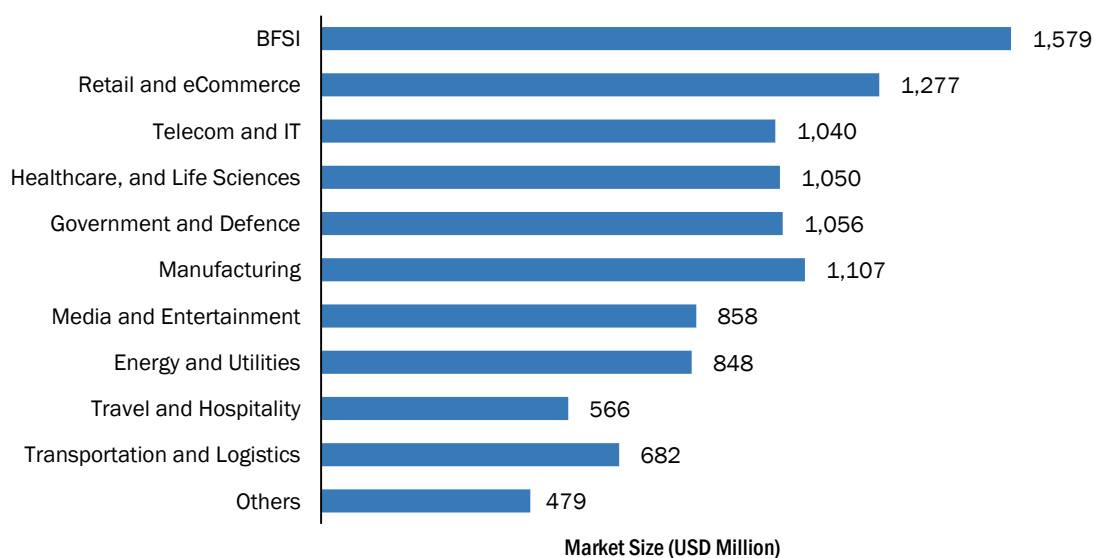
Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 19 LARGE ENTERPRISES SEGMENT EXPECTED TO ACCOUNT FOR A LARGER MARKET SHARE IN 2021



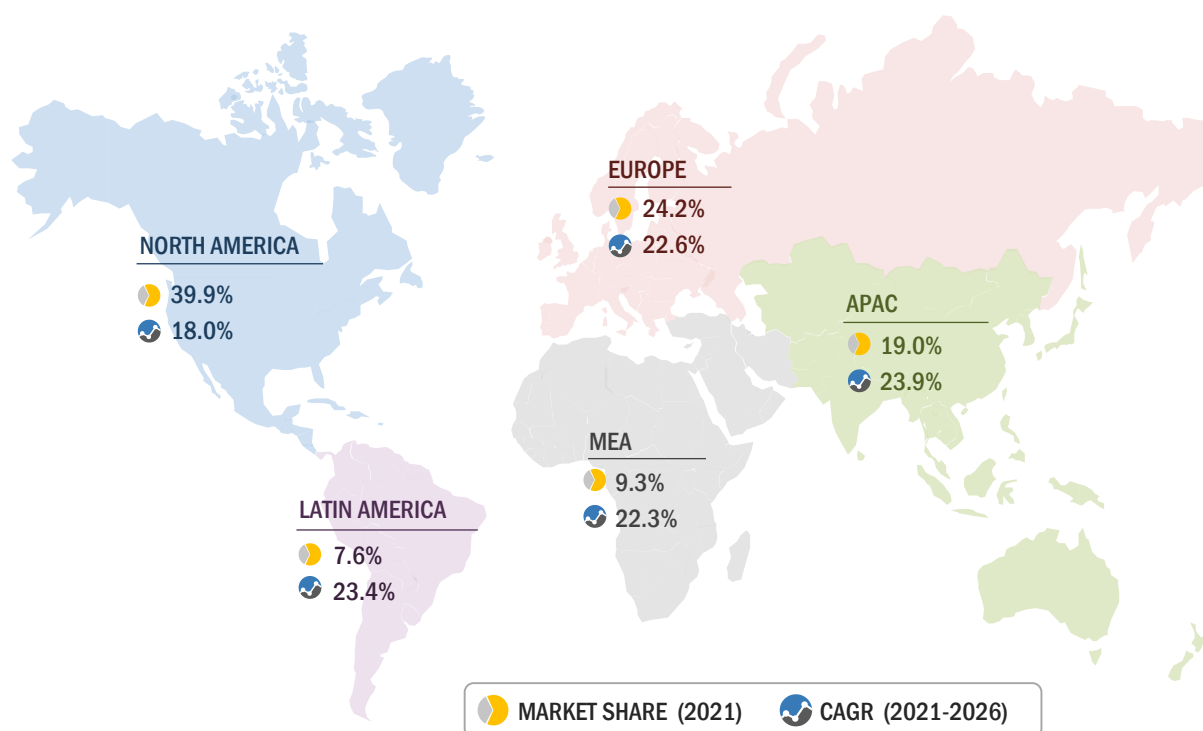
Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 20 BFSI VERTICAL TO ACCOUNT FOR THE LARGEST MARKET SHARE IN 2021



*Other verticals include research & education and real estate.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

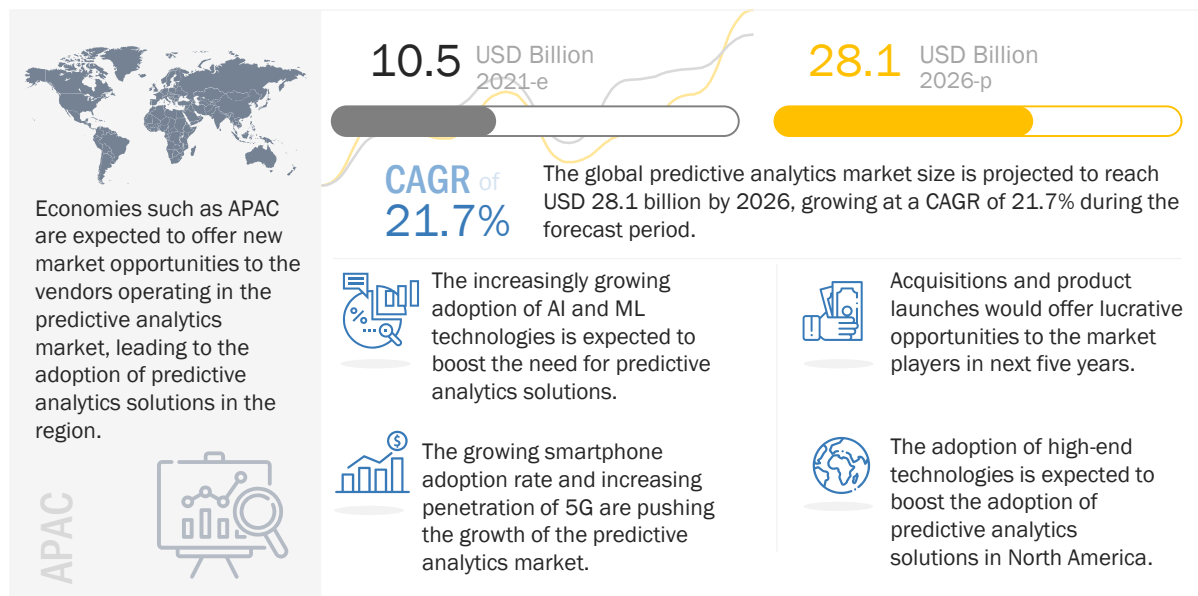
FIGURE 21 NORTH AMERICA TO ACCOUNT FOR THE LARGEST MARKET SHARE IN 2021


Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

4 PREMIUM INSIGHTS

4.1 ATTRACTIVE OPPORTUNITIES IN THE PREDICTIVE ANALYTICS MARKET

FIGURE 22 INCREASING ADOPTION OF AI AND ML TECHNOLOGIES TO BOOST THE MARKET GROWTH

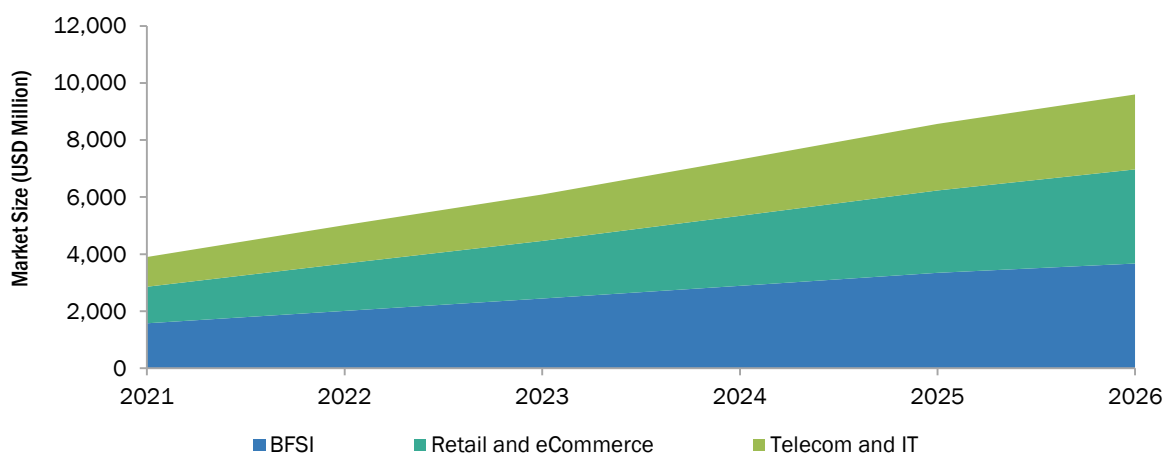


e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

4.2 PREDICTIVE ANALYTICS MARKET, BY VERTICAL

FIGURE 23 BFSI SEGMENT EXPECTED TO LEAD THE MARKET BY 2026



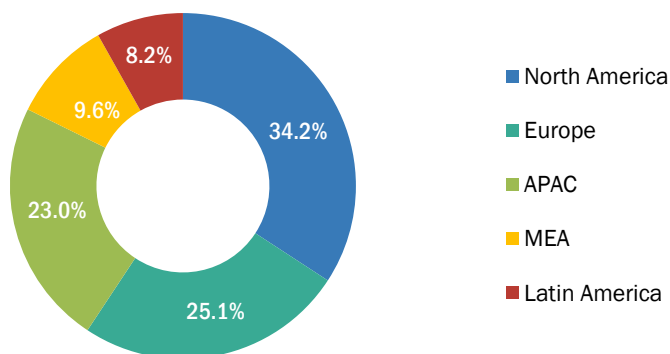
e: estimated; p: projected

Note: The market size of the top three verticals has been provided in the figure above.

Source: Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

4.3 PREDICTIVE ANALYTICS MARKET, BY REGION

FIGURE 24 NORTH AMERICA ACCOUNTED FOR THE LARGEST MARKET SHARE IN 2020

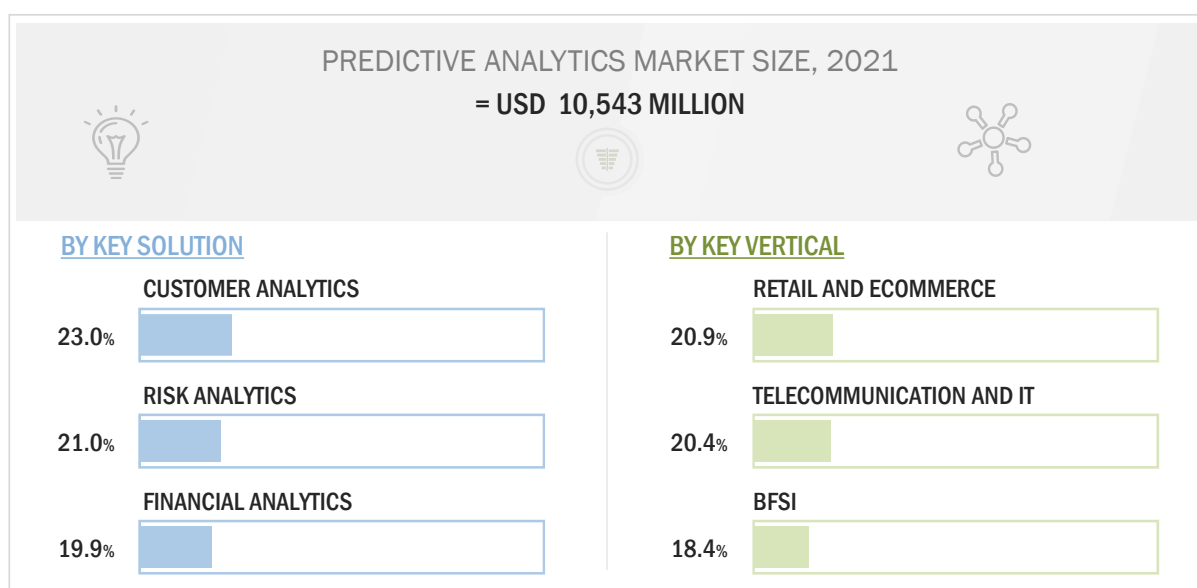


Note: The numbers in the figure above represent the market shares for 2021.

Source: Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

4.4 PREDICTIVE ANALYTICS MARKET, BY SOLUTIONS AND VERTICAL

FIGURE 25 CUSTOMER ANALYTICS AND TELECOM AND IT TO ACCOUNT FOR THE LARGEST RESPECTIVE MARKET SHARES IN 2021



Note:

The numbers in the figure above represent the market shares for 2021.

The market size for the top three solutions and verticals have been provided in the figure.

Source: Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

5 MARKET OVERVIEW AND INDUSTRY TRENDS

5.1 INTRODUCTION

The market overview chapter offers detailed information about the factors responsible for the growth of the predictive analytics market. It comprises market dynamics, such as drivers, restraints, opportunities, and challenges. It also includes the market evolution, use cases, case studies, regulatory compliance, value chain analysis, and future outlook related to the predictive analytics market. Globally, companies are rapidly adopting predictive analytics solutions and services to gain a competitive edge in the market. The key players in the predictive analytics market include IBM, SAS Institute, Microsoft, Google, SAP, AWS, and Oracle. These companies are continuously innovating their products and services to overcome large and complex data challenges by using statistical algorithms and ML techniques to forecast future outcomes based on historical data.

5.2 MARKET DYNAMICS

Predictive analytics solutions are widely adopted in North America and Europe, whereas the adoption rate is substantially increasing in APAC. The key factors driving the growth of this market include the need for digitalizing enterprise business functions to cater to the evolving customer preferences, the growing necessity to enhance operational efficiency in times of remote working, the rapid proliferation of mobile devices and 5G technology, and increase in the penetration of IoT-based devices. On the other hand, security and privacy concerns over confidential data hinder market growth. The demand for personalized mobile apps and websites and the adoption of disruptive technologies such as AI, ML, and big data are some of the major opportunities in the market. However, the market vendors face various challenges, including the lack of skilled personnel and issues related to IT modernization, which do not allow the market to grow rapidly to take advantage of the increasing demand for cloud services.

FIGURE 26 DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES:
PREDICTIVE ANALYTICS MARKET

 DRIVERS	▪ Rising adoption of big data and other related technologies ▪ Advent of machine learning and artificial intelligence ▪ Cost benefits of cloud-based predictive analytics solutions
 RESTRAINTS	▪ Changing regional data regulations leading to a time-consuming restructuring of predictive models ▪ Data security concerns
 OPPORTUNITIES	▪ Rising internet proliferation and growing usage of connected and integrated technologies ▪ Increasing demand for real-time analytical solutions to track and monitor COVID-19 spread ▪ Growing role of predictive analytics in achieving manufacturing excellence
 CHALLENGES	▪ Integration of data from data silos ▪ Ownership and privacy of collected data

Source: Secondary Research and MarketsandMarkets Analysis

5.2.1 DRIVERS

5.2.1.1 Rising adoption of big data and other related technologies

	Based on IBM's data, banks spend millions of dollars on the traditional on-boarding process. By using AI tools that are 80% automated, 90% of accuracy is achieved, resulting in the reductions (about 30% to 40%) of the original cost of the onboarding process.	PRIMARY INSIGHTS
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The use of advanced technologies, such as cloud, IoT, big data and analytics, mobility, and social media, has led to innovation and transformation, thereby stimulating growth in the business ecosystem. Digital technologies have transformed the legacy approach to business into a modern approach. For example, online services are now firmly established in the banking and financial sector, resulting in the proliferation of online activities and websites. According to a blog published by Usabilla (Netherlands) in June 2020, the top 100 organizations are 50% more likely to address customer empowerment as a key parameter for business growth. The blog also stated that 92% of leaders had employed advanced digital transformation strategies to enhance their customers' experience.

Today, data is recognized as a new resource. It requires efficient and effective utilization to ensure a competitive market edge. Businesses are interested in deriving insights from collected data for making better, real-time, and fact-based decisions. This demand for in-depth knowledge has accelerated the adoption of big data and related technologies. The exponential growth in data volume is due to the expansion of businesses worldwide, which is driving the rise in data volumes and sources. The

accumulation of big data in a single location has rapidly developed the evaluation capabilities of data science experts in every organization. Companies prefer to provide standalone solutions rather than combined solutions. This is eventually increasing the number of big data analytics start-ups, which are driving noteworthy innovations.

Big data has captured attention among businesses and consumers. It has not only become a major subject in technology and media, but has also made its way into many compliances, internal audit, and fraud risk management-related discussions. As an integral part of business operations in every industry, big data has influenced the adoption of predictive analytics worldwide. By leveraging the opportunities and challenges of big data predictive analytics, enterprises can optimize vital business processes, functions, and objectives. Predictive analytics enables organizations to meet stakeholder demands, manage data volumes, manage risks, improve process controls, and boost administrative performance by turning information into intelligence. Advanced predictive analytics solutions play a vital role in enabling an effective Intelligent Enterprise (IE) approach, which helps create a single view across the entire organization through the combination of standard reporting and data visualization.

5.2.1.2 Advent of machine learning and artificial intelligence



According to a recent survey from the Society of Actuaries (SOA) in August 2019, about 60% of healthcare executives were using predictive analytics within their organizations, while 20% of payers and providers planned to start using predictive analytics in the next year.

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Digitalization in the current day and age has exponentially transformed the landscape of predictive analytics, thereby encouraging predictive analytics vendors and service providers to adopt digitized systems for in-store activities. Organizations are increasingly focusing on the re-engineering of businesses with the help of the immense power of the internet and digital technology. The adoption of digital operations assists predictive analytics solution providers in finding newer ways to tackle the evolving challenges. Due to the digital transformation of retail and eCommerce businesses, operating costs have reduced and positively impacted the predictive analytics market growth. The advent of AI and ML has enhanced organizations' ability to gather information, analyze it, and make quick decisions with minimum human intervention. Organizations are also adopting AI systems to speed up investment decisions. For instance, Fukuoka Mutual Life Insurance has adopted AI to calculate payouts. AI systems analyze and suggest Governance, Risk Management, and Compliance (GRC)- related changes. Therefore, they help in monitoring and collecting the appropriate quality of data to meet regulatory standards. The introduction and implementation of ML and AI have helped providers in understanding customer perceptions by analyzing the data gathered from multiple sources. Enterprises have shifted their focus to deliver a unified customer experience across all channels and provide offers in real-time by analyzing customer buying patterns.

5.2.1.3 Cost benefits of cloud-based predictive analytics solutions

On-demand predictive analytics solutions reduce the cost of BI for organizations by reducing the upfront cost and total ownership cost of the analytics system. Cloud-based predictive analytics solutions enable business organizations to effectively utilize their financial resources as it compensates for the overall Capital Expenditure (CAPEX) and Operating Expenditure (OPEX) required for purchasing storage servers and other hardware equipment for the support and maintenance of analytics operations. As a result of these features, BI and business analytics vendors and new start-ups are offering predictive analytics solutions in cloud, thus driving market growth. Some technology initiatives such as Google's MapReduce framework for distributed computing on clusters of computers and the Apache Hadoop framework make it easy for analytics vendors to develop large-scale analytics engines that optimally utilize cloud. These analytics engines enable users to derive valuable business insights from structured as well as unstructured data.

5.2.2 RESTRAINTS

5.2.2.1 Changing regional data regulations leading to a time-consuming restructuring of predictive models

Predictive analytics delivers a decisive data interpretation and provides a larger picture to decision-makers to boost overall business performance. The modification of predictive models permits integration with both software and services, depending on the level and nature of the analysis. Professional services are necessary for customizing existing analytics solutions, which cater to particular data sets. As the predictive analytics concept is still in the nascent phase, the availability of skilled labor is also limited. This is another important factor restraining market growth.

Moreover, the changing regulations are considered one of the major factors slowing down the adoption of predictive analytics among organizations. For instance, data regulations may vary from region to region and country to country. General Data Protection Regulation (GDPR) enables the European Union (EU) to monitor and ensure the safety of EU citizens.

Compared to conventional solutions, the initial costs and time incurred for deploying predictive analytics solutions are higher, due to the involvement of BI and data analytics expertise. Though the deployment of predictive analytics solutions would reduce operating costs in later stages, the time consumed for managing and updating regulations is tedious. Therefore, the adoption of predictive analytics solutions is expected to witness a slower pace in small enterprises.

5.2.2.2 Data security concerns

The backbone of predictive analytics solutions, data, remains a critically important aspect that most organizations find difficult to manage. The inefficiency of managing exabytes and petabytes of data has increased the chances of security breaches and data losses. In today's competitive marketplace, marketing teams require real-time and secure data to deliver an outstanding customer experience. Organizations are gathering data through multiple touchpoints and measure them virtually. Such data is used in support and communication and may include a variety of data types. These data types comprise public information, big data, and small data collected from customers.

This data can include permissions, individual preferences, and updated contact information on products, services, and communication platforms. Hence, vendors need to ensure high-level data security to maintain customer trust. Recently, cybercriminals have gained access to widespread tools to obtain anything from passwords to secret questions and token-generated passwords. In such situations, marketing and IT teams need to work concurrently, providing each other with insights on when and how the data is gathered, processed, and used in operations.

Such potent cyber-attacks continue to hinder the widespread adoption of predictive analytics across data-intensive industrial sectors. However, this scenario is expected to gradually change throughout the forecast period as the benefits of technology adoption continue to outweigh the risks associated and with overall developments on the data security front.

5.2.3 OPPORTUNITIES

5.2.3.1 Rising internet proliferation and growing usage of connected and integrated technologies

The proliferation of the internet and the availability of various means to access the internet have led to a massive increase in data volumes generated. This would help in the advancement and expansion of high-speed internet services. Globalization and economic growth are also playing major roles in driving greater data generation worldwide. With the rise in touchpoints and the need for collecting data to understand consumer behavior, every touch by a consumer has become an important data point that can be processed to reveal user behavior. With the exponential rise in individual and organizational data, businesses are now deploying data scientists and analysts for processing collected data. Another factor accelerating adoption is the revenue-generating potential of predictive analytics. This is compelling firms to invest in predictive analytics.

The rise in connected and integrated technologies has provided a platform to predictive analytics solution vendors for leveraging the development and unprecedented growth of the internet. The eCommerce sector has modified the traditional shopping behavior of customers. Dedicated email campaigns, online/social media advertising, and cognitive customer analysis are key enablers driving sales and increasing customer loyalty. With connected devices coming to the forefront, retailers are focusing on the real-time analysis of customers' shopping behavior and market basket analysis for analyzing consumers' perception, which can be used for building tailor-made offers to increase customer retention. With the rise in the global IoT analytics demand in the retail vertical, the market is expected to offer unprecedented growth opportunities to predictive analytics vendors.

5.2.3.2 Increasing demand for real-time analytical solutions to track and monitor COVID-19 spread



According to State of BI & Analytics Report 2020, 55% of businesses were using data to improve efficiency, 47% to support customers, and 45% to predict future outcomes during COVID-19.

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During the uncertain situation of the COVID-19 pandemic, the healthcare vertical has sought to use big data and predictive analytics tools to better understand the virus and its spread. Predictive analytics has helped researchers worldwide build predictive analytics models that can track COVID-19 surges in different countries. Researchers from Binghamton University and the State University of New York have developed several predictive analytics models to examine COVID-19 trends and patterns. Hospitals and health systems have leveraged predictive models to gain insights into COVID-19 risks, disease outcomes, and the virus' potential impact on resources. Communities across countries have started relaxing social distancing orders; hence, it has become crucial for health systems to plan for changes in healthcare demands and surges in COVID-19 cases. The team at CommonSpirit Health has built predictive models using de-identified cell phone data, public health information, and data from the CommonSpirit Health system's care sites. Using these models, organizations can gain insights into the surges and dips of COVID-19 infection rates, allowing hospitals to plan better and work toward resuming their full spectrum of necessary services. Recently, a team from Florida Atlantic University (FAU) has utilized ML technology to build a COVID-19 knowledge base and risk assessment dashboard, which aims at addressing the discrepancies that exist in the context of the pandemic. Hence, using these predictive models, researchers can help organizations stay ahead of possible increases in COVID-19 patients by continuing to collect data from around the world to make predictive analytics models more accurate.

5.2.3.3 Growing role of predictive analytics in achieving manufacturing excellence

With predictive analytics, the quality management teams within a manufacturing setup can process large sets of information (historical) to reveal the underlying cause, thereby allowing early detection of issues and minimizing the probability of their occurrences in the future. For instance: product recall has traditionally remained a commonplace menace for consumer-oriented industries, such as automotive. As a result, forecasting tools and predictive analytics are being deployed to mitigate the risks of product recalls. The data can be used to find patterns and resolve quality issues in the nick of time to prevent them from happening altogether. This leads to both customer satisfaction and quality management at a cost-effective level.

5.2.4 CHALLENGES

5.2.4.1 Integration of data from data silos

Extracting value from data has become a key requirement for companies to successfully mitigate risks, target valuable customers, and evaluate business performance. In addition, monetizing these data assets requires the availability of a sufficient amount of data. However, data consolidation from distinct data sources into meaningful information can incite various new challenges for organizations, especially centralized business enterprises. Data exchange and data ecosystem deliver tools to analyze the collected data at a centralized location and help extract and cross-check business-critical components. The development of data exchanges and data ecosystems varies based on the assumptions made in the value of the data for each customer segment. Many predictive analytics providers offer unified data aggregation and data analytics platforms to help users successfully aggregate and analyze data from disparate data sources.

Furthermore, data exchange is a major concern for industries, such as BFSI and healthcare, which deal with people's confidential data. As the need for data exchange increases, it needs to be balanced with risk mitigation capabilities. With companies adopting predictive analytics software integrated with data security capabilities, data exchange is not expected to be a major issue in the near future.

5.2.4.2 Ownership and privacy of collected data

The market competition is increasing every day due to the introduction and expansion of new technologies, such as IoT, big data, and ML. These technologies have significantly contributed to the number of endpoints collecting data, thereby resulting in an increase in dynamic data that keeps changing/updating every second. For every organization that predicts customer behaviors by analyzing the collected sets of data, maintaining data privacy is a huge security mandate. Concerns related to the data ownership and privacy of collected data pose potential risks to the identity of individuals. Therefore, organizations are bound to follow certain limitations in data collection types, quantities, and methods. These limitations are likely to vary from nation to nation across the globe. Therefore, enterprises must stay abreast with the changing rules and data-related regulations.

Today, in the technology era, the advent of the internet has brought cybercrimes into the picture. Hackers target data storage for financial gains and other reasons. The breach of data impacts organizations' identity and security. Some governments hold enterprises responsible for damages triggered by the loss of confidential data. Therefore, enterprises ought to implement the latest security measures to ensure the privacy of collected data. While outsourcing data analysis to a third-party BI vendor, organizations must ensure that the BI vendor appropriately secures firms' data.

5.3 PREDICTIVE ANALYTICS MARKET: COVID-19 IMPACT

The spread of COVID-19 has generated a huge disruption in daily activities. It has forced people to follow social distancing policies. It also has temporarily halted business activities and restricted travel. Under such circumstances, the healthcare vertical has emerged as the biggest user of big data and predictive analytics to learn about the virus and its spreading patterns. The health infrastructure system has faced various challenges, including hospitals at peak capacity and the shortage of ventilators, masks, and other key sanitary materials. Hospitals and health systems have leveraged predictive models to gain useful insights on COVID-19 risks, disease outcomes, and the virus' potential impact on resources. The pandemic situation has also disrupted demand and supply cycles.

From a manufacturing standpoint, to remain in line with demand, manufacturers need real-time insights on demand and inventory status to make informed decisions. Predictive analytics has enabled manufacturers to determine the quantity of products to be produced and has also helped determine what should be the priority of the production line according to demand trends to ensure that production sustains. Technologies have helped drive businesses to the next level by providing the fuel for analytical tools to forecast, predict, and plan for issues before they occur.

Predictive analytics has enabled businesses to meet the increasing demand. The insights from predictive analytics have helped decision-makers anticipate specific future events. As per a study by the Council of Supply Chain Management Professionals, 93% of shippers and 98% of third-party logistics firms believe that data-driven decision-making is essential to supply chain activities. Moreover, 71% of the participants believe that big data improves quality and performance. Predictive modeling has provided organizations with transportation fleets to create future business insights with a significant degree of accuracy. With the help of sophisticated data analytics tools and modeling, these firms have now started using past and current project utilization data to reliably forecast budget trends milliseconds, days, or years into the future. Companies with transportation fleets that leverage the predictive modeling technology would increase their bottom-line efficiencies even further and amplify their competitive advantage post the pandemic.

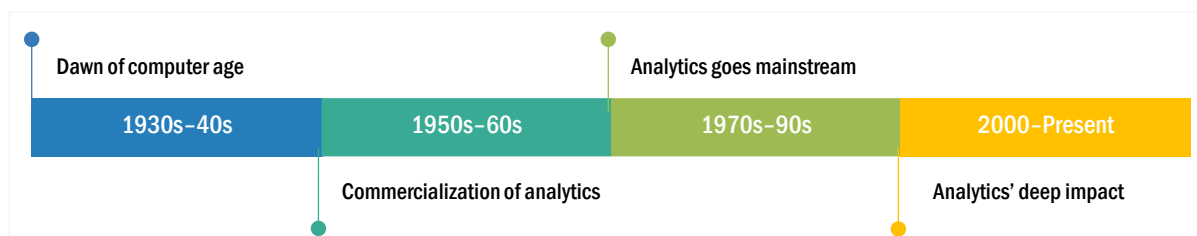
5.4 PREDICTIVE ANALYTICS: EVOLUTION

The analytics market evolution started with the digital support system and data warehouse, followed by mainstream analytics, the introduction of perspective and descriptive analytics, and finally, the introduction of predictive analytics. Organizations have adopted predictive analytics solutions to save time and costs associated with the traditional BI platforms and to generate valuable insights from past data for predicting the future. This evolution is marked by the development of analytical tools and the advantages associated with them for all potential users.

- **1930s–40s (dawn of computer age):** A British scientist, Alan Turing, developed a machine to decode German messages during World War II and laid the groundwork for modern computing. He also introduced the concept of AI. Later in 1944, the Manhattan Project team ran computer simulations to predict the behavior of nuclear chain reactions.
- **1950s–60s (commercialization of analytics):** In 1950, the ENIAC computer generated the first model to forecast the weather. In 1958, FICO applied predictive modeling to credit risk decisions. Later in 1966, Future SAS Institute started as a research project funded by the US Department of Agriculture.
- **1970s–1990s (analytics goes mainstream):** In 1973, the Black-Scholes model was created to predict the optimal price for stock options over time. In 1992, FICO deployed real-time analytics to fight against credit card frauds and, in 1998, Google applied algorithms to maximize result relevance in web searches.

- **2000–present (analytics' deep impact):** The widespread use of analytics began in early 2000 for dynamic ticket pricing, movie recommendations, shopping, and other purposes. The advent of big data and NLP took place to unlock the analytics value of unstructured data. Cloud-based analytic exchanges started, leading to collaboration between developers, businesses, researchers, scientists, and entrepreneurs.

FIGURE 27 EVOLUTION OF PREDICTIVE ANALYTICS



Source: Fico.com

5.5 PREDICTIVE ANALYTICS: ECOSYSTEM

Predictive analytics permits organizations to become more proactive and forward-looking. It assists in anticipating outcomes and behaviors based on the analysis of collected data.

FIGURE 28 PREDICTIVE ANALYTICS: ECOSYSTEM



Source: Secondary Literature and MarketsandMarkets Analysis

The process is described below:

- **Identify Problem:** The initial process includes defining the scope of projects and identifying data sets to be used.
- **Data Preparation:** This step comprises data mining by preparing data sets from multiple sources to obtain a complete view of customer interactions.
- **Data Exploration:** The process of inspecting and cleaning collected data.
- **Transform and Select:** This step deals with transforming sorted data and selecting it for further analysis.
- **Build Model:** The refined data is accumulated, and data models are created for discovering useful information.
- **Validate Model:** Once the data model is completed, it is validated on the basis of business rules.
- **Deploy Model:** The final step is to deploy the developed model in everyday decision-making to get desired outcomes.
- **Monitor Result:** Data models are governed and monitored to review the performance of developed models to ensure they are delivering expected results.

5.6 CASE STUDY ANALYSIS

This section comprises use cases across different verticals wherein predictive analytics offerings were deployed. Case studies include various predictive analytics solutions provided by different vendors and their applications.

TABLE 7 USE CASES: PROMINENT INDUSTRY-WISE INSTANCES

VERTICAL	USE CASE 1	USE CASE 2
BFSI	DBS is one of the leading banks in Asia. It was planning to create a data-driven organization where the staff can gain insights from billions of events and be at the forefront of innovation. The need for DBS bank to deliver a superior customer experience led DBS to become more data-driven and better predict customer needs across channels. However, the company's traditional technology stack for supporting advanced analytics was expensive to scale and not flexible enough to support this work.	PASHA Holding is a significant Azerbaijani corporation with more than ten companies in the banking, insurance, construction, technology, property development, and tourist industries in Azerbaijan, Georgia, and Turkey. PASHA Holding opted to build a unified MIS architecture on Alteryx with particular governance and consolidation standards and data visualization features to minimize the above pain points and optimize management and financial reporting processes.
Healthcare and Life Sciences	Clearsense is a smart data organization based in Jacksonville, Florida. It is re-imagining and simplifying data analytics to help healthcare organizations realize measurable value from their data. The company developed a secure, cloud-based healthcare data ecosystem that rapidly collected data from different sources to provide a real-time, SMARTView of any healthcare environment.	Body Biolytics wins the Under Armour Challenge by applying RapidMiner Predictive Analytics to biometric big data. Body Biolytics could process and analyze a large amount of wearable data rapidly, correctly, and intelligently with the help of RapidMiner's features and functionality.

Government and Defense	The HAN University of Applied Science wanted to improve the data literacy of its students by exploring the data analytics software and tools, to have access to a community of subject experts. HAN joined the Qlik Academic Program and provided access to Qlik software, support, and training via a remote learning platform, qualifications, and community support.	
Retail and eCommerce	Vantage Data is a retail-focused analytics company. It helps retail partners to plan their inventory and staffing for their retail locations. Vantage uses Alteryx to model several known factors (including the day of the week, season, time of the month, holidays, etc.) and now uses real-time weather forecasts to predict changing sales figures in stores, including sales figures across key categories.	Harmons is a supermarket business established in Utah. Harmons used Domo to prioritize orders so that customers could get what they wanted.
Telecom and IT	The seamless integration of RapidMiner's lightning-fast data science platform and QlikView's strong visualization capabilities provides a Customer Segmentation solution to drive revenue optimization. RapidMiner and Qlik provided a solution to connect, combine and analyze both structured and unstructured data from multiple data sources.	
Energy and Utilities	The client used the WNS solution to transform the collections process by embedding predictive analytics and changing the customer interaction strategy. The client intended to optimize its final debt collection processes to improve the recovery of receivables.	Schneider Electric developed Realift Rod Pump Control, a predictive Internet of Things (IoT) analytics solution based on Microsoft Azure Machine Learning service and Azure IoT Edge, to enable remote equipment maintenance, minimize maintenance costs, and increase worker and environment safety.
Media and Entertainment	DAZN, a subscription-based PaaS owned by Perform Group and dedicated to live and on-demand streaming of worldwide sports-related entertainment, leveraged AWS and Dataiku for enabling the non-technical staff to perform advanced customer segmentation, content attribution, and churn prediction.	ESPN required a mechanism to swiftly gather consumer feedback to recognize emergent issues with access to ESPN—and determine which services were impacted before subscriber churn occurred. After DOMO, service issue reports are now freely accessible and can be shared with anybody who wants to see them. There will be no more manual report production, Excel, or PowerPoint.
Travel and Hospitality	The Air France – KLM Group, an award-winning European airline, strives to provide high-quality customer service, according to the changing needs of the customers, using pervasive data intelligence for smoothly connecting more than 100 million passengers a year. Air France-KLM uses Teradata Vantage as the platform for pervasive data intelligence for simplicity, scalability, and performance needed to provide customers with a world-class travel experience.	Lufthansa is the largest German airline and the largest airline in Europe when its subsidiaries are included. The Lufthansa Industry Solutions team, a consulting group within Lufthansa, used RapidMiner to exploit a huge and diversified data collection to develop better arrival predictions than what was previously available.

Transportation and Logistics	Schneider, one of the largest truckload carriers in North America, found that its legacy technology was placing limitations on its quest to reinvent itself as a fully digitized company. Schneider used Oracle's Digital Logistics capabilities to provides customers and employees with relevant, up-to-date information, such as delivery schedules, as well as predictive analytics to assist with planning for future loads and orders.	Teradata Vantage helps the BMW Group manage moving parts from various suppliers in one of the world's most complicated supply chains. To keep automobiles rolling off the assembly lines, automakers must avoid costly supply chain disruptions. Vantage provides precise details about suppliers, price, and key performance indicators (KPIs) across all process partners to BMW Group.
Manufacturing	This case study focuses on its electronics manufacturing business, especially in Europe. The RapidMiner user group within the European electronics business is a data science team that is part of the organization focused on post-sales customer support. This electronics manufacturer used RapidMiner's ML capabilities for web and text mining to support its classification analyses.	The Volvo team used to spend an excessive amount of time gathering data and presenting it in an understandable and presentable manner. The two most widely used solutions at the moment are QlikView and Qlik Sense. They're adaptable enough that each user can find the best solution for their business problems.
Other Verticals*	Using Cloudera for analytics and machine learning, Rush University Medical Center receives new insights to enable individualized patient treatment programs and deliver precision medicine. Cloudera and MetiStream are collaborating with Rush University Medical Center to develop machine learning and provide analytics-as-a-service to hospital administrators, physicians, and researchers.	

*Other verticals include education and research and real estate.

Source: Company Website and Press Releases

5.6.1 BANKING, FINANCIAL SERVICES, AND INSURANCE

5.6.1.1 Case study 1: Streamlining the mortgage appraisal process

Description	DBS is one of the leading banks in Asia. It was planning to create a data-driven organization where the staff can gain insights from billions of events and be at the forefront of innovation.
Challenges	The need for the DBS bank to deliver a superior customer experience has led DBS to become more data-driven and better predict customer needs across channels. However, the company's traditional technology stack for supporting advanced analytics was expensive to scale and not flexible enough to support this work.
Solutions Offered	DBS built a central data team and enterprise data hub that enabled its staff to experiment more and be at the forefront of innovation to understand the customer experience and apply human-centered design to its services.
Benefits	The transformation of DBS to a data-driven organization improved its operations while reducing costs across the organization.

Source: Company Website

5.6.2 TELECOMMUNICATION AND IT

5.6.2.1 Case study 1: Customer segmentation

Description	The seamless integration of RapidMiner's lightning-fast data science platform and QlikView's strong visualization capabilities provide a Customer Segmentation solution to drive revenue optimization.
Challenges	One of the most important challenges IT companies are facing is customer segmentation due to multiple data sources and different business needs. Because IT has to prepare the data and modify the model to fit each business scenario manually, time and money are lost in the process.
Solutions Offered	RapidMiner and Qlik provided a solution to connect, combine, and analyze structured and unstructured data coming from multiple data sources. It enabled the company to create an automated segmentation system based on the users' specified parameters provided directly from the QlikView dashboard. The use of this QlikView connection enabled users to trigger a workflow directly from QlikView and evaluate different models based on those parameters.
Benefits	- Users can now visualize the customer base before and after segmentation in QlikView to quickly identify the most valued segments. - A greater understanding of customer-based results reduced the cost of sales and improved revenue opportunities, enabling them to immediately deliver accurate and timely decisions.

Source: Company Website

5.6.3 RETAIL AND ECOMMERCE

5.6.3.1 Case study 1: Using weather data to predict sales

Description	Vantage Data is a retail-focused analytics company. It helps retail partners to plan their inventory and staffing for their retail locations. Vantage uses Alteryx to model several known factors (including the day of the week, season, time of the month, holidays, etc.) and uses real-time weather forecasts to predict changing sales figures in stores, including sales figures across key categories.
Challenges	Vantage clients were looking to be more proactive with data, and they wanted to ensure that their resources (stock and personnel) were deployed forward in the most efficient manner possible.
Solutions Offered	The company used Designer to build the workflows; Alteryx Server is running regularly to provide updated patterns. While Alteryx Promote is not yet used, the company is developing a business case to have these models be managed and improved with promotion and using the API endpoint functionality to integrate predictions with source systems.
Benefits	Within the first few weeks of using the weather prediction model, the customers saw several key benefits, such as - deferring the planned markdown of winter stock days before a major cold front - reducing the stock out in leisure locations during a high temperature/good weather public holiday by planning an extra mid-day delivery in key locations. - adjusting the space allocation of summer products in stores in more temperate locations

Source: Company Website

5.6.4 HEALTHCARE AND LIFE SCIENCES

5.6.4.1 Case study 1: Improve healthcare operations through data analysis

Description	Clearsense is a smart data organization based in Jacksonville, Florida. It is re-imagining and simplifying data analytics to help healthcare organizations realize measurable value from their data. They have developed a secure, cloud-based healthcare data ecosystem that rapidly collects data from different sources to provide a real-time, SMARTView of any healthcare environment.
Challenges	Clinicians wanted to use ML and data science techniques to store and analyze data, but the necessary technology didn't exist. Healthcare practitioners understood the medical challenges that could be solved with data, but they were not data architects. They lacked medical expertise on topics such as electronic medical records, signs of patient distress, or hospital billing and reimbursement processes.
Solutions Offered	The Clearsense team was the first company to deliver SMART, real-time streaming data to its healthcare customers. Its Inception product makes data available for critical decisions in three areas, including clinical decisions, financial decisions, and operational decisions. Presently, inception customers have access to all data sources ingested with Hortonworks Data Flow (HDF), stored in HDP and delivered to the decision point.
Benefits	The solution enabled Clearsense to offer healthcare predictive analytics to smaller, rural, and underserved providers that wouldn't be able to afford a major up-front investment with a proprietary vendor selling a software license.

Source: Company Website

5.6.5 MANUFACTURING

5.6.5.1 Case study 1: Improving customer support in electronic manufacturing

Description	The company is one of the leading manufacturers of electronic products for the consumer and professional markets. Its diversified business includes consumer and professional electronics, gaming, entertainment, and financial services. This case study focuses on its electronics manufacturing business, especially in Europe. The RapidMiner user group within the European electronics business is a data science team that is part of the organization focused on post-sales customer support.
Challenges	The company understood the basic statistics related to customer support, such as how many people called, how long people stayed on the phone, how many people visited the support website, etc. But it was more difficult for the team to determine why people were calling or whether the customer support team could provide solutions for customers when they called.
Solutions Offered	This electronics manufacturer used RapidMiner's ML capabilities for web and text mining to support its classification analyses. It helped the team to understand more deeply what's going on with customer support calls. The team was able to do an even more powerful analysis with RapidMiner and ML. This included anomaly detection, long-tail problem identification, and forecasting. These predictive analyses became a new approach to the problem for the company.
Benefits	The team can now track the precision and accuracy of its predictions, which are strong and getting even stronger. RapidMiner's solution enabled the company to track the bottom-line impact of its analytical insights, using the same business KPIs, such as customer support, marketing, and product development, as its partner organizations use.

Source: Company Website

5.6.5.2 Case study 2: Improvements in products, services, and customer satisfaction through advanced analytics and AI

Description	KIA Motors America was facing significant issues in streamlining its vehicle servicing division to address a rising number of customer queries.
Challenges	A KIA dealership technician, on finding an issue or difficulty in diagnosing a problem, needed to contact the internal technical assistance hotline, which then generated a techline report based on information assimilated across dealerships and agents. This report used to take a couple of days to be created, involving a tedious process of combining data from a CRM system, phone records, and survey responses, resulting in significant delays in addressing the issue.
Solutions Offered	SAS Visual Analytics as part of all vehicle servicing division
Benefits	- Automated report generation providing clear visibility into agent performance - Reduction in production time to 30 minutes - Enabling techline agents to better assist KIA technicians and dealership customer

Source: Company Website

5.6.6 ENERGY AND UTILITIES

5.6.6.1 Case study 1: Increase debt collection

Description	The client is one of the leading energy and utility companies. The client had a dire business need to re-balance its energy final debt portfolio. On the one hand, its debt recovery rate was 4 percent compared to 14 percent achieved by its competitors; on the other hand, the commissions charged by the client's debt collection agencies were as high as 50 percent of the collected amount, which resulted in high operational costs.
Challenges	The client intended to optimize its final debt collection processes to improve the recovery of receivables. The client also wanted to formulate focused debt management strategies for different customer segments to manage customer write-offs more effectively and decrease operational costs.
Solutions Offered	The client used the WNS solution to transform the collections process by embedding predictive analytics and changing the customer interaction strategy. The solution's key aspects included the Propensity-to-Pay predictive data model, customer classification based on scores, customer segment prioritization, cost-benefit analysis, and performance monitoring of pilot strategies.
Benefits	- Debt collection increased by 50 percent within three months - The modified process recorded an 8 percent rise in conversion rates compared to the standard process - Operational expenses decreased by 20 percent

Source: Company Website

5.6.7 TRANSPORTATION AND LOGISTICS

5.6.7.1 Case study 1: Visualize and predict the status of the fleet

Description	Schneider, one of the largest truckload carriers in North America, found that its legacy technology was placing limitations on its quest to reinvent itself as a fully digitized company.
Challenge	Harnessing the data to visualize and predict the status of the fleet before a negative event can disrupt operations.
Solutions Offered	Schneider used Oracle's Digital Logistics capabilities to provide customers and employees with relevant, up-to-date information, such as delivery schedules, as well as predictive analytics to assist with planning for future loads and orders. Drivers received step-by-step assignments, and managers were notified of deviations so they could correct them quickly. The solution also helped in detecting driver behavior patterns for immediate intervention or future coaching.
Benefits	It enabled Schneider to supply predictive analytics for delivering superior customer experiences and avoiding disruption that resulted in precise Estimated Time of Arrival (ETA), so that recipients downstream can be proactively notified with push-alerts from chatbots and enabled preventive repair using data from on-board diagnostics to deliver advance warnings of impending equipment failure, allowing operators and fleet managers to take corrective action.

Source: Company Website

5.6.8 MEDIA AND ENTERTAINMENT

5.6.8.1 Case study 1: Scaling a small data team using machine learning

Description	DAZN, a subscription-based PaaS owned by Perform Group and dedicated to live and on-demand streaming of worldwide sports-related entertainment, leveraged AWS and Dataiku for enabling the non-technical staff to perform advanced customer segmentation, content attribution, and churn prediction.
Challenges	DAZN wanted a fast and low-maintenance way for enabling their small data team to handle predictive analytics and machine learning projects at scale to continue growing their businesses in existing and new markets. It also wanted data analysts, not necessarily technical or experienced in machine learning, who can contribute to impactful data projects.
Solutions Offered	- Content Attribution – for determining the fixtures that drive sales and enable contextual information on key fixtures in each market. - Propensity Modelling – for identifying customers that churn and enabling improved customer retention. - Survival Analysis – for understanding customer stickiness and enabling calculation of the expected revenues for understanding customer return on investment. - Natural Language Processing – Using NLP on social networks to perform market research. - Advanced Customer Segmentation – to identify user behaviors, particularly regarding content and devices on which customers use the product.
Benefits	The solutions enabled AWS and Dataiku to shift the data culture at DAZN and highlight the innovations in advanced analytics and machine learning throughout the company.

Source: Company Website

5.6.9 GOVERNMENT AND DEFENCE

5.6.9.1 Case study 1: HAN University to equip students thrive in a digital future

Description	The HAN University of Applied Science wanted to improve the data literacy of its students by exploring the data analytics software and tools, to have access to a community of subject experts.
Challenge	▪ Encouraging the students to access various BI tools ▪ Enabling the academics to integrate study materials into coursework ▪ Provide additional learning materials with no cost to the university
Solutions Offered	HAN joined the Qlik Academic Program and provided access to Qlik software, support, and training via a remote learning platform, qualifications, and community support.
Benefits	▪ Ensured access to various BI tools and expertise the data analytics best practices ▪ Allowed HAN to promote BI skills within all its coursework ▪ Provided students with certifications and digital badges that can be shared on resumes and social networks

Source: Company Website

5.6.10 TRAVEL AND HOSPITALITY

5.6.10.1 Case study 1: Optimizing customer experience with Teradata Vantage

Description	The Air France–KLM Group, an award-winning European airline, strives to provide high-quality customer service, according to the changing needs of the customers, using pervasive data intelligence for smoothly connecting 100M+ passengers a year.
Challenge	Air France–KLM wanted to quickly identify pricing and promotion opportunities, minimize and manage churn, and optimize web and call center experiences for enhancing the customer experience and increasing customer satisfaction at every touchpoint.
Solutions Offered	Air France–KLM provides greater customer context for each stage of the travel from pre-booking to departures, arrivals, and baggage claim, also to customer incident management via call center interactions, all by having the available customer data. The airlines make customer recommendations based not only on customer profiles but also at each stage of the customer's journey. Air France-KLM uses Teradata Vantage as the platform for pervasive data intelligence for simplicity, scalability, and performance needed to provide customers with a world-class travel experience.
Benefits	The airline continues to improve its famous high-quality customer service according to the evolving needs of its customers. Air France-KLM Group is smoothly connecting over 100 million passengers a year and increasing customer satisfaction and enhancing customer experience by scaling air travel experience by just tapping into all the data, all the time.

Source: Company Website

5.7 PATENT ANALYSIS

5.7.1 METHODOLOGY

The patent analysis offers crucial insights into patent filing, business interests, and patenting activities yearly and country-wise. It is a technique used to understand the information contained in patents. This study defines patents claiming inventions linked to the predictive analytics market. For this study, a well-planned strategy has been devised. Patents are categorized and analyzed using predictive analytics. This keyword is used for titles or abstracts to understand patent filing profiles. This section aims at providing an overview of patenting activities. Sources used for this analysis are The Lens—Patent, Scholarly Search, and Patentscope-WIPO.

5.7.2 DOCUMENT TYPE

TABLE 8 PATENTS FILED, 2018–2021

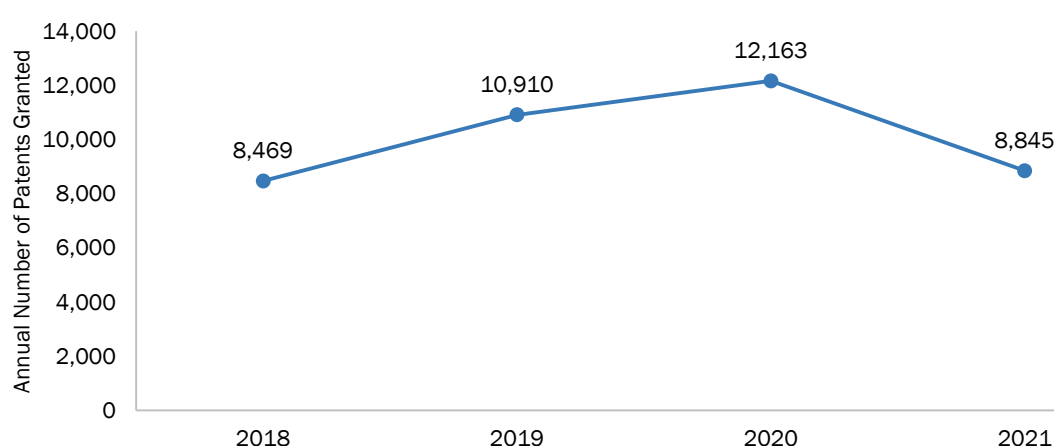
DOCUMENT TYPE	NO. OF PATENTS
Granted Patents	40,387
Patent Applications	65,864
Total	106,251

Source: The Lens— Patent, and Scholarly Search

5.7.3 INNOVATION AND PATENT APPLICATIONS

Analysis has been performed on patents filed and granted in the past three years specific to predictive analytics. An increasing trend has been observed in the number of patent grants in the past three years, with a total of 12,163 patents been granted in 2020. IBM Corporation, University of California, and General Electric are the top three companies that have applied for the maximum number of patents in the past three years.

FIGURE 29 TOTAL NUMBER OF PATENTS GRANTED IN A YEAR, 2018–2021

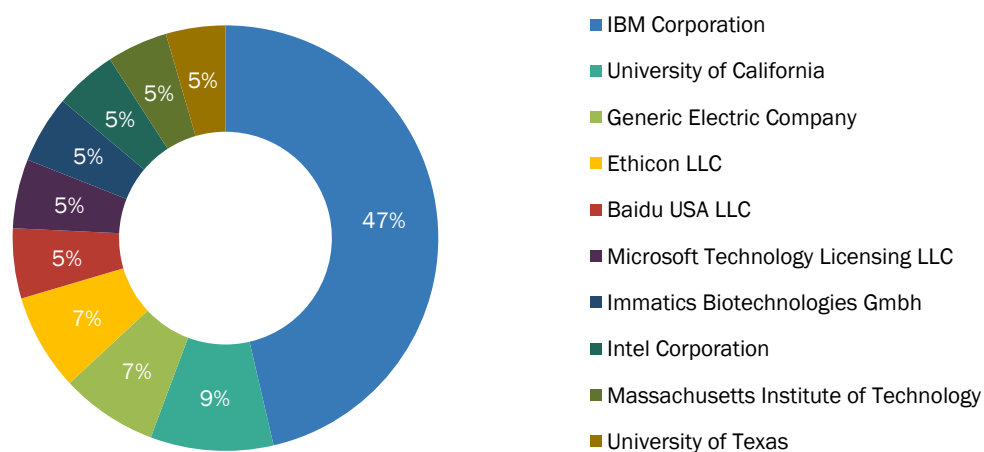


Source: Lens— Patent and Scholarly Search

5.7.3.1 Top applicants

The top applicant's analysis helps know which organization possesses the most patents relative to the predictive analytics inventions. This, in turn, will help collaborators identify inventors and vice versa for the joint development of products.

FIGURE 30 TOP TEN COMPANIES WITH THE HIGHEST NUMBER OF PATENT APPLICATIONS, 2018–2021



Source: The Lens—Patent and Scholarly Search

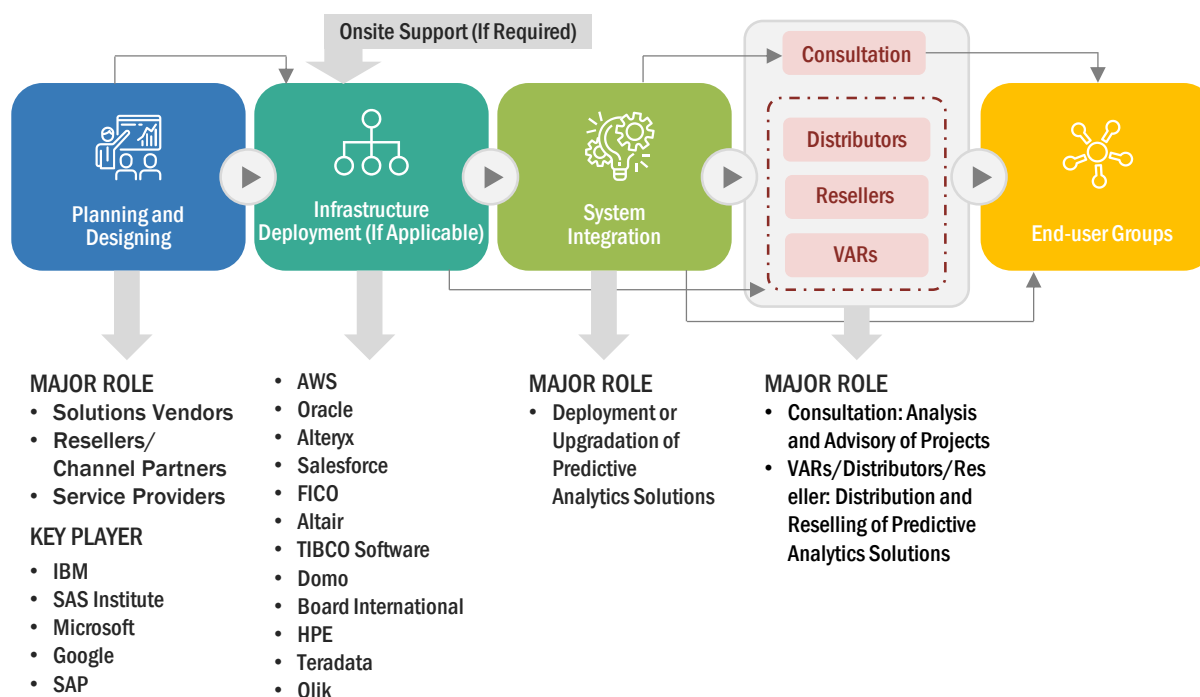
TABLE 9 TOP TEN PATENT OWNERS (US) IN THE PREDICTIVE ANALYTICS MARKET, 2018–2021

TOP TEN PATENT OWNERS (US) IN THE LAST THREE YEARS	NO. OF PATENT APPLICATIONS
IBM Corporation	3,574
General Electric Company	412
University of California	369
Immatics Biotechnologies GmbH	351
Microsoft Technology Licensing LLC	303
Baidu USA LLC	289
Cilag GmbH International	272
Intel Corporation	268
Ethicon LLC	267
Oracle International Corporation	257

Source: Lens—Patent and Scholarly Search

5.8 SUPPLY/VALUE CHAIN ANALYSIS

FIGURE 31 SUPPLY/VALUE CHAIN ANALYSIS



Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

The value chain of the predictive analytics market highlights the delivery methods of unique solutions/services to end users. These delivery methods are devised through collaborations between vendors and service providers. Companies independently offer both predictive analytics solutions and services to help organizations discover insights from the data generated from various sources. In the planning and designing phase, customers' requirements are considered, and predictive analytics solutions are customized based on these requirements. In the infrastructure deployment phase, all elements required in customers' premises are installed or made available through cloud. The system integration phase deals with deploying integrated solutions or upgrading existing predictive analytics solutions. The predictive analytics market comprises different solution modules of predictive analytics; these solution modules can be provided as a complete suite by companies or can be integrated with deployed products offered by third-party vendors.

IBM, SAS Institute, Microsoft, Google, SAP, AWS, Oracle, Alteryx, Salesforce, FICO, Altair, TIBCO Software, Domo, Board International, Teradata, HPE, and Qlik, are some well-established solution vendors in the predictive analytics market. Various verticals play a major role in this market, including BFSI, retail and eCommerce, government and defense, transportation and logistics, manufacturing, healthcare and life sciences, and energy and utilities.

TABLE 10 PREDICTIVE ANALYTICS MARKET: VALUE/SUPPLY CHAIN

COMPANY	ROLE IN ECOSYSTEM
Microsoft	Technology provider
IBM	Technology provider
SAP	Technology provider
SAS Institute	Technology provider
Oracle	Value-added service provider
Google	Technology provider
Salesforce	Solution designers and developers/consultants
AWS	Technology provider
HPE	Solution designers and developers/consultants
Teradata	Technology provider
Alteryx	Value-added service provider
FICO	Technology provider
Altair	Solution designers and developers/consultants
DOMO	Solution designers and developers/consultants
Cloudera	Technology provider
Board International	Value-added service provider
Tibco Software	System integrator
Hitachi Vantara	Value-added service provider
Qlik	Solution designers and developers/consultants
Happiest Minds	Solution designers and developers/consultants
Dataiku	Solution designers and developers/consultants
RapidMiner	Value-added service provider
Biofourmis	Solution designers and developers/consultants
In-med Prognostics	Value-added service provider
Aito.ai	Value-added service provider
Symend	Solution designers and developers/consultants
Onward Health	Solution designers and developers/consultants
Unioncrate	Solution designers and developers/consultants
CyberLabs	Value-added service provider
Actify Data Labs	Value-added service provider

Amlgo Labs	Solution designers and developers/consultants
Verimos	Solution designers and developers/consultants

Source: Company Website, Annual Reports, and SEC Filings

5.9 PRICING MODEL ANALYSIS

Predictive analytics solutions and service providers suggest various pricing models. Some vendors have several fixed subscription plans, while others provide flexible pricing that depends on the business size and industry and annual sales volume. Providers may only charge per transaction. Some companies share pricing information on requests.

The pricing of solutions and services is subscription-based and tailor-based. Most predictive analytics solutions offer free trial period days of subscription. The subscription is renewed yearly/monthly. Service providers may offer tiered pricing based on the number of users and volume discounts.

Predictive analytics solutions and services can be priced as per the method below:

- Pay-As-You-Go model – It is one of the pricing models for cloud services that encompass both subscription-based and consumption-based models. Pay-as-you-go models work differently as customers pay a one-time cost for products or services to get access to them. Then, if they need the products or services again, they will pay again when they choose. Cloudera Data Platform (CDP) offers public cloud subscriptions and on-premises private cloud subscriptions with Data Hub priced at USD 0.04/Cloudera Compute Unit (CCU) hourly rate and ML at USD 0.17/Cloudera Compute Unit (CCU) hourly rate. CDP private cloud comes in plus edition at USD 400/CCU + USD 25/TB annually.
- Subscription-based pricing model – It is one of the pricing methods where customers can purchase solutions or services from vendors for a specific period, monthly or annually, with a defined value. Oracle Analytics Cloud enterprise edition at USD 80/User/Month and professional edition at USD 16/User/Month. SAP Analytics Cloud offers flexible licensing options to fulfill an organization's needs in three categories:
 - 30-day free trial
 - Business Intelligence starting from USD 35 /User/Month
 - Planning: price given upon request
- Free Trial Period – Various companies offer a free trial period-based subscription, after which users can subscribe to the paid services of their choice. TIBCO Cloud Spotfire offers a subscription option of five types:
 - 30-day free trial
 - Analyst: USD 125/month or USD 1250/year
 - Business Author: USD 65/month or 650/year
 - Consumer: USD 25/month or 250/year
 - Library storage: USD 25/month or USD 250/year for 25 GB

5.10 PORTER'S FIVE FORCES ANALYSIS

FIGURE 32 PREDICTIVE ANALYTICS MARKET: PORTER'S FIVE FORCES ANALYSIS

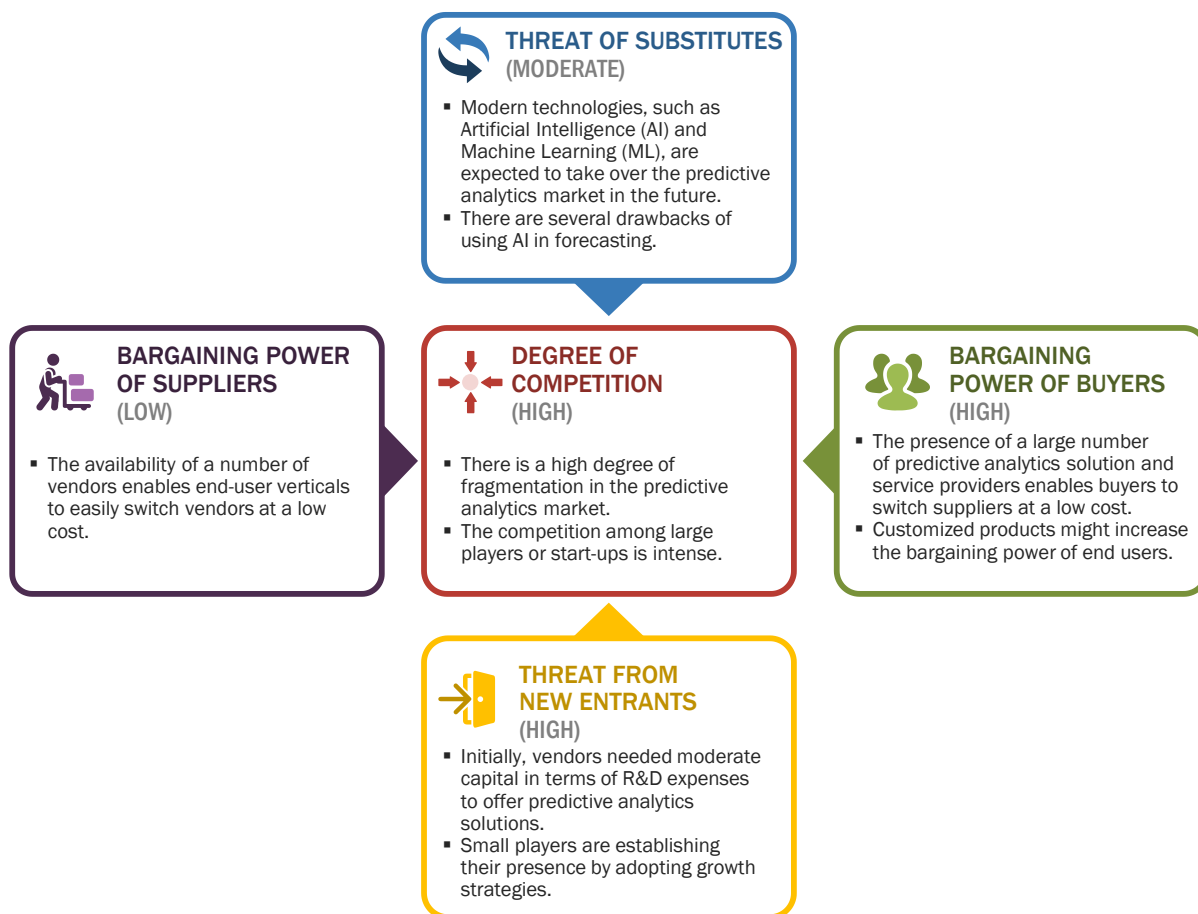


TABLE 11 PREDICTIVE ANALYTICS MARKET: PORTER'S FIVE FORCES ANALYSIS

PORTER'S FIVE FORCES	IMPACT
Degree of Competition	High
Bargaining Power of Suppliers	Low
Bargaining Power of Buyers	High
Threat of Substitutes	Moderate
Threat of New Entrants	High

5.10.1 THREAT OF NEW ENTRANTS

The threat of new entrants refers to the force of new potential competitors who can capture the market share of the existing companies in any market. Currently, the threat from new entrants in the predictive analytics market is high, as there is high to moderate R&D expenditure, and moderate to low initial investment is required. Several global and domestic players have an established base in the market. These market players have a global presence and offer various predictive analytics solutions for various applications. Various companies enter this market by adopting strategies, such as acquisitions, partnerships, new product releases, and integrating applications. The growing demand for predictive analytics solutions in emerging applications enables start-ups to set up their presence in the market. Hence, the threat of new entrants in the predictive analytics market is high.

5.10.2 THREAT OF SUBSTITUTES

The threat of substitutes refers to the possibility of alternative products replacing the existing ones in the market. Switching to substitutes will not be much expensive, especially for bigger corporations who have standardized their systems with specific predictive analytics solutions regardless of an update. Some of the predictive analytics business applications available in the market include functionalities such as Artificial Intelligence (AI), Machine Learning (ML), Data Mining, and Statistical Modelling. These advanced functionalities enable efficient prediction and analysis. An increase in innovation would translate to an increase in a better experience, which further would be worthy for the users. Loyal customers may approve or be able to afford the new prices, which would, in turn, provide better features. Hence, the threat of substitutes in the predictive analytics market is moderate.

5.10.3 BARGAINING POWER OF SUPPLIERS

The bargaining power of suppliers refers to the extent to which the solutions and services suppliers/providers pressurize end users of predictive analytics solutions and services. The suppliers of predictive analytics solutions and services can influence the prices of the solutions and services subscription-based pricing form. There are many suppliers or low switching costs between rival suppliers, enabling users of predictive analytics solutions and services to switch suppliers/providers at a low cost. As a result, the bargaining power of suppliers is low.

5.10.4 BARGAINING POWER OF BUYERS

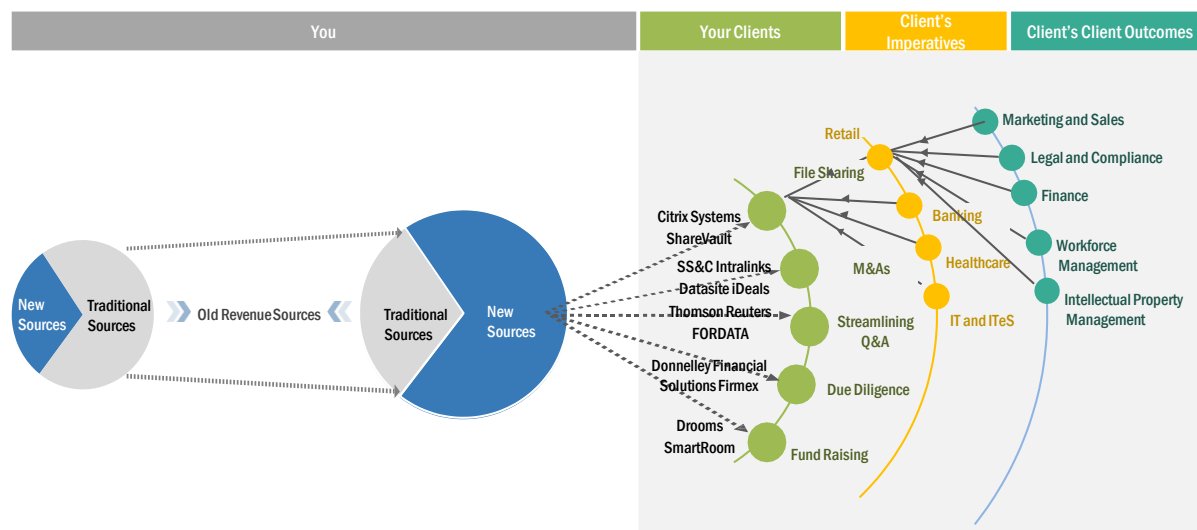
The bargaining power of buyers refers to the pressure that buyers exert on manufacturers of the predictive analytics market. The customer base for the predictive analytics market is expanding day by day due to the presence of established players and upcoming start-ups in this market. Several unorganized vendors provide predictive analytics solutions and services at a low rate as compared to other major players in the market. This increases the bargaining power of buyers in the predictive analytics market. Therefore, the bargaining power of buyers in this market is high.

5.10.5 INTENSITY OF COMPETITIVE RIVALRY

The competitive rivalry within the industry refers to the extent to which market players exert pressure on each other. The presence of various well-established players has intensified the level of competitive rivalry in the predictive analytics market. Some of these players are Microsoft, Google, Oracle, IBM, Salesforce, HPE. These predictive analytics solution providers have a strong foothold in the local and global markets due to their strong product and pricing portfolio, integration capabilities, and well-established strategic alliances. Besides these, there are several manufacturers in the unorganized sector, which collectively hold a prominent share in the market. Thus, the intensity of competitive rivalry in the market is high.

5.11 TRENDS/DISRUPTIONS IMPACTING BUYERS/CLIENTS IN THE PREDICTIVE ANALYTICS MARKET

FIGURE 33 PREDICTIVE ANALYTICS MARKET: TRENDS/DISRUPTIONS IMPACTING BUYERS/CLIENTS



Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

The predictive analytics market is estimated to grow at a CAGR of 21.7% during the forecast period. High security and authorized user access coupled with a cheaper and faster solution of predictive analytics are expected to drive the predictive analytics market. The unprecedented and simultaneous advances in technologies, such as AI, ML, blockchain, additive manufacturing, nanotechnology, biotechnology, energy storage, and quantum computing, are blurring traditional boundaries and creating new business models. These technological developments have led organizations to use predictive analytics to store documents securely to make predictive analytics valuable for facilitating important financial transactions, such as M&A, fundraising, and IPOs.

5.12 TECHNOLOGY ANALYSIS

The predictive analytics market is undergoing large-scale digital transformation, owing to the integration of predictive analytics technology with other technologies such as IoT, AI, ML, 5G, and blockchain. AI advancements, coupled with uninterrupted connectivity and real-time communication, enable the exponential growth of businesses across the globe. One of the technology drivers that has impacted the market is the emergence of analytics, which plays a pivotal role in the analytics industry. Technologies such as AI, 5G, and blockchain can catalyze the growth of new businesses or operating models, help new entrants disrupt the market, and enable incumbents to reach new performance levels.

5.12.1 AI AND PREDICTIVE ANALYTICS

Big data is a major driver for organizations to enhance their sustainability in a competitive business world. As more data is generated and stored, the need for more efficient, effective, and precise processes is increased. Predictive analytics is among those processes. It is the process of using data mining, statistics, and modeling to make predictions. The predictive analytics solution mines and analyzes historical data patterns to predict future outcomes by extracting information from data sets to determine patterns and trends. When predictive analytics is paired with computational power, it allows businesses to identify their potential customers by using personalized data collected over time. AI algorithms are increasingly integrating the ability to identify key emotions and produce insights that make prospecting more effective.

for potential buyers. Predictive analytics is driving the AI market by bringing the ability to go beyond the understanding of the historical data in customer intelligence. A combination of AI and predictive analytics is used in several areas, such as social media analysis, weather forecasting, and healthcare. RapidMiner has developed a system to perform sentiment analysis for customer comments in over 150,000 text-based forms in several different languages, including 50,000 tweets and Facebook posts, by working with AI and data science engineers at PayPal.

5.12.2 BLOCKCHAIN AND PREDICTIVE ANALYTICS

Data science is all about the analysis of data to extract useful insights. When data science is paired with blockchain, it makes the data even more useful. Blockchain and data science play an important role in industries that heavily rely on data, such as finance, retail and eCommerce, and telecommunications and IT. Performing predictive analysis when trading crypto markets is made far more accurate and precise by combining these two technologies. Governments increasingly rely on actionable insights derived from large amounts of data to help them improve their decision-making processes and outcomes. The predictive analysts of cryptocurrencies can use this technology for harnessing big data to make more informed forecasts for the future of cryptocurrencies. Predictive analytics is playing a major role in changing the dynamic of blockchain.

5.12.3 ML AND PREDICTIVE ANALYTICS

Nowadays, companies have been increasingly using ML-based predictive analytics. Previously, ML and predictive analytics seemed to be two different and unrelated concepts, but the demand for effective data analytics has brought ML algorithms to combine with predictive analytics. Today, predictive analytics extensively uses ML for data modeling due to its ability to accurately process vast amounts of data and recognize patterns. Predictive analytics and ML go together as predictive models that typically include ML algorithms. These models can be trained over time to respond to new data or values, delivering the results for businesses. Predictive modeling largely overlaps with the field of ML. When ML and AI are coupled with BI tools, such as Domo and Tableau, business executives can make sense of big data.

5.12.4 IOT AND PREDICTIVE ANALYTICS

Predictive analytics are changing the way manufacturers provide maintenance services and opening new possibilities for revenue streams. In the era of digital transformation, equipment is becoming more automated and complex; equipment owners are increasingly looking for more effective maintenance services to manage and maintain their valuable assets. With predictive analytics, manufacturers can offer customers superior maintenance services that increase uptime, improve product quality, and reduce costs. Software AG provides predictive analytics technology for manufacturers in its IoT predictive maintenance solution. It brings together IoT, streaming analytics, and process analytics into an integrated predictive maintenance solution. This predictive analytics solution directly taps into the power of IoT and data from sensors and actuators embedded inside equipment to be maintained.

5.12.5 5G AND PREDICTIVE ANALYTICS

Due to high bandwidth, low latency, and the ability to support higher end-point densities, 5G would allow companies to collect more data, quickly analyze it, make better predictions, and rapidly take action. In the coming days, the world will witness smarter AI, self-driving cars, and smart factories with data analytics and 5G. Therefore, it becomes imperative to focus on the connection between the two technologies. 5G is the key to drive ML integration with data analysis. As organizations collect complex data, predictive analytics can have an enormous impact on decision-making. Analytics in 5G would evolve to learn from context, predict what will happen next, and prescribe the best response.

5.13 REGULATORY IMPLICATIONS

5.13.1 INTRODUCTION

Privacy and data protection are integral parts of data analytics, as they involve the transmission of large chunks of data and personal databases. Numerous standards have been designed and developed worldwide by various organizations to help enterprises manage their businesses systematically and effectively. These standards are generally set by a recognized international body or industry groups to offer a common view of frameworks, processes, and practices, which organizations can easily conform to. Regions such as Europe and North America, where massive volumes of data are generated, have been following the standards introduced by these regulatory bodies. Various companies are also acknowledging the importance of complying with the safety-related data regulations, and as a consequence, are trying to comply with rules and regulations.

5.13.2 SARBANES-OXLEY ACT OF 2002

The Sarbanes-Oxley Act of 2002 was signed and brought into law by the US Congress in 2002 to protect investors from fraudulent accounting activities by organizations. The act made it mandatory for organizations to provide financial disclosures to prevent accounting fraud. Financial fraud scandals faced by Enron Corporation, Tyco International plc, and WorldCom were the main reasons for the implementation of this act. The act has also set compliance policies related to electronic records for IT departments. A few sections from the act are mentioned below:

- Section 302 states that the senior management must certify the accuracy of the financial statement.
- Section 404 states that all publicly traded companies must establish internal controls and procedures for financial reporting.
- Section 802 contains three rules that affect record keeping. The first deals with the distortion of original records, the second defines the period for storing records, and the third rule sets compliances for electronic records.

5.13.3 GENERAL DATA PROTECTION REGULATION

The General Data Protection Regulation (GDPR) (Regulation (EU) 2016/679) is a regulation by which the European Parliament, the Council of the EU, and the European Commission ensure the safety of EU citizens from privacy and data breaches in an ever-increasing, data-driven era. The GDPR extends the scope of the EU data protection law to all foreign organizations that are processing the data of EU citizens. It delivers the coordination of the data protection regulations throughout the EU, thereby making it easier for non-European companies to comply with the regulations. However, if the regulations are not complied with, the company may have to pay 4% of its worldwide turnover as a penalty.

5.13.4 BASEL

It is a city in Switzerland and is the headquarters of the Bank of International Settlement (BIS), which nurtures cooperation among central banks with the shared objective of achieving financial stability and common standards of banking regulations. BIS hosts the meetings of the senior officials of the central banks of the member countries (currently 27 nations). Basel Guidelines refer to the comprehensive administrative standards framed by a group of central banks called the Basel Committee on Banking Supervision (BCBS). The set of agreements mainly emphasizes on the risks to banks and financial systems and is called the Basel Accords. The purpose of the Basel Accords is to ensure that financial institutions have enough capital on account to meet obligations and absorb unexpected losses. Currently, the Basel 3 guidelines are accepted by the committee members. These guidelines were introduced in December 2010. The financial crisis of 2008 was the major reason for the introduction of these new norms.

5.14 PREDICTIVE ANALYTICS MODELS

Several distinct models and algorithms enable predictive analytical capabilities to be achieved, which can be applied to many use cases. Getting the most out of a predictive analytics solution and harnessing data to make informed decisions requires determining which predictive modeling techniques are ideal for the firm. Consider a retailer whose goal is to reduce customer attrition. They may not benefit from the same predictive analytics tools that a hospital uses to forecast the number of patients admitted to the emergency room in the next ten days.

5.14.1 CLASSIFICATION MODEL

In some ways, the classification model is the most basic of the various types of predictive analytics models. It categorizes data depending on what it has learned from historical data. Classification models are useful for answering yes or no questions and offering broad analysis that aids in making decisions. The categorization model's versatility—and the ease with which it can be retrained with new data—allows it to be used in various industries.

5.14.2 CLUSTERING MODEL

Based on related features, the clustering model divides data into separate, tiered smart groupings. If an eCommerce shoe firm wants to run customized marketing campaigns for its customers, it may sift through hundreds of thousands of data to come up with a strategy that is unique to each customer. However, they may easily divide clients into similar groups based on common features and design tactics for each group on a broader scale utilizing the clustering methodology. Other applications of this predictive modeling technique include categorizing loan applicants into "smart buckets" based on loan qualities, identifying high-crime zones in a city, and benchmarking SaaS customer data into groups to uncover global use patterns.

5.14.3 FORECAST MODEL

The forecast model, one of the most extensively used predictive analytics models, deals with metric value prediction, forecasting numeric value for new data based on prior data learnings. This model can be used wherever there is historical numerical data. A SaaS company, for instance, can forecast how many clients it will convert in a given week. Multiple input parameters are also considered by the forecast model.

5.14.4 TIME SERIES MODEL

The time series model comprises a series of data points with time as the input parameter. It develops a numerical meter based on the previous year's data and utilizes that metric to forecast the future three to six weeks' worth of data. The number of daily calls received in the previous three months, sales over the previous 20 quarters or the number of patients that showed up at a specific hospital in the previous six weeks are all instances of use cases for this model. It's a powerful approach to examine how a single parameter evolves over time with a level of precision that goes beyond basic averages. It also considers the seasons of the year and any events that may have an impact on the measure.

5.14.5 OUTLIERS MODEL

The outliers model focuses on data entries that are out of the ordinary in a dataset. It may spot unusual statistics on its own or in combination with other numbers and categories. The outlier model is especially effective in retail and finance for predictive analytics. When detecting fraudulent transactions, for instance, the model might include not just the quantity but also the location, time, purchasing history, and nature of the transaction.

6 PREDICTIVE ANALYTICS MARKET, BY COMPONENT

KEY FINDINGS

- The solutions segment is expected to grow from USD 7,876 million in 2021 to USD 20,953 million by 2026, at a CAGR of 21.6%, while the services segment is projected to grow from USD 2,667 million in 2021 to USD 7,191 million by 2026, at a higher CAGR of 21.9% during the forecast period.
- The risk analytics solution segment is expected to grow from USD 1,292 million in 2021 to USD 3,346 million by 2026, at a CAGR of 21.0%, while the sales analytics solution segment is projected to grow from USD 1,113 million in 2021 to USD 3,216 million by 2026, at the highest CAGR of 23.6% during the forecast period.
- The COVID-19 pandemic is impacting the adoption of predictive analytics solutions in verticals, such as transportation and logistics and manufacturing. By early 2021, considering the pandemic comes under control, advanced data analytics with real-time predictive capabilities would help bring business operations back on track and provide actionable futuristic intelligence, paving the way for the increased adoption of predictive analytics solutions.
- The demand for professional services is expected to rise due to a rise in tailored demands from customers. Customers are coming up with customization requirements in already installed predictive analytics solutions for enhancing overall performance. The customization requirements include enhanced real-time insights for precise alert systems, enhanced security measures, improved digital Customer Experience (CX), and the capability to extract the maximum potential from the rising enterprise data.

6.1 INTRODUCTION

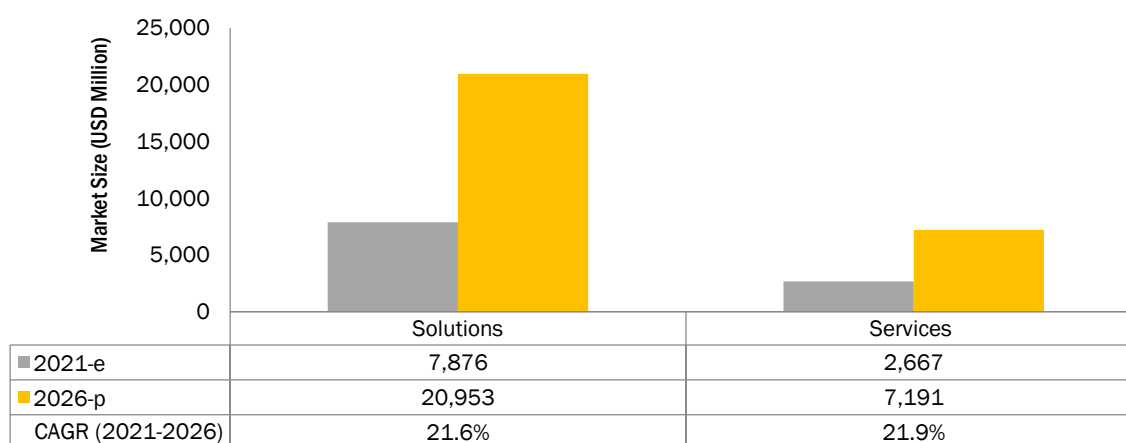
Predictive analytics derives insights from a wide range of internal repository databases, external data, and data sources to identify future risks and provide strategic insights for decision-making. Predictive analytics solutions and services provide a deeper and broader analysis of sudden damages and hard-to-predict events of low frequency but high severity. These solutions can help organizations reduce their total cost of risk and create more effective strategies. They enable organizations to proactively respond to changing business-related conditions, improve operations, and enhance customer interactions driven by real-time insights. IBM, Google, Alteryx, RapidMiner, SAP, SAS Institute, and Oracle are some of the leading vendors in the predictive analytics market.

This chapter includes predictive analytics solutions and services. The solutions included in this section are financial analytics, risk analytics, marketing analytics, sales analytics, customer analytics, web and social media analytics, supply chain analytics, network analytics, and others (HR analytics and legal analytics). The services provided in the predictive analytics market are professional services (deployment and integration, support and maintenance, and consulting) and managed services.

6.1.1 COMPONENTS: COVID-19 IMPACT

Due to the rapid global spread of COVID-19, the adoption of big data analytics tools is on the rise. The use of AL, ML, and natural language processing by researchers and developers has increased rapidly to track and contain COVID-19. Researchers at the Rensselaer Polytechnic Institute (RPI) are using big data solutions and analytics to effectively comprehend COVID-19 from different angles. Recently, the institute announced that it would offer government entities, research organizations, and verticals access to innovative AI tools, as well as experts in data and public health to help combat COVID-19. According to the State of BI & Analytics Report, 2020 - Special COVID-19 Edition published by Sisense in April 2020, the company surveyed 500 data professionals in the US, out of which 49% of respondents have reported an increase in analytics use and new opportunities across all departments post-COVID-19. Due to COVID-19, 55% of businesses have started using data to improve efficiency, 47% to support customers, and 45% to predict future outcomes. Several big organizations such as AWS and Google Cloud have offered researchers free access to open datasets and analytics tools to help them develop COVID-19 solutions faster. A team developed an AI tool at UTHealth. This has shown the need for stricter, immediate interventions in the Greater Houston area. At Stanford University, researchers have launched a data-driven model that predicts the possible outcomes of various intervention strategies.

FIGURE 34 SERVICES SEGMENT EXPECTED TO REGISTER A HIGHER CAGR DURING THE FORECAST PERIOD



e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 12 PREDICTIVE ANALYTICS MARKET SIZE, BY COMPONENT, 2015–2020 (USD MILLION)

Component	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Solutions	2,463	2,887	3,420	4,087	4,964	5,394	17.0%
Services	821	965	1,146	1,373	1,672	1,822	17.3%
Total	3,285	3,851	4,566	5,460	6,637	7,216	17.0%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 13 PREDICTIVE ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Solutions	7,876	10,288	12,648	15,413	18,293	20,953	21.6%
Services	2,667	3,493	4,306	5,262	6,261	7,191	21.9%
Total	10,543	13,781	16,954	20,675	24,554	28,144	21.7%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

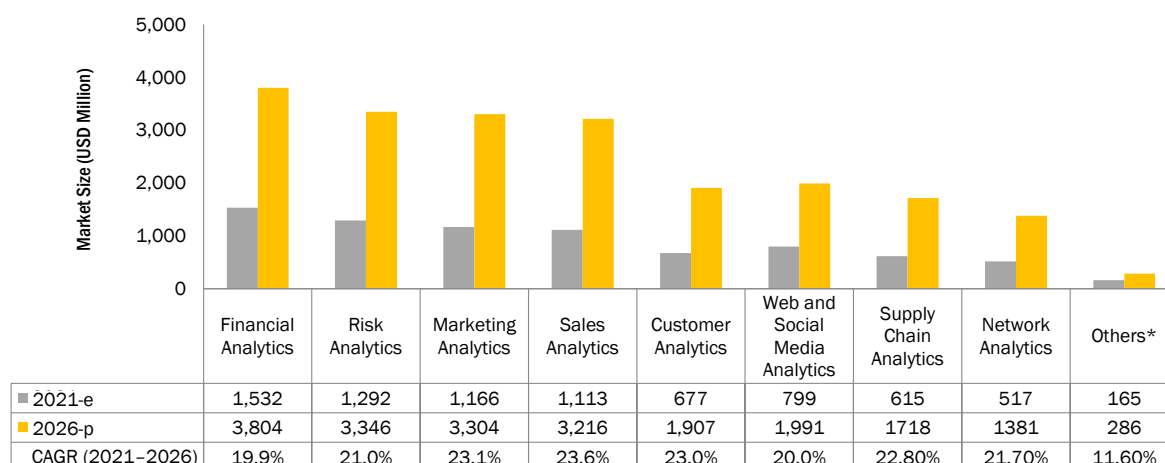
By component, the solutions segment is estimated to account for a larger market size in 2021 because of self-service capabilities. Analytics is applicable across verticals that generate continuous data. In many cases, the best type of data analytics for a company depends on its stage of development. Post the pandemic, organizations would look out for predictive analytics solutions to analyze data in different business processes, regression analysis, historical trend analysis and forecasting, clustering and segmentation, and standardized reporting. Most of the solution providers have realized this and started implementing their solutions with these features to help their clients. As per a survey from an analytics leader, SAS, and the Global Association of Risk Professionals (GARP), 81% of risk professionals in the financial services industry have already seen benefits from AI technologies. According to the survey, these risk professionals have not yet used any form of AI, while 84% plan to use machine learning and natural language processing in the next three years.

The services segment is expected to grow at a higher CAGR during the forecast period due to factors such as the requirement of technical expertise for product upgradation, maintenance, training, and consulting. These factors play an important role during and after the implementation of predictive analytics solutions. The demand for professional services, such as deployment & integration and consulting, is rising with the tailor-made features for predictive analytics solutions during the COVID-19 outbreak.

6.2 SOLUTIONS

The adoption of predictive analytics solutions has increased as they assist users in extracting relevant information from past data, which helps identify trends and predict future outcomes. Predictive analytics solutions help organizations improve their decision-making process by determining trends with the help of data mining, regression, and clustering. The major adoption of these solutions is in the developed countries of North America and Europe due to increased smart projects and IoT connectivity data. These solutions consist of software applications, which can be used to develop various business applications that include fraud detection, cyber risk management, consumer behavior analysis, and workforce management. Predictive analytics vendors provide solutions that help companies in data collection and analysis to generate meaningful insights. The developments and changes in the market are included, along with insights on major companies in the market. Major players offering predictive analytics solutions are IBM, Oracle, SAP, SAS Institute, Alteryx, Altair, and RapidMiner.

FIGURE 35 SALES ANALYTICS SEGMENT EXPECTED TO REGISTER THE HIGHEST CAGR DURING THE FORECAST PERIOD



e: estimated; p: projected

*Other solutions include HR analytics and legal analytics.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 14 PREDICTIVE ANALYTICS MARKET SIZE, BY SOLUTION, 2015-2020 (USD MILLION)

Solution	2015	2016	2017	2018	2019	2020	CAGR (2015-2020)
Financial Analytics	514	596	697	824	989	1,062	15.6%
Risk Analytics	418	487	573	682	823	890	16.3%
Marketing Analytics	338	402	482	583	717	789	18.4%
Sales Analytics	319	379	456	553	681	751	18.7%
Customer Analytics	197	234	280	339	417	458	18.4%
Web and Social Media Analytics	270	313	366	431	517	555	15.4%
Supply Chain Analytics	181	214	256	309	380	417	18.1%
Network Analytics	161	188	223	267	325	354	17.1%
Other Solutions*	65	74	85	98	115	119	12.8%
Total	2,463	2,887	3,420	4,087	4,964	5,394	17.0%

*Other solutions include HR analytics and legal analytics.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 15 PREDICTIVE ANALYTICS MARKET SIZE, BY SOLUTION, 2021–2026 (USD MILLION)

Solution	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Financial Analytics	1,532	1,977	2,401	2,891	3,390	3,804	19.9%
Risk Analytics	1,292	1,679	2,053	2,488	2,937	3,346	21.0%
Marketing Analytics	1,166	1,543	1,921	2,370	2,848	3,304	23.1%
Sales Analytics	1,113	1,475	1,840	2,276	2,740	3,216	23.6%
Customer Analytics	677	895	1,113	1,372	1,646	1,907	23.0%
Web and Social Media Analytics	799	1,030	1,250	1,503	1,761	1,991	20.0%
Supply Chain Analytics	614	811	1,007	1,239	1,485	1,718	22.8%
Network Analytics	517	676	832	1,014	1,205	1,382	21.7%
Other Solutions*	165	202	231	260	280	286	11.7%
Total	7,876	10,288	12,648	15,413	18,293	20,953	21.6%

e: estimated; p: projected

*Other solutions include HR analytics and legal analytics.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

6.2.1 FINANCIAL ANALYTICS

Financial analytics leads to ad-hoc analysis, which assists companies in having all workable solutions for their business-specific questions and forecasting past, present, and possible future financial scenarios. In the current competitive business scenarios, companies need more than accurate financial statements and reports from their financial analytics. They need more forward-looking, predictive insights to help them shape impactful business strategies and improve day-to-day decision-making in real-time. The financial analytics solution in predictive analytics can assist companies by combining internal financial information and operational data with external information, such as social media, demographics, and customer trading habits, which may address critical business questions with unprecedented ease, speed, and accuracy. In addition, financial analytics helps them implement BI and analytical tools to derive precise insights from financial data needed to make business decisions. IBM, SAP, SAS Institute, and Oracle are a few analytics leaders offering financial analytics to customers worldwide.

TABLE 16 FINANCIAL ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	239	268	311	359	412	434	12.6%
Europe	123	143	168	198	238	256	15.7%
APAC	71	87	107	132	175	196	22.4%
MEA	46	57	63	75	91	98	16.2%
Latin America	34	41	49	59	72	78	18.0%
Total	514	596	697	824	989	1,062	15.6%

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 17 FINANCIAL ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	612	779	928	1,070	1,189	1,301	16.3%
Europe	370	477	579	695	847	955	20.9%
APAC	291	386	483	624	759	875	24.6%
MEA	142	185	226	273	322	363	20.6%
Latin America	116	150	186	228	273	310	21.6%
Total	1,532	1,977	2,401	2,891	3,390	3,804	19.9%

e: estimated; p: projected

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 18 FINANCIAL ANALYTICS MARKET SIZE, BY APPLICATION, 2015–2020 (USD MILLION)

Application	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Fraud Detection	219	246	280	320	372	386	12.1%
Profitability Management	102	123	150	184	231	257	20.4%
Governance, Risk, and Compliance (GRC) Management	131	154	182	217	264	287	16.9%
Other Financial Analytics Applications*	63	73	86	102	122	132	15.9%
Total	514	596	697	824	989	1,062	15.6%

*Other financial analytics applications include asset and liability management and budgetary control management.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 19 FINANCIAL ANALYTICS MARKET SIZE, BY APPLICATION, 2021–2026 (USD MILLION)

Application	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Fraud Detection	539	672	788	914	1,030	1,117	15.7%
Profitability Management	384	513	644	799	968	1,113	23.7%
Governance, Risk, and Compliance (GRC) Management	418	546	670	815	966	1,095	21.2%
Other Financial Analytics Applications*	190	246	300	362	425	478	20.2%
Total	1,532	1,977	2,401	2,891	3,390	3,804	19.9%

e: estimated; p: projected

*Other financial analytics applications include asset and liability management and budgetary control management.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

6.2.1.1 Fraud detection

Fraud attempts have seen a drastic increase in recent times, thus making fraud detection more important and applicable across all the major verticals. These verticals include BFSI, retail, healthcare, manufacturing, government, and law enforcement agencies. Fraud detection through its real-time, near real-time, and batch analysis of all user activities, assists companies in safeguarding customer and enterprise information, assets, accounts, and transactions. For fraud detection, predictive analytics uses background server-based processes to examine users' current behavior patterns and compares them with their past buying patterns. Data mining technologies such as decision trees, ML, AI, cluster analysis, and neural networks generate a predictive model to understand fraudulent activities across networks.

6.2.1.2 Profitability management

Profitability management is a predictive analytics application that assists companies in building a predictive model for their business processes to accurately determine products, processes, and customer care and retain the cost against the sales price to maintain desired profitability. Profitability management lets companies make informed decisions, reduce operational costs, streamline processes, and deliver revenue growth. It also finds profitability drivers, such as profit centers, products, divisions, and regions, and supports sales, marketing, operations, and engineering departments, to manage their financial performance.

6.2.1.3 Governance, risk, and compliance management

Governance, risk, and compliance management applications assist companies in developing a framework to ensure issues that are increasingly pervading the IT space are addressed. Governance, risk, and compliance management have become an integral part of successful large organizations as it has been observed that many companies fall prey to heavy penalties because of non-compliance or due to the exploitation of confidential information by internal or external threat sources. The implementation of governance, risk, and compliance management applications may cost a substantial amount of time and money in the initial phase, but they provide a great value proposition for organizations and their clients in the long run. It is necessary that governance and compliance work together and are aligned toward a common set of goals and objectives, which organizations plan to achieve. Governance policies provide a strategic sight for organizations to align their business processes and compliance toward a common goal; risk management performs the task of safeguarding enterprises from internal and external threats and creates a value proposition for stakeholders. The compliance part provides a certain set of rules and regulations that need to be followed to protect public and organization interests.

6.2.1.4 Other financial analytics applications

Other applications include asset and liability management and budgetary control management. These applications provide asset managers, business controllers, and cost center managers a complete view of asset life cycle and budgetary control management, which is done by budgetary control analytics. Budget department managers use this analysis to manage and monitor budgets and actual spending and predict future operational costs. This analysis helps develop interactive dashboards on the BI platform. It presents a precise budgetary analysis across timelines so that it is easy to understand and interpret budgets and actual spending.

6.2.2 RISK ANALYTICS

Risk analytics mainly covers enterprise risks, such as financial risk and operational risks. It also covers the risks associated with an organization's network efficiency. Organizations use the risk analytics solution to enhance their risk intelligence capabilities to fight risk exposures. By using advanced analytical frameworks, companies can avoid, address, or quickly recover from major risk events. The risk analytics technique measures, quantifies, and predicts risks from past data. It provides organizations with valuable insights to make effective decisions and increase revenue while reducing risk costs. The risk analytics process includes identifying risks, categorizing them according to priorities, and quantifying them based on historical data, information from firms, and third-party establishments. Organizations are prone to various types of risks, such as credit risk, hedging risk, governance risk, fraud risk, and technology risk, which can be detrimental to their businesses in terms of loss of revenue and goodwill in the market. The risk analytics solution in predictive analytics facilitates enterprises to establish a baseline for measuring risks across verticals, such as banking and financial services, insurance, healthcare and life sciences, and retail and eCommerce, by incorporating all the facets of risks together into a single unified system that provides key decision-makers with adequate clarity in identifying, viewing, understanding, and managing risks.

TABLE 20 RISK ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	194	219	255	297	343	363	13.3%
Europe	100	117	138	164	199	215	16.5%
APAC	58	71	88	109	146	164	23.2%
MEA	38	46	52	62	76	82	17.0%
Latin America	28	33	40	49	60	65	18.8%
Total	418	487	573	682	823	890	16.3%

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 21 RISK ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	516	661	793	921	1,030	1,144	17.3%
Europe	312	405	495	599	734	840	21.9%
APAC	246	327	413	537	658	770	25.7%
MEA	120	157	193	235	279	320	21.6%
Latin America	98	128	159	197	236	273	22.7%
Total	1,292	1,679	2,053	2,488	2,937	3,346	21.0%

e: estimated; p: projected

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 22 RISK ANALYTICS MARKET SIZE, BY APPLICATION, 2015–2020 (USD MILLION)

Application	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Cyber Risk Management	177	206	242	286	345	371	15.9%
Operational Risk Management	79	91	106	136	148	158	14.9%
Credit and Market Risk Management	119	140	167	201	246	268	17.7%
Other Risk Analytics Applications*	42	50	59	70	85	92	16.8%
Total	418	487	573	694	823	890	16.3%

*Other risk analytics applications include reputational risk, environmental risk, third-party risk, and economic risk.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 23 RISK ANALYTICS MARKET SIZE, BY APPLICATION, 2021–2026 (USD MILLION)

Application	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Cyber Risk Management	537	696	848	1,024	1,205	1,368	20.5%
Operational Risk Management	226	290	351	421	493	556	19.7%
Credit and Market Risk Management	394	518	639	781	930	1,068	22.0%
Other Risk Analytics Applications*	134	175	215	262	310	354	21.4%
Total	1,292	1,679	2,053	2,488	2,937	3,346	21.0%

e: estimated; p: projected

*Other risk analytics applications include reputational risk, environmental risk, third party risk, and economic risk.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

6.2.2.1 Cyber risk management

Risk analytics enables organizations to mitigate cyber risks. Globally, losses arising from fraudulent cases due to the lack of cybersecurity are increasing at a phenomenal rate, making it very important for companies to properly plan cyber risk management in line with their business operational goals. Risk analytics services for cyber risk prevention are evolving from financial loss, disruption, or damage to the reputation of organizations due to the failure of IT systems. The government, BFSI, and healthcare verticals are the key adopters of risk analytics tools in the cyber risk space. Cyber risk analytics helps government organizations detect frauds and optimize investigators' efficiency. The implementation of risk modeling techniques into analytics enables them to identify probable frauds, the severity of frauds, and priorities for investigations based on the impact rate of each fraud. Nowadays, consumers expect a safe and transparent online shopping experience; thus, risk analytics solutions provide an edge by transparently assessing the fraudulent risk of a transaction in real-time during authentication.

6.2.2.2 Operational risk management

Operational risk management assists companies in improving their risk processes, controlling and optimizing capital allocation, meeting regulatory requirements, and mitigating the frequency and severity of risk events. Companies in verticals such as manufacturing, transportation and logistics, and energy and utilities are focusing on operational efficiency by using innovative technologies to avoid operational risks. These verticals are taking initiatives for improving their core functions, such as demand planning, revenue and workforce management, transit and pricing analysis, and the optimization of marketing and risk management functions. Companies are implementing fraud and risk analytics for analyzing all potential risks, such as operational, financial, as well as traveling. Risk analytics predicts the probability of equipment and vehicle failure, accidents, and corporate and financial frauds through the analysis of historical and present data.

6.2.2.3 Credit and market risk management

Credit and market risk management applications offer simulation-based risk measures at the required depth for middle office risk analysis. These applications can help organizations proactively manage regulatory credit risks using a single, end-to-end risk management environment. A transparent framework and integrated data aggregation, analytics, and reporting capabilities provide enterprises with open, flexible, and extensible means to measure and manage credit risks in a constantly changing regulatory environment. Better credit and market risk management also present an opportunity to improve the overall performance and secure a competitive advantage. Credit and market risk analytics helps companies retain their customers, control fees, and decrease delinquencies.

6.2.2.4 Other risk analytics applications

Organizations are often surrounded by risks that might affect their numerous factors, in addition to the primary operational area; these risks comprise reputational risk, environmental risk, third-party risk, and economic risk. Environmental risks are concerned with manufacturing plants and chemical factories, where chemical risk assessment is carried out to ensure that human health and the environment are least affected. The third-party risk management approach consists of understanding the level and type of risk assessments to be performed by a third party. It is directly related to the scope of work of third parties. Economic risks are caused by the instability of macroeconomic factors, such as exchange rates, government regulations, and political factors, which might affect investments, specifically in foreign countries. A risk that impacts either positively or negatively on a project is termed a project risk. The occurrence of losses related to any particular investment is called an investment risk. Interest rate risk, business risk, taxability risk, and inflationary risk are some of the investment risks. There can be situations wherein organizations may face losses in terms of brand reputation (which can be attributed to the declining revenue, rising operational costs, and defaming stakeholders' value in the market), which can adversely affect organizations.

6.2.3 MARKETING ANALYTICS

Marketing analytics is important for measuring, managing, and analyzing marketing performance to maximize the effectiveness of marketing campaigns and optimize Return on Investment (ROI). It helps marketers to be more efficient in executing their ideas and lowering marketing costs. Marketing analytics can offer profound insights into customer preferences and trends. It gathers data from across all marketing channels and consolidates it into a common marketing view. From this common view, companies can extract analytical results that can provide invaluable assistance to drive their marketing efforts.

TABLE 24 MARKETING ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	157	181	215	254	299	322	15.4%
Europe	81	96	116	140	173	190	18.6%
APAC	47	59	74	93	127	146	25.4%
MEA	30	38	44	53	66	73	19.1%
Latin America	22	27	34	42	52	58	20.9%
Total	338	402	482	583	717	789	18.4%

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 25 MARKETING ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	466	608	742	877	999	1,130	19.4%
Europe	282	372	463	570	712	829	24.1%
APAC	222	301	386	512	638	760	27.9%
MEA	108	144	181	224	271	316	23.8%
Latin America	89	117	149	187	229	269	24.9%
Total	1,166	1,543	1,921	2,370	2,848	3,304	23.1%

e: estimated; p: projected

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 26 MARKETING ANALYTICS MARKET SIZE, BY APPLICATION, 2015–2020 (USD MILLION)

Application	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Predictive Modeling	154	178	209	246	295	317	15.6%
Yield Management	64	79	98	123	157	177	22.4%
Product or Service Development Strategies	88	106	128	157	196	218	19.9%
Other Marketing Analytics Applications*	32	39	47	57	70	77	18.9%
Total	338	402	482	583	717	789	18.4%

*Other marketing analytics applications include demand forecasting and contact optimization.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 27 MARKETING ANALYTICS MARKET SIZE, BY APPLICATION, 2021–2026 (USD MILLION)

Application	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Predictive Modeling	456	586	709	849	990	1,113	19.6%
Yield Management	270	368	472	598	739	880	26.7%
Product or Service Development Strategies	326	437	550	687	835	979	24.6%
Other Marketing Analytics Applications*	114	152	190	235	284	331	23.6%
Total	1,166	1,543	1,921	2,370	2,848	3,304	23.1%

e: estimated; p: projected

*Other marketing analytics applications include demand forecasting and contact optimization.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

6.2.3.1 Predictive modeling

Predictive modeling is a marketing technique that uses existing and historical customer data to analyze and predict future behaviors. It helps marketers target the right customers and understand what products or services they are looking for in markets. Predictive modeling allows marketers to use this knowledge to wisely manage their marketing expenditure. It is a form of data-mining technology that works by analyzing historical and current data and generating a model to predict future outcomes.

6.2.3.2 Yield management

Yield management is the marketing pricing strategy used by companies to increase their revenue based on understanding, anticipating, and influencing consumer behaviors with limited resources. The objective of yield management is to align timing, pricing, and customer buying patterns to achieve higher sales and ROI. It helps control inventory to sell it to the right customers at the right time for the right price based on the buying patterns of customers.

6.2.3.3 Product and service development strategies

It is essential to analyze the market data for both launching new products or services and refining existing products or services to satisfy customers. Marketing analytics can help understand customer preferences and needs worldwide; analyze the data from various channels, including social media, emails, and mobile devices; and help marketers understand regions and products or services. They also help them develop product development strategies based on demands and customer preferences.

6.2.3.4 Other marketing analytics applications

Other applications of marketing analytics consist of demand forecasting and contact optimization. Demand forecasting assists companies in forecasting future demands for their products and services based on past events and customer demands. It also assists in forecasting prevailing trends. Contact optimization utilizes predictive analytics solutions to mail scores to customers, saturation measurements between marketing events, and optimization techniques to minimize contacts, maximize profitability, and enhance the customer experience with a minimal impact on companies' revenue. It finds the optimal contact strategy for each customer across channels, brands, locations, and time zones. These applications are heavily interconnected and help in optimizing customer contacts across the enterprise and getting the right offer, to the right customer, at the right time through the right channel and distributors.

6.2.4 SALES ANALYTICS

The sales analytics solution refers to detailed analyzed qualitative and quantitative data insights provided to organizations from compiled sales data, which identifies, understands, and predicts sales trends and sales results while helping users understand shifting trends and improvement points. This solution provides fact-based comprehensive analytical insights on data. It helps sales executives to identify cross-sell and up-sell opportunities for their existing clientele, analyze pipeline opportunities, generate business from new clientele, maximize value from their Customer Relationship Management (CRM) applications, and analyze customer spending trends. Sales analytics includes the features of predictive analytics patterns to leverage varying customer behaviors and determine actionable sales targets. This solution provides methods to identify, comprehend, model, track, and augment an organization's sales performance through predictive models. It is incorporated with BI tools that provide rich virtualization of data in a graphical view. Sales analytics also helps sales representatives track customer and product performance at every stage. It also assists them in generating leads and closing deals.

TABLE 28 SALES ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	148	171	203	241	284	307	15.6%
Europe	77	91	110	133	164	181	18.8%
APAC	44	55	70	89	121	139	25.6%
MEA	29	36	41	51	63	69	19.3%
Latin America	21	26	32	40	50	55	21.2%
Total	319	379	456	553	681	751	18.7%

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 29 SALES ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	445	581	711	842	961	1,100	19.9%
Europe	269	356	444	547	685	807	24.6%
APAC	211	288	370	492	614	740	28.5%
MEA	104	138	173	215	260	307	24.3%
Latin America	85	112	143	180	221	262	25.4%
Total	1,113	1,475	1,840	2,276	2,740	3,216	23.6%

e: estimated; p: projected

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 30 SALES ANALYTICS MARKET SIZE, BY APPLICATION, 2015–2020 (USD MILLION)

Application	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Sales Life Cycle Management	153	179	212	253	306	332	16.8%
Sales Rep Efficiency Optimization	107	130	159	196	246	276	20.8%
Other Sales Analytics Applications*	59	71	85	104	129	143	19.3%
Total	319	379	456	553	681	751	18.7%

*Other sales analytics applications include territory management and incentive compensation management.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 31 SALES ANALYTICS MARKET SIZE, BY APPLICATION, 2021–2026 (USD MILLION)

Application	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Sales Life Cycle Management	483	630	772	936	1,107	1,275	21.4%
Sales Rep Efficiency Optimization	417	562	713	898	1,099	1,311	25.7%
Other Sales Analytics Applications*	213	283	355	441	534	630	24.3%
Total	1,113	1,475	1,840	2,276	2,740	3,216	23.6%

e: estimated; p: projected

*Other sales analytics applications include territory management and incentive compensation management.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

6.2.4.1 Sales life cycle management

A sales life cycle starts with product awareness through lead generation and ends with sales. Each step in the sales life cycle involves various stages that are useful to understand where exactly the prospects or leads are in sales pipelines. The sales analytics solution helps organizations efficiently manage the sales life cycle and drive new business. It also helps provide insights into new leads, conversion history, and the average value per marketing-qualified and sales-qualified lead. The results from the solution help make the sales life cycle more efficient and generate more revenue. Sales life cycle management assists companies in predicting new future revenue opportunities and changing consumer trends.

6.2.4.2 Sales rep efficiency management

Sales rep efficiency management assists companies in overcoming sales reps' major concerns over misrepresenting hundreds of opportunities. As a result, they often end up investing too much time in low-quality opportunities and lesser time in valuable opportunities. The sales analytics solution helps monitor the nature of these leads and prioritize them. Sales managers can optimize the efficiency of their sales team by investing less time in pursuing lower-value opportunities.

6.2.4.3 Other sales analytics applications

Other applications of sales analytics include territory management and incentive compensation management. The incentive compensation management application is designed to provide organizations with a comprehensive suite of tools to manage, create, and distribute incentive data, compensation plans, sales reports, calculations, and dashboards. This application lets organizations have accurate compensation calculations, generate transaction registers, forecast incentives, and reduce errors of data inaccuracies in spreadsheets. The territory management application provides organizations with a channelized pattern to align their diversified sales teams. The alignment of sales personnel in territory management is made across regions, product lines, and customer groups. This application comprehensively helps organizations seamlessly align their customers under different territories to assist sales teams in handling client requirements and queries.

6.2.5 CUSTOMER ANALYTICS

Customer analytics is an advanced and systematic analytical process, also known as customer intelligence, in which customer data is used by companies to better understand customer behaviors and make strategic business decisions through customer segmentation and predictive analytics. It is a systematic analysis of customer insights, which assist organizations in identifying and targeting customers for marketing programs from their behavioral prediction. It helps organizations retain the existing customers, maximize the customer life cycle value, and gain customer loyalty. Data optimization is done by combining processes and technologies, such as behavioral analysis, sentiment analysis over social media, predictive modeling, segmentation, and data visualization and reporting. The analysis of voluminous data offers a competitive advantage to organizations.

TABLE 32 CUSTOMER ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	92	105	125	148	174	187	15.3%
Europe	47	56	67	82	101	111	18.5%
APAC	27	34	43	54	74	85	25.3%
MEA	18	22	26	31	38	42	19.0%
Latin America	13	16	20	24	30	34	20.9%
Total	197	234	280	339	417	458	18.4%

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 33 CUSTOMER ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	271	353	430	508	577	652	19.2%
Europe	164	216	268	330	411	479	24.0%
APAC	129	175	224	296	369	439	27.8%
MEA	63	84	105	130	156	182	23.7%
Latin America	51	68	86	108	133	155	24.7%
Total	677	895	1,113	1,372	1,646	1,907	23.0%

e: estimated; p: projected

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 34 CUSTOMER ANALYTICS MARKET SIZE, BY APPLICATION, 2015–2020 (USD MILLION)

Application	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Customer Segmentation and Clustering	48	59	72	88	111	124	20.6%
Customer Behavior Analysis	96	110	129	151	180	192	14.9%
Monitoring Customer Loyalty and Satisfaction	36	44	55	70	89	102	23.3%
Other Customer Analytics Applications*	17	21	25	30	38	41	19.0%
Total	197	234	280	339	417	458	18.4%

*Other customer analytics applications include operational efficiency management and campaign performance management.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 35 CUSTOMER ANALYTICS MARKET SIZE, BY APPLICATION, 2021–2026 (USD MILLION)

Application	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Customer Segmentation and Clustering	186	251	317	398	486	572	25.2%
Customer Behavior Analysis	274	350	420	499	577	642	18.6%
Monitoring Customer Loyalty and Satisfaction	156	213	274	348	431	514	27.0%
Other Customer Analytics Applications*	62	82	102	127	153	178	23.7%
Total	677	895	1,113	1,372	1,646	1,907	23.0%

e: estimated; p: projected

*Other customer analytics applications include operational efficiency management and campaign performance management.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

6.2.5.1 Customer segmentation and clustering

Organizations often have a very large amount of customer data, which may be different. Hence, they want to segment it into different clusters. They often want to separately analyze each segment as customers may have different preferences and different behavioral patterns. In such situations, customer analytics can be used to identify similarities or differences between those observations. Customer analytics can also be used to identify consumer segments or competitive sets of products. It is often necessary to split customer data into segments and perform any subsequent analysis within each segment to develop segment-specific insights. Clustering and segmentation techniques are used to group data observations in a few segments so that data within any segment is similar, while data across segments is different. Cluster analysis often requires a lot of contextual knowledge and creativity, which statistical tools can provide.

6.2.5.2 Customer behavior analysis

In predictive analytics, customer behavior analysis is the study of individuals, groups, or organizations and the processes companies select, secure, use, and dispose of products, services, experiences, or ideas to satisfy customer needs and wants. Most of the business decisions are based on purchasing behaviors and customer preferences. These business decisions rely on the assumptions, knowledge, and analysis of customer behaviors. Understanding and analyzing customer behaviors are essential for managers these days. It helps them provide value to customers, create competitive advantages, and apply marketing strategies based on purchasing behaviors.

6.2.5.3 Monitoring customer loyalty and satisfaction

Monitoring customer loyalty and satisfaction application are very important for organizations these days. Customers may deal with companies/products through various channels, including in-store, website, mobile app, promotional offer, Interactive Voice Response (IVR), and contact centers. The application is essential to improve customer experiences while decreasing service costs. Companies can deliver consistent customer experiences across channels. Every interaction, happening over any communication channel, is an opportunity for businesses to improve productivity and gain customer loyalty. Customer loyalty is increased by predicting and preventing customer churn.

6.2.5.4 Other customer analytics applications

Other applications in customer analytics consist of operational efficiency management and campaign performance management. In predictive analytics, customer analytics applications help users with operational efficiency by providing both data analysis and line-of-business users with self-service analytics so that they can deliver deeper insights in hours. Customer analytics applications enhance operational efficiency by delivering products or services to customers in the most cost-effective manner possible by ensuring high-quality products, services, and support.

6.2.6 WEB AND SOCIAL MEDIA ANALYTICS

The web and social media analytics solution analyzes entire web and social media data. This analyzed data helps in measuring, collecting, analyzing, and reporting the internet data for understanding and optimizing web usage. This solution assists in monitoring traction and engagement with online customer messages, while it also assists in exploring challenges and potential controversies around aligning marketing strategies. The web and social media are becoming a source of valuable data in the form of customer feedback through comments and prospects' views on companies, their products and services, market trends, and breaking news. Nowadays, large enterprises, as well as SMBs, rely largely on the digital data gathered through their web portals and various social media channels, including Facebook, LinkedIn, and Twitter. This is due to the rising need of digital marketers, advertisers, and publishers to segment their premium buyers from other customers, track the overall visitor traffic on the website and social media pages, manage ad campaigns, and improve complete web and social media performance. Web and social media analytics caters to all these requirements and helps enterprises reform their business strategies and add to online sales.

TABLE 36 WEB AND SOCIAL MEDIA ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	126	141	163	188	215	226	12.5%
Europe	65	75	88	104	125	134	15.6%
APAC	38	46	56	69	92	102	22.2%
MEA	24	30	33	39	48	51	16.1%
Latin America	18	21	26	31	38	41	17.9%
Total	270	313	366	431	517	555	15.4%

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 37 WEB AND SOCIAL MEDIA ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	319	406	483	556	617	681	16.4%
Europe	193	249	301	362	440	500	21.0%
APAC	152	201	251	325	394	458	24.7%
MEA	74	96	117	142	167	190	20.7%
Latin America	61	78	97	119	142	162	21.7%
Total	799	1,030	1,250	1,503	1,761	1,991	20.0%

e: estimated; p: projected

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 38 WEB AND SOCIAL MEDIA ANALYTICS MARKET SIZE, BY APPLICATION, 2015–2020 (USD MILLION)

Application	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Social Media Management	106	120	137	159	186	197	13.2%
Search Engine Optimization	91	107	127	153	187	202	17.4%
Performance Monitoring	50	58	68	81	98	105	16.1%
Competitor Benchmarking	24	28	33	39	47	50	16.0%
Total	270	313	366	431	517	555	15.4%

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 39 WEB AND SOCIAL MEDIA ANALYTICS MARKET SIZE, BY APPLICATION, 2021–2026 (USD MILLION)

Application	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Social Media Management	278	352	419	494	567	628	17.7%
Search Engine Optimization	296	386	475	579	687	786	21.6%
Performance Monitoring	153	198	241	292	343	390	20.6%
Competitor Benchmarking	73	94	115	139	164	186	20.7%
Total	799	1,030	1,250	1,503	1,761	1,991	20.0%

e: estimated; p: projected

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

6.2.6.1 Social media management

Social media channels facilitate consumers with opportunities to proactively connect with others and enable brands to engage with consumers in a two-way relationship. This interaction and communication over social media platforms generate large amounts of data that can be analyzed to gain actionable intelligence about content and campaign performance. Organizations can gain a competitive advantage by employing web and social media analytics solutions to discover, sort, sift through, and analyze the mountains of data present in these social media platforms. This helps them obtain capabilities to recognize patterns in customer sentiment and measure their marketing and branding efficiencies.

6.2.6.2 Search engine optimization

Appropriate site architectures are critical considerations when it comes to the online marketing of websites to attract potential visitors. In general, by tracking and ranking, any website can be advanced and modified to generate high-quality services to improve its rank among the competitors in the market. Search engine optimization has proved to be an effective and efficient tool in the predictive analytics market as it saves time and effort by providing the exact ranking of each keyword input in multiple search engines. By tracking the rank of each keyword and website traffic over any period, it becomes an easy task to modify websites. Search engine optimization points to website areas that need improvement to get a satisfactory ranking report to increase rankings.

6.2.6.3 Performance monitoring

Performance monitoring is one of the major applications of the web and social media analytics solution in the predictive analytics market. It lets companies record servers' performance and identify anomalies, which can compromise customer transactions and website browsing. It is noted that server performance has a direct influence on the results of marketing campaigns. For example, customers expect the eCommerce checkout process to be carried out fast after clicking the submit button, but they do not mind waiting for a long time to download or play demo videos on websites. If servers do not perform well, a negative impression is created in the minds of customers, and this reduces the retention rate of customers. Hence, the web and social media analytics solution is deployed to track the performance of websites.

6.2.6.4 Competitor benchmarking

Competitor benchmarking can be defined as the process of comparing firms' practices and performance measures with their most successful competitors. Benchmarking is a useful process to optimize organizations' strategies on social media and build new strategies inspired by competition. Analyzing other brands having similar products and target groups helps to know the standing of companies and their brands and what needs to be developed and improved. The purpose of competitor benchmarking is to gain a level of insights and create a marketing strategy based on competitor insights. Through social media benchmarking analysis, companies can know the best practices of social media and elements that provoke weaker interactions on networks.

6.2.7 SUPPLY CHAIN ANALYTICS

Supply chain analytics consists of the complete value chain, sourcing, manufacturing, distribution, and logistics. It aims at improving operational efficiency and effectiveness by enabling data-driven decisions at strategic, operational, and tactical levels. It assists companies in delivering revenue growth, improving margins, managing working capital in a better way, and enhancing overall control points across the supply chain. Nowadays, companies across verticals witness petabytes of data generation in terms of volume through data sources, such as ERP, CRM, Content Management System (CMS), Warehouse Management System (WMS), and Transportation Management System (TMS), across the entire supply chain, from the procurement and sourcing of raw materials to the manufacturing, distribution, and logistics of refined goods. Thus, businesses need to extract useful insights from this huge data that helps business owners make effective strategic decisions.

TABLE 40 SUPPLY CHAIN ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	84	97	114	135	158	170	15.1%
Europe	43	51	62	75	92	101	18.3%
APAC	25	31	39	50	67	77	25.1%
MEA	16	20	23	28	35	39	18.8%
Latin America	12	15	18	22	28	31	20.6%
Total	181	214	256	309	380	417	18.1%

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 41 SUPPLY CHAIN ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	245	319	389	458	521	588	19.1%
Europe	148	196	243	298	371	431	23.8%
APAC	117	158	202	268	333	395	27.6%
MEA	57	76	95	117	141	164	23.5%
Latin America	47	62	78	98	120	140	24.6%
Total	614	811	1,007	1,239	1,485	1,718	22.8%

e: estimated; p: projected

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 42 SUPPLY CHAIN ANALYTICS MARKET SIZE, BY APPLICATION, 2015–2020 (USD MILLION)

Application	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Distribution and Logistics Optimization	89	103	120	140	166	176	14.5%
Inventory Management	41	49	60	73	91	102	20.3%
Manufacturing Analysis	29	36	46	58	75	87	24.5%
Other Supply Chain Analytics Applications*	22	26	32	38	47	52	18.6%
Total	181	214	256	309	380	417	18.1%

*Other supply chain analytics applications include vendor performance management and spend analysis.

Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 43 SUPPLY CHAIN ANALYTICS MARKET SIZE, BY APPLICATION, 2021–2026 (USD MILLION)

Application	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Distribution and Logistics Optimization	250	317	378	446	512	566	17.7%
Inventory Management	154	207	262	328	401	472	25.2%
Manufacturing Analysis	134	185	240	307	383	460	28.0%
Other Supply Chain Analytics Applications*	77	102	127	157	189	220	23.3%
Total	614	811	1,007	1,239	1,485	1,718	22.8%

e: estimated; p: projected

*Other supply chain analytics applications include vendor performance management and spend analysis.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

6.2.7.1 Distribution and logistics optimization

Supply chain analytics is used by freight forwarders, shippers, and Third-Party Logistics (TPL) providers. The distribution and logistics optimization application assist shippers and TPLs in collecting and analyzing customers' supply chain data. The analysis of this customer-centric data is done with the help of high-end transportation and logistics analytics solutions to unravel numerous valuable insights for shippers and TPLs. These solutions ultimately help them reduce freight costs, which are measured on a 'per mile' basis. They also help shippers and TPLs utilize their assets fully across the entire logistics value chain, from carrier sourcing to load planning and management.

6.2.7.2 Inventory management

The supply chain analytics solution helps organizations optimize their inventory by effectively positioning raw materials, work-in-progress, and finished goods inventory throughout supply chain networks. It helps companies forecast demand better, reduce inventory levels with a minimal effect on sales or profits, and better sync production with the distribution. The analysis of data collected using various sources, including enterprise applications and sensors, uses an approach to ensure a lower total cost. The analysis considers working capital investment, service performance, and operational costs, including variable handling costs, inventory carrying costs, transportation costs, and backorder, stock-out, and lost-sales penalties.

6.2.7.3 Manufacturing analysis

The amount of data generated in the production department of businesses in verticals such as healthcare and life sciences, and manufacturing is massive. Change is the new constant as far as the manufacturing data is concerned. This is due to continuously changing market dynamics. The supply chain analytics solution helps numerous business organizations working in and across different verticals to collect and analyze the changing manufacturing data in real-time. This solution assists business executives in analyzing comprehensive production processes, measuring the performance of the production processes, and maintaining and enhancing the quality standards of produced commodities. It cumulatively helps business organizations in reducing their overall production costs and increasing production efficiencies.

6.2.7.4 Other supply chain analytics applications

Other applications considered in the supply chain analytics solution are vendor performance management and spend analysis. Predictive analytics provides companies with the risk model-based vendor performance management and analysis application, which is based on external data and ongoing vendor performance for efficiently evaluating and monitoring vendor performance to mitigate the future risk of vendor failure. It offers users spend analysis to identify anomalies in their direct and indirect spend through machine data analytics. This application uses analytical models on spend data to find anomalies that indicate potential savings, such as contract non-compliance, rate mismatches, duplicate invoices, and incorrect demand specifications.

6.2.8 NETWORK ANALYTICS

The network analytics solution assists companies in identifying and resolving most of the IT infrastructure issues before they impact operational performance, which might include memory usage, CPU usage, routing events, Syslog analytics, packet loss, performance thresholds, and latency variations. It helps organizations monitor and analyze data running across networks. This analysis is then used strategically for optimizing network performance and assists in taking proactive measures before any network mishap takes place. The number of connected devices across the world would grow enormously in the coming years with the increasing adoption of IoT, which, in turn, would be responsible for increasing the complexities of networks. To tackle this situation and make a mark in the dynamic telecom environment, the adoption of network analytic solutions is expected to rise significantly in the coming years.

TABLE 44 NETWORK ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	75	85	100	117	135	144	14.1%
Europe	39	45	54	64	78	85	17.2%
APAC	22	27	34	43	58	65	24.0%
MEA	14	18	20	24	30	33	17.7%
Latin America	11	13	16	19	24	26	19.5%
Total	161	188	223	267	325	354	17.1%

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 45 NETWORK ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	206	266	321	375	423	473	18.0%
Europe	125	163	200	244	301	347	22.7%
APAC	98	132	167	219	270	318	26.5%
MEA	48	63	78	96	114	132	22.4%
Latin America	39	51	64	80	97	113	23.5%
Total	517	676	832	1,014	1,205	1,382	21.7%

e: estimated; p: projected

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 46 NETWORK ANALYTICS MARKET SIZE, BY APPLICATION, 2015–2020 (USD MILLION)

Application	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Intelligent Network Optimization	42	51	62	77	96	106	20.1%
Traffic Management	100	116	136	160	192	206	15.6%
Other Network Analytics Applications*	18	21	25	31	37	41	17.8%
Total	161	188	223	267	325	354	17.1%

*Other network analytics applications include managed network security and configuration change management.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 47 NETWORK ANALYTICS MARKET SIZE, BY APPLICATION, 2021–2026 (USD MILLION)

Application	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Intelligent Network Optimization	158	209	262	325	392	456	23.7%
Traffic Management	299	387	471	569	670	760	20.5%
Other Network Analytics Applications*	60	79	98	120	144	166	22.4%
Total	517	676	832	1,014	1,205	1,382	21.7%

e: estimated; p: projected

*Other network analytics applications include managed network security and configuration change management.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

6.2.8.1 Intelligent network optimization

An intelligent network provides specific technical capabilities or services outside the conventional network standard spectrum. This term is often linked with telecom networks, as recent innovations have expanded telecom capabilities beyond its original primary function of facilitating phone calls. The network analytics solution helps organizations optimize the intelligent network.

6.2.8.2 Traffic management

With the increasing adoption of network bandwidth for multiple modes of communication, it is necessary for telecom companies to manage network traffic to avoid network congestion. Customers always expect to receive seamless network signals for continuous connectivity. The network analytics solution helps these companies to manage network traffic by analyzing network data in real-time.

6.2.8.3 Other Network Analytics Applications

Other network analytics applications consist of managed network security and configuration change management. Managed network security is segregated into identity and access management, Intrusion Detection Systems (IDS)/Intrusion Prevention Systems (IPS), antivirus/bot/malware, and other threats. It refers to taking preventive measures for protecting network infrastructures from unauthorized access, intrusions, or the misuse of networking components. It protects from several threats, including Distributed Denial of Service (DDoS) attacks, Virtual Local Area Network (VLAN) hopping, Domain Name System (DNS) spoofing, and Address Resolution Protocol (ARP) spoofing. Network configuration change management is the process of identifying and documenting network device configurations. The implementation of network configuration change management reduces network downtime, optimizes network security, and automates network changes, thereby minimizing configuration errors.

6.2.9 OTHER PREDICTIVE ANALYTICS SOLUTIONS

Other predictive analytics solutions include HR analytics and legal analytics. The HR analytics solution enables organizations to generate meaningful insights from workforce data to better predict the need for resources using various indicators, including performance, vacancy, and the movement of people. It helps organizations set workforce capacity and capability goals and track progress to meet demands across geographies. HR analytics enables organizations to align HR metrics with strategic business goals. Legal analytics assists companies in extracting actionable knowledge from data to assist in-house legal leaders and decision-makers in matter forecasting, process improvement, legal strategy, comparative legal costs, billing optimization, resource management, and financial operations when the legal environment becomes complex.

TABLE 48 OTHER SOLUTIONS: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	30	33	38	43	48	49	9.9%
Europe	16	18	21	24	28	29	13.0%
APAC	9	11	13	16	20	22	19.5%
MEA	6	7	8	9	11	11	13.5%
Latin America	4	5	6	7	8	9	15.2%
Total	65	74	85	98	115	119	12.8%

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 49 OTHER SOLUTIONS: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	66	80	89	96	98	98	8.3%
Europe	40	49	56	62	70	72	12.6%
APAC	31	39	47	56	63	66	16.0%
MEA	15	19	22	25	27	27	12.3%
Latin America	13	15	18	21	23	23	13.3%
Total	165	202	231	260	280	286	11.7%

e: estimated; p: projected

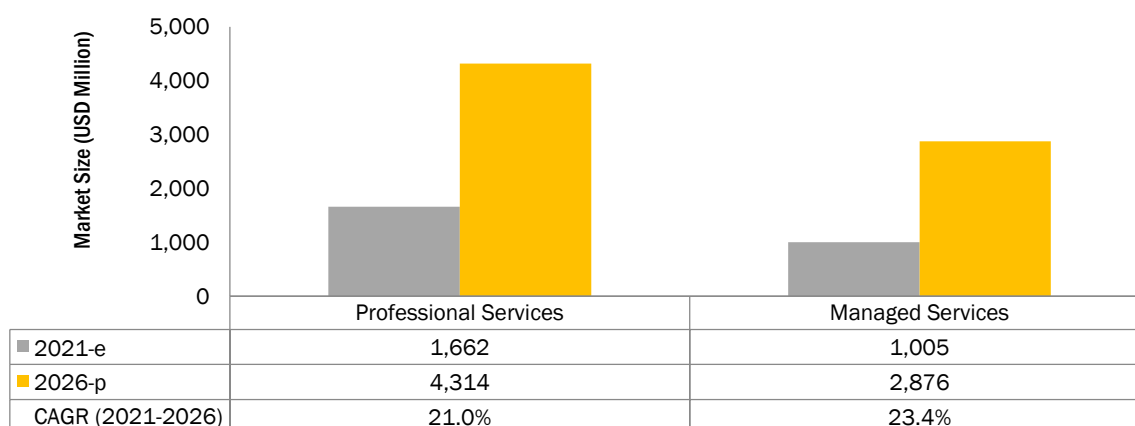
Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

6.3 SERVICES

This section covers the services offered in the predictive analytics market. It includes managed services and professional services (consulting and deployment & integration). The amount of data generated is increasing day-by-day due to factors, such as the increasing adoption of IoT-enabled technologies across verticals, which has encouraged organizations to adopt predictive analytics services. The adoption of services helps organizations enhance their productivity by providing support for delivering solutions and associated training. In today's marketplace, services are an essential part of the growing trend of IT technologies that include cloud and mobile solutions. Managed services take care of software operations and enable customers to focus on core business functions. Enterprises are continuously seeking services through consulting service providers to deploy efficient predictive analytics solutions. Training and support services provide information about the functioning of solutions so that companies can use them in the right manner and gain maximum benefits from them. The managed services subsegment is expected to grow at a higher CAGR during the forecast period.

FIGURE 36 MANAGED SERVICES SEGMENT EXPECTED TO GROW AT A HIGHER CAGR DURING THE FORECAST PERIOD



e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 50 PREDICTIVE ANALYTICS MARKET SIZE, BY SERVICE, 2015–2020 (USD MILLION)

Service	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Professional Services	538	625	733	873	1,056	1,143	16.3%
Managed Services	283	340	413	500	616	679	19.1%
Total	821	965	1,146	1,373	1,672	1,822	17.3%

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 51 PREDICTIVE ANALYTICS MARKET SIZE, BY SERVICE, 2021–2026 (USD MILLION)

Service	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Professional Services	1,662	2,162	2,646	3,210	3,788	4,314	21.0%
Managed Services	1,005	1,332	1,660	2,052	2,473	2,876	23.4%
Total	2,667	3,493	4,306	5,262	6,261	7,191	21.9%

e: estimated; p: projected

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

By service, the professional services segment accounted for a larger market size in 2020. Factors such as pre-sales and post-sales support and the lack of technical skills and capabilities needed for assistance during the upgradation of software drive the adoption of predictive analytics services. The trend of adopting managed services is rising among organizations, as these services help achieve business goals with minimal capital investment and time as they are managed by third-party vendors. Thus, the managed services subsegment is expected to grow at the highest CAGR during the forecast period.

6.3.1 MANAGED SERVICES

Managed services are offered by third-party players; they monitor and maintain software operations. These services are viewed as significant, as they are specifically related to client experience; enterprises cannot bargain on this variable as it helps them maintain their market position. Every technology domain requires well-delivered managed services. The services offered must fit perfectly into clients' environments. Technical expertise, service consistency, and flexibility must be provided by vendors, regardless of the geographical location of clients. It has become difficult for companies to focus on core business processes and support various other functions, resulting in the increasing significance of managed services in the predictive analytics market. All the pre- and post-deployment questions and needs of clients are taken care of under the managed services unit.

TABLE 52 MANAGED SERVICES: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	132	153	184	218	257	277	16.1%
Europe	68	82	99	120	149	164	19.3%
APAC	39	50	63	80	109	125	26.1%
MEA	25	32	38	46	57	63	19.8%
Latin America	19	23	29	36	45	50	21.6%
Total	283	340	413	500	616	679	19.1%

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 53 MANAGED SERVICES: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	402	525	642	759	867	984	19.6%
Europe	243	321	400	494	618	722	24.4%
APAC	191	260	334	443	554	662	28.2%
MEA	93	125	156	194	235	275	24.1%
Latin America	76	101	129	162	199	234	25.1%
Total	1,005	1,332	1,660	2,052	2,473	2,876	23.4%

e: estimated; p: projected

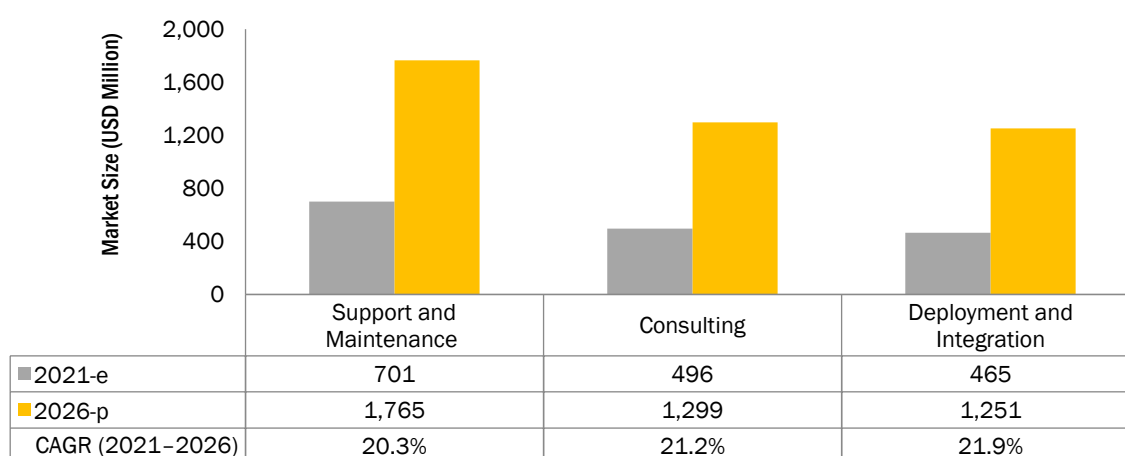
Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

6.3.2 PROFESSIONAL SERVICES

Professional services are categorized into three types: deployment and integration, support and maintenance, and consulting. These services are post-sales services, which are delivered to customers after the purchase of products. Customized, knowledge-based services offered to clients fall under professional services. Sometimes, new technology solutions have to be merged with the existing legacy IT infrastructures so that critical applications can be upgraded and better ways can be implemented to manage and store data. These services constitute an integral part in deploying solutions on board, imparting training, and handling and maintaining software solutions. Professional services include designing, planning, upgrading, and hosting consulting services that are offered to clients. Companies offering these services encompass consultants, solution experts, and dedicated project management teams, which specialize in the design and delivery of critical decision support software, tools, and services. The growth of professional services is mainly governed by the increasing complexity of operations and the growing deployment of BI solutions.

FIGURE 37 DEPLOYMENT AND INTEGRATION SEGMENT EXPECTED TO GROW AT THE HIGHEST CAGR DURING THE FORECAST PERIOD



e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 54 PROFESSIONAL SERVICES: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015-2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015-2020)
North America	250	282	327	381	440	467	13.3%
Europe	129	150	176	210	255	276	16.4%
APAC	75	91	112	140	187	211	23.1%
MEA	48	59	67	80	97	106	16.9%
Latin America	36	43	51	62	77	84	18.7%
Total	538	625	733	873	1,056	1,143	16.3%

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 55 PROFESSIONAL SERVICES: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	664	852	1,023	1,187	1,328	1,476	17.3%
Europe	401	522	638	772	946	1,083	22.0%
APAC	316	422	532	693	849	992	25.7%
MEA	155	202	249	303	360	412	21.7%
Latin America	126	164	205	254	305	352	22.7%
Total	1,662	2,162	2,646	3,210	3,788	4,314	21.0%

e: estimated; p: projected

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 56 PROFESSIONAL SERVICE: PREDICTIVE ANALYTICS MARKET SIZE, BY TYPE, 2015–2020 (USD MILLION)

Type	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Consulting	174	201	236	282	312	340	14.3%
Support and Maintenance	228	262	305	358	458	487	16.4%
Deployment and Integration	136	162	193	233	287	316	18.4%
Total	538	625	733	873	1,056	1,143	16.3%

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 57 PROFESSIONAL SERVICES: PREDICTIVE ANALYTICS MARKET SIZE, BY TYPE, 2021–2026 (USD MILLION)

Type	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Consulting	496	648	796	968	1,140	1,299	21.2%
Support and Maintenance	701	904	1,098	1,325	1,557	1,765	20.3%
Deployment and Integration	465	610	752	917	1,091	1,251	21.9%
Total	1,662	2,162	2,646	3,210	3,788	4,314	21.0%

e: estimated; p: projected

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

6.3.2.1 Support and Maintenance

Support and maintenance services comprise functional support, technical support, and database support that companies render to the end-customer while installing and maintaining the software. These services help companies understand the market trends, the changing business conditions, the clients' requirements or the service inconvenience, and aid in creating marketing and corporate branding campaigns. Support and maintenance services include the support provided during and post-installation and maintenance exercises. These services are fundamentally accommodated for software support, which can be offered from remote spots by means of web conferencing, phones, onsite help, and ticketing frameworks. Comprehensive training should incorporate all the requisites of theory and practice that guide the workforce in properly implementing the software.

TABLE 58 SUPPORT AND MAINTENANCE MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	106	118	136	156	191	199	13.4%
Europe	55	63	73	86	110	118	16.5%
APAC	32	38	47	57	81	90	23.2%
MEA	21	25	28	33	42	45	17.0%
Latin America	15	18	21	26	33	36	18.8%
Total	228	262	305	358	458	487	16.4%

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 59 SUPPORT AND MAINTENANCE MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	280	356	424	490	546	604	16.6%
Europe	169	218	265	319	389	443	21.2%
APAC	133	176	221	286	349	406	25.0%
MEA	65	84	103	125	148	169	20.9%
Latin America	53	69	85	105	125	144	22.0%
Total	701	904	1,098	1,325	1,557	1,765	20.3%

e: estimated; p: projected

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

6.3.2.2 Consulting

Consulting services are required by organizations for enhanced strategic outlook, improved performance efficiencies, and business operations transformation into more efficient and cost-effective operations. Some of the key features and benefits of consulting services are lowering risks, reducing complexities, and rising ROI, as they can be customized, easily applicable, and deliver maximum product assurance. Consulting services mainly steer around the critical issues and opportunities related to strategy, marketing, operations, technology, merger and acquisition, and corporate finance. These services handhold enterprises to increase effectiveness, boost performance, reduce costs, and enhance resilience. They are poised to grow at a steady speed as enterprises have started understanding the power of data analysis, which helps them to select appropriate technologies and vendors based on requirements and IT budget.

TABLE 60 CONSULTING MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	81	90	105	123	130	139	11.3%
Europe	42	48	57	68	75	82	14.4%
APAC	24	29	36	45	55	63	21.0%
MEA	16	19	21	26	29	31	14.9%
Latin America	12	14	16	20	23	25	16.7%
Total	174	201	236	282	312	340	14.3%

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 61 CONSULTING MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	198	255	308	358	400	444	17.5%
Europe	120	156	192	233	285	326	22.1%
APAC	94	126	160	209	255	299	25.9%
MEA	46	61	75	91	108	124	21.9%
Latin America	38	49	62	76	92	106	22.9%
Total	496	648	796	968	1,140	1,299	21.2%

e: estimated; p: projected

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

6.3.2.3 Deployment and integration

Predictive analytics solutions need to be integrated with the existing systems; therefore, they require the right level of connectors and back-end integration capabilities. System integration services begin with collecting customer requirements and then deploying, integrating, testing, and rolling out solutions. The deployment of predictive models provides options to deploy the analytical results for automating the decision-making process. It poses many challenges as organizations lack the integrated technical infrastructure for deploying solutions between various departments and business units. The validation and deployment of predictive models are time-consuming activities, which take months depending on business scenarios.

TABLE 62 DEPLOYMENT AND INTEGRATION: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	63	73	86	102	119	129	15.4%
Europe	33	39	46	56	69	76	18.6%
APAC	19	24	30	37	51	58	25.4%
MEA	12	15	18	21	26	29	19.1%
Latin America	9	11	13	17	21	23	20.9%
Total	136	162	193	233	287	316	18.4%

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 63 DEPLOYMENT AND INTEGRATION: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	186	240	291	339	383	428	18.2%
Europe	112	147	181	221	273	314	22.8%
APAC	88	119	151	198	244	288	26.7%
MEA	43	57	71	87	104	119	22.6%
Latin America	35	46	58	72	88	102	23.6%
Total	465	610	752	917	1,091	1,251	21.9%

e: estimated; p: projected

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

7 PREDICTIVE ANALYTICS MARKET, BY DEPLOYMENT MODE

KEY FINDINGS

- The on-premises segment is projected to grow from USD 4,795 million in 2021 to USD 12,027 million by 2026, at a CAGR of 20.2%, while the cloud segment is projected to grow at a higher CAGR of 23.7% during the forecast period.
- Highly secure data encryption and complete data visibility and control features are responsible for the high adoption of on-premise predictive analytics solutions.
- The high CAGR of the cloud segment can be attributed to the availability of easy deployment options, scalability, and minimal requirements of capital and time. These factors support the current lockdown scenario of COVID-19 as social distancing and lack of workforce impact the verticals. The public cloud segment has the highest adoption, as it offers benefits such as scalability and cost-effectiveness. These benefits are expected to be instrumental in propelling the growth of the overall predictive analytics market.

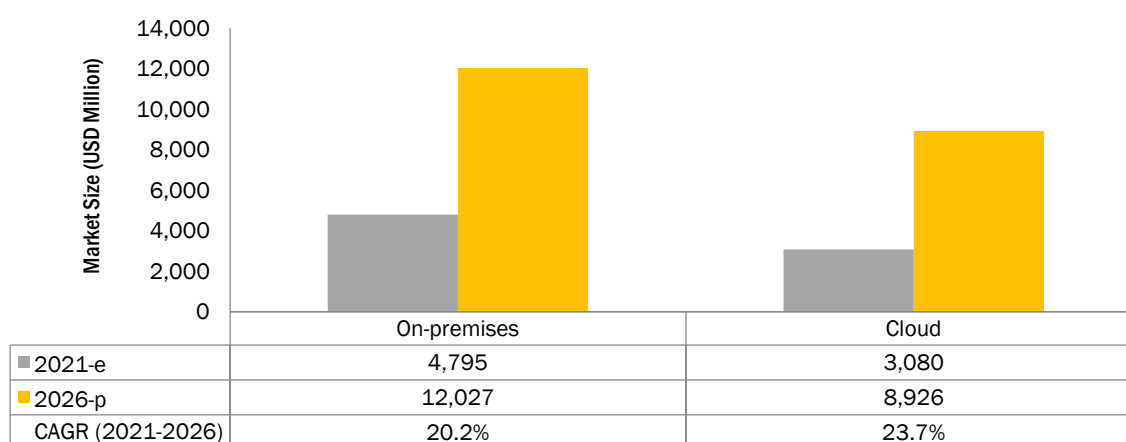
7.1 INTRODUCTION

The deployment modes in the predictive analytics market include on-premises and cloud. Predictive analytics solutions are deployed in various enterprises. Enterprises adopt cloud models due to inadequate resources. These models enable organizations to leverage the benefits of real-time predictive analytics. Despite high latency, cloud deployment models are preferred by verticals to maintain and predict future outcomes. The cloud segment is projected to grow in the near future.

7.1.1 DEPLOYMENT MODES: COVID-19 IMPACT

The COVID-19 pandemic has thrown everything into disarray on a massive scale. The epidemic, on the other hand, has turned out to be a boon for the cloud computing industry. Cloud platforms can assist in quickly launch new digital customer experiences and provide analytics at a lower cost than traditional technology platforms. When governments worldwide instituted a lockdown to contain the virus in early 2020, businesses were forced to relocate to a remote working environment from their traditional brick-and-mortar setup to maintain business continuity. As a result of this quick transformation, cloud-based solutions are in high demand. Today, cloud-based analytics platforms are used by 47% of businesses, according to an industry expert. COVID-19 has influenced the move to hybrid cloud systems, not just because customers demand it but also because the previous solutions (totally cloud-based or entirely on-premises) were failing. The COVID-19 outbreak is forcing numerous companies to review their IT strategies as a top priority. Several organizations emphasize the importance of having the right cloud-based collaboration tools, enabling employees to work remotely to limit the pandemic impact on productivity.

FIGURE 38 CLOUD SEGMENT TO ACCOUNT FOR A LARGER MARKET SIZE DURING THE FORECAST PERIOD



e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 64 PREDICTIVE ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2015–2020 (USD MILLION)

Deployment Mode	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
On-premises	1,606	1,868	2,197	2,595	3,112	3,331	15.7%
Cloud	857	1,019	1,222	1,492	1,852	2,063	19.2%
Total	2,463	2,887	3,420	4,087	4,964	5,394	17.0%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 65 PREDICTIVE ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2021–2026 (USD MILLION)

Deployment Mode	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
On-premises	4,795	6,159	7,563	9,094	10,646	12,027	20.2%
Cloud	3,080	4,128	5,085	6,319	7,646	8,926	23.7%
Total	7,876	10,288	12,648	15,413	18,293	20,953	21.6%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

The table above highlights the growth of the predictive analytics market by deployment mode. The market size of the cloud deployment mode is estimated to be the largest and projected to have the highest CAGR during the forecast period. The cloud-based deployment helps businesses more efficiently process and report data findings, enhance collaboration, and enable decision-makers to get faster access to business intelligence leading to its higher adoption in the predictive analytics market.

7.2 CLOUD

7.2.1 REDUCED OPERATIONAL COST AND HIGHER SCALABILITY TO ENABLE GROWTH IN CLOUD-BASED DEPLOYMENTS

The cloud deployment mode provides multiple benefits, such as reduced operational costs, simple deployments, and higher scalability. Due to this, the cloud deployment of predictive analytics solutions is projected to grow with the increasing awareness of the benefits of cloud-based solutions. Predictive analytics solution providers focus on the development of robust cloud-based predictive solutions for their clients, as many organizations have migrated to either private or public cloud. Additionally, providers such as IBM and Qual Metrix focus on the development of cloud software that would incorporate on-premise applications in the cloud. The cloud technology provides additional flexibility for business operations and real-time deployment ease to companies that implement predictive analytics. Cloud solutions make it easy for users to apply predictive capabilities to organizations and help maintain a competitive edge by eliminating the administrative roadblocks of supporting infrastructure. They also allow organizations to focus on improving their competencies. Predictive cloud solutions are also significantly gaining traction due to their cost-effective benefits and global availability.

TABLE 66 CLOUD: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	399	459	544	650	771	842	16.1%
Europe	206	245	294	359	447	498	19.3%
APAC	119	149	187	239	329	381	26.2%
MEA	77	97	111	136	170	191	19.9%
Latin America	57	69	86	107	135	151	21.7%
Total	857	1,019	1,222	1,492	1,852	2,063	19.2%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 67 CLOUD: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	1,231	1,627	1,965	2,338	2,681	3,053	19.9%
Europe	744	997	1,226	1,520	1,910	2,240	24.7%
APAC	585	805	1,022	1,365	1,713	2,053	28.5%
MEA	286	386	478	597	726	852	24.4%
Latin America	234	314	394	499	616	727	25.5%
Total	3,080	4,128	5,085	6,319	7,646	8,926	23.7%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

7.3 ON-PREMISES

7.3.1 WITH THE GROWING THREAT OF DATA THEFTS, SOME ORGANIZATIONS PREFER PREDICTIVE ANALYTICS SOLUTIONS TO REMAIN ON-PREMISES

Data-sensitive enterprises prefer the on-premises deployment model. Another important factor is the real-time availability of data for extracting insights; hence, despite the speedy developments of cloud-based solutions, predictive analytics solutions are expected to remain on-premises in some regions. However, the increasing dominance of the cloud in the coming years and the rising cost of support and maintenance of on-premises solutions are expected to drive the adoption of hosted solutions. The on-premises deployment of predictive analytics solutions provides organizations with significant improvements for data validation, standardization, and robust matching as done locally with monitoring. However, the emergence of predictive cloud solutions has impacted the growth of on-premise solutions. This can be validated by the high growth of the cloud segment during the forecast period.

Despite the security and confidentiality factor, organizations prefer on-premises predictive analytics solutions. Hence, many vendors continue to invest in the on-premises deployment model along with developing new cloud business models for predictive analytics solutions.

TABLE 68 ON-PREMISES: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	747	842	979	1,131	1,296	1,360	12.7%
Europe	385	449	529	625	751	804	15.8%
APAC	223	273	336	416	552	615	22.5%
MEA	145	177	200	237	286	308	16.3%
Latin America	106	127	154	185	227	244	18.1%
Total	1,606	1,868	2,197	2,595	3,112	3,331	15.7%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 69 ON-PREMISES: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	1,916	2,427	2,923	3,364	3,733	4,113	16.5%
Europe	1,158	1,487	1,823	2,187	2,660	3,019	21.1%
APAC	911	1,201	1,520	1,964	2,385	2,766	24.9%
MEA	446	576	711	859	1,011	1,149	20.8%
Latin America	364	469	586	718	857	980	21.9%
Total	4,795	6,159	7,563	9,094	10,646	12,027	20.2%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

8 PREDICTIVE ANALYTICS MARKET, BY ORGANIZATION SIZE

KEY FINDINGS

- The market size of the large enterprises segment is projected to grow from USD 7,117 million in 2021 to USD 17,900 million by 2026, at a CAGR of 20.3%, while the SMEs segment is projected to grow at a higher CAGR of 24.5% during the forecast period.
- Large enterprises have already started the implementation of predictive analytics solutions and are keen to implement advanced technologies to monitor and detect probable failures beforehand to avoid unexpected delays in operations. Predictive analytics solutions help businesses achieve better results by leveraging clients' historical enterprise data and predicting the impacts of customer engagement.
- The growing market share of SMEs across verticals, such as healthcare and life sciences, manufacturing, retail and eCommerce, BFSI, and energy and utilities, and their increasing focus on adopting technologically advanced solutions and services to get incredibly detailed insights into all aspects of operations would drive the growth of the predictive analytics market.

8.1 INTRODUCTION

The predictive analytics market has been segmented by organization size into large enterprises and SMEs. The study of SMEs and large enterprises is important as it has a significant impact on the adoption of predictive analytics solutions across regions. The market share of large enterprises is higher; however, the market for SMEs is expected to register a higher CAGR during the forecast period.

The organization size is identified based on the number of employees working in an organization as per global standards given below:

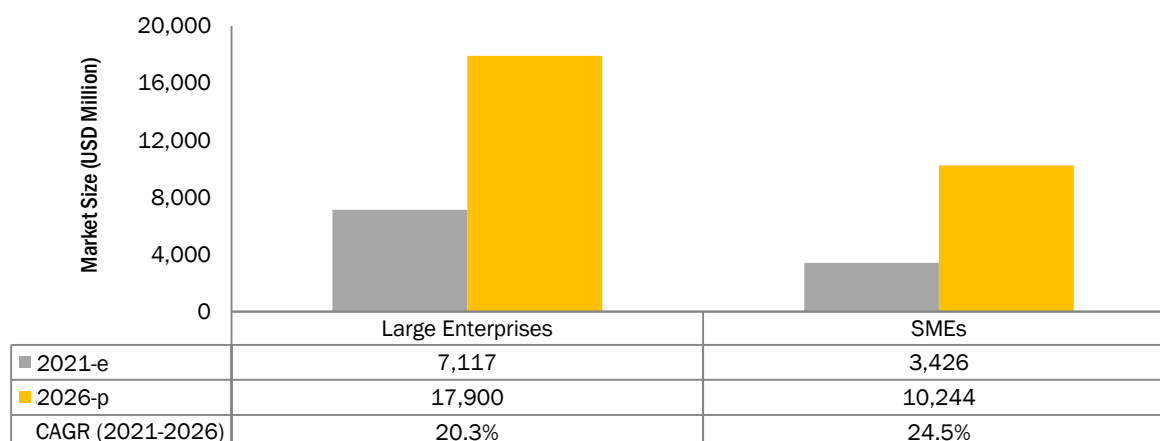
- SMEs: Less than 1,000 employees
- Large enterprises: More than 1,000 employees

8.1.1 ORGANIZATION SIZE: COVID-19 IMPACT

In the new normal, analytics may help organizations improve customer interaction, increase sales revenue, save costs, and improve compliance. While most businesses recognize the importance of analytics in key decision-making, they frequently face a challenge in deciding whether to establish an analytics team in-house or outsource some of their analytics work. It can be difficult and time-consuming to put together a fully effective analytics team within a corporation. To acquire faster and more effective insights, organizations should consider cooperating with a specialist managed analytics services provider. Hospitals worldwide have implemented AI to support medical professionals and treat infected patients to effectively monitor and manage the COVID-19 pandemic. Some AI-powered solutions such as camera-embedded facial scanners have been deployed that screen hospital visitors and analyze facial attributes and thermal scans to determine a visitor's fever conditions. Chinese technologists developed a no-contact infrared sensor system to distinguish individuals with fever, even in masses.

The COVID-19 pandemic has hit global trade at an unprecedented speed. Multinational companies were witnessing initial supply shocks; then, they started facing demand shocks, as many countries ordered people to stay at home. Government, businesses, and individual customers were struggling to procure basic products and materials. According to the data released by TradeShift, domestic and international trade transactions have suffered a drop of 56% since the mid of February. The US, UK, and Europe faced a combined initial drop of 26% at the beginning of April. As stated in the Price Waterhouse Cooper (PwC) April CFO Pulse survey, 77% of CFOs implement cost-containment measures. According to State of BI & Analytics Report 2020 - Special COVID-19 Edition published by Sisense in April 2020, SMEs shifted to adopt new use cases for their data in response to the crisis. They lead in the use of analytics across departments, with 68% of small businesses use analytics in operations, 56% in finance, 50% in sales, and 45% in the product. Salesforce has developed the Emergency Response Management for Public Sector solution that helps increase patient access to care with digital self-service, deliver smart, personalized patient care during crisis situations, and understand the risk of infection to people at various locations through manual contact tracing. Accenture has collaborated with SAP to develop the supply chain scenario planning-as-a-service that uses SAP Integrated Business Planning (IBP) and Accenture services to stabilize and build a stronger and more resilient supply chain. This service would help meet demands and address supply chain challenges.

FIGURE 39 SMALL AND MEDIUM-SIZED ENTERPRISES SEGMENT TO GROW AT A HIGHER CAGR DURING THE FORECAST PERIOD



e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 70 PREDICTIVE ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2015–2020 (USD MILLION)

Organization size	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Large Enterprises	2,365	2,742	3,213	3,794	4,560	4,910	15.7%
SMEs	920	1,109	1,353	1,666	2,077	2,306	20.2%
Total	3,285	3,851	4,566	5,460	6,637	7,216	17.0%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 71 PREDICTIVE ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2021–2026 (USD MILLION)

Organization size	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Large Enterprises	7,117	9,202	11,193	13,480	15,813	17,900	20.3%
SMEs	3,426	4,579	5,761	7,195	8,741	10,244	24.5%
Total	10,543	13,781	16,954	20,675	24,554	28,144	21.7%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

8.2 LARGE ENTERPRISES

8.2.1 EVER-INCREASING ADOPTION OF THE CLOUD AND LATEST TECHNOLOGIES TO DRIVE PREDICTIVE ANALYTICS ACROSS LARGE ENTERPRISES

The large enterprises considered in the report are organizations with an employee size of more than or equal to 1,000. The adoption of predictive analytics solutions and services among large enterprises is high due to the ever-increasing adoption of the cloud, and the trend is expected to continue during the forecast period. Large enterprises are heavily investing in advanced technology to increase the company's overall productivity and efficiency. Large enterprises accumulate huge chunks of data that can be attributed to the widespread client base. In large enterprises, data plays a major role in evaluating the overall performance of organizations. Good quality data has always been a key enabler for organizations to focus on enhancing their core competencies, ultimately leading them to have a competitive edge over their competitors. Predictive analytics carries extreme potential for large organizations as poor performances may cast them with the loss of clients, ultimately affecting their brand name. Moreover, large organizations are obligated to adhere to stringent compliance regulations such as General Data Protection Regulation (GDPR) and other banking-related regulations that include Basel III and Dodd-Frank Wall Street Reform and Consumer Protection Act; these regulations drive the adoption of predictive analytics among large enterprises.

TABLE 72 LARGE ENTERPRISES: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	1,100	1,236	1,431	1,654	1,899	2,005	12.8%
Europe	568	659	773	914	1,100	1,185	15.9%
APAC	328	400	492	608	809	906	22.5%
MEA	213	261	292	347	419	454	16.4%
Latin America	156	187	225	271	332	359	18.2%
Total	2,365	2,742	3,213	3,794	4,560	4,910	15.7%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 73 LARGE ENTERPRISES: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	2,843	3,626	4,326	4,987	5,545	6,122	16.6%
Europe	1,719	2,222	2,698	3,242	3,951	4,493	21.2%
APAC	1,352	1,794	2,250	2,912	3,542	4,117	24.9%
MEA	662	860	1,052	1,274	1,502	1,709	20.9%
Latin America	541	700	867	1,065	1,273	1,459	21.9%
Total	7,117	9,202	11,193	13,480	15,813	17,900	20.3%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

8.3 SMALL AND MEDIUM-SIZED ENTERPRISES

8.3.1 ROBUST CLOUD-BASED DEPLOYMENTS TO LEAD SMALL AND MEDIUM-SIZED ENTERPRISES, RECORDING A HIGHER GROWTH RATE

SMEs considered in the report are organizations with an employee size of 1–999. The management of SMEs is concerned related to the expenditures of deploying on-premises predictive analytics solutions and contracting skilled staff and IT teams for their maintenance. SMEs are leaning toward cloud-based solutions to overcome these issues and can effectively lower their operational costs by implementing analytics infrastructure in the cloud, a major factor driving the adoption of predictive analytics among SMEs. Predictive analytics players are making use of the cloud deployment model for either introducing dedicated cloud platforms or associating it with other cloud service organizations to provide cost-effective predictive analytics solutions to SMEs. Companies such as Oracle, IBM, and SAP are incorporating predictive analytics into their cloud offerings for automating business processes and operations across supply chains to enhance customer experiences. SMEs are making use of such offerings to improve their business outcomes. Predictive analytics finds its applications across many other verticals as data has emerged as an important strategic asset for businesses development. With the arrival of in-memory analytics and advanced visualization techniques, organizations can uncover the true potential of their gathered data.

TABLE 74 SMALL AND MEDIUM-SIZED ENTERPRISES: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	428	500	603	726	865	942	17.1%
Europe	221	266	326	401	501	557	20.3%
APAC	128	162	207	267	368	425	27.2%
MEA	83	105	123	152	191	213	20.8%
Latin America	61	75	95	119	151	169	22.7%
Total	920	1,109	1,353	1,666	2,077	2,306	20.2%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 75 SMALL AND MEDIUM-SIZED ENTERPRISES: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	1,369	1,804	2,227	2,662	3,065	3,504	20.7%
Europe	827	1,105	1,388	1,731	2,184	2,571	25.5%
APAC	651	893	1,158	1,554	1,958	2,356	29.3%
MEA	319	428	542	680	830	978	25.2%
Latin America	260	348	446	568	704	835	26.2%
Total	3,426	4,579	5,761	7,195	8,741	10,244	24.5%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

9 PREDICTIVE ANALYTICS MARKET, BY VERTICAL

KEY FINDINGS

- The BFSI vertical is projected to grow from USD 1,579 million in 2021 to USD 3,672 million by 2026, at a CAGR of 18.4% during the forecast period.
- The growth of the BFSI vertical can be attributed to the dynamic and competitive nature of the BFSI vertical, characterized by changing consumer demands, evolving macroeconomic landscapes, and evolving regulatory frameworks. As the BFSI vertical is shifting toward digitization, the number of cybercrimes is also increasing. Predictive analytics solutions can enable the vertical to reduce the number of frauds using advanced algorithms and big data tools to detect possible frauds and take preventive actions.
- During the COVID-19 pandemic, the BFSI vertical is using analytics solutions for numerous applications, such as predictive modeling, customer retention and fraud detection, and recommendation engine. Among these applications, predictive modeling would be used massively by BFSI organizations during the COVID-19 crisis as these organizations need to deal with critical and sensitive data and minimize risk and account management costs. Predictive modeling would gain more popularity after the pandemic for trading insights; organizations and individuals would use it to identify the most profitable deal in buying and selling stocks.
- The healthcare and life sciences vertical is projected to grow at the CAGR of 24.0% during the forecast period. During the COVID-19 pandemic, researchers and hospitals are leveraging predictive models to gain insights into COVID-19 risks, disease outcomes, and the virus' potential impact on resources. Predictive analytics has enabled hospitals and health systems to optimize care delivery for COVID-19 and other patients. It has helped them forecast patient volume, bed capacity, ventilator availability, and other metrics.

9.1 INTRODUCTION

Decisions can no longer be made on the basis of historical data or data within companies' repositories. They are made by understanding and analyzing the in-motion data, which is significantly important for predicting future events and trends. In today's challenging market conditions, it is important for organizations across verticals to have forward-looking analytics solutions and services, such as predictive analytics, for better decision-making. Industries are emphasizing encouraging the data-driven culture for decision-making. With the explosion of data across verticals, the need to have a robust predictive analytics tool has emerged out strongly. However, most organizations are flooded with data from every transaction and interaction. Predictive analytics can enable businesses to make quicker but fact-based decisions to enhance customer satisfaction and engage customers for a longer period.

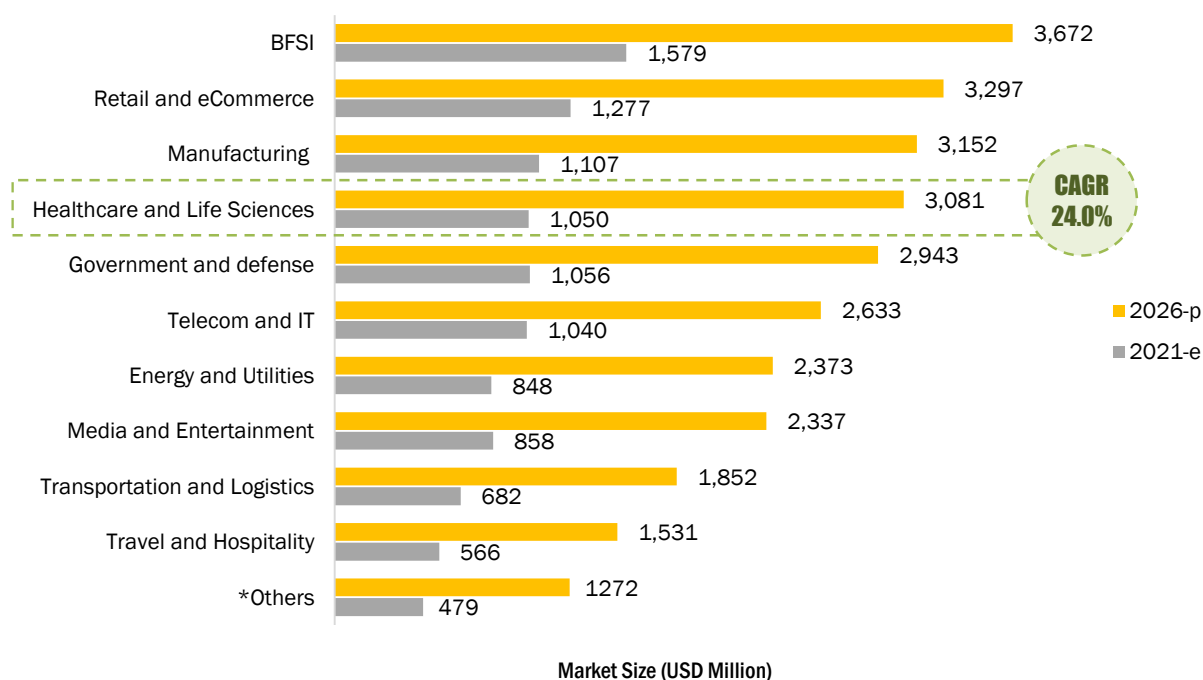
Various instances in the chapter clarify the importance of predictive analytics in various verticals. Moreover, predictive analytics would be embraced by all the verticals to cater to the basic need to optimize and improve business performance. The chapter covers the detailed analysis of the predictive analytics market by vertical. The predictive analytics market by vertical is segmented into BFSI, retail and eCommerce, manufacturing, government and defense, healthcare and life sciences, energy and utilities, transportation and logistics, and telecommunications and IT, media and entertainment, travel and hospitality, and others, which include, research and education and real estate.

9.1.1 VERTICALS: COVID-19 IMPACT

- Accenture and Microsoft worked in tandem with the Government of India's Digital India Corporation (MyGov) to equip its 1.3 billion citizens with a Responsible AI tool that provides accurate and useful information on the COVID-19 pandemic. Harnessing the capabilities of Microsoft's Power Virtual Agent low-code solution, the team moved fast and used Microsoft Azure to deploy the solution in days. The agent, referred to as MyGov Saathi, can handle up to 300,000 users per day and 20,000 concurrent users per minute. Several companies, such as Qure.ai, a Mumbai-based start-up, and TCS have used AI in a chest X-ray to diagnose COVID-19. The AI developed by Qure.ai also helps in identifying the extent of infection affecting the lungs.
- There has been drastic growth in the demand for robotic and remote-control applications. The AR/VR technology would also have a huge impact on remote work, entertainment, and services due to COVID-19. Facebook has launched the Shops feature that enables businesses to display and sell their products on Facebook as the COVID-19 pandemic has forced several brick and mortar stores to shut down. Facebook enables creating digital storefronts for free so that customers can directly buy products from Facebook or retailers' website.
- To address the COVID-19 impact on the BFSI vertical, the adoption of digital technology enablers has become a must. These technology enablers include video banking facilities, AI-supported tools, and conversational platforms for dealing with the surging call volumes and business process automation and ensuring the availability of digital banking services.
- Universities and colleges are shifting toward the blended learning model, wherein face-to-face delivery, along with an online model, would become a new norm.
- Since most of the insurers and agents, brokers, and financial advisors are working from home, they would face risk management and logistics challenges as COVID-19 would disrupt their client servicing.
- According to Perusahaan Listrik Negara (PLN), the state-owned power generation and transmission firm, the consumption of coal by power firms in Indonesia is expected to be 95.6 million tons in 2020, which is less than 97 million tons recorded in 2019.

- According to the FICCI-EY study on the economic impact of the pandemic on the healthcare vertical conducted in June 2020, the COVID-19 outbreak had resulted in a 70-80 % drop in footfalls and test volumes and a 50-70 % drop in revenue at the end of March. The study also revealed that the vertical is expected to witness short-term operating losses to the tune of Rs 14,000 to 24,000 crore for a quarter. The onset of such losses has caused cash balances to slump, even for large corporate hospitals. Many small hospitals and nursing homes in tier II and III cities have already shut down their facilities due to liquidity challenges.
- Currently, the healthcare segment across the globe is struggling to deal with the imbalance in supply and demand as global medical supply chains are in a weak state due to the shutdown of manufacturing plants. However, as the healthcare spending for COVID-19 is increasing exponentially, research companies that are producing test kits, developing vaccines, and manufacturing medical equipment would be benefitted the most.
- The current situation is influencing many sectors across the world. As a result, businesses are focusing on implementing their business continuity plans, figuring out how to effectively serve their customers, and ensuring their personnel is safe throughout this crisis. The virus is causing widespread disruption in business and consumer behavior.

FIGURE 40 HEALTHCARE AND LIFE SCIENCES VERTICAL TO REGISTER THE HIGHEST CAGR DURING THE FORECAST PERIOD



*Other verticals include research & education and real estate.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 76 PREDICTIVE ANALYTICS MARKET SIZE, BY VERTICAL, 2015–2020 (USD MILLION)

Vertical	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
BFSI	578	663	765	889	1,050	1,110	13.9%
Retail and eCommerce	422	484	569	676	815	879	15.8%
Telecom and IT	361	417	484	568	677	722	14.8%
Healthcare and Life Sciences	279	337	411	505	630	704	20.3%
Government and Defense	306	364	438	531	654	719	18.6%
Manufacturing	328	389	466	562	690	750	18.0%
Media and Entertainment	263	309	368	442	538	587	17.4%
Energy and Utilities	238	284	343	418	518	574	19.3%
Travel and Hospitality	164	195	234	283	347	383	18.4%
Transportation and Logistics	197	235	281	339	415	461	18.5%
*Other Verticals	148	175	208	248	302	328	17.3%
Total	3,285	3,851	4,566	5,460	6,637	7,216	17.0%

*Other verticals include research & education and real estate.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 77 PREDICTIVE ANALYTICS MARKET SIZE, BY VERTICAL, 2021–2026 (USD MILLION)

Vertical	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
BFSI	1,579	2,012	2,441	2,896	3,345	3,672	18.4%
Retail and eCommerce	1,277	1,659	2,020	2,450	2,893	3,297	20.9%
Telecom and IT	1,040	1,343	1,635	1,976	2,327	2,633	20.4%
Healthcare and Life Sciences	1,050	1,403	1,761	2,187	2,647	3,081	24.0%
Government and Defense	1,056	1,397	1,713	2,113	2,536	2,943	22.7%
Manufacturing	1,107	1,461	1,814	2,233	2,676	3,152	23.3%
Media and Entertainment	858	1,128	1,391	1,699	2,021	2,337	22.2%
Energy and Utilities	848	1,122	1,393	1,713	2,049	2,373	22.8%
Travel and Hospitality	566	738	913	1,118	1,333	1,531	22.0%
Transportation and Logistics	682	893	1,105	1,355	1,617	1,852	22.1%
*Other Verticals	479	625	768	936	1,111	1,272	21.6%
Total	10,543	13,781	16,954	20,675	24,554	28,144	21.7%

e: estimated; p: projected

*Other verticals include research & education and real estate.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

9.2 BANKING, FINANCIAL SERVICES, AND INSURANCE

9.2.1 PREDICTIVE ANALYTICS SOLUTIONS IMPROVE BUSINESS PERFORMANCE, REDUCE COST, AND CHURNING

The banking and financial sector is one of the highly competitive verticals. It is constantly under pressure to reduce costs and improve its profit margin. With the increasing competition and technology changes, the banking and financial sector is primarily working toward enhancing its strategies for predictive analytics solutions. The sector is flooded with large volumes of internal and external data and is adopting analytics solutions to mitigate different types of risks. Predictive analytics solutions can help automate various processes, such as lending and compliance management, for banking and financial service providers. Predictive analytics solutions are becoming crucial to proactively analyze risk patterns, thereby taking actions to mitigate them. They enable account managers and product leaders to obtain a complete view of customer accounts, including personal and commercial accounts, to better assess the risks associated with individual customer activities and portfolios. Analytics is applied in all areas of insurance organizations. They are used in various areas such as claims, marketing, customer experience, finance, and human resources.

BFSIs are using real-time analysis to predict which customer segments are most inclined to acquire a particular product and at what point in the customer journey cross-selling is almost certain to occur. Customer churn can be avoided with the use of these insights. For example, a European bank employed a number of retention methods aimed at inactive clients to combat a dwindling customer base, but with little success. Then it moved on to ML algorithms that are deployed in predictive analytics to anticipate which clients who are now engaged are essential to minimize its business with the bank. As a result of this new understanding, a targeted campaign was launched, which reduced churn by 15%. Predictive analytics helps insurers in evaluating and assessing the risks associated with an insurance policy. Banking and financial service providers are expected to adopt predictive analytics solutions to improve business performance, optimize risk-return outcomes, and reduce compliance costs. The vertical also needs to cascade complex control requirements and effectively handle costs across supply and services networks. Organizations in the BFSI vertical are running the risk of default in payment or repayment and volatility across the trading market. Risk analytics, financial analytics, and customer analytics are some of the major predictive analytics solutions that are utilized by companies operating in the BFSI vertical.

Additionally, banking and financial services organizations have to face frauds at different levels, which vary from purchases made by stolen credit cards to money laundering, first-party frauds, and insurance claims. Predictive analytics also complements organizations' current transaction monitoring systems by tracking frauds before they occur.

TABLE 78 BANKING, FINANCIAL SERVICES, AND INSURANCE: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	269	299	341	388	437	453	11.0%
Europe	139	159	184	214	253	268	14.1%
APAC	80	97	117	142	186	205	20.6%
MEA	52	63	70	81	97	103	14.6%
Latin America	38	45	54	64	77	81	16.3%
Total	578	663	765	889	1,050	1,110	13.9%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 79 **BANKING, FINANCIAL SERVICES, AND INSURANCE: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)**

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	631	793	943	1,071	1,173	1,256	14.8%
Europe	381	486	588	696	836	922	19.3%
APAC	300	392	491	625	749	845	23.0%
MEA	147	188	229	274	318	351	19.0%
Latin America	120	153	189	229	269	299	20.1%
Total	1,579	2,012	2,441	2,896	3,345	3,672	18.4%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

9.3 RETAIL AND ECOMMERCE

9.3.1 ADOPTION OF PREDICTIVE ANALYTICS SOLUTIONS BOOSTS BUSINESS DECISION PERFORMANCE AND PROFIT MARGINS

The increasing digitalization and the growing use of connected devices have transformed the way the retail industry is operating. Predictive analytics can help the retail and eCommerce vertical increase its profitability by offering actionable insights related to changing customer behavior and patterns. Challenges associated with increased globalization, the complexity of supply chains, and the growing competitive environment have also fueled the demand for predictive analytics solutions. The growing complexity of regulatory and compliance requirements, such as the Consumer Product Safety Improvement Act (CPSIA) and Fair Packaging and Labeling Act (FPLA), is another driver for the adoption of predictive analytics solutions by the retail and eCommerce vertical. Hence, the sector can integrate predictive analytics solutions into its operations to cut costs, track regulatory requirements, and enhance profit margins. As the industry is turning towards a more digitized approach, retailers are implementing advanced analytics practices for addressing the major business challenges and maximizing benefits. Predictive models enable businesses to determine the intent of their customers. These models assist retailers in meeting market demands. Predictive analytics monitors historical product pricing, customer interests, competitor pricing, and inventory to deliver the best possible prices to gain maximum profits. For instance, in Amazon's marketplace, sellers that use algorithmic pricing are benefitted through better visibility, sales, and timely customer feedback. According to the retail intelligence company, Upstream Commerce, an automated predictive and dynamic pricing tool, delivers 20% additional net profit.

Predictive analytics is a key solution that helps retailers and eCommerce businesses build smarter market strategies and make faster decisions. It helps significantly enhance business operations, which include improved customer engagements, the identification of better target prospects, price optimization, and predictive inventory management. With the added ability to deliver more effective marketing, higher profits, and more controlled operations, predictive analytics has become a critical component to both eCommerce and retail businesses. Predictive analytics enables businesses to make better decisions and achieve better results in their daily operations. For example, a European firm changed its retailer-specific assortments and planograms by using predictive analytics to analyze consumer data to find which Stock Keeping Unit (SKUs) is sold well in certain retail formats and which SKUs to swap in and out to best serve customer preferences.

TABLE 80 RETAIL AND ECOMMERCE: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	196	218	253	294	339	359	12.8%
Europe	101	116	137	163	197	212	16.0%
APAC	59	71	87	108	145	162	22.6%
MEA	38	46	52	62	75	81	16.5%
Latin America	28	33	40	48	59	64	18.2%
Total	422	484	569	676	815	879	15.8%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 81 RETAIL AND ECOMMERCE: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	510	654	781	906	1,014	1,127	17.2%
Europe	308	401	487	589	723	827	21.8%
APAC	243	324	406	529	648	758	25.6%
MEA	119	155	190	232	275	315	21.5%
Latin America	97	126	157	194	233	269	22.6%
Total	1,277	1,659	2,020	2,450	2,893	3,297	20.9%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

9.4 TELECOM AND IT

9.4.1 USAGE OF SMARTPHONES, COMPLEXITY IN THE TELECOM INDUSTRY, AND GROWTH IN SERVICE PROVIDERS RAPIDLY DRIVE THE MARKET

The telecom and IT sector is highly competitive and continuously focuses on delivering enhanced services to acquire new customers and retain the existing ones. Factors such as the growing complexity of the telecom industry and the growth in the number of service providers have created the demand to adopt predictive analytics solutions. The fast use of smartphones and other connected mobile devices has increased the volume of data flowing across telecom networks. Operators must process, store, and extract information from the given data. Predictive analytics can help them maximize profits by optimizing network consumption, enhancing customer experience, and improving scalability.

The telecom and IT sector has been affected by digital disruption. The increasing use of connected devices, IoT, and implementation of 5G have forced service providers to focus on adopting innovative business models and develop new and innovative solutions to achieve competitive advantage. Telecommunication companies are thriving to be innovative and maximize their revenue by implementing the right tools at the right place to harness massive volume, type, format, and velocity of data to leverage actionable insights from the data. Telecommunication companies manage terabytes of data stored in silos and scattered across the organization. Thus, for faster and easier processing of useful data, telcos are searching for a data-driven advanced analytics solution to achieve accurate real-time insights through data mining and

predictive analytics. The telecommunications and IT vertical has witnessed rapid growth in the past decade, undergoing various transformations. With its large size, the industry exhibits complex contracts, interdependencies of services, disparate spending, and minimal resources and time.

The major advantages of predictive analytics to telecom companies are a rise in sales, improved risk management, a decrease in operational costs, and customer behavior analysis. Predictive analytics tools can help in identifying potential threats that are prevailing in the industry to avoid losses. Predictive analytics is utilized across all telecommunication enterprises to smoothen the supply chain, understand targets, and craft marketing campaigns. For instance, network optimization has become an essential part of the telecommunications industry. Predictive analytics facilitates better customer experiences. Optimizing a cellular network involves making numerous decisions supported by predictive analytics based on historical data analysis. This optimization helps make businesses stronger, prone to lower risks, and achieve better outcomes.

TABLE 82 TELECOM AND IT: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	168	188	216	248	282	295	11.9%
Europe	87	100	116	137	163	174	15.0%
APAC	50	61	74	91	120	133	21.6%
MEA	33	40	44	52	62	67	15.5%
Latin America	24	28	34	41	49	53	17.2%
Total	361	417	484	568	677	722	14.8%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 83 TELECOM AND IT: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	415	529	632	731	816	901	16.7%
Europe	251	324	394	475	581	661	21.4%
APAC	198	262	329	427	521	606	25.1%
MEA	97	126	154	187	221	251	21.1%
Latin America	79	102	127	156	187	215	22.1%
Total	1,040	1,343	1,635	1,976	2,327	2,633	20.4%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

9.5 TRANSPORTATION AND LOGISTICS

9.5.1 BUSINESS TRANSFORMATION AND REDUCED COSTS DRIVE GROWTH OF PREDICTIVE ANALYTICS

The transportation and logistics business is undergoing massive transformations. Digital transformations, changing consumer expectations, evolving company models, and changing customer expectations are all examples of this. The key to thriving is to fine-tune strategies to stay competitive in the face of market upheavals. Predictive analytics solutions are being used by agile firms in the sector to improve logistics management using data-backed insights. Predictive analytics in transportation and logistics can help suggest actions that can reduce costs and increase the chance to deliver freight on time. These recommendations may be a change in a lane, a mode of transportation, shipping dates, or even the type of vehicle used. This can save time and enable the organization to be nimbler to changing conditions. The utilization of predictive analysis in transport and logistics operations management can be useful to make business transformations, specifically in terms of cost-efficiency. For instance, road freight and transportation and vehicle fleets are outnumbered, and it is a difficult job to monitor and maintain these vehicles despite the necessity of constant maintenance. Advanced analytics instead of preventive maintenance enables companies to perform predictive maintenance. Predictive analytics solutions gather vehicle data from sensors, which are then examined to identify components that are most likely to break or underperform. This enables technicians to make necessary repair/replace decisions and avoid expensive post-damage repairs.

By linking historical data with consumer profiles, economic indicators, and geo-localized market data, logistics and transportation providers can forecast demand with increasing accuracy. This helps anticipate daily volumes, optimize delivery routes, and allocate resources to efficiently deliver services, ultimately enhancing customer satisfaction. For instance, ArcelorMittal, a global steelmaker that operates as two interdependent companies, ArcelorMittal Mines Canada and ArcelorMittal Infrastructure Canada, optimizes its supply-chain logistics by implementing predictive analytics. To remove logistics bottlenecks and improve resiliency, ArcelorMittal uses predictive analytics solutions to transform and correlate data from its distinct systems for creating intelligent insights for supporting both strategic decisions and everyday business decisions to augment the use of the company's equipment and transportation infrastructure.

TABLE 84 TRANSPORTATION AND LOGISTICS: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	92	106	125	148	173	188	15.5%
Europe	47	56	68	82	100	111	18.7%
APAC	27	34	43	54	74	85	25.5%
MEA	18	22	26	31	38	43	19.2%
Latin America	13	16	20	24	30	34	21.0%
Total	197	235	281	339	415	461	18.5%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 85 TRANSPORTATION AND LOGISTICS: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	273	352	427	501	567	633	18.4%
Europe	165	215	266	326	404	465	23.1%
APAC	130	174	222	293	362	426	26.9%
MEA	63	83	104	128	154	177	22.8%
Latin America	52	68	86	107	130	151	23.8%
Total	682	893	1,105	1,355	1,617	1,852	22.1%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

9.6 MEDIA AND ENTERTAINMENT

9.6.1 RISING CONTENT CONSUMPTION AND DIGITAL ENTERTAINMENT ADOPTION TO GENERATE DEMAND FOR PREDICTIVE ANALYTICS

The current situation has thrown the world's economy into disarray, causing several businesses to change and become more focused on digital-first experiences. Online content consumption has almost doubled in the last year, and digital entertainment adoption has continued to accelerate throughout all age groups. In this new digital era, the only option for media and entertainment companies to expand their user base, retain consumers, and increase brand and product loyalty is to use data to influence their product and content strategy. Predictive analytics is required for media and entertainment products to understand audience preferences and behavior for modifying their content development strategy, efficiently employing different channels, and improving any proprietary channel. Predictive analytics solutions are being used by media and entertainment organizations to turn data into goods that their customers adore.

TABLE 86 MEDIA AND ENTERTAINMENT: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	122	139	164	193	224	240	14.4%
Europe	63	74	89	106	130	142	17.6%
APAC	36	45	56	71	95	108	24.3%
MEA	24	29	33	40	50	54	18.1%
Latin America	17	21	26	32	39	43	19.9%
Total	263	309	368	442	538	587	17.4%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 87 MEDIA AND ENTERTAINMENT: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	343	445	538	628	709	799	18.4%
Europe	207	272	335	409	505	587	23.1%
APAC	163	220	280	367	453	538	26.9%
MEA	80	106	131	161	192	223	22.8%
Latin America	65	86	108	134	163	190	23.9%
Total	858	1,128	1,391	1,699	2,021	2,337	22.2%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

9.7 MANUFACTURING

9.7.1 ADOPTION OF PREDICTIVE ANALYTICS BOOSTS PERFORMANCE AND IMPROVES DECISION-MAKING

Predictive analytics is a fundamental part of the manufacturing sector. Several business trends, including globalization, vertical integration, and margin pressure, are prompting manufacturers to look for new and novel methods to use predictive analytics to improve decision-making and performance. The manufacturing vertical has taken a long time to embrace analytics. Applying analytics to a wide variety of business domains, from finance and supply chain to sales and marketing and product creation, is currently affecting other sectors. Manufacturing companies depend on a wide range of external suppliers that provide materials, technology, data, consulting, and professional services. Since products are often at the hub of a manufacturing company, the first phase of analytics is frequently focused on improving product development. Common focus areas include supply chain optimization, sales and marketing budget management, warranty cost reduction, and general financial management. Innovative insights in these areas can have a significant influence on business performance and can provide a significant return on an organization's analytics investment. Analytics can offer new business models based on the sale of services rather than merely products. Predictive analytics plays an important part in manufacturing, planning, warranty analysis, and supply chain intelligence, among others. Customers demand quality products at lower prices with faster delivery; hence, it is vital for manufacturers to deliver a differentiated experience throughout the entire customer life cycle. Through the implementation of predictive analytics, manufacturers can predict the tendency of prospects to purchase and the level to which they would accept given offers. By analyzing group behavior, manufacturers can plan logistics and operations that are based on the demand for a particular type of product in a region.

With the advent of technologies, such as the Industrial Internet of Things (IIoT), predictive analytics is becoming more intelligent and empowering the manufacturing vertical. Along with the rising demand, manufactures are also using the capabilities of predictive analytics to enhance their production abilities. Countries such as the US and China have used predictive analytics to improve manufacturing at a global scale. There will be prominent use of predictive analytics solutions in the manufacturing vertical for predictive maintenance, asset management, and remote monitoring applications. The adoption of predictive analytics would help IT departments improve individual operations and manage a huge volume of databases across entire organizations. As companies are becoming more versatile, IT departments are facing challenges in the management of assets. This increases the responsibility of managing assets, which drives the IT industry to adopt predictive analytics solutions

TABLE 88 MANUFACTURING: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	153	175	207	245	288	306	14.9%
Europe	79	93	112	135	166	181	18.1%
APAC	46	57	71	90	122	138	24.9%
MEA	30	37	42	51	64	69	18.6%
Latin America	22	26	33	40	50	55	20.4%
Total	328	389	466	562	690	750	18.0%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 89 MANUFACTURING: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	442	576	701	826	938	1,078	19.5%
Europe	267	353	437	537	669	791	24.2%
APAC	210	285	365	482	600	725	28.1%
MEA	103	137	171	211	254	301	23.9%
Latin America	84	111	141	176	215	257	25.0%
Total	1,107	1,461	1,814	2,233	2,676	3,152	23.3%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

9.8 HEALTHCARE AND LIFE SCIENCES

9.8.1 REAL-TIME DECISION-MAKING TO PROVIDE NEW INNOVATIVE SOLUTIONS AND DELIVER PROPER INSIGHTS FOR PATIENTS

Pressures on healthcare and life sciences companies worldwide to cut costs, enhance coordination and outcomes, give more with less, and be more patient-centric are growing. This is driving the growth of the predictive analytics industry. Randomized clinical trials are costly to run and ineffective in detecting rare occurrences, treatment effects that are heterogeneous, or long-term outcomes. To find such connections, life sciences companies use predictive analytics in healthcare. Asserting causal relationships, on the other hand, can be challenging since data can be easily misread to make unrelated elements appear interdependent. Predictive analytics helps in decision-making by providing real-time information that can support decisions and deliver actionable insights for the vertical. It offers insights that can aid in enhancing the efforts put into patient care, enhance services, and improve existing procedures. Predictive analytics helps in providing new innovative solutions and treatments that can be properly tracked, measured, and analyzed in the vertical. The oldest group of organizations delivering predictive analytics services in the healthcare and life sciences industries are IBM and SAS. These companies, which were founded long before the dot-com bubble, have formed relationships with a huge number of Fortune 500 companies. They use these connections to provide predictive analytics services. Real-time Electronic Health Records (EHRs) data analytics has helped a hospital located in Texas, US, cut patient readmission by 5%, indicating

the importance of real-time analytics in the healthcare vertical. Predictive analytics solutions enable organizations to consolidate data at a centralized location, categorize it, and maintain it in easy-to-understand formats, resulting in enhanced customer experiences. “EHRs offer a treasure trove of data to support suicide risk detection,” says Kaiser Permanente. In a 2018 study conducted by Kaiser Permanente (KP) and Mental Health Research Network (MHRN), the combination of EHR data and a standard depression questionnaire accurately identified individuals who had attempted suicide. Using a predictive algorithm, the team found that suicide attempts and successes were 200 times more likely among the top one percent of patients flagged.

TABLE 90 HEALTHCARE AND LIFE SCIENCES: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	130	152	183	220	263	287	17.2%
Europe	67	81	99	122	152	170	20.4%
APAC	39	49	63	81	112	130	27.4%
MEA	25	32	37	46	58	65	21.0%
Latin America	18	23	29	36	46	52	22.8%
Total	279	337	411	505	630	704	20.3%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 91 HEALTHCARE AND LIFE SCIENCES: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	419	553	681	809	928	1,054	20.2%
Europe	254	339	424	526	661	773	25.0%
APAC	199	274	354	472	593	709	28.9%
MEA	98	131	166	207	251	294	24.7%
Latin America	80	107	136	173	213	251	25.8%
Total	1,050	1,403	1,761	2,187	2,647	3,081	24.0%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

9.9 TRAVEL AND HOSPITALITY

9.9.1 NEED TO DETERMINE NEW PROSPECTS AND OFFER STANDARD SERVICES FOR BETTER CUSTOMER ENGAGEMENT TO DRIVE PREDICTIVE ANALYTICS SOLUTIONS

Travel and hospitality is one of the most competitive verticals; the players operating in this vertical are maximizing the implementation of the efficient use of collected data for facilitating predictive analytics to determine new prospects and offer standard services for better customer engagement. The proper analysis of the data generated in this vertical (either through customers or traffic management) can offer significant insights into customer behavior, which can be used for better traffic management. Singapore, for instance, uses data on local traffic conditions in real-time to determine prices for road tolls. This provides an incentive for drivers to avoid driving during the most congested periods and optimizes the use of road networks.

TABLE 92 TRAVEL AND HOSPITALITY: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	76	88	104	123	145	156	15.4%
Europe	39	47	56	68	84	92	18.6%
APAC	23	28	36	45	62	71	25.4%
MEA	15	19	21	26	32	35	19.1%
Latin America	11	13	16	20	25	28	20.9%
Total	164	195	234	283	347	383	18.4%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 93 TRAVEL AND HOSPITALITY: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	226	291	353	414	467	523	18.3%
Europe	137	178	220	269	333	384	23.0%
APAC	108	144	183	241	298	352	26.8%
MEA	53	69	86	106	127	146	22.7%
Latin America	43	56	71	88	107	125	23.7%
Total	566	738	913	1,118	1,333	1,531	22.0%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

9.10 ENERGY AND UTILITIES

9.10.1 DEMAND FOR INTELLIGENT PRODUCTION, DISTRIBUTION, AND CONSUMPTION OF ENERGY AND UTILITIES DRIVE PREDICTIVE ANALYTICS SOLUTIONS

There is a good level of adoption of predictive analytics solutions in the energy and utilities vertical. Faced with fluctuating demand and the rising consumer interest in renewable energy, energy and utility firms are looking at alternative energy sources to distribute new services to customers and discover new revenue streams. Consumers now have more options than ever before, thanks to deregulation, which has forced providers to cut costs and prices effectively. Predictive analytics is important in this vertical, which enables intelligent production, distribution, and consumption of energy and utilities. Predictive analytics helps monitor vast volumes of real-time data to estimate the generation and consumption patterns, predict shortage in energy supply and future demand by studying patterns, and help optimize the energy supply and demand cycle. Predictive analytics makes it simple for energy analysts to prepare, combine, analyze data from various sources, and execute analytics. Factors such as changing political conditions and the economic environment create an opportunity for vendors to deliver predictive analytics solutions to help energy and utility service providers tackle challenges related to the internal and external environment. One of the major drivers to plan for the future is to recognize unseen patterns and trends to anticipate fluctuations. Predictive analytics can assist in finding both new and historical trends, which can be combined with other correlated factors to create a plan, helping organizations to become proactive to cope up with market demands and trends. Organizations operating in the energy and utilities vertical are required to forecast demand and load, respectively. In the case of utilities, it is mandatory to submit an accurate requirement to forecast load at regular intervals.

A huge chunk of data is generated from oil wells, utility grids, gas grids, smart grids, and other sensors. This terabyte of data from both structured and unstructured sources is needed to be analyzed and get real-time insights; hence, the vertical is in search of advanced analytical tools to get actionable insights. Considering the transformation in the energy and utilities vertical, there is a huge possibility of facing unprecedented challenges, such as the rising cost of operations, changing regulations, rising environmental concerns, and meeting the changing consumer expectations. Cutting excess costs and managing resources are some of the prominent factors driving the adoption of advanced tools in respective verticals. The deployment of technologies such as Advanced Metering Infrastructure (AMI) and Supervisory Control and Data Acquisition (SCADA) systems is also helping in improving the amount and quality of data that the utilities vertical supplies and distributes.

TABLE 94 ENERGY AND UTILITIES: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	111	128	153	182	216	234	16.2%
Europe	57	68	83	101	125	138	19.4%
APAC	33	41	53	67	92	106	26.3%
MEA	21	27	31	38	48	53	19.9%
Latin America	16	19	24	30	38	42	21.8%
Total	238	284	343	418	518	574	19.3%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 95 ENERGY AND UTILITIES: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	339	442	538	634	718	812	19.1%
Europe	205	271	336	412	512	596	23.8%
APAC	161	219	280	370	459	546	27.6%
MEA	79	105	131	162	195	227	23.5%
Latin America	64	85	108	135	165	193	24.6%
Total	848	1,122	1,393	1,713	2,049	2,373	22.8%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

9.11 GOVERNMENT AND DEFENSE

9.11.1 DEEP ANALYSIS AND FORECASTING OF TRENDS LEAD TO THE GROWTH OF PREDICTIVE ANALYTICS SOLUTIONS

With analytics solutions across the full range of government duties, including public safety, defense, national intelligence, social programs, cybersecurity, and finance and operations, predictive analytics enables governments throughout the world to care for and ensure citizens' safety. For the defense industry, predictive analytics provides a thorough analysis and forecasting of trends, firms, contracts, platforms, and semiconductor content. Predictive analytics is used by leading defense and intelligence agencies to explore high-level dangers and even anticipate future threats. It does not need defense businesses to break new ground and has been used to influence decision-making in other government areas, including the intelligence community. In the private sector, corporations such as Google, Amazon, and Facebook have aided the adoption of advanced predictive analytics by the general public. Using predictive and other advanced analytics to successfully drive budget decision-making can help defense organizations save money. Predictive analytics can be used to identify people, equipment, and maintenance reductions that will have the least impact on readiness. For instance, a company might wish to conserve money by postponing maintenance on a rarely flown plane.

TABLE 96 GOVERNMENT AND DEFENSE: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	142	164	195	231	272	293	15.6%
Europe	73	87	105	128	158	173	18.8%
APAC	42	53	67	85	116	133	25.6%
MEA	28	35	40	49	60	66	19.3%
Latin America	20	25	31	38	48	53	21.1%
Total	306	364	438	531	654	719	18.6%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 97 GOVERNMENT AND DEFENSE: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	422	550	662	782	889	1,007	19.0%
Europe	255	337	413	508	634	739	23.7%
APAC	201	272	344	457	568	677	27.5%
MEA	98	131	161	200	241	281	23.4%
Latin America	80	106	133	167	204	240	24.5%
Total	1,056	1,397	1,713	2,113	2,536	2,943	22.7%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

9.12 OTHER VERTICALS

Other verticals in the predictive analytics market include research, education, and real estate. Although these verticals lag in the adoption of predictive analytics, there lies a significant opportunity for these verticals to embrace advanced analytics technologies. The increasing competition due to rapid globalization is one of the factors driving the adoption of predictive analytics solutions in these verticals.

Based on the data presented, predictive analytics makes predictions. These forecasts can be prepared for educational institutions and organizations based on enrolment, facility usage, class involvement, and other factors. Educational institutions are under increased pressure to retain and develop students after shifting their focus from raising semester enrolment numbers to increasing the rate of successful student graduations. The pressure for students who join Higher Education to acquire a degree, as well as the students and their families' anxiety about successfully graduating owing to the financial commitment involved, have fueled this shift in attitude. With this in mind, predictive analytics can be utilized to give students more visibility into their educational achievement, allowing them to react more quickly to their progress.

According to one industry analyst, 2.5 quintillion bytes of data were generated per day in 2018. It is now significantly higher. The power of predictive analytics was unlocked as users moved away from traditional analytics methodologies and began to employ emerging technologies and new and unorthodox data sources. Both real estate agents and consumers gain from this new method of working. Because of its remarkable ability to give quick and precise actionable insights, predictive analytics is redefining the future of the real estate market.

This is largely due to the emergence of big data and the various ways in which customers may now tap into the hidden power of real estate data. Organizations expect predictive analytics to play an even bigger role in altering the real estate market in the future.

TABLE 98 OTHER VERTICALS: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015–2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
North America	69	79	93	108	126	134	14.3%
Europe	35	42	50	60	73	79	17.4%
APAC	21	26	32	40	54	61	24.2%
MEA	13	17	19	23	28	30	18.0%
Latin America	10	12	15	18	22	24	19.8%
Total	148	175	208	248	302	328	17.3%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 99 OTHER VERTICALS: PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
North America	191	246	297	346	390	435	17.9%
Europe	116	151	185	225	278	319	22.5%
APAC	91	122	154	202	249	293	26.3%
MEA	45	58	72	88	106	122	22.2%
Latin America	36	48	60	74	89	104	23.3%
Total	479	625	768	936	1,111	1,272	21.6%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

10 PREDICTIVE ANALYTICS MARKET, BY REGION

KEY FINDINGS

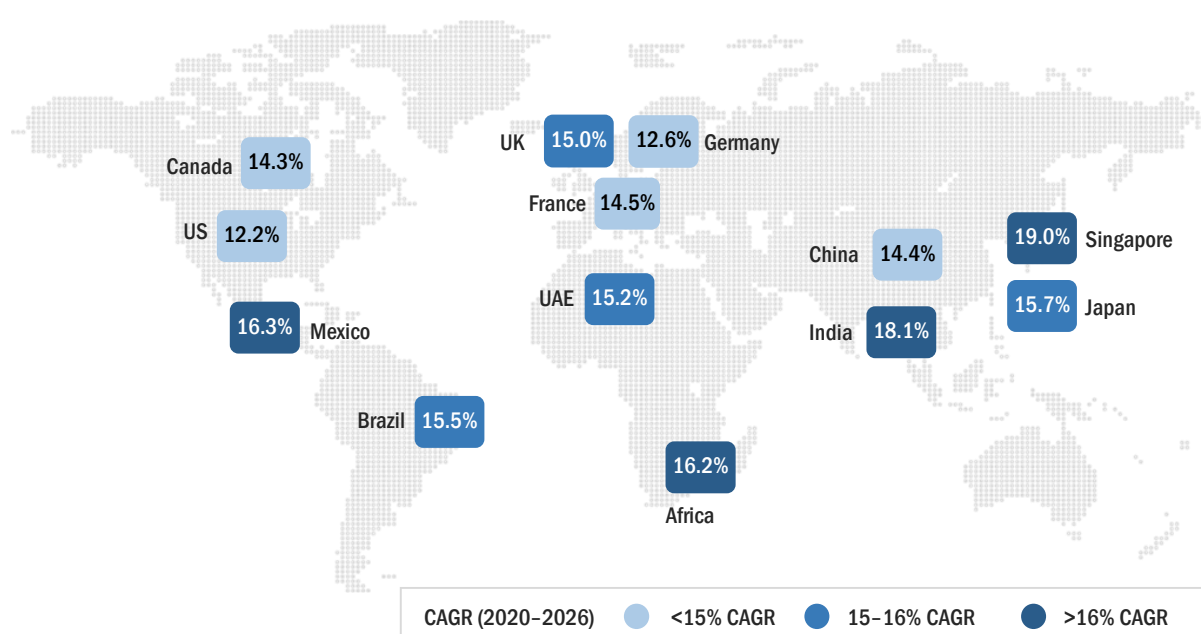
- North America is estimated to account for the highest share and is projected to grow from USD 4,212 million in 2021 to USD 9,625 million by 2026, at a CAGR of 18.0% during the forecast period. This can be attributed to the plummeting costs of predictive analytics solutions, the increasing amount of data across verticals, and rising spend by companies on adopting real-time analytics solutions.
- The increasing developments of real-time solutions by various vendors across the US and the growing demand for predictive analytics tools are expected to contribute to the growth of the US market.
- APAC is projected to grow at the highest CAGR of 26.4% during the forecast period. The increasing volumes of data, advancements in AI and big data technologies, growing concerns about data integrity, and rising demand for useful insights are major factors driving the predictive analytics market across APAC. Countries such as China, ANZ, and Japan are focusing on enhancing the management of data to enable data-based business decisions and enhance business processes.
- MEA is projected to grow at a CAGR of 22.3% due to various government initiatives in Saudi Arabia related to the adoption of predictive analytics technologies for moving toward digital transformation.

10.1 INTRODUCTION

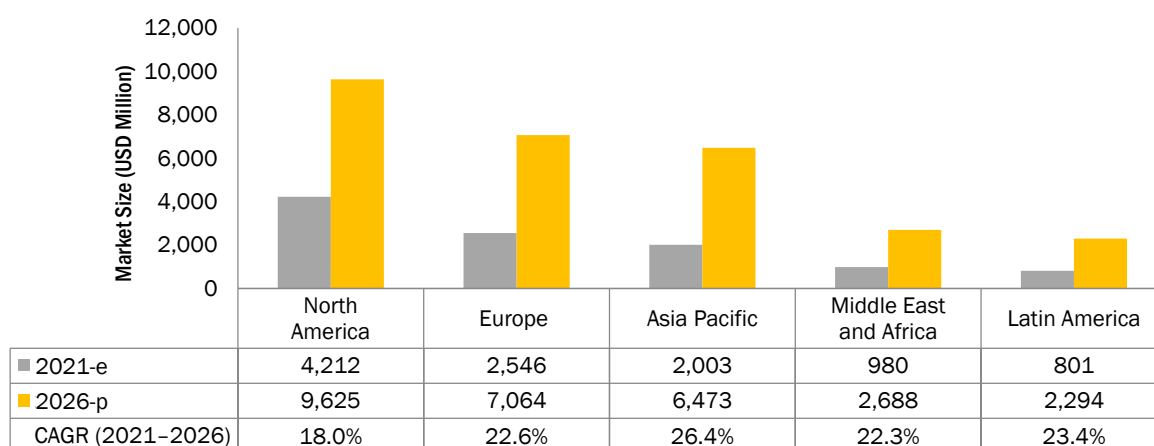
The predictive analytics market has been segmented into five major regions: North America, Europe, APAC, Latin America, and MEA. The increasing business interest toward advanced analytics for future estimation, compelling business necessity to preserve a competitive edge and the growing adoption of big data and other associated technologies are expected to drive the overall growth of the predictive analytics market across all the regions. The analysis demonstrates that the spending on analytics due to the rising IoT data and generation of insights across all the major industry verticals is increasing across the world, among which North America is expected to be the highest market shareholder, followed by Europe. Additionally, the growth of the manufacturing, energy and utilities, and retail and eCommerce industry verticals is subject to economic conditions and infrastructure developments across the regions.

In North America, predictive analytics solutions and services are considered to be highly effective across most organizations and verticals. Europe is gradually advancing toward incorporating predictive analytics with its market. APAC and Latin America, respectively, are expected to witness a substantial rise in the adoption of predictive analytics software and services, while the MEA region is more cautious in adopting these solutions.

FIGURE 41 JAPAN AND AUSTRALIA & NEW ZEALAND TO ACCOUNT FOR HIGHEST CAGRS DURING FORECAST PERIOD



Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 42 ASIA PACIFIC TO ACCOUNT FOR THE HIGHEST CAGR DURING FORECAST PERIOD

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 100 PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2015-2020 (USD MILLION)

Region	2015	2016	2017	2018	2019	2020	CAGR (2015-2020)
North America	1,528	1,736	2,033	2,380	2,765	2,947	14.0%
Europe	788	925	1,098	1,315	1,601	1,742	17.2%
APAC	456	562	699	875	1,177	1,331	23.9%
MEA	296	366	415	500	611	667	17.7%
Latin America	217	262	320	390	484	528	19.5%
Total	3,285	3,851	4,566	5,460	6,637	7,216	17.0%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 101 PREDICTIVE ANALYTICS MARKET SIZE, BY REGION, 2021-2026 (USD MILLION)

Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021-2026)
North America	4,212	5,430	6,553	7,649	8,610	9,625	18.0%
Europe	2,546	3,327	4,086	4,973	6,135	7,064	22.6%
APAC	2,003	2,687	3,408	4,466	5,500	6,473	26.4%
MEA	980	1,289	1,594	1,954	2,333	2,688	22.3%
Latin America	801	1,048	1,314	1,633	1,977	2,294	23.4%
Total	10,543	13,781	16,954	20,675	24,554	28,144	21.7%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

In North America, predictive analytics solutions and services are highly effective in most organizations and verticals due to the increasing importance of data tracking and data-driven insights. Europe is gradually advancing toward incorporating predictive analytics within its market. APAC is showing a substantial rise in the adoption of predictive analytics solutions and services during the forecast period, while Latin America and MEA are slowly picking up predictive analytics due to its benefits for various industries to get user insights.

10.2 NORTH AMERICA

The adoption of predictive analytics solutions is expected to be the highest in North America, as compared to the other regions. This is mainly because of the presence of various developed economies, such as Canada and the US, and due to the focus on innovations obtained from R&D and various technologies. North America, a hub of data generation in large quantities, is the largest market for predictive analytics solutions, with a share of 37.0% in the global market. The high penetration of predictive analytics in multiple verticals, such as manufacturing, healthcare, transportation, and logistics, is driving the growth of this market in this region. The North American region has been extremely open toward the adoption of any new and innovative technologies and is expected to provide market growth opportunities to predictive analytics vendors, as it is expected to witness an exponential growth of IT device-based generated data and stringent laws and policies for safeguarding the data. Consequently, the predictive analytics components are said to be showing strong positive trends in the region, as several companies and industries are adopting predictive analytics software and services at various levels as a part of their growth strategies to strive in the market and increase the efficiency of IoT data for enhancing their overall productivity. Many predictive analytics giants, such as IBM, Oracle, SAS Institute, and RapidMiner, offer predictive analytics software that includes sensor data analytics, IoT gateway analytics, and network management.

North America has been the most promising region across various verticals, such as the BFSI, manufacturing, retail and eCommerce, healthcare and life sciences, and telecommunications and IT. The government sector has also joined the race to become a technologically advanced industry vertical, serving a large customer base. The US Air Force uses predictive maintenance technology to predict when an aircraft may break down and then carry out preventive replacements and maintenance to avoid it. This reduces aircraft downtime and helps minimize future repair costs. Some of the biggest retail giants are also located in this region. In terms of adoption, North America contributes significantly to the market and is expected to grow further. The US is said to have adopted predictive analytics to a great extent.

10.2.1 NORTH AMERICA: COVID-19 IMPACT

- The long-term impact of COVID-19 can be significant revenue losses, while the short-term impact would result in the shutting down of various manufacturing and utility plants due to lockdown.
- HACERA, an enterprise blockchain startup, collaborated with IBM and Oracle to launch the MiPasa app to assist data integration and analytics for the COVID-19 situation.
- According to a survey conducted by ThomasNet.com in June 2020, which reveals the Coronavirus' Impacts on North American Manufacturers, 47% of the US manufacturers stated that they are now seeking domestic sources of supply. 53% of manufacturers report the recent fluctuations in oil prices will affect business. Many respondents said that this would have a positive impact, noting the shifts would cause a reduction in transport costs and prices for raw materials, such as plastic. Over 55% of the manufacturers expect their revenue to decline as a result of the COVID-19 outbreak, and 31% expect it to remain the same. This is a stark departure from February survey results, which found that only 34% anticipated their revenue to decline, and 53% expected it to remain the same.

- Owing to the COVID-19 impact, manufacturing companies in the US would see a tremendous opportunity through the adoption of emerging technologies (IoT, AI, and ML) to reduce risk failure in equipment's working ability.
- According to statistics released by the White House US, in the pessimistic scenario, it can be expected that the US may see 240,000 fatalities due to this pandemic.
- During the COVID-19 pandemic, technologies such as AI and robotics have boosted the company's growth by creating opportunities for innovation, but the pandemic has reduced manpower efforts, which may affect the working culture in the region.
- Google and Apple have collaborated to enable the use of Bluetooth technology to help government and health agencies reduce the spread of COVID-19.
- There has been a downfall in the travel and hospitality industry due to nationwide lockdowns.
- Email frauds increased by 667% in March 2020 alone, according to an industry expert. Users have become three times more vulnerable to phishing assaults, according to an industry expert. From a security standpoint, the key advantage of predictive analytics is that they enable users to skip all these channels and retain their data and communication within the platform's bounds and channels.
- In March 2020, ResilientGrid launched the COVID-19 Quick Start Remote Visualization program designed to extend a utilities' important operations center data quickly and efficiently to their remote employees.

TABLE 102 NORTH AMERICA: PROMINENT PLAYERS

NORTH AMERICA: PREDICTIVE ANALYTICS PROMINENT PLAYERS	
IBM	Microsoft
SAS Institute	SAP
Oracle	

10.2.2 NORTH AMERICA: REGULATIONS

10.2.2.1 Personal Information Protection and Electronic Documents Act (PIPEDA)

The PIPEDA is Canada's key private-sector privacy law. Compliance with PIPEDA is essential for private sector organizations operating in Canada. The following organizations and people were deemed to fall under PIPEDA:

- A daycare center, even though it received government funding
- A non-profit that was engaged in commercial activities
- A public-sector doctor undertaking an examination for an insurance company

10.2.2.2 Gramm-Leach-Bliley (GLB) Act

The GLB Act was developed in 1999 by the US federal government. It is also known as the Financial Services Modernization Act. The major objective of this act was to maintain the security of private information pertaining to every individual that is handled by financial institutes. It includes provisions to protect consumers' personal financial information held by banks, securities companies, and insurance companies. Under the jurisdiction of the Federal Trade Commission (FTC), these financial institutions are required to design, implement, and maintain an information security program to protect the privacy and integrity of customer data.

10.2.2.3 Health Insurance Portability and Accountability Act (HIPAA) of 1996

The HIPAA was signed and brought into law by President Bill Clinton in 1996. The Act was passed onto US legislation to safeguard and protect medical information by maintaining data privacy. The five sections included in the act are as follows:

- HIPAA Title I protects the health insurance cover of individuals and prohibits group health plans from denying coverage to any individual.
- HIPAA Title II establishes national standards for processing electronic healthcare transactions.
- HIPAA Title III includes provisions related to tax and guidelines for medical care.
- HIPAA Title IV includes provisions for individuals with pre-existing conditions and those seeking continued coverage.
- HIPAA Title V includes provisions on company-owned life insurance and treatment of those who lose their US citizenship for income tax purposes.

10.2.2.4 Federal Information Security Management Act (FISMA)

The FISMA of 2002 enforces stringent standards to ensure the security, confidentiality, and integrity of US federal property and information. It sets the requirements for each federal agency to create, document, and implement programs that ensure security for the agencies' data and the systems that support the agencies' operations and assets. This extends to state governments that manage federal programs, contractors, and other non-governmental organizations working with and funded by the federal government. The examples of federal programs that states manage are Medicare, Medicaid, and unemployment insurance.

10.2.2.5 Federal Information Processing Standards (FIPS)

Created by the National Institute of Standards and Technology's (NIST's) Computer Security Division, the FIPS established a data security and computer system standard that organizations must adhere to per the FISMA of 2002.

To become FIPS compliant, a US government agency or contractor's computer systems must meet the requirements outlined in the FIPS publications numbered 140, 180, 186, 197, 198, 199, 200, 201, and 202.

- FIPS 140 covers cryptographic module and testing requirements in both hardware and software.
- FIPS 180 specifies how organizations can be FIPS compliant when using secure hash algorithms for computing a condensed message.
- FIPS 186 is a group of algorithms for generating digital signatures.
- FIPS 197 is a standard that created the Advanced Encryption Standard, a publicly accessible cipher approved by the National Security Agency (NSA) for top secret information.

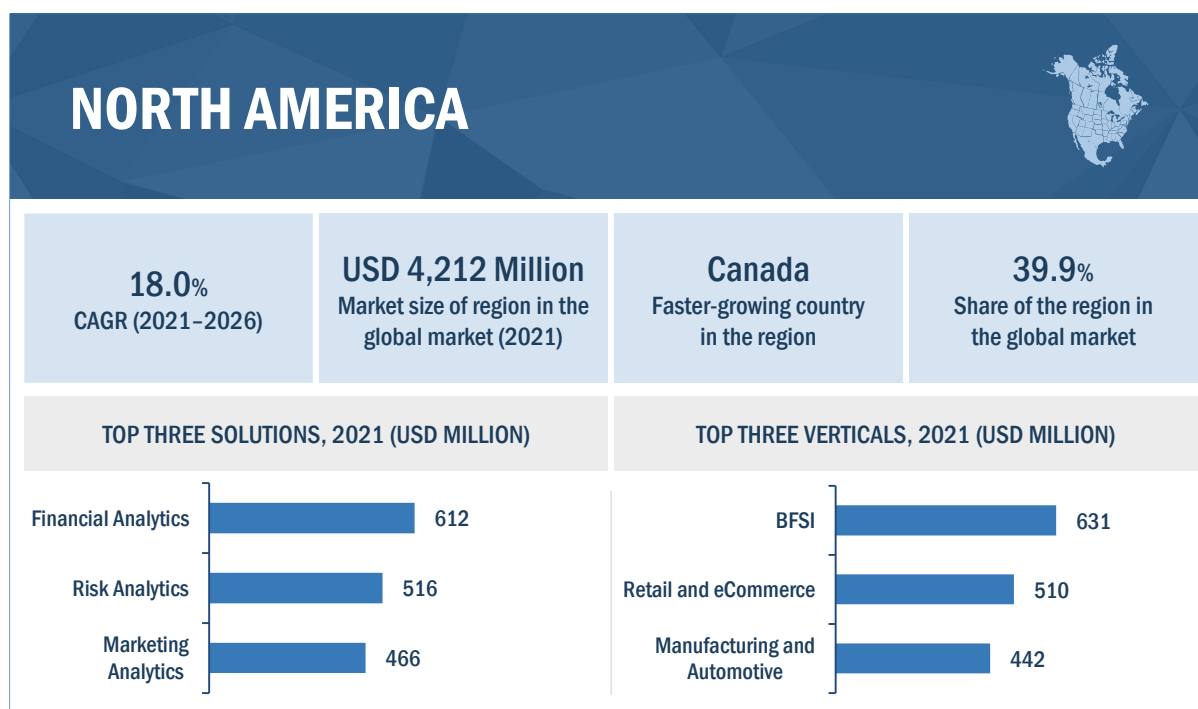
- FIPS 198 is about a mechanism for message authentication that utilizes cryptographic hash functions.
- FIPS 199 standardizes how federal agencies categorize and secure information.
- FIPS 200 is a standard that helps federal agencies with risk management through levels of information security based on risk levels.
- FIPS 201 specifies the standard for common identification for federal employees and contractors.
- FIPS 202 gives the specifications for the Secure Hash Algorithm-3 (SHA-3). It consists of four cryptographic hash functions and two extendable-output functions.

10.2.2.6 California Consumer Privacy Act (CSPA)

The CSPA of 2018 gives consumers more control over the personal information that businesses collect related to them. This landmark law secures new privacy rights for consumers in California, which are as follows:

- The right to know about the personal information a business collects about them and how it is used and shared
- The right to delete personal information collected from consumers
- The right to opt-out of the sale of their personal information
- The right to non-discrimination for exercising their CCPA rights

FIGURE 43 NORTH AMERICA: MARKET SNAPSHOT



Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 103 NORTH AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY COMPONENT, 2015–2020 (USD MILLION)

Component	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Solutions	1,146	1,301	1,523	1,782	2,068	2,203	14.0%
Services	382	435	510	599	697	744	14.3%
Total	1,528	1,736	2,033	2,380	2,765	2,947	14.0%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 104 NORTH AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Solutions	3,146	4,053	4,888	5,702	6,414	7,166	17.9%
Services	1,066	1,376	1,664	1,947	2,196	2,459	18.2%
Total	4,212	5,430	6,553	7,649	8,610	9,625	18.0%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 105 NORTH AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY SOLUTION, 2015–2020 (USD MILLION)

Solution	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Financial Analytics	239	268	311	359	412	434	12.6%
Risk Analytics	194	219	255	297	343	363	13.3%
Marketing Analytics	157	181	215	254	299	322	15.4%
Sales Analytics	148	171	203	241	284	307	15.6%
Customer Analytics	92	105	125	148	174	187	15.3%
Web and Social Media Analytics	126	141	163	188	215	226	12.5%
Supply Chain Analytics	84	97	114	135	158	170	15.1%
Network Analytics	75	85	100	117	135	144	14.1%
Other Solutions*	30	33	38	43	48	49	9.9%
Total	1,146	1,301	1,523	1,782	2,068	2,203	14.0%

*Other solutions include HR analytics and legal analytics.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 106 NORTH AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY SOLUTION, 2021–2026 (USD MILLION)

Solution	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Financial Analytics	612	779	928	1,070	1,189	1,301	16.3%
Risk Analytics	516	661	793	921	1,030	1,144	17.3%
Marketing Analytics	466	608	742	877	999	1,130	19.4%
Sales Analytics	445	581	711	842	961	1,100	19.9%
Customer Analytics	271	353	430	508	577	652	19.2%
Web and Social Media Analytics	319	406	483	556	617	681	16.4%
Supply Chain Analytics	245	319	389	458	521	588	19.1%
Network Analytics	206	266	321	375	423	473	18.0%
Other Solutions*	66	80	89	96	98	98	8.3%
Total	3,146	4,053	4,888	5,702	6,414	7,166	17.9%

e: estimated; p: projected

*Other solutions include HR analytics and legal analytics.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 107 NORTH AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY SERVICE, 2015–2020 (USD MILLION)

Service	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Professional Services	250	282	327	381	440	467	13.3%
Managed Services	132	153	184	218	257	277	16.1%
Total	382	435	510	599	697	744	14.3%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 108 NORTH AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY SERVICE, 2021–2026 (USD MILLION)

Service	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Professional Services	664	852	1,023	1,187	1,328	1,476	17.3%
Managed Services	402	525	642	759	867	984	19.6%
Total	1,066	1,376	1,664	1,947	2,196	2,459	18.2%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 109 NORTH AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2015–2020 (USD MILLION)

Professional Service	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Consulting	81	90	105	123	130	139	11.3%
Deployment and Integration	63	73	86	102	119	129	15.4%
Support and Maintenance	106	118	136	156	191	199	13.4%
Total	250	282	327	381	440	467	13.3%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 110 NORTH AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2021–2026 (USD MILLION)

Professional Service	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Consulting	198	255	308	358	400	444	17.5%
Deployment and Integration	186	240	291	339	383	428	18.2%
Support and Maintenance	280	356	424	490	546	604	16.6%
Total	664	852	1,023	1,187	1,328	1,476	17.3%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 111 NORTH AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2015–2020 (USD MILLION)

Deployment Mode	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
On-premises	747	842	979	1,131	1,296	1,360	12.7%
Cloud	399	459	544	650	771	842	16.1%
Total	1,146	1,301	1,523	1,782	2,068	2,203	14.0%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 112 NORTH AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2021–2026 (USD MILLION)

Deployment Mode	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
On-premises	1,916	2,427	2,923	3,364	3,733	4,113	16.5%
Cloud	1,231	1,627	1,965	2,338	2,681	3,053	19.9%
Total	3,146	4,053	4,888	5,702	6,414	7,166	17.9%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 113 NORTH AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2015–2020 (USD MILLION)

Organization Size	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Large Enterprises	1,100	1,236	1,431	1,654	1,899	2,005	12.8%
SMEs	428	500	603	726	865	942	17.1%
Total	1,528	1,736	2,033	2,380	2,765	2,947	14.0%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 114 NORTH AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2021–2026 (USD MILLION)

Organization Size	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Large Enterprises	2,843	3,626	4,326	4,987	5,545	6,122	16.6%
SMEs	1,369	1,804	2,227	2,662	3,065	3,504	20.7%
Total	4,212	5,430	6,553	7,649	8,610	9,625	18.0%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 115 NORTH AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY VERTICAL, 2015–2020 (USD MILLION)

Vertical	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
BFSI	269	299	341	388	437	453	11.0%
Retail and eCommerce	196	218	253	294	339	359	12.8%
Telecom and IT	168	188	216	248	282	295	11.9%
Healthcare and Life Sciences	130	152	183	220	263	287	17.2%
Government and Public Sector	142	164	195	231	272	293	15.6%
Manufacturing and Automotive	153	175	207	245	288	306	14.9%
Media and Entertainment	122	139	164	193	224	240	14.4%
Energy and Utilities	111	128	153	182	216	234	16.2%
Travel and Hospitality	76	88	104	123	145	156	15.4%
Transportation and Logistics	92	106	125	148	173	188	15.5%
Other Verticals*	69	79	93	108	126	134	14.3%
Total	1,528	1,736	2,033	2,380	2,765	2,947	14.0%

*Other verticals include education & research and real estate.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 116 NORTH AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY VERTICAL, 2021–2026 (USD MILLION)

Vertical	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
BFSI	631	793	943	1,071	1,173	1,256	14.8%
Retail and eCommerce	510	654	781	906	1,014	1,127	17.2%
Telecom and IT	415	529	632	731	816	901	16.7%
Healthcare and Life Sciences	419	553	681	809	928	1,054	20.2%
Government and Public Sector	422	550	662	782	889	1,007	19.0%
Manufacturing and Automotive	442	576	701	826	938	1,078	19.5%
Media and Entertainment	343	445	538	628	709	799	18.4%
Energy and Utilities	339	442	538	634	718	812	19.1%
Travel and Hospitality	226	291	353	414	467	523	18.3%
Transportation and Logistics	273	352	427	501	567	633	18.4%
Other Verticals*	191	246	297	346	390	435	17.9%
Total	4,212	5,430	6,553	7,649	8,610	9,625	18.0%

e: estimated; p: projected

*Other verticals include education & research and real estate.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 117 NORTH AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY COUNTRY, 2015–2020 (USD MILLION)

Country	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
US	1,146	1,301	1,523	1,782	2,068	2,203	14.0%
Canada	382	435	510	599	697	744	14.3%
Total	1,528	1,736	2,033	2,380	2,765	2,947	14.0%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 118 NORTH AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY COUNTRY, 2021–2026 (USD MILLION)

Country	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
US	3,146	4,053	4,888	5,702	6,414	7,166	17.9%
Canada	1,066	1,376	1,664	1,947	2,196	2,459	18.2%
Total	4,212	5,430	6,553	7,649	8,610	9,625	18.0%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

10.2.3 UNITED STATES

10.2.3.1 Rapid adoption of digitally innovative solutions and presence of tech giants to drive market growth

According to a recent report by an industry expert, the US accounted for the highest tech market share in the world, representing 32% of the total share, or approximately USD 1.7 trillion in 2020. The US is the major contributor to the North American region in data generation and technological adoption. The presence of leading global players has resulted in the early adoption of these solutions among companies and is the major factor that has consistently kept the country a leader in every business aspect. The major verticals contributing to the high adoption of predictive analytics include healthcare, BFSI, retail, manufacturing, and government and defense. The US government provides support by promoting an ideal environment for research and innovation; this has led developers to invest aggressively in new solutions that effectively integrate the functionalities of various established tools. The Defense Advanced Research Projects Agency (DARPA), in collaboration with commercial technology experts, developed a platform that monitors dubious online advertisements and infers connections between them and child trafficking rings. Moreover, data-related regulations are the other prominent factors contributing to the adoption of predictive analytics solutions across various verticals. As communities across the country continue to relax social distancing orders, it will be crucial for health systems to plan for changes in healthcare demands and a surge in the number of COVID-19 patients. The team at CommonSpirit Health built predictive models using de-identified cell phone data, public health information, and data from the system's own care sites.

According to a research study in the US, during the COVID-19 epidemic, 50% of individuals predicted that cloud usage would be more than expected. Vendors in the country are investing in these technologies to provide better functioning of predictive analytics solutions to the users to make timely decisions and increase the way of making faster business decisions for increasing business revenue.

10.2.4 CANADA

10.2.4.1 Startup ecosystem and transformation into data-driven organizations to drive the growth of predictive analytics in Canada

The startup ecosystem of Canada is strong, and Canada's startups bring technologies to the market quickly. Canadian organizations are expected to adopt the predictive analytics software aggressively during the forecast period. Although the country is not on par with the US in the adoption of predictive analytics solutions, various Canadian enterprises are focusing on embracing predictive analytics technology. The country has also set up an institution, Canada.ai, which would provide information related to advancements in AI and ML research to various startups and enterprises. With updated regulations, businesses are now facing big penalties as a result of non-compliance with laws.

In many cases, every year, companies are adjusting their strategies for their existing database solutions and trying to transform into data-driven organizations. Moreover, the country has mitigated financial losses and risks, and credit card providers and other financial institutions have implemented fraud monitoring solutions, such as Predictive Analytics Technologies (PAT), as a part of the fraud prevention programs, to reduce credit card fraud. These factors are some of the contributors that are driving the adoption of predictive analytics solutions in Canada. Despite the dearth of direct information on the predictive analytics market in Canada, the country, similar to the rest of North America and the world, is on its path to comprehensive adoption and use of predictive analytics.

10.3 EUROPE

Achievements of Europe in science and technology have been significant. R&D is an integral part of the European economy. The robust economic infrastructure and the high internet penetration make Europe one of the top markets for cloud-based services, which show a high contribution in the predictive analytics market. The SMEs in the UK, Germany, and France are investing in incorporating analytical platforms to leverage benefits, in turn, reducing unwanted costs. As the economic turmoil continues in Europe, enterprises are likely to undertake a stronger cost containment stance sooner. The major countries considered in the region for analysis are the UK, Germany, France, and Rest of Europe.

Along with these countries, several other European countries are also incorporating predictive analytics solutions and services into their existing IT portfolios to deliver quick and actionable executive insights and keep pace with competitors from other regions. The cloud industry in the UK has an established base of business customers who understand the cloud value proposition. Enterprises in the region have strong technical expertise, along with bigger Information Technology (IT) budgets to enable innovative predictive analytics solutions. The European brands are adopting digital transformation through migration from legacy systems to virtual environments at a slow pace. New technologies and business models are being introduced rapidly in the European region due to the fast-changing business requirements. Continuous innovations and increased spending on the adoption of newer technologies are the results of a widespread shift in customer demands and dynamic changes in the business models of today's data-driven industry.

The low cost of analytics components, cloud adoption, and the increasing usage of the web and higher speed broadband connectivity across the region are some of the driving factors expected to drive the adoption of predictive analytics in this region. Though Europe's technology domain has been witnessing a slower growth compared to North America, the recent investments in this region by American technology vendors indicate rising opportunities locally. The deployment of AI has become critical for the success of SMEs in the EU. In industrial sectors, AI solutions are becoming ever more important as they help to optimize production processes, predict machinery failures, and develop more efficient smart services. European industry can also harness big data and the smart use of ICT to enhance productivity and performance and pave the way for innovation.

10.3.1 EUROPE: COVID-19 IMPACT

- Owing to COVID-19, Europe has been one of the worst-affected regions. As a result, key European industries have been impacted. However, software companies, particularly those in the predictive analytics sector, have made certain adjustments to stay afloat during the crisis.
- COVID-19 would have a short-term impact on manufacturing companies. Post the lockdown period, companies will witness new opportunities in digital transformation in Italy.
- As a result of the coronavirus interruption, not all mergers and acquisitions in the region have been abandoned. Many have shut down in the last few months. The social and economic issues that COVID-19 has produced may, however, trigger provisions in the instruments that underpin and enact these agreements.
- The EU announced a space program with over 30 satellites through its Earth Observation component, Copernicus, and its satellite navigation system, Galileo, to respond during COVID-19.
- According to the simulations carried out by a JRC research team, the disposable income of European households would fall by 5.9% on average in 2020 due to the COVID-19 pandemic with no policy measures were put in place to alleviate the impact.
- The EC launched a data portal for scientists studying the SARS-CoV-2 virus to accelerate access to data sets and tools.

- According to the data published by HawkEye 360, the maritime activity of Italian ports has witnessed a drop by about 51% due to the lockdown in the country.
- A German startup developed a corona-bot that helps in clarifying whether a person is showing the symptoms of the SARS-CoV-2 infection.
- Travel and tourism services have been among the first and most affected sectors, from both demand and supply sides. The standstill in tourism will affect small and less diversified countries in the Western Balkans, Turkey, and the Caucasus.
- The UK government switched to Google-Apple technology for its new virus app for England and Wales after scrapping a centralized version because of technical issues.
- According to a recent World Bank report, the Europe and Central Asia regions, including the Western Balkans, South Caucasus, and Central Asia, are expected to be the worst-hit regions in the world in terms of remittances with a projected slump of 27.5% in 2020.
- The COVID-19 pandemic largely affected the higher education sector as on-campus activities were suspended by many universities, and restrictions were put on immigration.

TABLE 119 EUROPE: PROMINENT PLAYERS

EUROPE: PREDICTIVE ANALYTICS PROMINENT PLAYERS	
Teradata	HPE
Salesforce	Alteryx

10.3.2 EUROPE: REGULATIONS

10.3.2.1 General Data Protection Regulation (GDPR)

GDPR (Regulation [EU] 2016/679) is a regulation by which the European Parliament, the Council of the European Union, and the European Commission ensure the safety of the EU citizens from privacy and data breaches. GDPR extends the scope of the EU data protection law for all foreign organizations that process the data of EU citizens. It delivers the coordination of data protection regulations throughout the EU, thereby making it easier for non-European companies to comply with the regulations. However, if the regulations are not complied with, companies may have to pay four percent of their worldwide turnover as a penalty. Key privacy and data protection requirements of GDPR include the following:

- Confidentiality of collected data to ensure data protection
- Safe transfer of data across borders
- Seeking consent of subjects for data processing
- Providing data breach notifications

GDPR would impose a consistent data security law for all EU members and the companies marketing goods and services to EU residents, whatever be the location.

10.3.2.2 European Committee for Standardization (CEN)

The European CEN is a government body that works along with ISO and was established to develop the IoT standards in the region. It imposes rules and regulations for integrating the data collected from sensors with the existing RFID-driven barcode systems.

10.3.2.3 EU's General Court

The EU's General Court (General Court) combines the protections applicable to information requests issued by the European Commission (EC) to those in dawn raids by requiring the EC to review potentially crucial documents only in ad-hoc predictive analytics and followed by the investigated party's counsel. The EC's power to request information from companies derives from Article 18(2) and (3) of Regulation No 1/2003, pursuant to which the EC can either send a simple request for information or adopt a decision requiring companies to provide information. In both cases, the EC has to inform the companies about (i) the legal basis for the request, (ii) the relevant time limits, and (iii) the penalties that apply for non-compliance. These penalties can amount to 1% of the group company's total worldwide annual turnover or can consist of daily penalties amounting to 5% of the daily turnover in the case of a formal information request by decision.

TABLE 120 EUROPE: PREDICTIVE ANALYTICS MARKET SIZE, BY COMPONENT, 2015–2020 (USD MILLION)

Component	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Solutions	591	693	823	985	1,197	1,302	17.1%
Services	197	232	276	331	403	440	17.4%
Total	788	925	1,098	1,315	1,601	1,742	17.2%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 121 EUROPE: PREDICTIVE ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Solutions	1,902	2,484	3,048	3,707	4,570	5,259	22.6%
Services	644	843	1,038	1,266	1,564	1,805	22.9%
Total	2,546	3,327	4,086	4,973	6,135	7,064	22.6%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 122 EUROPE: PREDICTIVE ANALYTICS MARKET SIZE, BY SOLUTION, 2015–2020 (USD MILLION)

Solution	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Financial Analytics	123	143	168	198	238	256	15.7%
Risk Analytics	100	117	138	164	199	215	16.5%
Marketing Analytics	81	96	116	140	173	190	18.6%
Sales Analytics	77	91	110	133	164	181	18.8%
Customer Analytics	47	56	67	82	101	111	18.5%
Web and Social Media Analytics	65	75	88	104	125	134	15.6%
Supply Chain Analytics	43	51	62	75	92	101	18.3%
Network Analytics	39	45	54	64	78	85	17.2%
Other Solutions*	16	18	21	24	28	29	13.0%
Total	591	693	823	985	1,197	1,302	17.1%

*Other solutions include HR analytics and legal analytics.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 123 EUROPE: PREDICTIVE ANALYTICS MARKET SIZE, BY SOLUTION, 2021–2026 (USD MILLION)

Solution	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Financial Analytics	370	477	579	695	847	955	20.9%
Risk Analytics	312	405	495	599	734	840	21.9%
Marketing Analytics	282	372	463	570	712	829	24.1%
Sales Analytics	269	356	444	547	685	807	24.6%
Customer Analytics	164	216	268	330	411	479	24.0%
Web and Social Media Analytics	193	249	301	362	440	500	21.0%
Supply Chain Analytics	148	196	243	298	371	431	23.8%
Network Analytics	125	163	200	244	301	347	22.7%
Other Solutions*	40	49	56	62	70	72	12.6%
Total	1,902	2,484	3,048	3,707	4,570	5,259	22.6%

e: estimated; p: projected

*Other solutions include HR analytics and legal analytics.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 124 EUROPE: PREDICTIVE ANALYTICS MARKET SIZE, BY SERVICE, 2015–2020 (USD MILLION)

Service	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Professional Services	129	150	176	210	255	276	16.4%
Managed Services	68	82	99	120	149	164	19.3%
Total	197	232	276	331	403	440	17.4%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 125 EUROPE: PREDICTIVE ANALYTICS MARKET SIZE, BY SERVICE, 2021–2026 (USD MILLION)

Service	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Professional Services	401	522	638	772	946	1,083	22.0%
Managed Services	243	321	400	494	618	722	24.4%
Total	644	843	1,038	1,266	1,564	1,805	22.9%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 126 EUROPE: PREDICTIVE ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2015–2020 (USD MILLION)

Professional Service	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Consulting	42	48	57	68	75	82	14.4%
Support and Maintenance	55	63	73	86	110	118	16.5%
Deployment and Integration	33	39	46	56	69	76	18.6%
Total	129	150	176	210	255	276	16.4%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 127 EUROPE: PREDICTIVE ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2021–2026 (USD MILLION)

Professional Service	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Consulting	120	156	192	233	285	326	22.1%
Support and Maintenance	169	218	265	319	389	443	21.2%
Deployment and Integration	112	147	181	221	273	314	22.8%
Total	401	522	638	772	946	1,083	22.0%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 128 EUROPE: PREDICTIVE ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2015–2020 (USD MILLION)

Deployment Mode	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
On-premises	385	449	529	625	751	804	15.8%
Cloud	206	245	294	359	447	498	19.3%
Total	591	693	823	985	1,197	1,302	17.1%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 129 EUROPE: PREDICTIVE ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2021–2026 (USD MILLION)

Deployment Mode	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
On-premises	1,158	1,487	1,823	2,187	2,660	3,019	21.1%
Cloud	744	997	1,226	1,520	1,910	2,240	24.7%
Total	1,902	2,484	3,048	3,707	4,570	5,259	22.6%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 130 EUROPE: PREDICTIVE ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2015–2020 (USD MILLION)

Organization Size	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Large Enterprises	568	659	773	914	1,100	1,185	15.9%
SMEs	221	266	326	401	501	557	20.3%
Total	788	925	1,098	1,315	1,601	1,742	17.2%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 131 EUROPE: PREDICTIVE ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2021–2026 (USD MILLION)

Organization Size	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Large Enterprises	1,719	2,222	2,698	3,242	3,951	4,493	21.2%
SMEs	827	1,105	1,388	1,731	2,184	2,571	25.5%
Total	2,546	3,327	4,086	4,973	6,135	7,064	22.6%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 132 EUROPE: PREDICTIVE ANALYTICS MARKET SIZE, BY VERTICAL, 2015–2020 (USD MILLION)

Vertical	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
BFSI	139	159	184	214	253	268	14.1%
Retail and eCommerce	101	116	137	163	197	212	16.0%
Telecom and IT	87	100	116	137	163	174	15.0%
Healthcare and Life Sciences	67	81	99	122	152	170	20.4%
Government and Public Sector	73	87	105	128	158	173	18.8%
Manufacturing and Automotive	79	93	112	135	166	181	18.1%
Media and Entertainment	63	74	89	106	130	142	17.6%
Energy and Utilities	57	68	83	101	125	138	19.4%
Travel and Hospitality	39	47	56	68	84	92	18.6%
Transportation and Logistics	47	56	68	82	100	111	18.7%
Other Verticals*	35	42	50	60	73	79	17.4%
Total	788	925	1,098	1,315	1,601	1,742	17.2%

*Other verticals include education & research and real estate.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 133 EUROPE: PREDICTIVE ANALYTICS MARKET SIZE, BY VERTICAL, 2021–2026 (USD MILLION)

Vertical	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
BFSI	381	486	588	696	836	922	19.3%
Retail and eCommerce	308	401	487	589	723	827	21.8%
Telecom and IT	251	324	394	475	581	661	21.4%
Healthcare and Life Sciences	254	339	424	526	661	773	25.0%
Government and Public Sector	255	337	413	508	634	739	23.7%
Manufacturing and Automotive	267	353	437	537	669	791	24.2%
Media and Entertainment	207	272	335	409	505	587	23.1%
Energy and Utilities	205	271	336	412	512	596	23.8%
Travel and Hospitality	137	178	220	269	333	384	23.0%
Transportation and Logistics	165	215	266	326	404	465	23.1%
Other Verticals*	116	151	185	225	278	319	22.5%
Total	2,546	3,327	4,086	4,973	6,135	7,064	22.6%

e: estimated; p: projected

*Other verticals include education & research and real estate.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 134 EUROPE: PREDICTIVE ANALYTICS MARKET SIZE, BY COUNTRY, 2015–2020 (USD MILLION)

Country	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
UK	287	333	390	461	554	595	15.7%
Germany	166	195	233	280	343	375	17.7%
France	186	221	266	322	397	437	18.6%
Rest of Europe*	150	176	210	252	307	335	17.5%
Total	788	925	1,098	1,315	1,601	1,742	17.2%

*Rest of Europe includes Spain, Sweden, Switzerland, and Denmark.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 135 EUROPE: PREDICTIVE ANALYTICS MARKET SIZE, BY COUNTRY, 2021–2026 (USD MILLION)

Country	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
UK	858	1,106	1,381	1,658	2,018	2,314	21.9%
Germany	550	722	891	1,089	1,350	1,561	23.2%
France	647	855	1,022	1,258	1,571	1,808	22.8%
Rest of Europe*	491	644	793	967	1,196	1,381	23.0%
Total	2,546	3,327	4,086	4,973	6,135	7,064	22.6%

e: estimated; p: projected

*Rest of Europe includes Spain, Sweden, Switzerland, and Denmark

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

10.3.3 UNITED KINGDOM

10.3.3.1 Advanced IT infrastructure, technologies, and continued transition toward online services to drive growth in the UK

The UK is one of the strongest economies in Europe. The predictive analytics market in the UK is becoming relentless because of the examination of high-volume information, the speed of information growth, and the assortment of information utilized for viable business decisions. Real-time information is imperative for verticals, such as the BFSI and healthcare, for the proficient processing of unstructured information. Apart from this, utilizing advanced predictive analytics solutions is expected to drive market growth in the UK. Additionally, the implementation of predictive analytics solutions by universities is expected to propel market growth in this region. The predictive analytics market growth is expected to be driven by the continued transition toward online services due to the relative ease and speed they can be accessed. The increasing user access and simplicity stem from technological advances.

The savings offered by moving online, in terms of money and time, will also be a significant factor for this transition. In addition, the need for risk management and the adoption of cloud-based technology by large, medium, and small businesses are propelling this region forward. Following the COVID-19 lockdown in the UK, giving customers with consistently good digital experience has been vital, regardless of the industry. The UK and European policymakers are encouraging the governments of the countries to increase their data capture and adopt predictive analytics for pushing the digital economy, which are the driving factors for the predictive analytics market in the region. A resolution was passed by the European Parliament to

ensure a 'data-driven economy,' which would aggressively assert the implementation of advanced analytics for business houses and government verticals.

10.3.4 GERMANY

10.3.4.1 Government initiatives for technological developments in manufacturing to drive the growth of these solutions in Germany

Germany is one of the world's leading industrial countries. It is one of the most active players in adopting innovative digital technologies across industry verticals and is one of the leading countries for investments around data management solutions in the European region. The German government plays an important role in strengthening the country's position in the manufacturing vertical. The High-Tech Strategy 2020 describes Germany's priorities in the field of technology. To increase digitalization, the German government would spend approximately USD 4 billion on AI products and services by 2025. Predictive analytics helps manufacturers share new designs or product data.

In May 2021, the World Health Organization (WHO) and the German Federal Republic have announced the creation of a new worldwide center for pandemic and epidemic intelligence, data, surveillance, and analytics innovation. Predictive analytics has numerous applications in the manufacturing industry, including forecasting, text mining for product development, predictive maintenance and logistics, risk management and prevention, and stream mining and real-time prediction. Various government-funded organizations, such as The German Research Center for Artificial Intelligence (DFKI) and non-profit Open Knowledge Foundation Germany, have partnered with corporations, such as Siemens and AGT International, in an initiative called the Big Project.

10.3.5 FRANCE

10.3.5.1 Heavy R&D investments; digitalization; and strong hold of retail, aerospace and defense, and manufacturing verticals to drive the growth in France

France has widespread internet penetration. According to reports, France has consistently been one of the top R&D patent producers, and its strong hold in retail, aerospace and defense, and manufacturing verticals has resulted in a significant rise in the predictive analytics market by companies operating in the country. According to a survey, reshaping government services through digitalization could save the French government USD 14.4 billion annually within five years. As part of this support plan, the government plans to spend a significant amount of money on all things digital, including startup investment, infrastructure investment, and digital transformation. This is expected to further drive the adoption of predictive analytics, which would enable the government to create a better economic impact and provide users with a better experience. France is already one of the leading countries for smart grids and smart cities, which has contributed to the adoption of the predictive analytics market. In May 2018, Alteryx expanded its presence by opening offices in France, which is expected to help vendors in the country to experience faster business decisions with the use of the modern Alteryx analytics platform. Despite the French government's significant steps to help businesses and workers, including a record amount of stimulus, the latest projections suggest that France's GDP would fall by 8% this year. Simultaneously, the crisis is expected to present unique chances for consolidation and, possibly, a more efficient and resilient M&A strategy.

10.3.6 REST OF EUROPE

The major countries adopting predictive analytics solutions and services include Spain, Sweden, Switzerland, Denmark, and the remaining countries. These countries are witnessing signs of growth in the predictive analytics market. Enterprises in this region are strongly reinforced by governments with the needed support for bringing in innovations and technological advancements. European countries have recognized the role of technologies in economic and social development and creating opportunities for international businesses. With the increasing need to gather important information from unstructured data, companies rely more on big data analytics. Moreover, the consistently changing dynamic landscape has enforced organizations to search for fresh methods for differentiation. Organizations are significantly innovating solutions to gain customer loyalty. The prominence of local players in delivering best-in-class solutions has influenced the adoption of predictive analysis among local retailers, thereby strengthening their position in the market.

Predictive analytics has lately emerged as a boon for offering different levels of transformation in managing a business. The advent of the cloud in this region is adding to the adoption of cloud-based solutions, which is said to be fueling the demand for predictive analytics solutions in this region. For instance, Spanish Public Administrations Network (SARA) has been appointed as a strategic initiative and a forum for providing cloud services to government agencies.

10.4 ASIA PACIFIC

APAC majorly comprises Australia, New Zealand, China, India, and Japan. The region is expected to experience extensive growth opportunities in the overall predictive analytics market over the next few years. The technology industry, including mobile, social networking, cloud, and IoT, in the region, is rapidly developing and disrupting. Digital leaders are strategizing to optimize operations and enhance customer experiences. Companies operating in APAC continue to focus on improving customer services to drive market competitiveness and revenue growth. The rapid growth of the predictive analytics market in the APAC region can be attributed to the massive growth in business deals and transactions, such as mergers and acquisitions, joint ventures, and fundraising, across all the industry verticals in the region. The rapid expansion of domestic enterprises, as well as the development of infrastructure, are some of the important factors expected to drive the growth of the predictive analytics market in APAC. Fast-growing countries, such as China, India, Singapore, Malaysia, and Australia, are rapidly moving toward cloud adoption. The APAC region has started to grow in the overall predictive analytics market and is expected to witness the highest growth rate in the next few years. However, the rising awareness about the increasing business productivity, supplemented with the competently designed predictive analytics solutions offered by the vendors present in this region, has helped APAC to become a highly potential region. This region is also witnessing an increasing demand for cloud-driven predictive analytics solutions, thereby resulting in increasing investments and technological advancements across the manufacturing industry. Analytics is gaining traction in countries such as South Korea, Australia, Japan, and New Zealand. Players in the IT sector, such as IBM, have currently formed collaborations with the Government of India to help develop smart cities.

In May 2019, SAS Institute launched a cloud offering in an AWS data center in Singapore to accelerate cloud adoption in APAC.

10.4.1 ASIA PACIFIC: COVID-19 IMPACT

- During the COVID-19 pandemic, the use of predictive analytics has increased since it helps reduce communication and due diligence delays while also enabling safe digital information transfer. Many firms have sought digital solutions that enable online remote access to essential data as a result of the developing culture of "work from home" in this region during the COVID-19 pandemic.
- The uncertainty that had been ushered in might probably stay for several years. However, upheaval brings with it a huge opportunity to not only restart but to construct new paths. Predictive analytics experts that succeed will be devising and implementing a plan that accelerates change, increases resilience, and maximizes business impact.
- It is vital for data-smart organizations across APAC to generate improved efficiencies, and better data utilization is essential to this during the pandemic. This, in turn, will help unlock the value of data at a profound level and generate actionable insights at scale, enabling them to save money. Businesses are quickly digitizing, and the epidemic has hastened this tendency. As a result, predictive analytics teams are in high demand, as are the insights they give.
- Businesses are concerned related to client loyalty, optimism, confidence, and trust as a result of the chaotic circumstances created by the COVID-19 epidemic. As a result, several businesses may use predictive analytics to develop ways to assist clients in this difficult scenario, which will positively impact APAC following the pandemic.
- According to Microsoft, APAC's GDP stands to gain an extra USD 387 billion by 2021 and grow by 1% annually if the region's manufacturing sector embraces digital transformation.
- The rising demand for predictive analytics, which is cloud-driven and cloud-supported, has resulted in the increase in demand for analytics solutions in the APAC region, thereby resulting in increasing investments and technological advancements across various industries.
- The increasing adoption of AI and deep learning algorithms contribute to the growth of the predictive analytics market in India.
- The rising usage of social media platforms, such as Facebook, LinkedIn, and websites, in Japan, would drive the adoption of predictive analytics solutions by vendors to better analyze customer behavior and enhance the customer experience.
- Government policy support and technological integration across China would drive the growth of the predictive analytics market.
- The region is witnessing an increase in competition and the widespread presence of smart devices and connected machines, resulting in increasing investments and technological advancements across various verticals.
- Strong economy and transformation in the IT sector would drive the growth of the predictive analytics market in China.

TABLE 136 ASIA PACIFIC: PROMINENT PLAYERS

ASIA PACIFIC: PREDICTIVE ANALYTICS PROMINENT PLAYERS	
AWS	IBM
SAS	Happiest Minds
FICO	

10.4.2 ASIA PACIFIC: REGULATIONS

10.4.2.1 Personal Data Protection Act (PDPA)

PDPA establishes a data protection law that comprises various rules governing the collection, use, disclosure, and care of personal data in Singapore. It recognizes both the rights of individuals to protect their personal data, including rights of access and correction, and the need of organizations to collect, use, or disclose personal data for legitimate and reasonable purposes. The PDPA provides a national Do Not Call (DNC) registry for establishments. The DNC Registry enables individuals to register their Singaporean telephone numbers to opt out of receiving marketing phone calls and mobile text messages, such as Short Message Service (SMS) or Multimedia Messaging Service (MMS), and faxes from organizations.

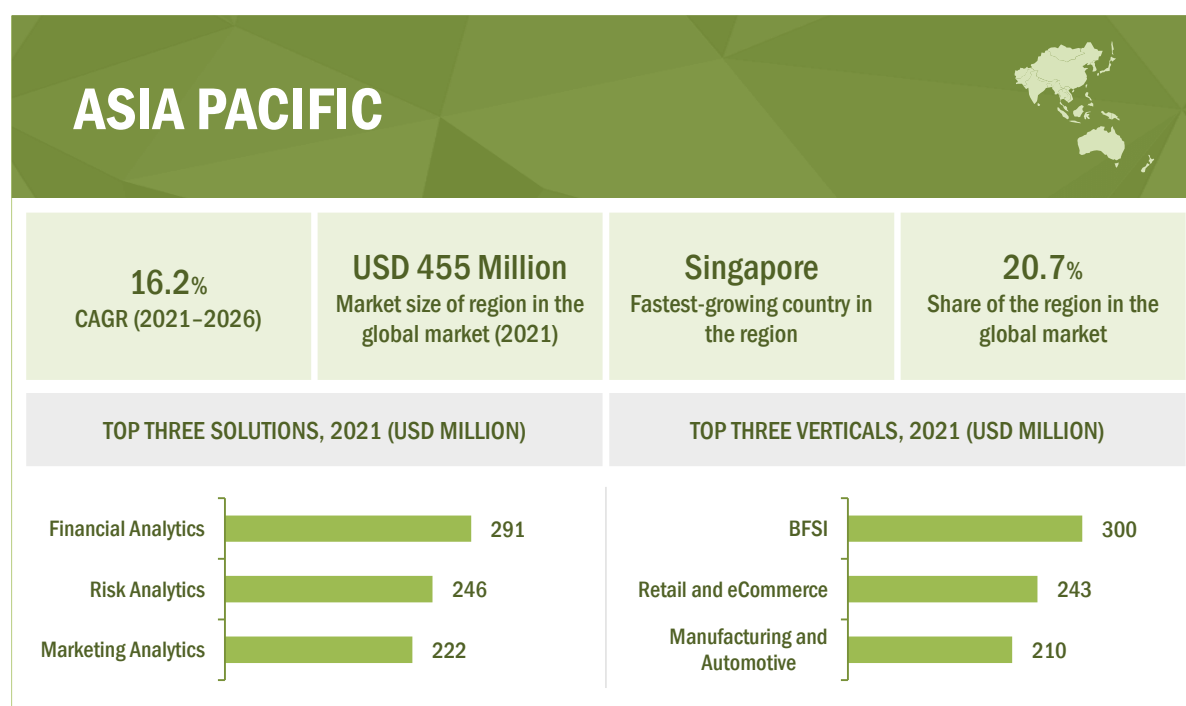
The PDPA covers personal data stored in electronic and non-electronic forms. The data protection provisions in PDPA (parts III to VI) generally do not apply to the following:

- Any individual acting on a personal or domestic basis
- Any employee acting in the course of their employment with an organization
- Any public agency or an organization in the course of acting on behalf of a public agency in relation to the collection, use, or disclosure of the personal data; companies can refer to the Personal Data Protection (Statutory Bodies) Notification 2013 for the list of specified public agencies
- Business contact information: This refers to an individual's name, position name or title, business telephone number, business address, business electronic mail address or business fax number, and any other similar information related to the individual, not provided by the individual solely for their personal purposes

These rules act as a baseline law operating as part of the law of Singapore. They do not supersede the existing statutes, such as the Banking Act and Insurance Act, but work in conjunction with them and the common law.

10.4.2.2 International Organization for Standardization (ISO) 27001

The ISO 27001 compliance provides a risk-based framework of policies and procedures for establishing, implementing, operating, monitoring, reviewing, maintaining, and improving the Information Security Management System (ISMS). The procedures and policies include defining a security policy and the scope of ISMS, conducting risk assessments, managing identified risks, selecting control objectives and controls to be implemented, and preparing a statement of applicability. ISO 27001 guidelines cover various areas, such as access control, audit, incident response, and system and information integrity. The key 114 controls required as a part of ISO 27001 compliance are security policy; organization of information security; asset management; HR security; physical and environmental security; communications and operations management; access control; information systems acquisition, development, and maintenance; information security incident management; business continuity management; and compliance.

FIGURE 44 ASIA PACIFIC: MARKET SNAPSHOT

*Rest of APAC includes Singapore, South Korea, and Malaysia.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 137 ASIA PACIFIC: PREDICTIVE ANALYTICS MARKET SIZE, BY COMPONENT, 2015–2020 (USD MILLION)

Component	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Solutions	342	421	523	655	881	995	23.8%
Services	114	141	175	220	297	336	24.2%
Total	456	562	699	875	1,177	1,331	23.9%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 138 ASIA PACIFIC: PREDICTIVE ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Solutions	1,496	2,006	2,542	3,329	4,098	4,819	26.4%
Services	507	681	866	1,137	1,403	1,654	26.7%
Total	2,003	2,687	3,408	4,466	5,500	6,473	26.4%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 139 ASIA PACIFIC: PREDICTIVE ANALYTICS MARKET SIZE, BY SOLUTION, 2015–2020 (USD MILLION)

Solution	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Financial Analytics	71	87	107	132	175	196	22.4%
Risk Analytics	58	71	88	109	146	164	23.2%
Marketing Analytics	47	59	74	93	127	146	25.4%
Sales Analytics	44	55	70	89	121	139	25.6%
Customer Analytics	27	34	43	54	74	85	25.3%
Web and Social Media Analytics	38	46	56	69	92	102	22.2%
Supply Chain Analytics	25	31	39	50	67	77	25.1%
Network Analytics	22	27	34	43	58	65	24.0%
Other Solutions*	9	11	13	16	20	22	19.5%
Total	342	421	523	655	881	995	23.8%

*Other solutions include HR analytics and legal analytics.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 140 ASIA PACIFIC: PREDICTIVE ANALYTICS MARKET SIZE, BY SOLUTION, 2021–2026 (USD MILLION)

Solution	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Financial Analytics	291	386	483	624	759	875	24.6%
Risk Analytics	246	327	413	537	658	770	25.7%
Marketing Analytics	222	301	386	512	638	760	27.9%
Sales Analytics	211	288	370	492	614	740	28.5%
Customer Analytics	129	175	224	296	369	439	27.8%
Web and Social Media Analytics	152	201	251	325	394	458	24.7%
Supply Chain Analytics	117	158	202	268	333	395	27.6%
Network Analytics	98	132	167	219	270	318	26.5%
Other Solutions*	31	39	47	56	63	66	16.0%
Total	1,496	2,006	2,542	3,329	4,098	4,819	26.4%

e: estimated; p: projected

*Other solutions include HR analytics and legal analytics.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 141 ASIA PACIFIC: PREDICTIVE ANALYTICS MARKET SIZE, BY SERVICE, 2015–2020 (USD MILLION)

Service	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Professional Services	75	91	112	140	187	211	23.1%
Managed Services	39	50	63	80	109	125	26.1%
Total	114	141	175	220	297	336	24.2%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 142 ASIA PACIFIC: PREDICTIVE ANALYTICS MARKET SIZE, BY SERVICE, 2021–2026 (USD MILLION)

Service	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Professional Services	316	422	532	693	849	992	25.7%
Managed Services	191	260	334	443	554	662	28.2%
Total	507	681	866	1,137	1,403	1,654	26.7%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 143 ASIA PACIFIC: PREDICTIVE ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2015–2020 (USD MILLION)

Professional Service	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Consulting	24	29	36	45	55	63	21.0%
Support and Maintenance	32	38	47	57	81	90	23.2%
Deployment and Integration	19	24	30	37	51	58	25.4%
Total	75	91	112	140	187	211	23.1%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 144 ASIA PACIFIC: PREDICTIVE ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2021–2026 (USD MILLION)

Professional Service	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Consulting	94	126	160	209	255	299	25.9%
Support and Maintenance	133	176	221	286	349	406	25.0%
Deployment and Integration	88	119	151	198	244	288	26.7%
Total	316	422	532	693	849	992	25.7%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 145 ASIA PACIFIC: PREDICTIVE ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2015–2020 (USD MILLION)

Deployment Mode	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
On-premises	223	273	336	416	552	615	22.5%
Cloud	119	149	187	239	329	381	26.2%
Total	342	421	523	655	881	995	23.8%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 146 ASIA PACIFIC: PREDICTIVE ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2021–2026 (USD MILLION)

Deployment Mode	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
On-premises	911	1,201	1,520	1,964	2,385	2,766	24.9%
Cloud	585	805	1,022	1,365	1,713	2,053	28.5%
Total	1,496	2,006	2,542	3,329	4,098	4,819	26.4%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 147 ASIA PACIFIC: PREDICTIVE ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2015–2020 (USD MILLION)

Organization Size	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Large Enterprises	328	400	492	608	809	906	22.5%
SMEs	128	162	207	267	368	425	27.2%
Total	456	562	699	875	1,177	1,331	23.9%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 148 ASIA PACIFIC: PREDICTIVE ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2021–2026 (USD MILLION)

Organization Size	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Large Enterprises	1,352	1,794	2,250	2,912	3,542	4,117	24.9%
SMEs	651	893	1,158	1,554	1,958	2,356	29.3%
Total	2,003	2,687	3,408	4,466	5,500	6,473	26.4%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 149 ASIA PACIFIC: PREDICTIVE ANALYTICS MARKET SIZE, BY VERTICAL, 2015–2020 (USD MILLION)

Vertical	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
BFSI	80	97	117	142	186	205	20.6%
Retail and eCommerce	59	71	87	108	145	162	22.6%
Telecom and IT	50	61	74	91	120	133	21.6%
Healthcare and Life Sciences	39	49	63	81	112	130	27.4%
Government and Public Sector	42	53	67	85	116	133	25.6%
Manufacturing and Automotive	46	57	71	90	122	138	24.9%
Media and Entertainment	36	45	56	71	95	108	24.3%
Energy and Utilities	33	41	53	67	92	106	26.3%
Travel and Hospitality	23	28	36	45	62	71	25.4%
Transportation and Logistics	27	34	43	54	74	85	25.5%
Other Verticals*	21	26	32	40	54	61	24.2%
Total	456	562	699	875	1,177	1,331	23.9%

*Other verticals include education & research and real estate.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 150 ASIA PACIFIC: PREDICTIVE ANALYTICS MARKET SIZE, BY VERTICAL, 2021–2026 (USD MILLION)

Vertical	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
BFSI	300	392	491	625	749	845	23.0%
Retail and eCommerce	243	324	406	529	648	758	25.6%
Telecom and IT	198	262	329	427	521	606	25.1%
Healthcare and Life Sciences	199	274	354	472	593	709	28.9%
Government and Public Sector	201	272	344	457	568	677	27.5%
Manufacturing and Automotive	210	285	365	482	600	725	28.1%
Media and Entertainment	163	220	280	367	453	538	26.9%
Energy and Utilities	161	219	280	370	459	546	27.6%
Travel and Hospitality	108	144	183	241	298	352	26.8%
Transportation and Logistics	130	174	222	293	362	426	26.9%
Other Verticals*	91	122	154	202	249	293	26.3%
Total	2,003	2,687	3,408	4,466	5,500	6,473	26.4%

e: estimated; p: projected

*Other verticals include education & research and real estate.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 151 ASIA PACIFIC: PREDICTIVE ANALYTICS MARKET SIZE, BY COUNTRY/REGION, 2015–2020 (USD MILLION)

Country/Region	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
China	165	201	247	306	407	456	22.5%
Japan	96	119	148	186	252	286	24.5%
India	46	57	71	90	122	138	24.9%
ANZ	63	78	98	125	170	195	25.5%
Rest of APAC*	87	107	133	168	226	256	24.2%
Total	456	562	699	875	1,177	1,331	23.9%

*Rest of APAC includes Singapore, South Korea, and Malaysia.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 152 ASIA PACIFIC: PREDICTIVE ANALYTICS MARKET SIZE, BY COUNTRY/REGION, 2021–2026 (USD MILLION)

Country/Region	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
China	677	896	1,122	1,450	1,762	2,044	24.7%
Japan	433	583	743	978	1,210	1,431	27.0%
India	210	285	365	482	600	712	27.6%
ANZ	297	403	518	687	856	1,021	28.0%
Rest of APAC*	387	520	661	869	1,073	1,265	26.8%
Total	2,003	2,687	3,408	4,466	5,500	6,473	26.4%

e: estimated; p: projected

*Rest of APAC includes Singapore, South Korea, and Malaysia.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

10.4.3 CHINA

10.4.3.1 The use of predictive analytics services for driving policy support and technological investments to fuel its adoption in China

China has shown an increased interest in driving policy support and technological investments in the APAC region. The country maintains a strong manufacturing base and is a leading global exporter of manufactured goods. The Chinese government initiatives for R&D and investments from global players are the key drivers of the country's growing predictive analytics market. Industry players across China are using predictive analytics solutions to enhance their marketing and sales business functions. Building new infrastructure has become a top development priority for China, referring to infrastructure that is digital, smart, and innovative. Chinese companies have exported surveillance tools to over 63 nations around the world, as well as a developing model of authoritarian rule in the digital sphere as a result of investments in internet infrastructure and information and communications technology. This so-called digital dictatorship of the CCP has been thoroughly documented across continents. However, such attempts are based on AI and predictive analytics, and they give a plethora of data collecting options for Chinese data surveillance platforms. As part of its post-COVID-19 relief package, China is making plans to construct new digital infrastructure across the country, including building 5G networks, AI, IoT, intercity high-speed rail, and setting up R&D institutions that give rise to government support for private sector participation. Traditional

industries, such as healthcare and life sciences, education, retail and eCommerce, manufacturing, services, and logistics, will be shifting toward digital business models that will adapt to the radical digital transformation or application of new technologies. Owing to the increased number of SMEs in the country, the demand for cloud-based predictive analytics software and services is expected to increase in China.

10.4.4 JAPAN

10.4.4.1 Rise of innovative technologies and collaboration between governments and businesses for digital transformation in Japan

Japan is widely known for innovations, such as high-quality production, just-in-time manufacturing, world-leading commercial electronics, extensive use of robots, solar power, and LED lighting. The collaboration between governments and businesses has transformed Japan into one of the world's most successful economies, both in terms of overall productivity and personal incomes. Japan is focusing on embracing newer technologies. Japan is a highly developed economy and is well-known for its advancements in ML, big data, and other cutting-edge technologies that enhance its productivity and process optimization. The entry of international businesses into the country is driving high data usage due to the increased usage of applications and platforms, such as data analytics, big data, and AI. Japan has a fast-growing economy, with its rapidly growing security and financial markets. The evolution of big data technology has helped Japan mitigate the risks by predicting the possibility of earthquakes. It has provided around 90% accuracy in recent times, and the companies that are examining satellite data are able to forecast an earthquake 30 days prior to the occurrence. Moreover, safety in the production facility has emerged as a major concern in the manufacturing industry, and machine breakdowns have caused severe production losses to the industry. These are the other factors expected to drive the adoption of the predictive analytics market in Japan.

The rising adoption can be attributed to the forward-looking vision of the companies that are thriving for better customer experience and satisfaction. In February 2020, the Japanese government agreed to a deal with AWS to help in moving HR systems and document management tools in the cloud environment by 2025. Being a developed economy with impressive growth in telecom and other sectors, Japan is expected to rapidly adopt predictive analytics solutions and services. The Japanese government and the private sector are making huge investments in AI technologies to increase competitiveness in Japan.

10.4.5 INDIA

10.4.5.1 Local entrepreneurs with a shift in technologies to provide support to Indian companies in the predictive analytics market

During the last decade, India has witnessed a marked shift where technologies have moved beyond supporting new business models and operational effectiveness for shaping business performance. When it comes to enterprises, India is an expanding area, and hence, it proves to be fertile for the propagation of predictive analytics services. Many foreign investors across the globe are actively seeking business with Indian companies. India is a rich ground for the dissemination of analytics services, with several local entrepreneurs working with huge global corporations and functioning as outsource supports for more established markets. According to a survey conducted by Opus research in October 2019, India showcases a strong interest in new technologies and believes that AI and analytics can relieve the workforce from repetitive, uncreative tasks, with a significant 82% of respondents agreeing with the statement.

10.4.6 AUSTRALIA & NEW ZEALAND

10.4.6.1 Adoption of advanced technologies and increase in young startups to increase the demand for predictive analytics

Australia continually produces new technology, from Wi-Fi to the bionic ear, the creation of huge technological businesses, to the growth of young startups. It is now viewed as an analytics hub. Such Australian technology enterprises frequently serve a global market, which has grown to perceive Australian cities as less competitive and crowded technology hubs than other major global technology hubs such as Silicon Valley and London. Australian enterprises are boosting their big data analytics capability by investing more in tools and services to bolster business processes and create new products and services. Australian vendors are adopting predictive analytics solutions with the adoption of the cloud environment, which is expected to increase the security and agility of data across the organization. In May 2019, a bank in ANZ changed its way of deriving insights from data to help its institutional customers make better strategic business decisions. The company has leveraged Google Cloud data analytics capabilities that will help financial institutions generate meaningful data insights quickly, helping customers make informed business decisions on issues, such as liquidity, risk, and cash management, or strategic calls, including store locations, inventory, and market positioning. The Massachusetts Institute of Technology (MIT) has announced that a Living Lab will be established at Adelaide's Lot Fourteen innovation hub. They assist users in gaining insight into important business challenges and making decisions from your data by giving timely data analysis via real-time reporting and visual representations.

10.4.7 REST OF ASIA PACIFIC

The Rest of APAC is also expected to be a major market for predictive analytics solutions and services, as countries such as Singapore, South Korea, and Malaysia are expected to play leading roles. Factors such as the availability of a skilled workforce at reasonable costs, development of infrastructure, technological capabilities, and government support to harness the market opportunities have contributed to the growth of the predictive analytics market in the region. The Rest of APAC is witnessing collaborations and partnerships and attracting investments from global organizations to promote R&D in big data, AI, and IoT. For instance, in January 2019, Rolls Royce announced its partnerships with Indian startups through its R2 Data Labs. The company would mentor and provide technical support to startups that specialize in IoT, AI, advanced analytics, quantum computing, and blockchain. This shift is driving the adoption of predictive analytics solutions and services in the region. The increasing government spending would boost IoT and ML infrastructure, which acts as a major factor driving the growth of the predictive analytics market in South Korea. According to a report from internet metrics company Ookla, internet speed in Singapore is the fastest in the world. This is mainly due to significant investments in infrastructure by various vendors in the country. A global technical company, HP, which is based in Singapore, opened its data-driven research campus for advanced manufacturing. Through this approach, the manufacturing ecosystems of HP have boosted 20% growth in its printer's supplies factory operation. The Rest of APAC is witnessing collaborations and partnerships for promoting R&D in data analytics, IoT, and AI.

10.5 MIDDLE EAST AND AFRICA

Predictive analytics vendors in the market are also trying to expand their footprints in MEA and are following various growth strategies to expand their customer base across the region. The countries in the Middle East are adopting technology advancements and investing in developing and building knowledge-based economies. Several organizations in the Middle East are leveraging these developments. The adoption of modern technologies has increased dramatically in MEA countries. The rapid digitalization in countries such as the UAE and South Africa has resulted in several linked devices in the area in recent years. Data security has become a prime concern for businesses in the United Arab Emirates (UAE) due to the availability of various financial services and their transactions; such factors are expected to boost the growth of the predictive analytics market in the coming years. Several government-led efforts are transforming the industrial sector's dynamics and helping reshape the landscape in significant economies, such as South Africa and the UAE. The UAE government is the first in the Middle East to announce its AI strategy and the first in the world to establish an AI Ministry. The government intends to invest in AI technologies in nine different areas. AI is widely being used in predictive analytics solutions, and this is driving the market.

The major drivers for the adoption are the rapid growth of infrastructure technologies and affordability of technologies specifically in customer- and data-centric verticals, such as BFSI, retail and eCommerce, and energy and utilities. Regional initiatives taken by the governments and the readiness of fast-growing SMEs in adopting emerging technologies are additional factors expected to drive the adoption of predictive analytics solutions and services in MEA.

The African governments are focusing on smart agriculture initiatives through the adoption of big data analytics and IoT solutions and services. For instance, smart irrigation, where IoT- and analytics-based measures could take into account on various parameters, including encompassing everything from moisture and humidity through to temperature and light intensity, and calculating the precise requirements for water, which will contribute to enhancing the irrigation efficiency and thus, reduce water losses. Predictive analytics solutions adoption will enable African farmers to track, analyze, and improve the quality of soil to avoid degradation and to minimize erosion, densification, salinization, acidification, and pollution by toxic elements.

10.5.1 MIDDLE EAST AND AFRICA: COVID-19 IMPACT

- A project has been developed by the Social Science Academy of Nigeria to leverage data science and AI tools for developing a better understanding of the impact of COVID-19 on gender-based violence and women support who are at the greatest risk.
- The York University joined forces with epidemiologists, modelers, physicists, statisticians, software engineers, and data scientists across Africa to integrate the power of predictive modeling and simulations with the capacity of a comprehensive COVID-19 monitoring dashboard that will be used to predict epidemic trends and inform decision-making and real-time management across Africa.
- The oil exporters of the Middle East are affected by the plummeting oil prices as a result of travel restrictions, following the public health crisis during the COVID-19 pandemic that may result in the downfall of the oil prices.
- A sudden drop in domestic and external demand for goods and products, especially crude oil and shutdown of production due to labor shortage, have been witnessed in the COVID-19 situation.
- The retail and eCommerce vertical is witnessing significant demand in the KSA as there is a surge in the online purchase of food products during the COVID-19 pandemic.
- Oil exporters are facing several problems due to travel restrictions and bans. Therefore, oil prices have fallen by over 50%. This has majorly impacted the economy of the Middle East.

- The UAE's Ministry of Economy has set up an interactive online tool to help companies address the impact of the COVID-19 pandemic, support business continuity, and promote sound business practices as part of efforts to shore up the private sector during the crisis.
- Governments in the Middle East are using real-time monitoring applications to spot quarantine breaches and backtracking metadata to figure out where they had been and whom they had contacted.
- South Africa has established the COVID-19 ZA South Africa Dashboard maintained by the Data Science for Social Impact Research Group at the University of Pretoria. Big data and AI have been majorly utilized in the healthcare sector for contact tracking and vaccine development to curb the effects of the COVID-19.

TABLE 153 MIDDLE EAST AND AFRICA: PROMINENT PLAYERS

MIDDLE EAST AND AFRICA: PREDICTIVE ANALYTICS PROMINENT PLAYERS	
Qlik	Happiest minds
Cloudera	Board International

10.5.2 MIDDLE EAST AND AFRICA: REGULATIONS

10.5.2.1 Israeli Privacy Protection Regulations (Data Security), 5777-2017

- The Israeli Privacy Protection Regulations (Data Security), 5777-2017, entered into force concurrently with the EU General Data Protection Regulation in May 2018. The regulations significantly expanded the privacy protection obligations that apply to most companies operating in Israel. These regulations impose detailed information security requirements upon any organization that is in possession of a database containing personal data, including data relating to employees, candidates for new positions, consumers, and other personal information.

10.5.2.2 GDPR Applicability in KSA

- From May 2018, KSA companies that process personal data of EU-based data subjects in the context of activities of a European establishment, offer goods or services to, or monitor the behavior of EU-based data subjects must comply with GDPR.
- Businesses in the KSA that fall within the scope of GDPR must appoint an EU representative located in one of the European countries of the individuals who are offered products or subject to behavioral monitoring. The representative acts on behalf of the companies in the KSA and may be addressed by any EU data protection supervisory authority and the data subject.
- The enforcement of the GDPR legislation leads to increased reliance on real-time analytics. Big data needs to be stored securely. GDPR enables customers to remove their data from one vendor's database and switch it to another vendor, enabling greater customer control over data.

10.5.2.3 Protection of Personal Information Act (POPIA)

- POPIA 4 of 2013 is the comprehensive data protection legislation enacted in South Africa. POPIA aims to give effect to the constitutional right to privacy while balancing this against competing rights and interests, particularly the right of access to information. POPIA was signed into law on November 19, 2013.
- The purpose of POPIA is to protect people from harm by protecting their personal information, to stop their money and their identity from being stolen, and generally to protect their privacy, which is a fundamental human right.
- The legislation affects any natural or juristic person who processes personal information, including large corporations and governments. The data protection laws of several other countries exempt SMEs, but not currently in South Africa.

TABLE 154 MIDDLE EAST AND AFRICA: PREDICTIVE ANALYTICS MARKET SIZE, BY COMPONENT, 2015–2020 (USD MILLION)

Component	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Solutions	222	274	311	374	457	499	17.6%
Services	74	92	104	126	154	169	17.9%
Total	296	366	415	500	611	667	17.7%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 155 MIDDLE EAST AND AFRICA: PREDICTIVE ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Solutions	732	962	1,189	1,457	1,738	2,001	22.3%
Services	248	327	405	497	595	687	22.6%
Total	980	1,289	1,594	1,954	2,333	2,688	22.3%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 156 MIDDLE EAST AND AFRICA: PREDICTIVE ANALYTICS MARKET SIZE, BY SOLUTION, 2015–2020 (USD MILLION)

Solution	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Financial Analytics	46	57	63	75	91	98	16.2%
Risk Analytics	38	46	52	62	76	82	17.0%
Marketing Analytics	30	38	44	53	66	73	19.1%
Sales Analytics	29	36	41	51	63	69	19.3%
Customer Analytics	18	22	26	31	38	42	19.0%
Web and Social Media Analytics	24	30	33	39	48	51	16.1%
Supply Chain Analytics	16	20	23	28	35	39	18.8%
Network Analytics	14	18	20	24	30	33	17.7%
Other Solutions*	6	7	8	9	11	11	13.5%
Total	222	274	311	374	457	499	17.6%

*Other solutions include HR analytics and legal analytics.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 157 MIDDLE EAST AND AFRICA: PREDICTIVE ANALYTICS MARKET SIZE, BY SOLUTION, 2021–2026 (USD MILLION)

Solution	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Financial Analytics	142	185	226	273	322	363	20.6%
Risk Analytics	120	157	193	235	279	320	21.6%
Marketing Analytics	108	144	181	224	271	316	23.8%
Sales Analytics	104	138	173	215	260	307	24.3%
Customer Analytics	63	84	105	130	156	182	23.7%
Web and Social Media Analytics	74	96	117	142	167	190	20.7%
Supply Chain Analytics	57	76	95	117	141	164	23.5%
Network Analytics	48	63	78	96	114	132	22.4%
Other Solutions*	15	19	22	25	27	27	12.3%
Total	732	962	1,189	1,457	1,738	2,001	22.3%

e: estimated; p: projected

*Other solutions include HR analytics and legal analytics.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 158 MIDDLE EAST AND AFRICA: PREDICTIVE ANALYTICS MARKET SIZE, BY SERVICE, 2015–2020 (USD MILLION)

Service	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Professional Services	48	59	67	80	97	106	16.9%
Managed Services	25	32	38	46	57	63	19.8%
Total	74	92	104	126	154	169	17.9%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 159 MIDDLE EAST AND AFRICA: PREDICTIVE ANALYTICS MARKET SIZE, BY SERVICE, 2021–2026 (USD MILLION)

Service	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Professional Services	155	202	249	303	360	412	21.7%
Managed Services	93	125	156	194	235	275	24.1%
Total	248	327	405	497	595	687	22.6%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 160 MIDDLE EAST AND AFRICA: PREDICTIVE ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2015–2020 (USD MILLION)

Professional Service	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Consulting	16	19	21	26	29	31	14.9%
Support and Maintenance	21	25	28	33	42	45	17.0%
Deployment and Integration	12	15	18	21	26	29	19.1%
Total	48	59	67	80	97	106	16.9%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 161 MIDDLE EAST AND AFRICA: PREDICTIVE ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2021–2026 (USD MILLION)

Professional Service	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Consulting	46	61	75	91	108	124	21.9%
Support and Maintenance	65	84	103	125	148	169	20.9%
Deployment and Integration	43	57	71	87	104	119	22.6%
Total	155	202	249	303	360	412	21.7%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 162 MIDDLE EAST AND AFRICA: PREDICTIVE ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2015–2020 (USD MILLION)

Deployment Mode	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
On-premises	145	177	200	237	286	308	16.3%
Cloud	77	97	111	136	170	191	19.9%
Total	222	274	311	374	457	499	17.6%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 163 MIDDLE EAST AND AFRICA: PREDICTIVE ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2021–2026 (USD MILLION)

Deployment Mode	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
On-premises	446	576	711	859	1,011	1,149	20.8%
Cloud	286	386	478	597	726	852	24.4%
Total	732	962	1,189	1,457	1,738	2,001	22.3%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 164 MIDDLE EAST AND AFRICA: PREDICTIVE ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2015–2020 (USD MILLION)

Organization Size	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Large Enterprises	213	261	292	347	419	454	16.4%
SMEs	83	105	123	152	191	213	20.8%
Total	296	366	415	500	611	667	17.7%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 165 MIDDLE EAST AND AFRICA: PREDICTIVE ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2021–2026 (USD MILLION)

Organization Size	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Large Enterprises	662	860	1,052	1,274	1,502	1,709	20.9%
SMEs	319	428	542	680	830	978	25.2%
Total	980	1,289	1,594	1,954	2,333	2,688	22.3%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 166 MIDDLE EAST AND AFRICA: PREDICTIVE ANALYTICS MARKET SIZE, BY VERTICAL, 2015–2020 (USD MILLION)

Vertical	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
BFSI	52	63	70	81	97	103	14.6%
Retail and eCommerce	38	46	52	62	75	81	16.5%
Telecom and IT	33	40	44	52	62	67	15.5%
Healthcare and Life Sciences	25	32	37	46	58	65	21.0%
Government and Public Sector	28	35	40	49	60	66	19.3%
Manufacturing and Automotive	30	37	42	51	64	69	18.6%
Media and Entertainment	24	29	33	40	50	54	18.1%
Energy and Utilities	21	27	31	38	48	53	19.9%
Travel and Hospitality	15	19	21	26	32	35	19.1%
Transportation and Logistics	18	22	26	31	38	43	19.2%
Other Verticals*	13	17	19	23	28	30	18.0%
Total	296	366	415	500	611	667	17.7%

*Other verticals include education & research and real estate.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 167 MIDDLE EAST AND AFRICA: PREDICTIVE ANALYTICS MARKET SIZE, BY VERTICAL, 2021–2026 (USD MILLION)

Vertical	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
BFSI	147	188	229	274	318	351	19.0%
Retail and eCommerce	119	155	190	232	275	315	21.5%
Telecom and IT	97	126	154	187	221	251	21.1%
Healthcare and Life Sciences	98	131	166	207	251	294	24.7%
Government and Public Sector	98	131	161	200	241	281	23.4%
Manufacturing and Automotive	103	137	171	211	254	301	23.9%
Media and Entertainment	80	106	131	161	192	223	22.8%
Energy and Utilities	79	105	131	162	195	227	23.5%
Travel and Hospitality	53	69	86	106	127	146	22.7%
Transportation and Logistics	63	83	104	128	154	177	22.8%
Other Verticals*	45	58	72	88	106	122	22.2%
Total	980	1,289	1,594	1,954	2,333	2,688	22.3%

e: estimated; p: projected

*Other verticals include education & research and real estate.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 168 MIDDLE EAST AND AFRICA: PREDICTIVE ANALYTICS MARKET SIZE, BY COUNTRY, 2015–2020 (USD MILLION)

Country	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
UAE	154	189	212	250	299	319	15.6%
South Africa	38	48	55	68	84	93	19.4%
Rest of MEA*	103	129	148	182	228	255	19.9%
Total	296	366	415	500	611	667	17.7%

*Rest of MEA includes Qatar, Saudi Arabia, Israel, and Kuwait.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 169 MIDDLE EAST AND AFRICA: PREDICTIVE ANALYTICS MARKET SIZE, BY COUNTRY, 2021–2026 (USD MILLION)

Country	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
UAE	458	587	721	864	1,008	1,125	19.7%
South Africa	139	185	232	288	349	413	24.4%
Rest of MEA*	383	517	641	801	975	1,150	24.6%
Total	980	1,289	1,594	1,954	2,333	2,688	22.3%

e: estimated; p: projected

*Rest of MEA includes Qatar, Saudi Arabia, Israel, and Kuwait.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

10.5.3 UAE

10.5.3.1 New investments, strategies, and products to drive the adoption of predictive analytics solutions in the region

The traditional business paradigm in the UAE has shifted away from oil, retail, and real estate to IT and digital innovations. The UAE has demonstrated its determination to stay ahead of the technological curve with the newest investments, strategies, and products in domains such as IoT, cloud, and eCommerce. They are prepared to embrace emerging technologies, such as predictive analytics, ML, and AI, by implementing effective plans and capital expenditures. With strategic and financial firms in the Middle East showing interest, UAE businesses are doing what businesses all over the world are doing: leveraging providers of predictive analytics to meet the requirements of the analytics process in an efficient and cost-effective manner. The UAE is committed to its Smart City Initiative 2021 in Dubai, aiming to make the city among the world's best-connected places. The Government of UAE is exploring the potential of AI, analytics, and IoT technologies to improve industry efficiency in terms of equipment expense and production costs. The UAE government is spending on digital transformation for utilities and is looking for reliable and secure solutions to manage assets efficiently and optimize their operations. The UAE is believed to have ambitious plans to build multiple smart cities and provide several online government services to its citizens. According to the report by the Dubai Technology Entrepreneur Campus, the UAE has witnessed the fastest growth in the coming years. According to a research study, remote health monitoring, smart buildings, and smart home concepts will see significant levels of investment over the next few years. The UAE government has introduced an initiative called Smart Dubai with an agreement between Dubai's Supreme Legislation Committee, the Department of Economic Development (DED), and the Dubai Land Department (DLD) to implement an IoT strategy that will help to create a digital world for future

government by 2021. According to Alteryx's 2019 Data + Digitization Report, almost two-third (62%) of UAE-based businesses classify themselves in the advanced stages of their analytic journeys. UAE businesses anticipate the biggest technology game-changers for data analytics success within the next five years to include self-service analytics (31%), augmented data discovery (27%), and explainable AI (27%).

10.5.4 SOUTH AFRICA

10.5.4.1 Growing digitalization to offer opportunities for deploying predictive analytics solutions in South Africa

South Africa has been lagging in technology adoption due to the lack of infrastructure, capital, human resources, and preventive government policies. However, the region has witnessed rapid growth in its IT industry in recent years, resulting in a major transformation of the FinTech sector in this region. In South Africa, there is a considerable need for training in ML, data science, and analytics. South Africa is a big market with numerous investment opportunities for multinationals. The growth of the country is attributed to various industry verticals, from manufacturing to technology, and energy and mining, which are supporting economic growth for the entire country. For instance, in 2018, China announced an investment worth USD 10 billion in the country. It has also invested USD 840 million in the BAIC vehicle plant at the Coega harbor.

Saudi Arabia is also committed to investing USD 10 billion in the energy sector, covering refineries, petrochemicals, and renewable energy. The African government is driving the country toward the implementation of a National Development Plan (NDP) and vision for 2023 across public and private sectors to improve the digital transformation of the organizations in the region. NDP is considered a blueprint document for African vendors. The aim of NDP is to eliminate poverty and reduce inequality by 2023 via the adoption of advanced technologies, such as big data analytics, IoT, AI, ML, social media, and cloud, that will enhance citizen engagement and drive operational efficiencies. Global industry players, such as IBM, Google, and Microsoft, have also made significant investments to conduct R&D activities for AI, analytics, and IoT in Africa. These are some indicators that favor the development of the predictive analytics market in South Africa. The country is stepping up to implement the National Development Plan (NDP) and Vision for 2030 across the public and private sectors. Digital transformation (DX) has become a key component of these development efforts. The South African government is investing in Industry 4.0 across the manufacturing, transportation and logistics, and energy and utility verticals, which would drive the growth of the predictive analytics market.

10.5.5 REST OF THE MIDDLE EAST AND AFRICA

Qatar, Saudi Arabia, Israel, and Kuwait are the major Middle Eastern countries expected to contribute to the predictive analytics market. These countries are expected to support the overall MEA technology market growth, while the UAE is expected to lead the technology adoption in MEA. The region comprises numerous technology startups and other SMEs, relying on advanced technologies to improve work efficiency and business productivity. The rise in the use of advanced solutions for the thorough and secured delivery of confidential information is a major driver for the advanced applications of predictive analytics in Middle Eastern countries. With the increasing globalization, there is a growing need for faster and synchronized delivery of information. This factor has fueled the demand for predictive analytics in the region.

10.6 LATIN AMERICA

With the current scenario, Latin America has started showing improvements despite the region's gradual adoption rate in terms of technologies and associated services. The startup setup landscape has evolved due to public and private investments, major infrastructure projects, and the entrepreneurial environment. However, the major shift toward digitalization in recent years and favorable government policies and initiatives are expected to change the scenario in the next few years and boost the local technology infrastructure and adoption of analytics across verticals. The Latin American region includes major countries, such as Brazil, Mexico, and Argentina. The region is slowly gaining traction in adopting major technologies, such as IoT, cloud, and AI. Due to the staggering growth of digitalized infrastructure across the region, the generation of large data volumes has become limited, which is expected to impact the growth of the predictive analytics market negatively. The countries considered for Latin America include Brazil, Mexico, and Rest of Latin America. Countries such as Brazil and Mexico have shown growth opportunities for predictive analytics vendors. Technological adoption is low in Latin America, but it is expected to grow in the coming years, primarily because many organizations invest in implementing various big data analytics solutions. For instance, IBM expanded its global cloud footprint by launching the IBM Cloud Multizone Region (MZR) in Latin America by late 2020. Brazil is among the countries with an enormous dependence on the manufacturing sector and is witnessing increasing internet adoption. However, the lack of a skilled workforce to support technologies is expected to hinder the market growth to some extent.

10.6.1 LATIN AMERICA: COVID-19 IMPACT

- During the pandemic, the major Latin American countries such as Brazil, Mexico, and Argentina are undergoing digital transformations, as evidenced by high mobile penetration; a growing IoT ecosystem; and soaring adoption of cloud computing, social media, and eGovernance platforms in these countries. As a result, the amount of digital data generated in the region is massive.
- Informal employment is prevalent in several Latin American countries. According to an industry expert, informal employment in Peru accounted for approximately 73% of the total employment in 2019 and contributed 19% to the country's GDP. Governments were compelled to choose between the economy and public health. Medical and biotech innovation is also a priority in Latin American governments. A huge number of specialists and research centers in Colombia, for example, are researching COVID-19 vaccines and therapies.
- Latin America's industrial sector is energy-intensive and highly dependent on China. China is one of the markets hit hard by COVID-19, and accounts for 30% of Brazilian exports, represents Argentina's second-largest trading partner, and is the main off-taker for Chile's mining industry.
- According to a recent survey conducted by DNA Human Capital in Chile, Colombia, and Peru, more than two-thirds (about 68.7%) of business executives believe COVID-19 would greatly affect the bottom-line of their companies.
- A recent report from UNICEF revealed that over 154 million (about 95%) students are temporarily out of school due to COVID-19 in Latin America and the Caribbean region.
- eCommerce companies, such as MercadoLibre, are also experiencing challenges in their business operations. In response to the pandemic, quarantine policies, and growing demand for basic necessities, the company announced fee waivers for its sellers of basic necessity products through the end of March. MercadoLibre reported 114% and 403% increase in its pharmacy and home laundry sales, respectively, in Mexico since the announcement.

- According to Telcos, mobile access is a powerful tool that can support the entire region to control the spread of the virus. Telefonica Brasil, on April 02, 2020, promised to offer cellphone observations to analyze the quarantined population to slow down the virus outbreak. In addition, they are offering big data capabilities and management to aggregated network data, mobility data, cloud data processing centers, as well as digital assistance capabilities for public administration and health institutions.

TABLE 170 **LATIN AMERICA: PROMINENT PLAYERS**

LATIN AMERICA: PREDICTIVE ANALYTICS PROMINENT PLAYERS	
CyberLabs	TIBCO Software
Hitachi Vantara	Qlik

10.6.2 LATIN AMERICA: REGULATIONS

10.6.2.1 Brazil Data Protection Law

- Brazil's data protection law, Lei Geral de Proteção de Dados (LGPD), came into force in August 2020. Before the LGPD, the data protection legislation in Brazil was sector-based and primarily regulated by the country's civil rights framework for the Internet (Internet Act) and Consumer Protection Code.
- The LGPD attempts to unify over 40 different statutes that currently govern personal data, both online and offline, by replacing certain regulations and supplementing others.
- The LGPD imposes detailed rules for the collection, use, processing, and storage of personal data. This new regulation replaces more than 40 different regulations that deal with data protection. The final version of the LGPD also introduced important revisions to the original version of the law, such as the creation of the National Data Protection Authority that will be responsible for overseeing and enforcing the LGPD.

10.6.2.2 Argentina Personal Data Protection Law No. 25.326

- Argentina was the first Latin American country to implement data protection laws and the first non-European country to be recognized by the European Commission as having adequate levels of data protection. The need to revisit the current legislation is a result of technological advances and the changing international landscape with the introduction of the GDPR since the Argentinian Personal Data Protection Act 2000 came into force.
- Argentina's Personal Data Protection Law No. 25.326 has guided how data is processed and protected. For the past year, lawmakers have debated a replacement bill that would more closely align with GDPR. Among other mandates, it requires that employers obtain consent from individuals before data is collected. The new bill expressly permits and regulates background checks and provides individuals with the right to access and request the deletion of their personal data. The replacement bill will go before the Senate, where it has a high likelihood of passing. If the bill passes into law, organizations will have a two-year period to prepare before enforcement.
- Argentina's new draft data protection bill proposes further changes to bring the country's data protection law in line with the GDPR. The bill acknowledges the right to be forgotten and the right to data portability. The other changes include stricter provisions in the area of cross-border transfers to countries with inadequate levels of data protection, new legal bases for data processing other than data subject consent, including legitimate interests, and new definitions of biometric and genetic data.

TABLE 171 LATIN AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY COMPONENT, 2015–2020 (USD MILLION)

Component	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Solutions	163	196	239	292	362	395	19.4%
Services	54	66	80	98	122	133	19.7%
Total	217	262	320	390	484	528	19.5%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 172 LATIN AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Solutions	599	783	980	1,218	1,473	1,708	23.3%
Services	203	266	334	416	504	586	23.7%
Total	801	1,048	1,314	1,633	1,977	2,294	23.4%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 173 LATIN AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY SOLUTION, 2015–2020 (USD MILLION)

Solution	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Financial Analytics	34	41	49	59	72	78	18.0%
Risk Analytics	28	33	40	49	60	65	18.8%
Marketing Analytics	22	27	34	42	52	58	20.9%
Sales Analytics	21	26	32	40	50	55	21.2%
Customer Analytics	13	16	20	24	30	34	20.9%
Web and Social Media Analytics	18	21	26	31	38	41	17.9%
Supply Chain Analytics	12	15	18	22	28	31	20.6%
Network Analytics	11	13	16	19	24	26	19.5%
Other Solutions*	4	5	6	7	8	9	15.2%
Total	163	196	239	292	362	395	19.4%

*Other solutions include HR analytics and legal analytics.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 174 LATIN AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY SOLUTION, 2021–2026 (USD MILLION)

Solution	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Financial Analytics	116	150	186	228	273	310	21.6%
Risk Analytics	98	128	159	197	236	273	22.7%
Marketing Analytics	89	117	149	187	229	269	24.9%
Sales Analytics	85	112	143	180	221	262	25.4%
Customer Analytics	51	68	86	108	133	155	24.7%
Web and Social Media Analytics	61	78	97	119	142	162	21.7%
Supply Chain Analytics	47	62	78	98	120	140	24.6%
Network Analytics	39	51	64	80	97	113	23.5%
Other Solutions*	13	15	18	21	23	23	13.3%
Total	599	783	980	1,218	1,473	1,708	23.3%

e: estimated; p: projected

*Other solutions include HR analytics and legal analytics.

Note: Numbers are rounded off to the nearest million, which may result in rounding-off errors.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 175 LATIN AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY SERVICE, 2015–2020 (USD MILLION)

Service	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Professional Services	36	43	51	62	77	84	18.7%
Managed Services	19	23	29	36	45	50	21.6%
Total	54	66	80	98	122	133	19.7%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 176 LATIN AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY SERVICE, 2021–2026 (USD MILLION)

Service	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Professional Services	126	164	205	254	305	352	22.7%
Managed Services	76	101	129	162	199	234	25.1%
Total	203	266	334	416	504	586	23.7%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 177 LATIN AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2015–2020 (USD MILLION)

Professional Service	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Consulting	12	14	16	20	23	25	16.7%
Support and Maintenance	15	18	21	26	33	36	18.8%
Deployment and Integration	9	11	13	17	21	23	20.9%
Total	36	43	51	62	77	84	18.7%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 178 LATIN AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2021–2026 (USD MILLION)

Professional Service	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Consulting	38	49	62	76	92	106	22.9%
Support and Maintenance	53	69	85	105	125	144	22.0%
Deployment and Integration	35	46	58	72	88	102	23.6%
Total	126	164	205	254	305	352	22.7%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 179 LATIN AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2015–2020 (USD MILLION)

Deployment Mode	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
On-premises	106	127	154	185	227	244	18.1%
Cloud	57	69	86	107	135	151	21.7%
Total	163	196	239	292	362	395	19.4%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 180 LATIN AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2021–2026 (USD MILLION)

Deployment Mode	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
On-premises	364	469	586	718	857	980	21.9%
Cloud	234	314	394	499	616	727	25.5%
Total	599	783	980	1,218	1,473	1,708	23.3%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 181 LATIN AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2015–2020 (USD MILLION)

Organization Size	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Large Enterprises	156	187	225	271	332	359	18.2%
SMEs	61	75	95	119	151	169	22.7%
Total	217	262	320	390	484	528	19.5%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 182 LATIN AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2021–2026 (USD MILLION)

Organization Size	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Large Enterprises	541	700	867	1,065	1,273	1,459	21.9%
SMEs	260	348	446	568	704	835	26.2%
Total	801	1,048	1,314	1,633	1,977	2,294	23.4%

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 183 LATIN AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY VERTICAL, 2015–2020 (USD MILLION)

Vertical	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
BFSI	38	45	54	64	77	81	16.3%
Retail and eCommerce	28	33	40	48	59	64	18.2%
Telecom and IT	24	28	34	41	49	53	17.2%
Healthcare and Life Sciences	18	23	29	36	46	52	22.8%
Government and Public Sector	20	25	31	38	48	53	21.1%
Manufacturing and Automotive	22	26	33	40	50	55	20.4%
Media and Entertainment	17	21	26	32	39	43	19.9%
Energy and Utilities	16	19	24	30	38	42	21.8%
Travel and Hospitality	11	13	16	20	25	28	20.9%
Transportation and Logistics	13	16	20	24	30	34	21.0%
Other Verticals*	10	12	15	18	22	24	19.8%
Total	217	262	320	390	484	528	19.5%

*Other verticals include education & research and real estate.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 184 **LATIN AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY VERTICAL, 2021–2026 (USD MILLION)**

Vertical	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
BFSI	120	153	189	229	269	299	20.1%
Retail and eCommerce	97	126	157	194	233	269	22.6%
Telecom and IT	79	102	127	156	187	215	22.1%
Healthcare and Life Sciences	80	107	136	173	213	251	25.8%
Government and Public Sector	80	106	133	167	204	240	24.5%
Manufacturing and Automotive	84	111	141	176	215	257	25.0%
Media and Entertainment	65	86	108	134	163	190	23.9%
Energy and Utilities	64	85	108	135	165	193	24.6%
Travel and Hospitality	43	56	71	88	107	125	23.7%
Transportation and Logistics	52	68	86	107	130	151	23.8%
Other Verticals*	36	48	60	74	89	104	23.3%
Total	801	1,048	1,314	1,633	1,977	2,294	23.4%

e: estimated; p: projected

*Other verticals include education & research and real estate.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 185 **LATIN AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY COUNTRY, 2015–2020 (USD MILLION)**

Country	2015	2016	2017	2018	2019	2020	CAGR (2015–2020)
Brazil	120	144	175	211	260	282	18.6%
Mexico	46	55	68	83	103	114	20.1%
Rest of Latin America*	51	63	77	96	120	133	21.0%
Total	217	262	320	390	484	528	19.5%

*Rest of Latin America includes Argentina, Columbia, Peru, and Chile.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 186 **LATIN AMERICA: PREDICTIVE ANALYTICS MARKET SIZE, BY COUNTRY, 2021–2026 (USD MILLION)**

Country	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Brazil	425	551	699	862	1,036	1,200	23.1%
Mexico	173	228	286	358	435	507	24.0%
Rest of Latin America*	204	269	328	413	506	587	23.6%
Total	801	1,048	1,314	1,633	1,977	2,294	23.4%

e: estimated; p: projected

*Rest of Latin America includes Argentina, Columbia, Peru, and Chile.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

10.6.3 BRAZIL

10.6.3.1 Increased use of the internet, technologies, and government organizations to increase the demand in Brazil

Brazil is one of the leading countries in Latin America in the adoption of predictive analytics solutions. Over the next five years, starting in 2020, the country is taking initiatives to achieve significant efficiency advantages in smart connected networks, digitization, and automated operations. According to Software.org, 21% to 24% of SMEs in Brazil expect digitization to aid in the development of more personalized products and services. The need to extract, visualize, and execute this intelligence in near-real-time, particularly during the COVID-19 epidemic, including measures to restrict its spread and assist firms to stay afloat, is fast becoming a mission-critical aim, boosting the predictive analytics market. Brazil is shaping the technological landscape of Latin America. Brazilian organizations are aware of the advantages of advanced analytics. According to the Dell EMC global survey, about 93% of the decision-makers in Brazil have an understanding of the value that big data can add to their businesses, out of which only 46% are planning to use these technologies due to budget constraints. However, due to the increasing social media networking and more businesses turning online in Brazil, there lies a need for understanding customer behavior in order to optimize processes and prices and make the products available.

Despite unfavorable economic scenarios, retailers are likely to profit from the rising eCommerce industry and platform in the region by adopting predictive analytics solutions and services to improve the overall customer experience and offer customized products and services.

10.6.4 MEXICO

10.6.4.1 Growth in trade, customer base, and government initiatives to fuel the growth of predictive analytics

In Mexico, the healthcare and life sciences industry vertical is said to be the second-largest in the whole of the Latin American region, after Brazil. Therefore, it is expected to offer several opportunities for the adoption of predictive analytics solutions and services in the marketing and sales business function. Cloud adoption in Mexico is also expected to increase due to major benefits, such as lower costs, improved innovations, and enhanced productivity. On February 26, Mexico's Federal Consumer Protection Agency (*Procuraduría Federal del Consumidor* - PROFECO) unveiled a new voluntary Ecommerce Ethics Code (*Código de Ética en Materia de Comercio Electrónico*), which can be adopted by companies doing business online in the country. Although the eCommerce Ethics Code is optional, it signifies the authority's baseline expectation for market participants and establishes the parameters and regulations for suppliers of goods and services who sell through digital platforms or virtual salesrooms. The growing predictive analytics trend in Mexico is owing to factors such as cost-effectiveness, growth in big data, the integration of mobile

technology, and the development of the social media landscape. The major companies, such as SAS Institute, SAP, Oracle, and IBM, are the key players in providing new business solutions in the predictive analytics market. Mexico's economy has moved into an open and diverse economy over the past 10 years. The stable economic growth indicates that the Mexican economy still has a high potential and will continue to grow dynamically in the future.

The increasing trade and rising customer base are expected to fuel the overall market growth in Mexico. Foreign trade comprises a larger percentage of Mexico's economy due to the benefits of free trade agreements. Government initiatives related to the development of Mexico are expected to fuel the predictive analytics market.

10.6.5 REST OF LATIN AMERICA

The Rest of Latin America includes Argentina, Colombia, Peru, and Chile. Various factors have limited the growth of these countries in the predictive analytics market. These factors are limited IT budgets, weak government regulations, and the lack of technological awareness and a skilled IT workforce. These countries contributed to the growth of the predictive analytics market in the region but are still behind Brazil and Mexico. In June 2019, the Argentine National Congress passed the Knowledge Economy Law to promote knowledge-based and digital economy investments, including software development and related activities, such as cloud computing, in addition to biotechnology, nanotechnology, aerospace, AI, robotics, and IoT, will be rewarded with tax incentives. The Government of Argentina has further implemented the SAP HANA platform to move toward a digital transformation that can help in the management of city resources, including streetlamps, parks, bus stops, drainage systems, buildings, and bridges that will provide real-time data insights. Colombia Data Analytics is a team of economists, statisticians, epidemiologists, pharmacists, mathematicians, clinician-scientists, and medical writers based in New York City, US, which aims to transform the healthcare sector in Colombia by achieving better healthcare services. Vendors in Peru and Chile are also in the initial phase of adopting predictive analytics solutions that will contribute toward the growth of the region.

Most of the verticals, such as BFSI, healthcare and life sciences, manufacturing, energy and utilities, retail and eCommerce, and transportation and logistics, have realized that the implementation of predictive analytics solutions shows its strength toward more advanced AI- and IoT-based solutions. In Peru, PwC's core focus is on mining and other heavy industries, where IoT-based big data analytics is in demand. Businesses throughout the rest of Latin America are eager to embrace AI-infused IoT, and they need proven solutions that can help businesses improve business operations and customer experience.

11 COMPETITIVE LANDSCAPE

11.1 OVERVIEW

The competitive landscape analyzes the growth strategies implemented by various major players in the predictive analytics market globally. The section includes the market evaluation framework, revenue analysis, and market ranking of the vendors in the predictive analytics market. It also highlights the strategies undertaken by the companies to expand their market shares and increase their predictive analytics technology offerings in the market.

11.2 KEY PLAYER STRATEGIES

TABLE 187 OVERVIEW OF STRATEGIES DEPLOYED BY KEY PLAYERS IN THE PREDICTIVE ANALYTICS MARKET

KEY PLAYER	PRODUCT TYPE	DEVELOPMENT	REGION
Microsoft	Solution	Microsoft partnered with Adaptive Biotechnologies to launch the ImmuneCODE database for sharing the immune response of the COVID-19 virus. The partnership aims at providing advanced solutions for diagnosing, treating, and preventing the virus and augmenting the existing research efforts that primarily focus on the biology of the virus.	The US contributes more than 50% of its company's revenue.
IBM	Solution	IBM expanded its cloud abilities in Latin America. These abilities will assist companies in shifting to the hybrid cloud model. It launched a new Multizone Region (MZR) for IBM cloud, which would help clients deploy workloads and applications in hybrid cloud environments.	North America contributes more than 40% to the company's revenue.
Oracle	Solution	Oracle announced to launch of 20 new Oracle Cloud regions. Oracle will launch 20 new Oracle Cloud regions by the end of 2020 for 36 Oracle Cloud Infrastructure regions, which include regions in new countries and dual, geographically separated regions.	North America contributes more than 50% of its revenue.
SAP	Investment	SAP established COVID-19 Emergency Fund (USD 3.4 Million) to support the urgent needs of the World Health Organization (WHO), Centers for Disease Control (CDC) and Prevention Foundation, and smaller nonprofits and social enterprises. These organizations work on the front lines to cater to local communities in crisis.	North America contributes more than 40% of its revenue.
SAS	Solution	SAS Institute enhanced its offerings of Software-as-a-Service (SaaS) across the UK public sector. It collaborated with AWS to make analytics available in cloud in the UK for public sector clients.	The US contributes more than 40% of its revenue.

Source: Company annual reports, Press Release, and MarketsandMarkets Analysis

The strategies of various predictive analytics players are assessed, along with their choices listed below:

- **Product Portfolio:** Key players in the predictive analytics market offer a comprehensive product portfolio. These players offer various predictive analytics and services to end users. This software may be in the form of API, platforms, and tools to make faster business decisions.
- **Regional Focus:** Players in the predictive analytics market have a well-diversified presence across the globe.
- **Inorganic plays/partnerships:** Key players have been primarily deploying inorganic development strategies. They are leveraging partnerships focused on technology enhancements to grow their revenues.

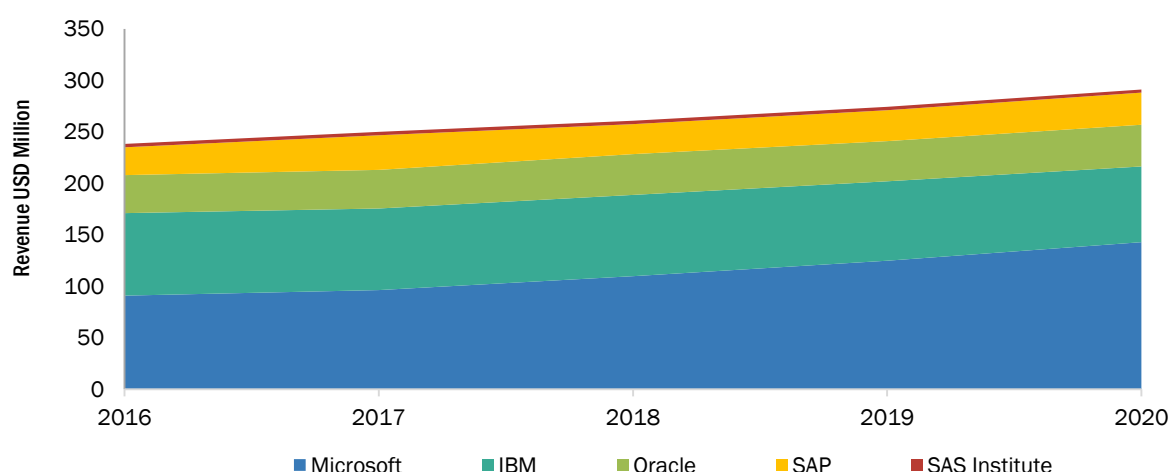
Considering the above considerations, MarketsandMarkets has concluded that different companies are deploying different strategies to win in the predictive analytics market. While SS&C Intralinks mostly focuses on North America, IBM, SAP, and SAS Institute have a diverse geographic revenue mix. In general, key companies are all strong in their home regions and explore geographic diversification alternatives to grow their businesses.

Microsoft, Oracle, IBM, SAP, and SAS Institute have been focusing on increasing their market shares through partnerships and acquisitions in the last few years. Companies are adopting growth strategies, but their focus is on inorganically expanding their presence.

11.3 REVENUE ANALYSIS

Players in the predictive analytics market were identified through secondary research. The share of each market player, in terms of value, was inferred from annual reports, presentations, and secondary research sources. Primary research interviews were conducted with key opinion leaders, such as CEOs, directors, industry experts, and other executives to validate revenue and market share.

FIGURE 45 REVENUE ANALYSIS FOR KEY COMPANIES IN THE PAST FIVE YEARS



Source: Company Website, Annual Reports, and SEC Filings

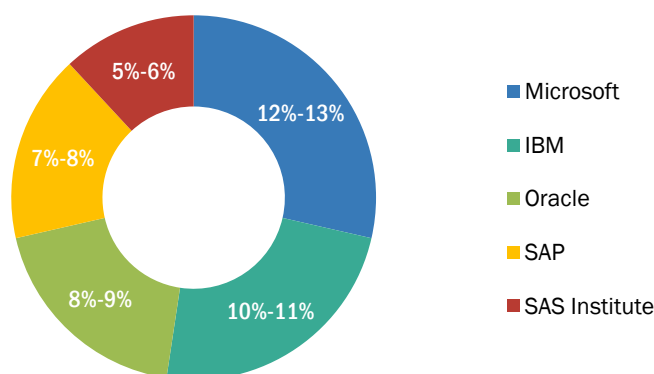
The global predictive analytics market is competitive with five public players: Microsoft, Oracle, IBM, SAP, and SAS Institute, holding less than 50% market share. From the figure provided above, it can be understood that the business revenue for most of the major players has slightly increased in the last two years, and it is expected to grow. Some of the reasons for the increase in revenue growth could be the customers moving toward predictive analytics after the pandemic, following the social distancing as the new normalcy.

11.4 MARKET SHARE ANALYSIS

In 2021, the predictive analytics market was competitive with the presence of several global and regional key market players, such as Microsoft, Oracle, IBM, SAP, and SAS Institute. These companies develop predictive analytics solutions and software based on extensive R&D, ensuring large market shares. The other players in this market are Google, Salesforce, AWS, HPE, Teradata, Alteryx, FICO, Altair, DOMO, Cloudera, Board International, Tibco Software, Hitachi Vantara, Qlik, Happiest Minds, Dataiku, RapidMiner, Biofourmis, In-med Prognostics, Aito.ai, Symend, Onward Health, Unioncrate, CyberLabs, Actify Data Labs, Amlgo Labs, and Verimos.

The degree of competition in the predictive analytics market is competitive in 2021, with the top five companies expecting to register a combined market share of less than 50%. This segment studies growth strategies such as product launches, partnerships, acquisitions, expansions, contracts, and collaborations adopted by market players from January 2016 to June 2021 to expand their global presence and increase their shares in the predictive analytics market. The chart below gives a detailed ranking of key market players. Factors such as the market presence of companies, their client/customer base, total revenue generated by them from the predictive analytics segment, and key developments undertaken by them have been considered for ranking the top five players in the predictive analytics market. The key manufacturers and technology providers that are a part of the predictive analytics market ecosystem have been considered under the company profiles section.

FIGURE 46 PREDICTIVE ANALYTICS MARKET SHARE ANALYSIS



Source: Company Website, Annual Reports, and SEC Filings

TABLE 188 PREDICTIVE ANALYTICS MARKET: DEGREE OF COMPETITION

DEGREE OF COMPETITION	COMPETITIVE
Total market share of top five players	45%–50%
Microsoft	12%–13%
IBM	10%–11%
Oracle	8%–9%
SAP	7%–8%
SAS Institute	5%–6%

THE DEGREE OF COMPETITION IS DEFINED AS BELOW:

Fragmented: When the top five players have a total market share < 25%

Competitive: When the top five players have a total of 25%–50% market share

Consolidated: When the top five players have a total market share > 50%

Note: The market share of the above companies is calculated based on the relevant segment revenues generated by companies.

Source: Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets

11.5 COMPANY EVALUATION QUADRANT

11.5.1 STARS

Vendors who fall into this category receive high scores for most of the evaluation criteria. They have a strong and established product portfolio and a strong market presence. They provide mature and reputable predictive analytics solutions and services and also have a strong business. IBM, Microsoft, SAP, SAS Institute, Google, Oracle, Salesforce, Board International, Teradata, Alteryx and Cloudera are the vendors who fall in this category.

11.5.2 EMERGING LEADERS

Emerging leaders are vendors who have demonstrated substantial product innovations as compared to their competitors. They have a focused product portfolio. However, they do not have a very strong growth strategy for their overall business. Rapidminer, Altair, Qlik, DOMO, and TIBCO Software are the vendors who fall in this category.

11.5.3 PERVASIVE PLAYERS

Pervasive players are established vendors with very strong business strategies. However, they are scored low in terms of the product portfolio. They focus on specific types of technologies related to products. The vendor who falls under this category is AWS, HPE, and FICO.

11.5.4 PARTICIPANTS

These are vendors with niche product offerings and are starting to gain their position in the market. They do not have strong business strategies as compared to other established vendors. They might be new entrants in the market and require some time before gaining significant traction. Vendors who fall under this category are Dataiku, Happiest Minds, and Hitachi Vantara.

FIGURE 47 KEY PREDICTIVE ANALYTICS MARKET PLAYERS, COMPANY EVALUATION MATRIX, 2021


Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

11.6 COMPETITIVE BENCHMARKING

TABLE 189 COMPANY SOLUTION FOOTPRINT (TOP 3)

COMPANY NAME	FINANCIAL ANALYTICS	MARKETING ANALYTICS	CUSTOMER ANALYTICS	TOTAL SCORE
Microsoft	Y	Y	N	3
IBM	N	Y	Y	3
SAP	Y	Y	Y	4
SAS Institute	Y	Y	N	3
Oracle	Y	Y	Y	4
Google	Y	Y	Y	4
Salesforce	Y	Y	Y	4
AWS	N	Y	Y	3

HPE	Y	Y	Y	4
Teradata	Y	Y	Y	3
Alteryx	Y	Y	Y	4
FICO	Y	Y	Y	3
Altair	Y	Y	N	3
DOMO	Y	N	Y	2
Cloudera	Y	Y	Y	3
Board International	Y	Y	Y	2
Tibco Software	N	Y	Y	2
Hitachi Vantara	Y	Y	Y	2
Qlik	Y	Y	Y	2
Happiest Minds	Y	Y	Y	2
Dataiku	Y	Y	Y	2
RapidMiner	N	Y	N	2
Biofourmis	N	N	N	1
In-med Prognostics	N	N	N	1
Aito.ai	Y	Y	Y	1
Symend	Y	N	Y	1
Onward Health	N	N	N	1
Unioncrate	Y	N	N	2
CyberLabs	N	N	N	1
Actify Data Labs	N	Y	Y	2
Amlgo Labs	Y	Y	Y	2
Verimos	Y	N	N	1

Source: Annual Reports, Press Releases, Investor Presentations, Primary Interviews, and MarketsandMarkets Analysis

TABLE 190 COMPANY REGION FOOTPRINT

COMPANY NAME	NORTH AMERICA	EUROPE	APAC	MIDDLE EAST AND AFRICA	LATIN AMERICA	TOTAL SCORE
Microsoft	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
IBM	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
SAP	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
SAS Institute	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
Oracle	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
Google	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
Salesforce	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
AWS	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
HPE	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
Teradata	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
Alteryx	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
FICO	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
Altair	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
DOMO	Yes (✓)	No	No	No	No	1
Cloudera	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
Board International	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
Tibco Software	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
Hitachi Vantara	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
Qlik	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	No	4
Happiest Minds	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	No	4
Dataiku	Yes (✓)	Yes (✓)	Yes (✓)	No	No	3
RapidMiner	Yes (✓)	Yes (✓)	No	No	No	2
Biofourmis	Yes (✓)	Yes (✓)	Yes (✓)	No	No	3
In-med Prognostics	No	No	Yes (✓)	No	No	1
Aito.ai	Yes (✓)	Yes (✓)	No	No	No	2
Symend	Yes (✓)	Yes (✓)	No	No	No	2
Onward Health	No	No	Yes (✓)	No	No	1
Unioncrate	Yes (✓)	Yes (✓)	No	No	No	2
CyberLabs	No	No	No	No	Yes (✓)	1

Actify Data Labs	No	No	Yes (✓)	No	No	1
Amlgo Labs	No	No	Yes (✓)	No	No	1
Verimos	Yes (✓)	No	No	No	No	1

Source: Annual Reports, Press Releases, Investor Presentations, Primary Interviews, and MarketsandMarkets Analysis

11.7 STARTUP/SME EVALUATION QUADRANT

The startup/SME evaluation matrix provides information related to the major players offering predictive analytics solutions and outlines findings and analysis on how well each market vendor performs within the pre-defined competitive leadership mapping criteria. The startups/SMEs selected for the matrix were founded after 2015. Vendor evaluations are based on two broad categories: strength of product portfolio and business strategy excellence. Each category carries various criteria based on which the vendors have been evaluated. The evaluation criteria considered under the strength of the product portfolios include breadth of offering, delivery (based on industries that the vendors cater to, deployment modes, and subscriptions), and support (based on pre and post-sales support services). The evaluation criteria considered under the business strategy excellence include reach (based on geographic presence), channel (based on channel partners that the vendors cater to), and inorganic growth (comprising partnerships, collaborations, and acquisitions).

11.7.1 PROGRESSIVE COMPANIES

Onward Health, Unioncrate, CyberLabs, Actify Data Labs, Amlgo Labs, and Cyber Labs are progressive companies in the predictive analytics market. They offer strong solution portfolios in the market. These vendors have been marking their presence by offering customized solutions as per requirements and adopting growth strategies to achieve advanced infrastructure in the predictive analytics market.

11.7.2 RESPONSIVE COMPANIES

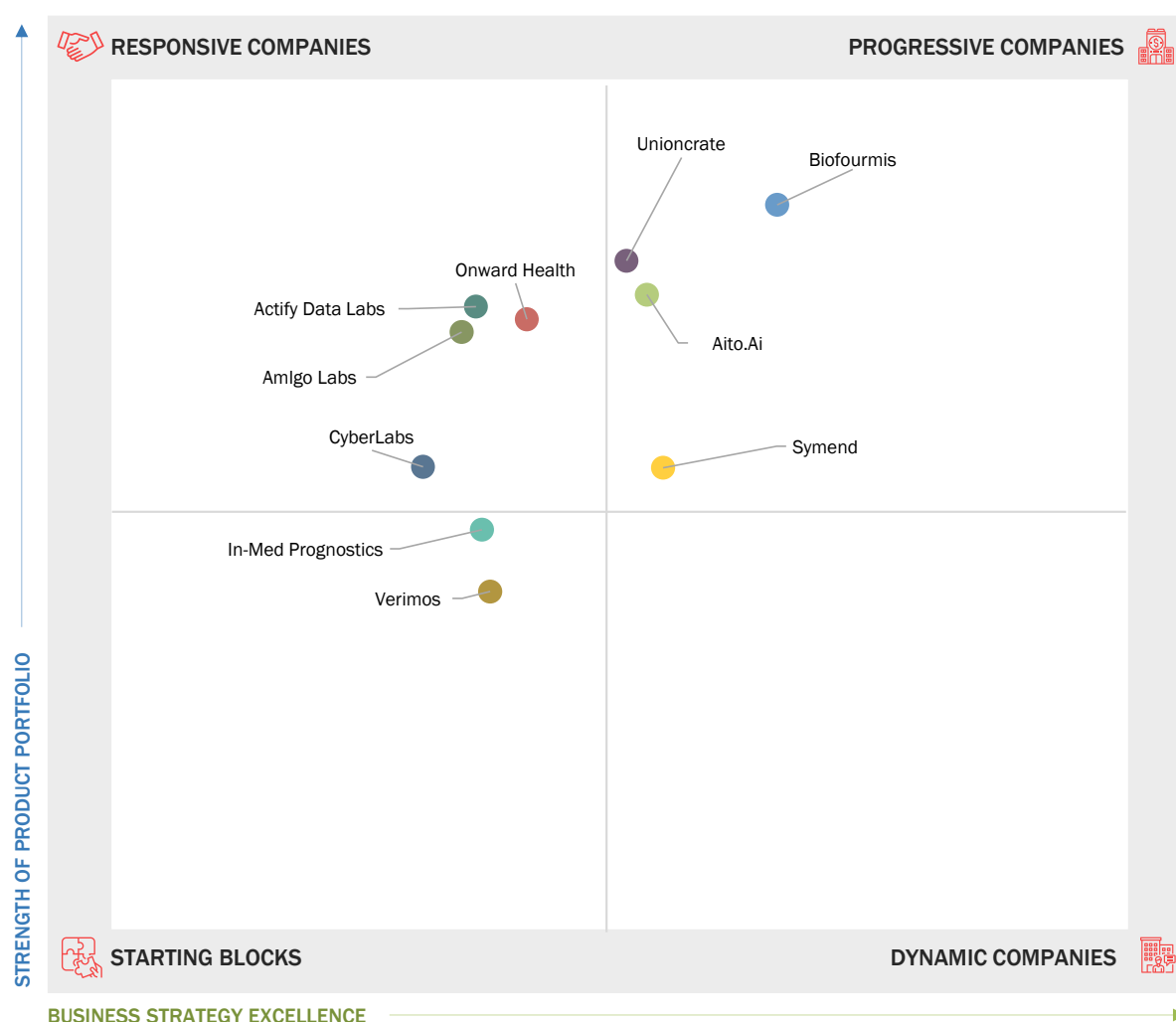
Biofourmis, Aito.ai, Symend, Onward Health, and Unioncrate are recognized as responsive companies in the predictive analytics market. It possesses innovative solutions to cater to future predictive analytics demands. They are concerned related to their product portfolio and have a robust potential to build strong business strategies to expand their business and stay at par with progressive companies. These vendors have been offering predictive analytics solutions to their customers for fulfilling their demands. Responsive companies have been at the forefront of the deployment of their solutions.

11.7.3 DYNAMIC COMPANIES

There are no dynamic companies in the predictive analytics market.

11.7.4 STARTING BLOCKS

In-med prognostics and Verimos are recognized as the starting blocks in the predictive analytics market. Most starting blocks have been undertaking multiple acquisitions to boost their sales capabilities across regions to offer integrated solutions and services to a wide range of clients.

FIGURE 48 STARTUP/SME PREDICTIVE ANALYTICS MARKET EVALUATION MATRIX, 2021

Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

11.8 COMPETITIVE SCENARIO

Key market developments include the recent developments of the key players in the global predictive analytics market. The market size analysis of the key industry players has been done based on various parameters, including the number of predictive analytics stakeholders, the depth of product offerings, and the revenue generated.

11.8.1 PRODUCT LAUNCHES

Companies have adopted the strategy of new product (solution) launches and enhancements to enhance their product portfolios and meet the growing demand for the predictive analytics market, globally. Leading companies, such as Alteryx, Oracle, IBM, SAS Institute, Oracle, Google, and Salesforce, have adopted this strategy.

TABLE 191 **PRODUCT LAUNCHES, JANUARY 2018–MAY 2021**

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2021-May	SAS Institute	Solution	SAS Viya	SAS will reinforce the foundation for data and analytic success by incorporating new data management solutions into its powerful, cloud-native SAS Viya platform.
2021-Apr	Oracle	Solution	SailGP	Oracle and SailGP, the world's finest sailing racing league, have expanded their successful predictive analytics platform in preparation for the second season. Oracle Cloud Infrastructure (OCI) is used by SailGP to supply real-time data to each of the league's eight national teams, as well as broadcast partners and fans globally.
2020-Jul	Google	Solution	BigQuery Omni	Google Cloud announced BigQuery Omni powered by Anthos for Multi-Cloud Analytics. BigQuery Omni provides enterprises with the flexibility they need to break down silos and create actionable business insights without having to pay expensive egress fees for moving data from other cloud providers to Google Cloud. It will help customers securely and cost-effectively access, analyze and find new insights empowering innovation in their business.
2020-Jun	SAS Institute	Solution	SAS Viya	SAS Institute released a new update for SAS Viya. The SAS Viya 4 release would constitute a new category of analytics for the cloud. The new category would help bring powerful analytics. This transformational release of cloud-native SAS Viya is the result of an investment worth USD 1 billion in AI.
2020-Apr	IBM	Solution		IBM released novel AI-powered technologies. IBM released novel AI-powered technologies to help the Health and Research Community accelerate the discovery of medical insights and treatments for COVID-19.
2019-Sep	Google	Solution	Cloud Storage Connector	Google introduced cloud storage connector for Predictive workloads. This solution allowed organizations to substitute their Hadoop Distributed File System (HDFS) with Google Cloud Storage. The Cloud Storage connector is an open-source Java client library that implements Hadoop Compatible FileSystem (HCFS) and runs inside Hadoop JVMs, which allows big-data processes, like Hadoop or Spark jobs, access to underlying data from Google Cloud Storage. The solution supports high-performance and archival use cases.
2019-Sep	Salesforce	Solution	Einstein Analytics	Salesforce launched Einstein Analytics for Financial Services. Einstein Analytics for Financial Services is a customizable analytics solution that provides AI-augmented business intelligence for wealth advisors, retail bankers, and managers. The solution would give a customized CRM experience for financial institutions.

2018-Aug	Alteryx	Solution	Visualytics	Alteryx announced a new version of the Alteryx Analytics Platform. Alteryx announced Visualytics that provides real-time, interactive visualizations across the Alteryx platform. Visualytics helps maximize the effectiveness of business-critical models by giving instant visual validation to data scientists on the health of their analysis.
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Source: Press Releases

11.8.2 DEALS

The inorganic strategies of acquisitions, partnerships, collaborations, and investments have been used by leading market players to broaden their customer base and enhance their technological capabilities. Key market players such as Teradata, Salesforce, SAP, TIBCO software, Cloudera, and Altair have adopted this strategy.

TABLE 192 DEALS, JANUARY 2018–MAY 2021

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION	DEAL SIZE
2021-Sep	Strategic Partnership	Teradata	Tesco PLC	Teradata stated that Tesco PLC is expanding its cooperation with Teradata by incorporating Vantage into its data platform. Tesco may use Vantage to eliminate silos and query all their data in real time, regardless of whether it is in the cloud, multiple clouds, on-premises, or any mix thereof, to gain a comprehensive view of their business.	NA
2021-Aug	Acquisition	Altair	S-FRAME	Altair announced the acquisition of S-FRAME Software, a structural analysis software platform used by engineers worldwide to evaluate a structure's ability to endure external loads (such as wind, water, and snow) and meet design code requirements.	NA
2021-Jan	Acquisition	TIBCO Software	ibi	TIBCO Software announced the successful completion of its acquisition of Information Builders, Inc. (ibi), which it announced in October. ibi's data management and analytics capabilities will be added to the superior TIBCO Connected Intelligence platform as a result of the acquisition. Following the completion of the acquisition, TIBCO software will concentrate on business alignment and resource mapping to ensure the continued health and support of its increased client base and global partner network. TIBCO's innovative products will provide ibi customers and partners with a slew of new modern solutions to address important business concerns.	NA
2020-Aug	Strategic Partnership	SAP	HPE	SAP partnered with HPE to deliver SAP HANA Enterprise Cloud with HPE GreenLake Cloud Services. This partnership will enable customers to keep their SAP software landscape and data on-premises.	NA

2020-Feb	Acquisition	Salesforce	Vlocity	Salesforce acquired Vlocity for USD 1.3 billion. Vlocity is one of the largest and fastest-growing Salesforce-native Independent Software Vendors (ISVs). Through this acquisition, Salesforce took the ownership of industry cloud.	USD 1.3 billion
2020-Apr	Collaboration	SAP	EY and Qualtrics	EY, SAP, and Qualtrics collaborated to provide resources to governments around the world for fighting the rapidly evolving COVID-19 pandemic. The collaboration would help organizations take immediate action in the fight against COVID-19. The solution would be available to 50 US states and more than 90 countries worldwide.	NA
2019-Jun	Strategic Partnership	Cloudera	IBM	Cloudera and IBM announced a partnership for bringing AI solutions and advanced data to organizations on the Apache Hadoop ecosystem. Due to this partnership, data-driven decisions will be accelerated.	NA
2018-December	Acquisition	Altair	Datawatch	Altair announced the acquisition of Datawatch. Altair acquired Datawatch that provides data intelligence to boost business analytics.	NA

Source: Press Releases

11.8.3 OTHERS

The other strategies include developments such as expansions and sales contracts. Key market players, such as Salesforce, HPE, and Cloudera, have adopted this strategy.

TABLE 193 OTHERS, 2018–2021

YEAR-MONTH	COMPANY NAME	DEVELOPMENT TYPE	DESCRIPTION
2021-Sep	Salesforce	Initiative	Salesforce announced new grants totaling USD 17.25 million to five school districts in the United States: San Francisco Unified School District, Oakland Unified School District, Chicago Public Schools, Indianapolis Public Schools, and the New York City Department of Education. Furthermore, USD 1.75 million was awarded to education nonprofits in the US and Europe.
2020-May	HPE	Initiative	HPE deployed four COVID-19 test labs and four outpatient department centers across India to help contain the COVID-19 pandemic. HPE's COVID-19 test labs are self-contained, IoT-enabled, and telemedicine-ready facilities designed to enable quick and clean testing of suspected individuals. These facilities would operate on the basis of the specifications provided by the Ministry of Science & Technology.
2018-Mar	Cloudera	Business Expansion	Cloudera announced the expansion of Budapest, Hungary, headquarters. Cloudera expanded the R&D team of Budapest to increase the Hungarian employee base supported by Hungarian Investment Promotion Agency (HIPA).

Source: Press Releases

12 COMPANY PROFILES

12.1 INTRODUCTION

The section provides a brief overview of the major players operating in the predictive analytics market. It comprises information related to solutions and services; key strengths, weaknesses, and competitive threats; and recent developments during the last three years of the top players in the predictive analytics market. It analyzes the organic and inorganic growth strategies adopted by various players in the market to expand their global presence and increase their market share. The major players of the predictive analytics market generate a significant portion of their revenue from North America, followed by Europe and APAC. The top 32 vendors have been evaluated in this evaluation matrix, including Microsoft, IBM, SAP, SAS Institute, Oracle, Google, Salesforce, AWS, HPE, Teradata, Alteryx, FICO, Altair, DOMO, Cloudera, Board International, Tibco Software, Hitachi Vantara, Qlik, Happiest Minds, Dataiku, RapidMiner, Biofourmis, In-med Prognostics, Aito.ai, Symend, Onward Health, Unioncrate, CyberLabs, Actify Data Labs, Amlgo Labs, and Verimos.

12.2 KEY PLAYERS

12.2.1 MICROSOFT

12.2.1.1 Business and financial overview

TABLE 194 MICROSOFT: BUSINESS OVERVIEW

COMPANY AT A GLANCE - 2021	
Year Founded	1975
Country	US
City	Seattle, WA
No. of Employees	18,100+
Overall Annual Revenue	USD 143 billion
R&D Expenditure	USD 19 billion
Ownership	Public (NASDAQ: MSFT)

Source: Company Website, Annual Reports, and SEC Filings

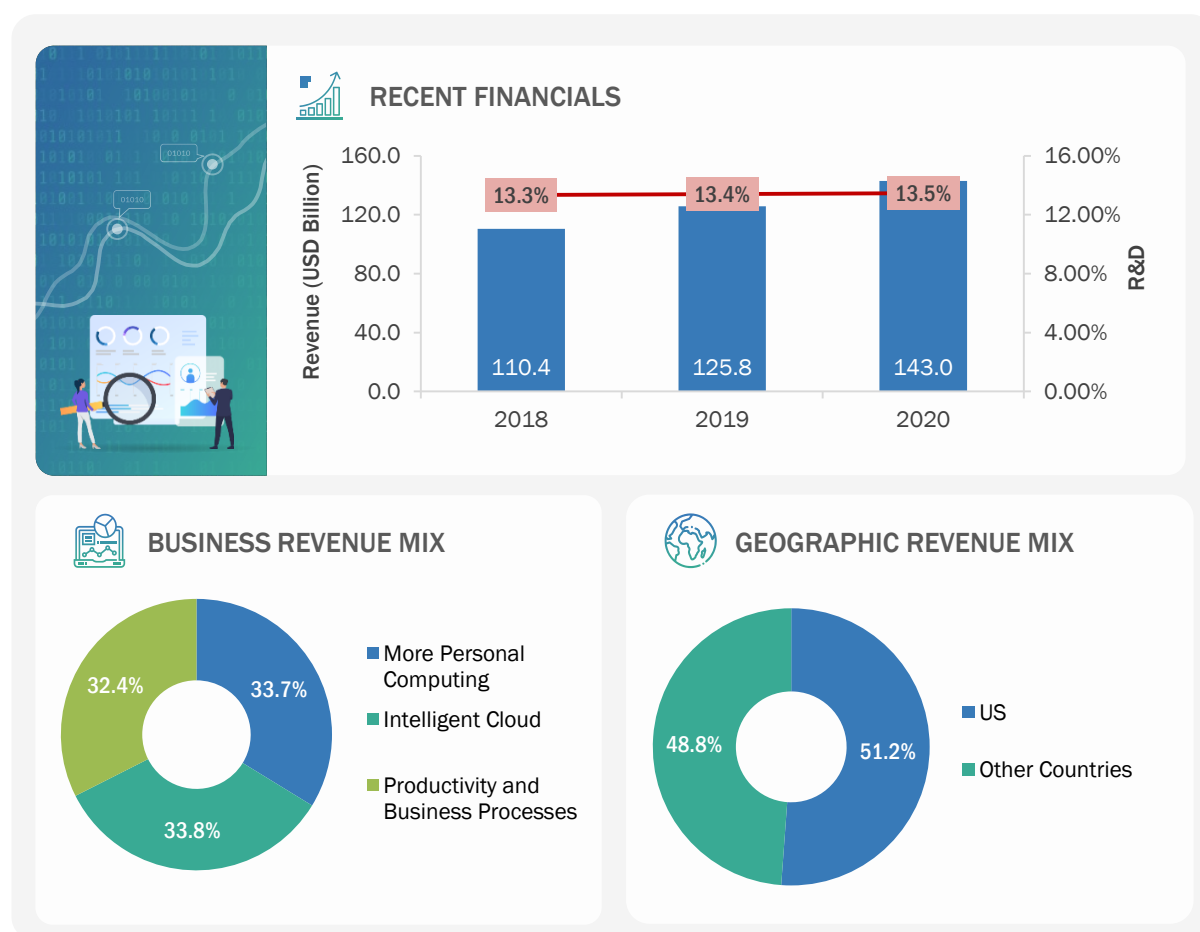
Microsoft develops software, services, devices, and solutions to compete in the intelligent cloud and intelligent edge verticals. With continuous investments in the cloud, Microsoft enables its customers to digitalize their business processes. The company's offerings include cloud-based solutions that provide customers with software, platforms, content, and deliver solution support and consulting services for its clients. Its product offerings include operating systems (OS), cross-device productivity applications, server applications, business solution applications, desktop and server management tools, software development tools, and video games. Microsoft's platforms and tools help drive the productivity of small businesses, the competitiveness of large businesses, and the efficiency of the public sector. The company's platform accelerates innovation across the spectrum of intelligent edge devices, from IoT sensors to gateway devices and edge hardware to build, manage, and secure edge workloads. Microsoft will invest USD 1

billion over the next four years in new technologies and innovative climate solutions. The company has also pledged a USD 50 million investment in AI for Earth to accelerate innovation.

The company is known to invest in data centers and other hybrid and edge infrastructure to enhance its services. It caters to a wide range of verticals, including finance and insurance, manufacturing and retail, energy and utilities, media and entertainment, public sector, healthcare, and IT and telecommunications. It has a geographical presence in more than 190 countries across North America, APAC, Latin America, MEA, and Europe.

In the predictive analytics market, Microsoft offers Azure ML that can be used to train, deploy, automate, manage, and track ML models. It supports various types of ML, including classical ML, deep learning, and supervised and unsupervised learning.

FIGURE 49 MICROSOFT: COMPANY SNAPSHOT



Note 1: The company's fiscal year ends on December 31.

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

12.2.1.2 Solutions offered

TABLE 195 MICROSOFT: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTION	DESCRIPTION	APPLICATION
Solution	Azure ML	It offers end-to-end life cycle support for ML. It contains tools that help to rapidly build and deploy ML models. It is a cloud-based service that enables detecting patterns while processing a huge amount of data to predict occurrences while processing new data. It gives built-in support for open-source tools and frameworks. These tools and frameworks allow ML model training and inferencing.	- ML model training - Inferencing - Predictive modeling - Manufacturing analysis - Product recommendation

Source: Company Website

12.2.1.3 Recent developments

TABLE 196 MICROSOFT: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2019-May	Microsoft	Solution	Azure ML	The Azure ML platform enabled intelligent applications, such as product recommendations in retail, load forecasting in energy production, image processing in healthcare, and predictive maintenance in manufacturing. The update released by Microsoft enabled enterprise-grade model deployment, management, and monitoring to simplify and accelerate the ML life cycle.
2018-Dec	Microsoft	Solution	Azure ML service	The Azure ML service contained advanced capabilities that would simplify and accelerate the process of building, training, and deploying ML models. It offers support for popular open-source frameworks, such as PyTorch, TensorFlow, and scikit-learn.

Source: Press Releases

TABLE 197 **MICROSOFT: DEALS**

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2020-Jun	Strategic partnership	Microsoft	Adaptive Biotechnologies	Microsoft partnered with Adaptive Biotechnologies to launch the ImmuneCODE database for sharing the immune response of the COVID-19 virus. The partnership aimed at providing advanced solutions for diagnosing, treating, and preventing the virus and augmenting the existing research efforts that primarily focus on the biology of the virus.
2020-Jun	Strategic partnership	Microsoft	SAS Institute	SAS Institute partnered with Microsoft. Through this partnership, Microsoft would offer cloud-based SAS industry solutions to its customers. As part of the partnership, both companies would migrate analytical products and the industry solutions of SAS Institute to Microsoft Azure as the preferred cloud provider for SAS Cloud.
2020-May	Collaboration	Microsoft	UnitedHealth Group	Microsoft collaborated with UnitedHealth Group to launch the ProtectWell protocol and app to support return-to-workplace planning and COVID-19 symptom screening. ProtectWell combined UnitedHealth Group's clinical and data analytics capabilities with Microsoft's technology leadership to help in the next phases of COVID-19 recovery efforts. The ProtectWell app was powered by Microsoft Azure, AI, and analytic solutions and took advantage of the Microsoft Healthcare Bot service being used around the world for AI-assisted COVID-19 symptom triaging.
2019-Oct	Acquisition	Microsoft	Mover	Microsoft acquired Mover, a Canadian startup that focused on cloud file migration. It supported migration from various cloud service providers, including Box, Dropbox, Egnyte, and Google Drive, to OneDrive and SharePoint.
2019-Jul	Acquisition	Microsoft	BlueTalon	Microsoft acquired BlueTalon, which provided Unified Data Access Control solutions for modern data platforms. It offered a customer-proven, data-centric solution for data access management and auditing across different systems.

Source: Press Releases

12.2.1.4 MNM view

12.2.1.4.1 Key strengths/right to win

Microsoft is a key player in the predictive analytics market. The growth of Microsoft largely depends on securely delivering continuous innovation and advancing leading productivity and collaboration tools and services, including Office, Dynamics, and LinkedIn. The company has been offering analytics solutions for a long time and has a global presence in this market, earning a 51% share from the US and the rest from other countries. The company's predictive analytics solutions help to deliver product recommendations in retail, load forecasting in energy production, image processing in healthcare, and predictive maintenance in manufacturing. The largest opportunities for the company are partnerships with various organizations on their digital transformation. Microsoft is uniquely positioned to become the strategic digital transformation platform.

12.2.1.4.2 Strategic choices made

Microsoft has been continuously investing in datacenters and other hybrid and edge infrastructure to support its services. The company plans to invest USD 5 billion in IoT and intelligent edge technologies and ecosystems by 2022. It would further advance customer and partner success to enhance its IoT journey. The Azure platform of Microsoft uniquely offers hybrid consistency, developer productivity, AI capabilities, and trusted security and compliance.

While the main R&D facilities of Microsoft are in Redmond, Washington, US, it also operates R&D facilities in the other parts of the US and around the world, including Canada, China, the Czech Republic, India, Ireland, Israel, and the UK. This global approach helps Microsoft to remain competitive in local markets and attracts new talent worldwide.

12.2.1.4.3 Weaknesses and competitive threats

Based on the geographic mix of 2020, it can be said that Microsoft is heavily dependent on the US market for most of its revenue as it constitutes a 51% share in the country's market. The Azure platform of Microsoft is facing diverse competition from companies offering platform-based ecosystems. These companies include Amazon, Google, IBM, Oracle, Salesforce, and SAP. The investments made by Microsoft in infrastructure and devices may continue to increase its operating costs and affect its operating margins.

The extent to which the COVID-19 pandemic is impacting businesses would depend on various evolving factors that cannot be reliably predicted. These factors include the duration and scope of the pandemic; governmental, business, and individuals' actions in response to the pandemic; and the impact on economic activities that include the possibility of recession or financial market instability. These factors may adversely impact consumer, business, and government spending on technology as well as customers' ability to pay for the products and services offered by Microsoft on an ongoing basis.

12.2.2 IBM

12.2.2.1 Business and financial overview

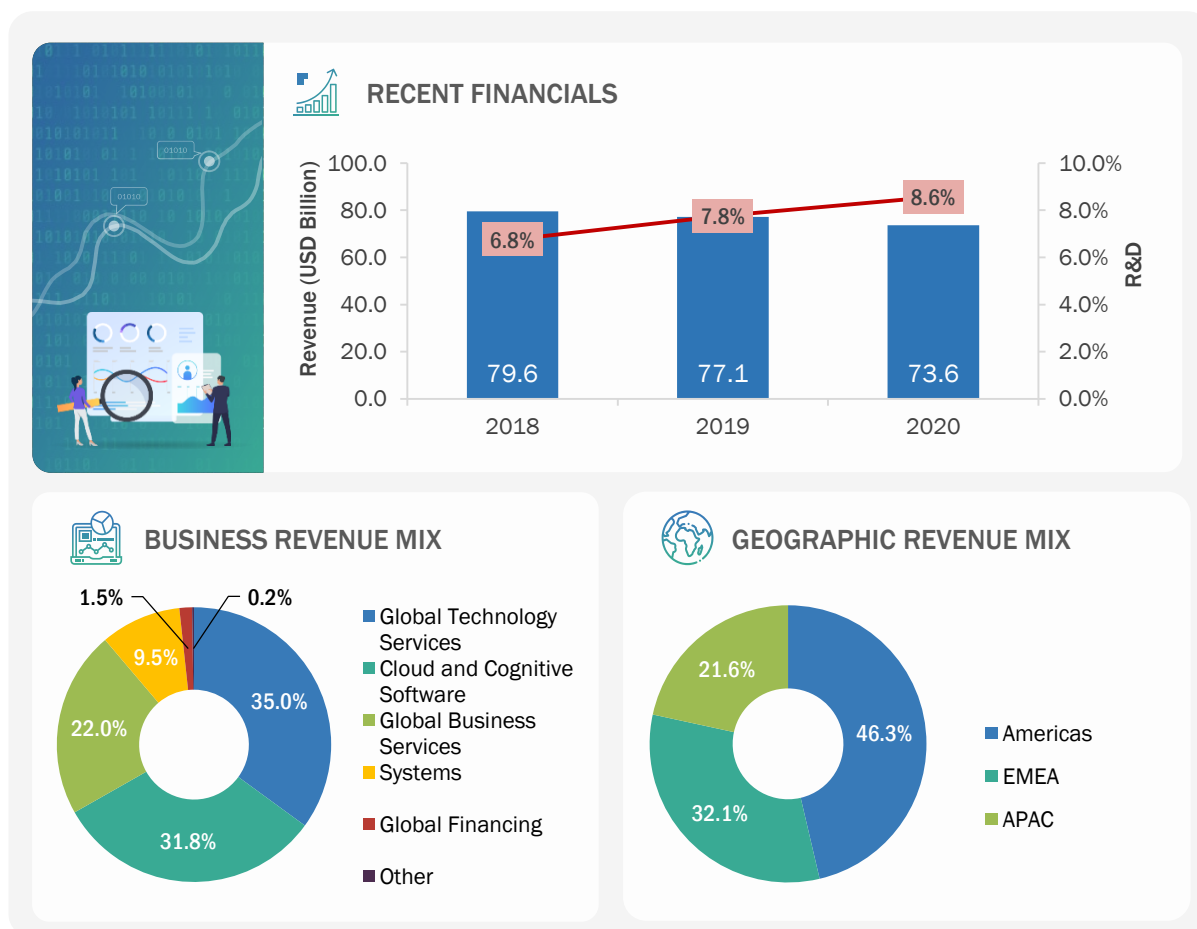
TABLE 198 IBM: BUSINESS OVERVIEW

COMPANY AT A GLANCE - 2021	
Year Founded	1911
Country	US
City	New York, NY
No. of Employees	430,000+
Overall Annual Revenue	USD 73.6 billion
R&D Expenditure	USD 6 billion
Ownership	Public (NYSE: IBM)

Source: Company Website, Annual Reports, and SEC Filings

IBM is a multinational technology and consulting corporation that offers infrastructure, hosting, and consulting services. The company operates through five major business segments: cloud and cognitive software, global business services, global technology services, systems, and global financing. It caters to various industry verticals that include aerospace and defense, education, healthcare, oil and gas, automotive, electronics, insurance, retail and consumer products, banking and finance, energy and utilities, life sciences, telecommunications, media and entertainment, chemicals, government, manufacturing, travel and transportation, construction, and metals and mining. The company has a robust presence in the Americas, Europe, MEA, and APAC and clients in more than 175 countries.

IBM offers IBM Predictive Analytics in the predictive analytics market. This solution helps discover actionable data-driven insights to drive customer engagement, reveal revenue-increasing opportunities, and outsmart frauds for protecting revenue and reputation.

FIGURE 50 IBM: COMPANY SNAPSHOT

Note 1: The company's fiscal year ends on December 31.

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

12.2.2.2 Solutions offered

TABLE 199 IBM: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTIONS	DESCRIPTION	APPLICATION
Solution	IBM Predictive Analytics	The solution unlocks data-driven insights that help businesses to make better decisions. IBM Predictive Analytics is a combination of advanced analytics capabilities spanning ad-hoc statistical analysis, predictive modeling, data mining, text analytics, entity analytics, optimization, real-time scoring, and ML, among others. The predictive analytic products offered under this solution are IBM Statistical Package for Social Sciences (SPSS) Modeler and IBM SPSS Statistics.	- Predictive modeling - Data mining - Text analytics - Entity analytics - Optimization - Real-time scoring

Source: Company Website

12.2.2.3 Recent developments

TABLE 200 IBM: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2020-Apr	IBM	Solution	Novel AI-powered technologies	IBM released novel AI-powered technologies to help the health and research community accelerate the discovery of medical insights and treatments for COVID-19. The solution was launched to help healthcare researchers, doctors, and scientists around the world accelerate COVID-19 drug discovery.
2019-Sep	IBM	Solution	Cloud Pak	IBM added new features to the Cloud Pak for Data solution. IBM added new DataOps features that will help clients get their business ready for AI. The new StoredIQ InstaScan offering would help enterprises streamline risk assessment, compliance, and AI projects.

Source: Press Releases

TABLE 201 IBM: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2020-Jul	Collaboration	IBM	Verizon	IBM collaborated with Verizon to work on 5G and edge computing innovation to strengthen the future of Industry 4.0. This collaboration would enable remote control robotics, near real-time cognitive video analysis, and plant automation. Both companies would jointly develop solutions to address worker safety, predictive maintenance, product quality, and production automation.
2020-Jul	Acquisition	IBM	WDG Automation	IBM acquired WDG Automation, a software provider of Robotic Process Automation (RPA). This acquisition would advance IBM's comprehensive AI-infused automation capabilities spanning business processes and IT operations. It would extend the ability of IBM services to transform the operations of clients as RPA, analytics, and AI bring more intelligence to enterprise workflows.
2020-Jun	Strategic partnership	IBM	SAP	IBM partnered with SAP. With this partnership, both companies would develop several new offerings designed to create a more predictable journey for businesses to become data-driven intelligent enterprises.
2020-Mar	Collaboration	IBM	White House Office of Science and Technology Policy and the US Department of Energy	IBM collaborated with the White House Office. In collaboration with the White House Office of Science and Technology Policy and the US Department of Energy, among others, IBM helped to launch COVID-19 High Performance Computing Consortium. The Consortium would bring more computing power—16 systems with more than 330 petaflops, 775,000 CPU cores, 34,000 GPUs, and many other features.

2018-Oct	Acquisition	IBM	Red Hat	IBM acquired Red Hat. This acquisition provided an open approach to the cloud, preventing the portability of data across many cloud platforms that will provide data security and cloud management consistency.
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Source: Press Releases

TABLE 202 IBM: OTHERS

MONTH-YEAR	COMPANY NAME	DEVELOPMENT TYPE	DESCRIPTION
2019-Aug	IBM	Business expansion	IBM expanded its cloud abilities in Latin America. These abilities would assist companies in shifting to the hybrid cloud model. It launched a new Multizone Region (MZR) for IBM cloud, which would help clients deploy workloads and applications in hybrid cloud environments.
2018-Jun	IBM	Business expansion	ExxonMobil, Bausch + Lomb, and Westpac shifted their workloads to IBM Cloud.

Source: Press Releases

12.2.2.4 MNM view

12.2.2.4.1 Key strengths/right to win

IBM creates value for clients by enabling new capabilities, leveraging products and solutions, and integrating technology and expertise in business processes with a broad ecosystem of partners and alliances. The company is one of the top vendors in the predictive analytics market due to its strong portfolio of solutions and services. IBM secured more than 1,800 patents in 2019. The data offerings of IBM help clients organize, collect, analyze, and embed their data into their workflow and span areas ranging from data management and discovery to reporting, governance, compliance, and risk management. One-third of IBM's research spending is focused on data, analytics, and cognitive computing. IBM has clients in more than 175 countries, with a presence in North America, Europe, APAC, MEA, and Latin America.

12.2.2.4.2 Strategic choices made

IBM has chosen to enhance its portfolio by launching new solutions, particularly in the cloud. This has resulted in IBM having a strong portfolio of solutions and services in the cloud. The investments made by the company in data and AI have strengthened its position of enterprise market leadership. It has invested USD 45 billion in R&D, forging the futures of cloud, AI, blockchain, and quantum computing. The company continues to invest, for a long-term opportunity, in key growth countries such as India and China and the regions of Eastern Europe, MEA, and Latin America to expand its product and service portfolios. In India, Vodafone-Idea, a joint venture of Vodafone Group, signed an IT outsourcing partnership with IBM to use IBM's hybrid cloud, analytics, and security capabilities for accelerating Vodafone-Idea's progression to an open, agile, and secure IT environment.

12.2.2.4.3 Weaknesses and competitive threats

IBM's revenue is highly dependent on the American market as the Americas contribute to 47% of its total revenue. As IBM is continuously investing in new technologies and key strategic areas for driving its revenue growth, the risk of failure of innovative initiatives is also increasing, which can impact the company's competitiveness. As the company has a global presence, its business and operations could be impacted by local legal, economic, political, and health conditions.

12.2.3 SAP

12.2.3.1 Business and financial overview

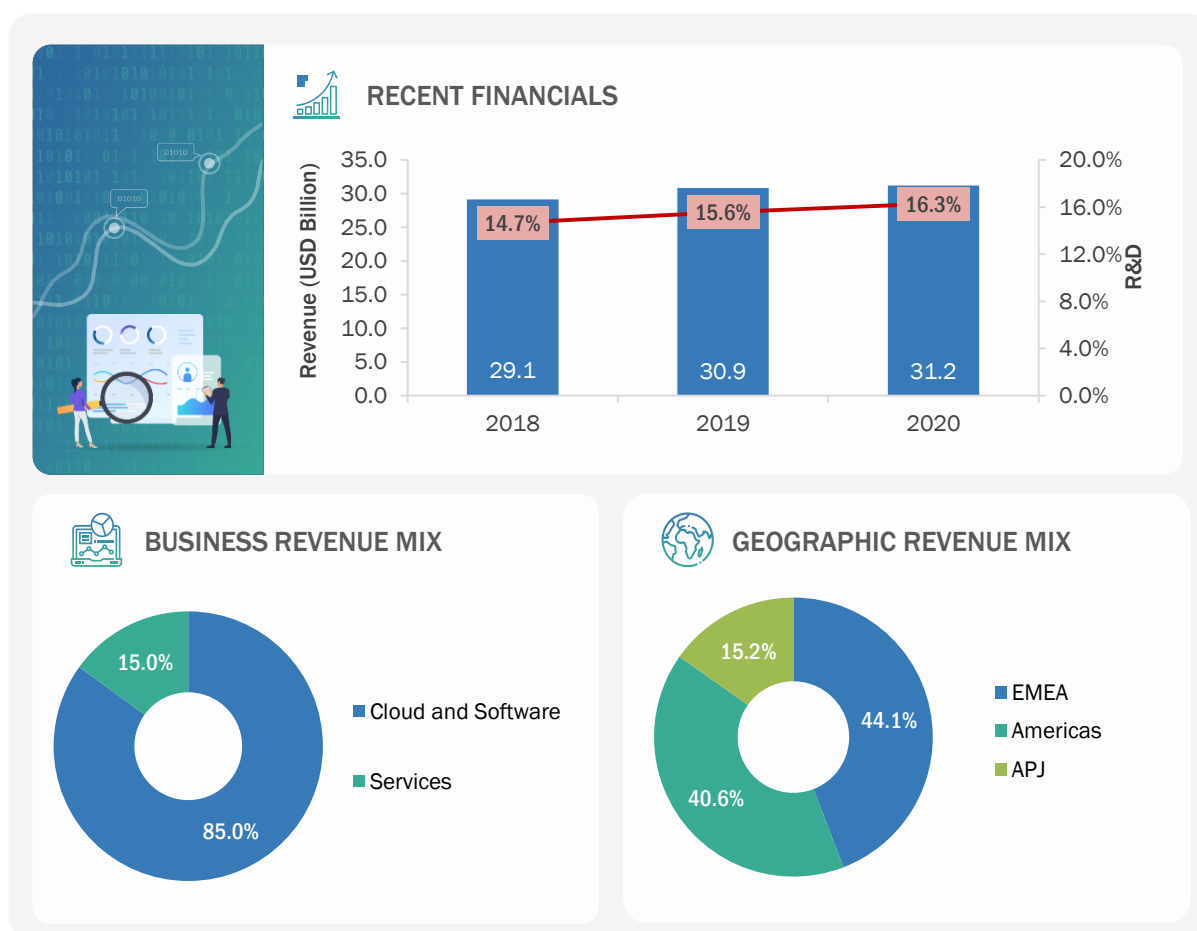
TABLE 203 SAP: BUSINESS OVERVIEW

COMPANY AT A GLANCE - 2021	
Year Founded	1972
Country	Germany
City	Walldorf
No. of Employees	102,430
Overall Annual Revenue	USD 31.2 billion
R&D Expenditure	USD 5.1 billion
Ownership	Public (NYSE: SAP)

Source: Company Website, Annual Reports, and SEC Filings

SAP is a leading provider of enterprise application solutions and services. It is also leading experience management, analytics, and BI company. Its solutions are compliant with GDPR. They enable enterprises to build intelligent AI- and ML-based software to unite human expertise with machine-generated insights. The company segments its diverse portfolio into applications, technology, and services; intelligent spend group; and Qualtrics. It works on an intelligent enterprise framework, which includes experience, intelligence, and operations business models. The company is known to offer the SAP HANA platform through its experience model of the intelligent enterprise framework, which allows both the transactional processing for data capture and retrieval and analytical processing for business intelligence and reporting. The company's software, technologies, and services address the three core elements of the intelligent enterprise: intelligent suite, business technology platform, and experience management platform for 25 industries and 12 lines of business. SAP's core business consists of selling software licenses and providing services, such as consulting, training, and maintenance. It has regional offices in 180 countries and caters to various industries, including energy and natural resources, financial services, consumer, discrete, and public services. SAP has around 437,000 customers worldwide, and its vast product portfolio covers Enterprise Resource Planning (ERP) and Finance, Customer Relationship Management (CRM) and Customer Experience, Network and Spend Management, Digital Supply Chain, HR and People Engagement, and Business Technology Platform. It has a geographical presence in the Americas, Europe, MEA, and APAC.

In the predictive analytics market, SAP offers the SAP Predictive Analytics solution. The solution inherits its data acquisition and data manipulation functionalities from SAP Lumira.

FIGURE 51 SAP: COMPANY SNAPSHOT

Note 1: The company's fiscal year ends on December 31.

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

12.2.3.2 Solutions offered

TABLE 204 SAP: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTIONS	DESCRIPTION	APPLICATION
Solution	SAP Predictive Analytics	It is used for statistical analytics, data mining, and predictive analytics. It enables building predictive models for discovering hidden insights and relationships in data to make accurate predictions about future occurrences. It helps gain unprecedented insights for customer acquisition through cross-sell, up-sell, and churn prevention. It takes the best action for every interaction across customer channels.	- Churn prevention - Predictive modeling - Customer behavior analysis
Solution	SAP BusinessObjects	This solution helps make informed decisions with access to real-time insights on a single platform. It allows expanding BI, predictive analytics, and planning capabilities with SAP Analytics Cloud.	- Performance monitoring - Real-time monitoring - Product and service development strategies

Source: Company Website

12.2.3.3 Recent developments

TABLE 205 SAP: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2020-Jun	SAP	Software	Corona Warn App	SAP launched the German government's Corona Warn App. The app was developed in collaboration between SAP and Deutsche Telekom as well as other partners. It was developed in an open-source mode, and the program code was continuously visible to the public on the development platform, GitHub.
2019-Oct	SAP	Solution	SAP announced enhancements for its business technology platform.	These enhancements reduced the gap between SAP HANA and analytics, enabling users to make smarter and faster business decisions. This approach helped customers get superior business results.
2019-Mar	SAP	Solution	SAP Analytics Cloud	SAP announced enhancements to the SAP Analytics Cloud solution, which included augmented analytics, BI, enterprise planning workflows, and data integration capabilities.
2018-Oct	SAP	Solution	SAP Cloud	SAP released new services for the SAP cloud platform to enable enterprises to easily connect demand and supply chains, fulfill security regulations and privacy norms, and provide a unique cloud environment.
2018-Mar	SAP	Software	SAP Predictive Analytics	The application edition of SAP Predictive Analytics software would help enterprises create and manage predictive ML models for applications running business activities, such as supply chain, finance and accounting, human capital management, and customer management.

Source: Press Releases

TABLE 206 SAP: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2020-Aug	Strategic partnership	SAP	HPE	SAP partnered with HPE to deliver SAP HANA Enterprise Cloud with HPE GreenLake Cloud Services. This partnership would enable customers to keep their SAP software landscape and data on-premises.
2020-Apr	Collaboration	SAP	EY and Qualtrics	EY, SAP, and Qualtrics collaborated to provide resources to governments around the world for combating the rapidly evolving COVID-19 pandemic. Qualtrics' and SAP's solutions would be implemented by EY's teams to help governments pre-screen and route potential COVID-19 patients, support healthcare workers, and manage the flow of information and resources to their citizens.

2018-Nov	Acquisition	SAP	Qualtrics	SAP signed an agreement to acquire Qualtrics, the global pioneer of the Experience Management (XM) software category, which enabled organizations to thrive in today's experience economy. SAP and Qualtrics together would accelerate the new XM category by combining experience data and operational data to power the experience economy.
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Source: Press Releases

TABLE 207 **SAP: OTHERS**

YEAR-MONTH	COMPANY NAME	DEVELOPMENT TYPE	DESCRIPTION
2020-Apr	SAP	Investment	SAP established COVID-19 Emergency Fund (USD 3.4 million) to support the urgent needs of the World Health Organization (WHO), Centers for Disease Control (CDC) and Prevention Foundation, and smaller nonprofits and social enterprises. These organizations work on the front lines to cater to local communities in crisis.

Source: Press Releases

12.2.3.4 MNM view

12.2.3.4.1 Key strengths/right to win

SAP's cloud and software revenue include cloud revenue and software licenses and support revenue. Cloud revenue and software revenue are the key revenue drivers of SAP as they affect the other revenue streams of SAP. SAP helps companies to enhance feedback with analytics to offer end-to-end customer experiences. Its strategy is to be a company powered by Intelligent Enterprise through Intelligent Suite of line of business (LoB) applications, Business Technology Platform, and the Experience Management (XM) platform. Its business analytics tools help customers to get insights, enabling them to adapt to businesses in real-time. It introduced its Business Technology Platform in 2019. The platform provides solutions across four technology areas: database and data management, analytics, application development, and integration and intelligent technologies.

12.2.3.4.2 Strategic choices made

To expand its business, the company is focusing on acquiring businesses, products, and technologies. The company is mainly focusing on organic investments in technology and innovations for ensuring the sustainable growth of its product portfolio. In January 2019, SAP acquired Qualtrics International Inc. to accelerate the new XM category by combining experience and operational data. SAP supports entrepreneurs through venture capital funds managed by Sapphire Ventures. Sapphire Ventures manages over USD 3.5 billion and has invested in more than 160 countries.

12.2.3.4.3 Weaknesses and competitive threats

As SAP operates globally in more than 180 countries and territories, there are numerous risks subject to international business operations. Uncertainty in the global economy, financial markets, social and political instability caused by state-based conflicts, terrorist attacks, and war could lead to SAP's business disruption and impose a negative impact on the business, financial position, profit, and cashflows of the company. As SAP expands into new countries and markets, including emerging and high-risk markets, risks may intensify due to the application of respective local laws and regulations.

12.2.4 ORACLE

12.2.4.1 Business and financial overview

TABLE 208 ORACLE: BUSINESS OVERVIEW

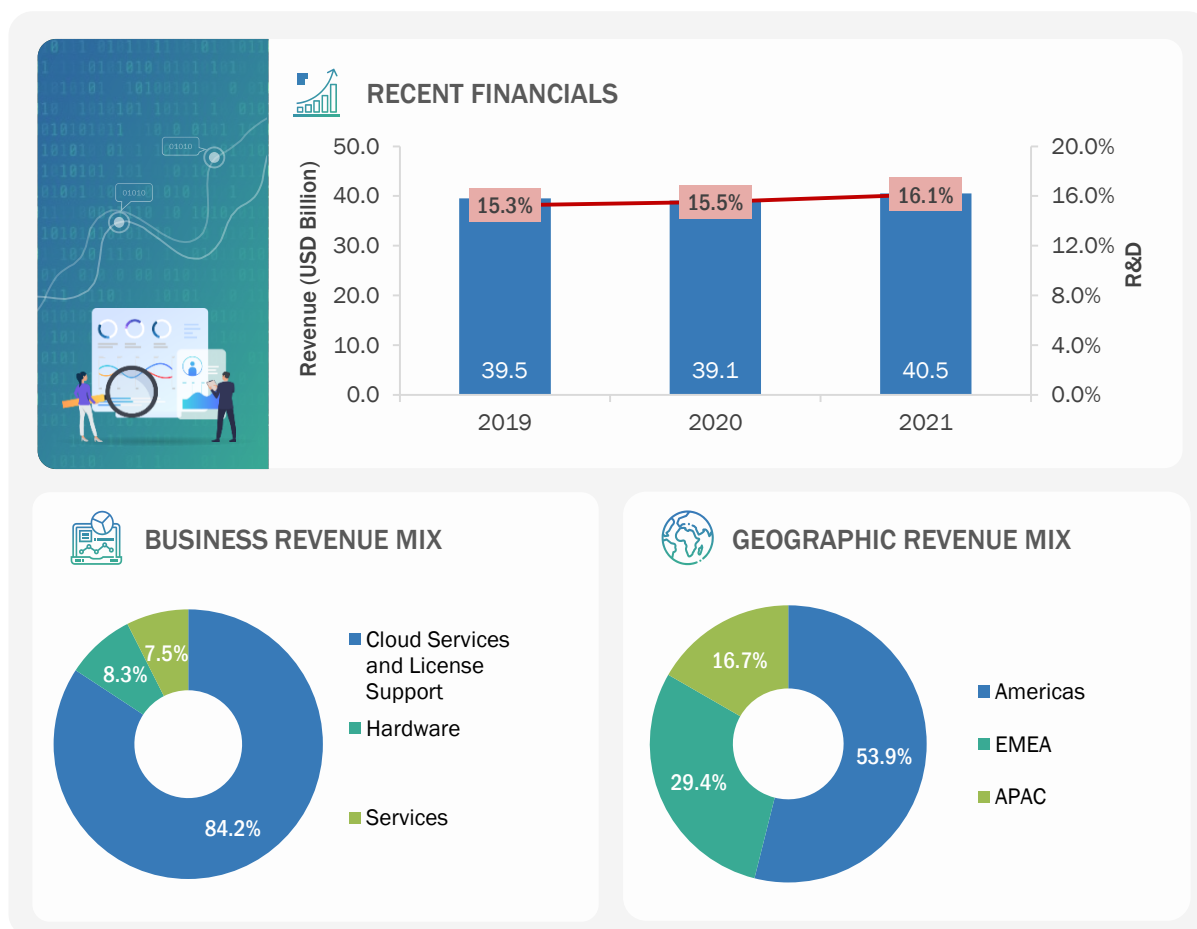
COMPANY AT A GLANCE - 2021	
Year Founded	1977
Country	US
City	Austin, TX
No. of Employees	135,000
Overall Annual Revenue	USD 40.5 billion
R&D Expenditure	USD 6.5 billion
Ownership	Public (NYSE: ORCL)

Source: Company Website, Annual Reports, and SEC Filings

Oracle is a global leader in delivering a broad spectrum of products, solutions, and services designed for meeting the requirements of corporate IT environments, such as platforms, applications, and infrastructure. Oracle's customers include businesses of various sizes, government agencies, educational institutions, and resellers. The company, directly and indirectly, sells its products and services through a worldwide sales force and Oracle Partner Network, respectively. It specializes in developing, manufacturing, and marketing hardware systems, databases and middleware software, and application software.

The company provides SaaS offerings designed to incorporate emerging technologies, such as IoT, AI, ML, and blockchain. It operates through three business segments, including cloud and license, hardware, and services, in more than 175 countries. It caters to 430,000 customers across industry verticals, such as banking, telecommunications, engineering and construction, financial services, healthcare, insurance, public sector, retail, and utilities. The company offers its solutions to customers present in more than 145 countries across the Americas, Europe, APAC, and MEA.

In the predictive analytics market, the company offers Oracle Data Mining (ODM) and Oracle Cloud Infrastructure Data Science Platform. ODM is a component of the Oracle Advanced Analytics Database Option and provides powerful data mining algorithms that help data analysts discover insights, make predictions, and leverage their Oracle data and investment

FIGURE 52 ORACLE: COMPANY SNAPSHOT

Note 1: The company's fiscal year ends on May 31.

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

12.2.4.2 Solutions offered

TABLE 209 ORACLE: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTIONS	DESCRIPTION	APPLICATION
Solution	Oracle Data Mining (ODM)	ODM helps build and apply predictive models in Oracle Database for predicting customer behavior, targeting best customers, developing customer profiles, identifying cross-selling opportunities, and detecting anomalies and potential frauds. Oracle Data Miner creates predictive models that can be integrated with applications to automate the discovery and distribution of new business intelligence predictions, patterns, and discoveries throughout the enterprise.	- Fraud detection - Profitability management - Customer segmentation and clustering - Customer behavior analysis
Solution	Oracle Cloud Infrastructure Data Science	With Oracle Cloud Infrastructure Data Science, enterprises can quickly leverage predictive analytics to drive positive business outcomes. The platform is a part of Oracle's Data and AI Platform that enables data scientists to easily build, train, and manage ML models on Oracle Cloud using Python, JupyterLab, and open-source ML libraries.	- Predictive modeling - Yield management - Product and service development strategies

12.2.4.3 Recent developments

TABLE 210 ORACLE: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2020-Mar	Oracle	Solution	Therapeutic Learning System	Oracle built and donated COVID-19 Therapeutic Learning System to the US government. Therapeutic Learning System helped physicians record patients' daily progress.
September-2019	Oracle	Solution	Oracle Analytics	Oracle announced the availability of its first Oracle Analytics for Applications offering. This offering was designed for Fusion cloud ERP applications used by companies to run financial processes. Oracle Analytics for Fusion ERP provides personalized analytics and improved cross-line-of-business analytics to line-of-business users and decision-makers.

Source: Press Releases

TABLE 211 ORACLE: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2021-Apr	Solution	Oracle	SailGP	Oracle and SailGP, the world's finest sailing racing league, have expanded their successful predictive analytics platform in preparation for the second season. Oracle Cloud Infrastructure (OCI) is used by SailGP to supply real-time data to each of the league's eight national teams, as well as broadcast partners and fans globally.
2018-May	Solution	Oracle	DataScience	DataScience platforms were used to organize work, easily access data and computing resources, and execute end-to-end model development workflows. This acquisition would enable customers to fully utilize ML due to the addition of a leading data science platform to Oracle Cloud.

Source: Press Releases

TABLE 212 ORACLE: OTHERS

YEAR-MONTH	COMPANY NAME	DEVELOPMENT TYPE	DESCRIPTION
2019-Sep	Oracle	Business expansion	Oracle announced the launch of 20 new Oracle Cloud regions by the end of 2020 for 36 Oracle Cloud Infrastructure regions, which included regions in new countries and dual, geographically separated regions in the US, the UK, EU, the UAE, Saudi Arabia, Canada, Brazil, Japan, South Korea, Australia, India, and Israel, and new government regions in the UK and Israel.

Source: Press Releases

12.2.4.4 MNM view

12.2.4.4.1 Key strengths/right to win

Oracle Cloud software-as-a-service and infrastructure-as-a-service offerings provide opportunities for Oracle to expand its cloud and license business and market them to small and medium-sized businesses and the non-IT lines of business purchasers. It offers a broad portfolio of offerings to address the big data requirements of organizations. These offerings include cloud-based services for data integration, data management, data science, and analytics and integrated ML. The company has been continuously supporting and partnering with organizations that provide critical medicines, research, and goods and services to combat the COVID-19 pandemic.

12.2.4.4.2 Strategic choices made

The investments made by Oracle in Oracle products and services through cloud and license, hardware and services businesses play an important role in its corporate strategy. In FY 2020, 2019, and 2018, the company invested USD 6.1 billion, USD 6.0 billion, and USD 6.1 billion, respectively, in R&D to enhance its existing portfolio of offerings and products and develop new technologies and services. As the world continues to navigate the risks and uncertainties associated with the COVID-19 pandemic, Oracle is committed to providing critical technologies, programs, and support to individuals and organizations to navigate, adjust, and continue their operations even in the pandemic situation.

12.2.4.4.3 Weaknesses and competitive threats

The conditions caused by the COVID-19 pandemic have adversely affected the willingness of customers to purchase Oracle's products and delayed prospective customers' purchasing decisions. In the fourth quarter of fiscal 2020, the company experienced more revenue declines as compared to the fourth quarter of fiscal 2019 and delayed payments from customers. If Oracle is unable to respond to and manage the impact of the COVID-19 pandemic, the business may be harmed. ML and AI are increasingly driving innovations in technology, but if they fail to operate as anticipated or the other products of the company do not perform as required, the business and reputation may be harmed.

12.2.5 SAS INSTITUTE

12.2.5.1 Business and financial overview

TABLE 213 SAS INSTITUTE: BUSINESS OVERVIEW

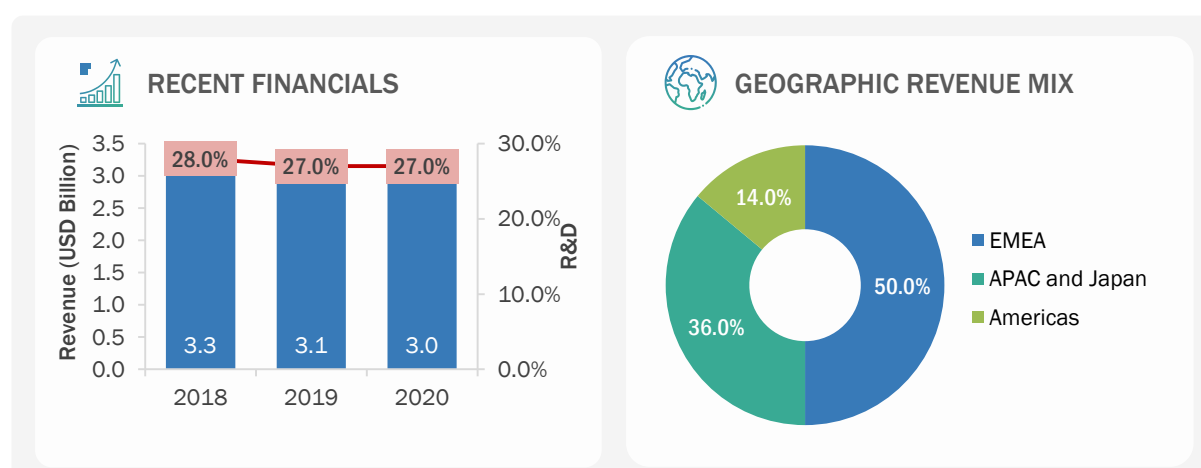
COMPANY AT A GLANCE - 2021	
Year Founded	1976
Country	US
City	Cary, NC
No. of Employees	13,939
Overall Annual Revenue	USD 3 billion
R&D Expenditure	USD 0.8 billion
Ownership	Private

Source: Company Website, Annual Reports, and SEC Filings

SAS Institute is the largest independent vendor in the predictive analytics market. The main objective of SAS Institute is to meet the customer requirements of each industry as well as provide new innovations in analytics. The company caters to customers across industries, such as retail, BFSI, government, healthcare, technology, and media and telecom. Cleveland Clinic has partnered with SAS Institute to create innovative models that help forecast patient volume, bed capacity, and medical equipment availability to address COVID-19 challenges.

In the predictive analytics market, the company offers the SAS Predictive Analytics solution. The solution helps turn data into timely insights for better and faster decision-making

FIGURE 53 SAS INSTITUTE: COMPANY SNAPSHOT



Note 1: The company's fiscal year ends on December 31.

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

12.2.5.2 Solutions offered

TABLE 214 SAS INSTITUTE: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTIONS	DESCRIPTION	APPLICATION
Solution	SAS Predictive Analytics	This solution helps derive accurate insights at the right time to increase the reach and value of data. It helps make evidence-based decisions. The different functionalities of SAS Predictive Analytics include data preparation, visualization, descriptive and predictive analysis, model deployment, and governance and operationalization.	- Governance, risk and compliance management - Predictive modeling
Solution	SAS Visual Analytics	The solution helps to visually explore all data, discover new patterns, and publish reports to the web and mobile devices. It enables building predictive models using the raw data and generate model outputs.	Risk analytics
Solution	SAS Visual Data Mining and ML	The solution solves complex problems faster with a single, integrated in-memory environment. It enables building and deploying own custom predictive modeling applications and Create score code for implementing predictive models.	Predictive modeling

Source: Company Website

12.2.5.3 Recent developments

TABLE 215 SAS INSTITUTE: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2021-May	SAS Institute	Solution	SAS Viya	SAS will reinforce the foundation for data and analytic success by incorporating new data management solutions into its powerful, cloud-native SAS Viya platform.
2020-Apr	SAS Institute	Solution	Online Learning Offerings	SAS launched free offerings on the online learning platform. These free offerings include SAS Academy for Data Science, SAS Learning Subscription, and customer intelligence resources.
2020-Mar	SAS Institute	Solution	COVID-19 dashboard	SAS Institute created a COVID-19 public dashboard using SAS Viya for visualizing key metrics and trends related to the COVID-19 pandemic.
2018-Jul	SAS Institute	Solution	SAS Viya	SAS Institute released new capabilities for SAS Viya. This new release was fueled by updated AI capabilities that would help data scientists and business users understand ML and AI models and adjust them for the required results. It would help organizations automate analytics using AI and enhance data privacy protection.

Source: Press Releases

TABLE 216 SAS INSTITUTE: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2021-Sep	Collaboration	SAS Institute	University of North Carolina	SAS and the University of North Carolina at Chapel Hill collaborated on the medication development process to avoid infectious disease threats such as COVID-19 from becoming pandemics. The collaboration was centered on the work of the University's Rapidly Emerging Antiviral Drug Development Initiative (READDI), developing broad-spectrum antiviral medications in case of future pandemics.
2019-Oct	Strategic partnership	SAS Institute	Red Hat	SAS Institute partnered with Red Hat to combine the analytics of SAS Institute with Red Hat OpenShift across hybrid cloud environments. Red Hat OpenShift was an enterprise Kubernetes platform offered by Red Hat.
2019-Apr	Strategic partnership	SAS Institute	Siemens	SAS Institute partnered with Siemens to meet the growing AI and ML demands. This partnership boosted productivity and decreased the operational risks of companies.
2018-Mar	Strategic partnership	SAS Institute	Data Republic	SAS Institute partnered with Data Republic to improve the data exchange platform services of Data Republic.

Source: Press Releases

TABLE 217 SAS INSTITUTE: OTHERS

YEAR-MONTH	COMPANY NAME	DEVELOPMENT TYPE	DESCRIPTION
2019-May	SAS Institute	Business expansion	SAS Institute launched a cloud offering in an AWS data center in Singapore.
2019-Nov	SAS Institute	Business expansion	SAS Institute enhanced its offerings of Software-as-a-Service (SaaS) across the UK public sector. It collaborated with AWS to make analytics available in the cloud in the UK for public sector clients.

Source: Press Releases

12.2.5.4 MNM view

12.2.5.4.1 Key strengths/right to win

SAS Institute has adapted to the demand for analytics in the cloud by introducing SAS Viya 4 to become a cloud-native company. The company helps customers operationalize their analytics and modernize the cloud to achieve the full potential of their analytics and AI investments and ultimately be successful in their digital transformation. SAS Platform is an engine that powers its analytics technology and handles the entire analytics life cycle, helping businesses operationalize analytics and get models into production faster.

12.2.5.4.2 Strategic choices made

SAS Institute has announced a USD 1 billion investment to advance AI technology, education, and services. A significant part of its annual revenue is reinvested in R&D to optimize the analytics life cycle for its customers. In 2019, SAS Institute announced AgTech, a new business unit to help farmers secure a safe and sustainable food supply for the growing world population. The company has invested USD 1 billion in AI over three years through software innovation, education, and expert services.

12.2.5.4.3 Weaknesses and competitive threats

The company's revenue is highly dependent on the American market as the Americas contribute to 49% of its total revenue. As SAS Institute highly depends on partnerships and business expansions, its business and operations could be impacted by local legal, economic, political, and health conditions.

12.2.6 GOOGLE

12.2.6.1 Business and financial overview

TABLE 218 GOOGLE: BUSINESS OVERVIEW

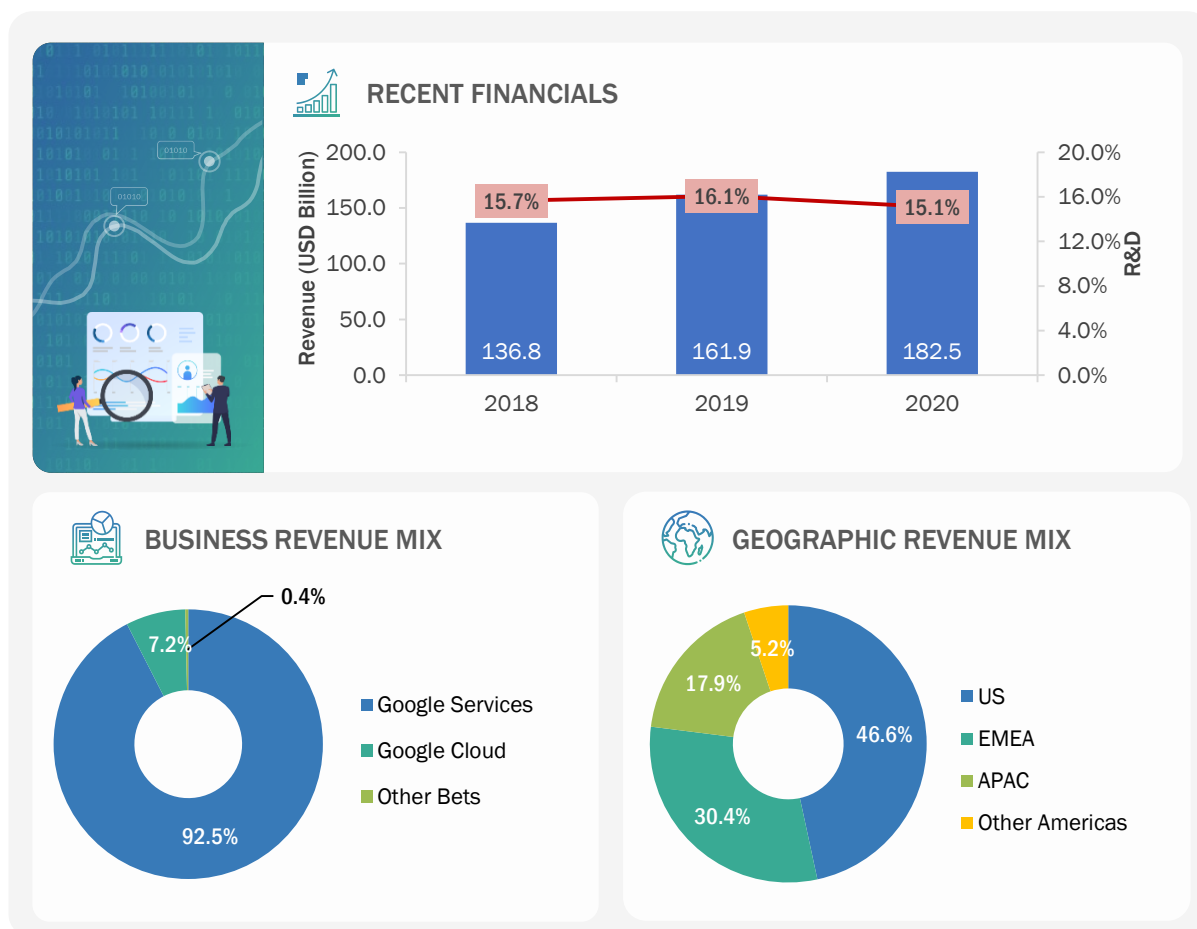
COMPANY AT A GLANCE - 2021	
Year Founded	1998
Country	US
City	Mountain View, CA
No. of Employees	135,300+
Overall Annual Revenue	USD 183 billion
R&D Expenditure	USD 28 billion
Ownership	Public (NYSE: GOOGLE)

Source: Company Website, Annual Reports, and SEC Filings

Google's company's parent organization, Alphabet, was formed by restructuring Google in October 2015. It became the parent company of Google and several former Google subsidiaries. Google's product innovations have increased the use of its services, and the company is one of the most recognized brands in the world. Its core products and platforms include Android, Chrome, Gmail, Google Drive, Google Maps, Google Play, Search, and YouTube. Every product has over one billion active users every month. As more and more digital experiences are being built in the cloud, its cloud products are helping enterprises of all sizes take advantage of the latest technologies and efficiently run their businesses. Google provides its big data solution through the Google Cloud Platform.

The company's global clientele base is spread across industry verticals, such as automotive, BFSI, retail and eCommerce, education, energy, engineering, entertainment, environment, food, beverage, government, healthcare, manufacturing, media, telecommunications and IT, transportation, and travel and hospitality. Google heavily invests in platforms, such as the Android mobile OS, Chrome browser, Chrome OS, and Daydream Virtual Reality (VR) platforms. It operates through 70 offices in more than 50 countries worldwide. It has a geographical presence in North America, Europe, APAC, MEA, and Latin America.

In the predictive analytics market, Google offers the Smart Analytics platform that is Google Cloud's fully managed serverless analytics platform. The platform enables analysts to discover predictive insights using simple Structured Query Language (SQL), built-in plug-and-play ML models, and automated workflows

FIGURE 54 GOOGLE: COMPANY SNAPSHOT

Note 1: The company's fiscal year ends on December 31.

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

12.2.6.2 Solutions offered

TABLE 219 GOOGLE: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTIONS	DESCRIPTION	APPLICATION
Solution	Smart Analytics Platform	Google Cloud's fully managed serverless analytics platform eliminates constraints of scale, performance, and cost to enable businesses to gain real-time insights, improve decision-making, and accelerate innovation.	- Governance, risk, and compliance management - Risk analytics - Sales life cycle management - Predictive modeling
		The Marketing Analytics solution offered under this platform helps collect all the data related to the customer journey and predict marketing and business outcomes. It also helps create personalized experiences for customers.	
Solution	Google AI Platform	Google AI Platform enables building, deploying, and managing ML models and makes it easy for developers, data scientists, and data engineers to streamline their ML workflows. Prediction done through the Google AI platform enables managing the infrastructure needed to run the ML models and makes it available for online and batch prediction requests.	Predictive Modeling

12.2.6.3 Recent developments

TABLE 220 GOOGLE: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2020-Jul	Google	Solution	BigQuery Omni	Google Cloud announced BigQuery Omni powered by Anthos for Multi-Cloud Analytics. BigQuery Omni provided enterprises with the flexibility needed to break down silos and create actionable business insights without having to pay expensive egress fees for moving data from other cloud providers to Google Cloud.
2019-Sep	Google	Solution	Cloud Storage Connector.	Google introduced a cloud storage connector for predictive workloads. This solution allowed organizations to substitute their Hadoop Distributed File System (HDFS) with Google Cloud Storage, an open-source Java client library that implemented Hadoop Compatible FileSystem (HCFS) and ran inside Hadoop Java Virtual Machines (JVMs). These allowed big-data processes such as Hadoop or Spark jobs and access to underlying data from Google Cloud Storage.

Source: Press Releases

TABLE 221 GOOGLE: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2020-Sep	Strategic partnership	Google	Defense Innovation Unit (DIU)	DIU chose Google Cloud for prototyping an AI-enabled digital pathology solution at selected Department of Defense (DoD) facilities. The solution would help US Military Health System with predictive cancer diagnoses. The goal of this project was to improve the accuracy of diagnoses by assisting physicians in handling a large volume of data while making diagnostic and treatment decisions.
2020-Aug	Strategic partnership	Google	Best Buy	Google Cloud and Best Buy expanded their partnership through a multi-year agreement named Google Cloud as the cloud provider for Best Buy's Enterprise Data Platform. Best Buy would leverage the cloud infrastructure and analytics services of Google Cloud to empower its data-driven retail strategy and deliver more personalized shopping experiences.
2020-Jul	Strategic partnership	Google	Orange	Google Cloud formed a strategic partnership with Orange for data, AI, and edge computing services. This partnership would accelerate the transformation of Orange's IT infrastructure and the development of future cloud services in edge computing.

2020-Apr	Collaboration	Google	Apple	Google and Apple collaborated to enable the use of Bluetooth technology to help government and healthcare agencies reduce the spread of COVID-19.
2019-Nov	Collaboration	Google	Vodafone	Vodafone collaborated with Google Cloud for infrastructure, data analytics, and ML. Vodafone's big data platform named Neuron would be powered by GCP to create new products, engage with customers, and leverage real-time analytics.

Source: Press Releases

12.2.7 SALESFORCE

12.2.7.1 Business and financial overview

TABLE 222 SALESFORCE: BUSINESS OVERVIEW

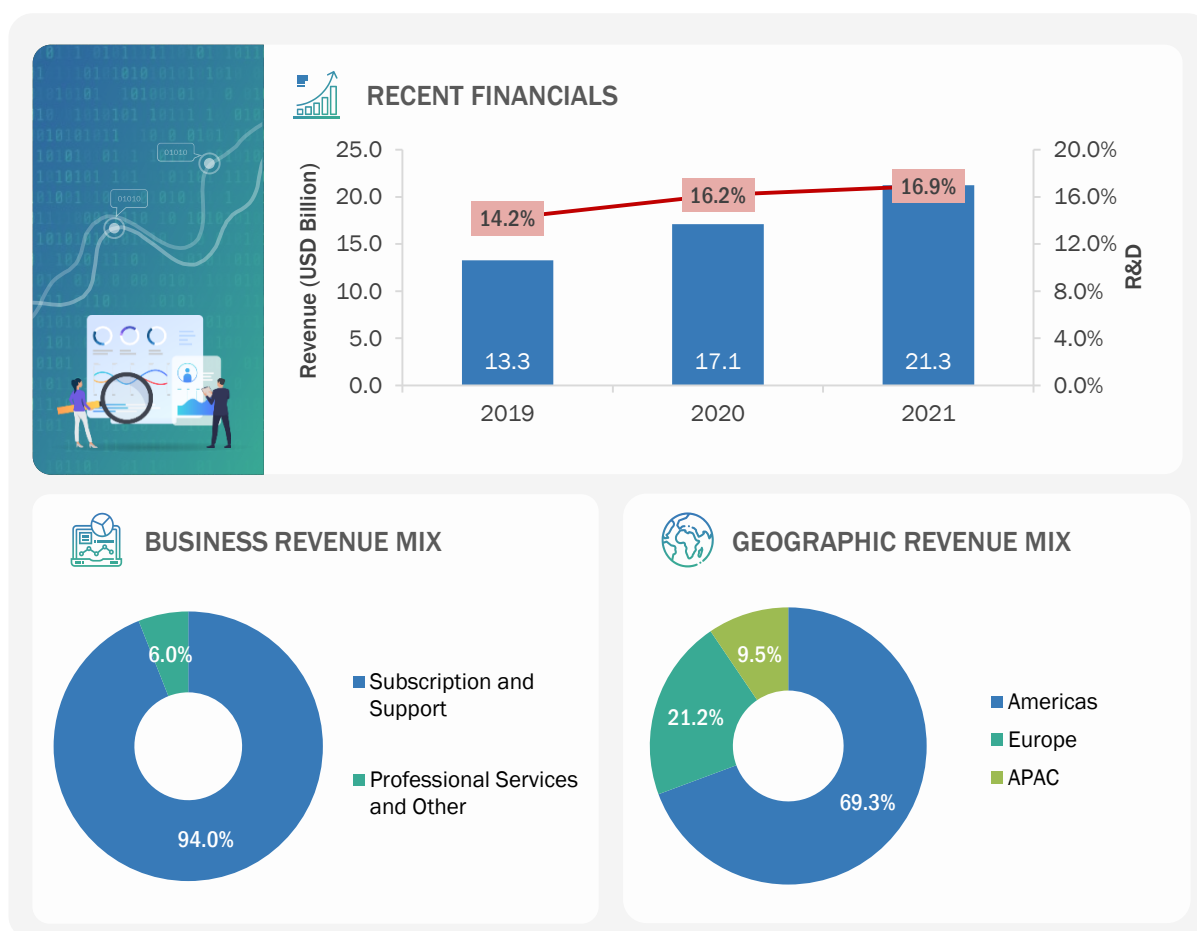
COMPANY AT A GLANCE - 2021	
Year Founded	1999
Country	US
City	San Francisco, CA
No. of Employees	56,606
Overall Annual Revenue	USD 21.3 billion
R&D Expenditure	USD 3.6 billion
Ownership	Public (NYSE: CRM)

Source: Company Website, Annual Reports, and SEC Filings

Salesforce is a Customer Relationship Management (CRM) solution and service provider company. Salesforce acquired Tableau, a leading provider of an analytics platform. This acquisition would strengthen Salesforce's position in the digital transformation space. The company's integrated platform provides a single shared view of every customer across departments, such as marketing, sales, commerce, and services. Salesforce has a community of over 10 million innovators, disruptors, and community shapers who are called Trailblazers.

The company offers a wide range of products and services across segments, such as sales, services, marketing, application, analytics, employee experience, trailblazers and reskilling, and enablement and collaboration. Most of these segments operate on a single trusted cloud platform. Salesforce's service offerings can be easily deployed through mobile devices and internet browsers and integrated with other platforms and enterprise applications. It offers a cross-cloud technology named Salesforce 360, which helps its clients have a single integrated, holistic customer profile for various departments. The organization caters to various industries, including BFSI, healthcare and life sciences, government, communications, retail, consumer goods, media, manufacturing, transportation, and hospitality, automotive, and education. It is present in North America, Europe, APAC, and MEA.

In the predictive analytics market, the company offers Salesforce Einstein Analytics. It provides a platform to directly connect with CRM data for generating dashboards and reports.

FIGURE 55 SALESFORCE: COMPANY SNAPSHOT

Note 1: The company's fiscal year ends on December 31.

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

12.2.7.2 Solutions offered

TABLE 223 SALESFORCE: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTIONS	DESCRIPTION	APPLICATION
Solution	Salesforce Einstein Analytics	- It is a cloud-based secure analytics platform developed for mid-sized businesses and large enterprises to implement the rapid exploration of data. It contains Einstein Discovery that provides AI-powered recommendations, answers, and explanations. Einstein Analytics offers purpose-built templates for various use cases in businesses, such as sales analytics, service analytics, field service analytics, change analytics, fundraising analytics, pipeline analytics, quoting analytics, snapshot analytics, financial advisor analytics, and adoption analytics. - The solution is native to the Salesforce platform and helps spot trends faster, predict outcomes, and get recommendations.	Fraud detection and profitability management

Solution	Predictive Analysis in Tableau	Tableau contains out-of-the-box stats and predictive technologies, which help data experts codify theses and uncover latent variables. It offers the following benefits:	
		- See all the facets of IT organizations - Monitor in real-time - Allocate resources efficiently - Deploy securely across the enterprise	- Predictive modeling - Yield management

Source: Company Website

12.2.7.3 Recent developments

TABLE 224 SALESFORCE: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2020-Mar	Salesforce	Solution	Salesforce Care	Salesforce launched Salesforce Care for the manufacturing vertical to help manufacturers pivot their operations and reallocate resources to the production of COVID-19 relief and response supplies, such as Personal Protective Equipment (PPE), ventilators, virus testing equipment, and other response supplies.
2019-Nov	Salesforce	Solution	Anypoint Runtime Fabric	Salesforce's MuleSoft announced an update for Anypoint Runtime Fabric. The company would extend the solution to run it on Google Cloud. This would enable Google Cloud customers to easily manage their business-critical applications within their existing infrastructure, in the cloud or a hybrid environment. Anypoint Runtime Fabric contained a new Google BigQuery connector in Anypoint Exchange that would bring together data from virtually any system to build a complete view of the customer.
2019-Jul	Salesforce	Solution	Einstein Analytics	Einstein Analytics was launched as Salesforce's proprietary predictive analytics and AI platform that powered 6.5 billion predictions daily. Einstein Analytics for Partners was designed to be a solution that would tap AI to identify and prioritize leads. The solution delivered predictive recommendations by mining past leads and opportunity performance data.
2019-May	Salesforce	Solution	Einstein Analytics for Financial Services	Einstein Analytics for Financial Services was a customizable analytics solution that would provide AI-augmented business intelligence for wealth advisors, retail bankers, and managers. The solution would give a customized CRM experience for financial institutions.

Source: Press Releases

TABLE 225 SALESFORCE: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2020-Sep	Acquisition	Salesforce	Mobify	Salesforce signed an agreement to acquire Mobify. Through this acquisition, Mobify would provide Salesforce Commerce Cloud with a modern storefront solution that would help brands customize their commerce experiences faster and more frequently, delivering an enhanced shopping experience across channels. It would increase conversion and revenue to drive success in the digital-first economy.
2020-Feb	Acquisition	Salesforce	Vlocity	Salesforce acquired Vlocity for USD 1.3 billion. Vlocity was one of the largest and fastest-growing Salesforce-native Independent Software Vendors (ISVs). Through this acquisition, Salesforce took ownership of the industry cloud.
2019-Aug	Acquisition	Salesforce	Tableau	With this acquisition, Salesforce aimed at competing with Microsoft in the area of power BI and data visualization space.

Source: Press Releases

TABLE 226 SALESFORCE: OTHERS

YEAR-MONTH	COMPANY NAME	DEVELOPMENT TYPE	DESCRIPTION
2021-Sep	Salesforce	Initiative	Salesforce announced new grants totaling USD 17.25 million to five school districts in the US: San Francisco Unified School District, Oakland Unified School District, Chicago Public Schools, Indianapolis Public Schools, and the New York City Department of Education. Furthermore, USD 1.75 million was awarded to education nonprofits in the US and Europe.

Source: Press Releases

12.2.8 AWS

12.2.8.1 Business and financial overview

TABLE 227 AWS: BUSINESS OVERVIEW

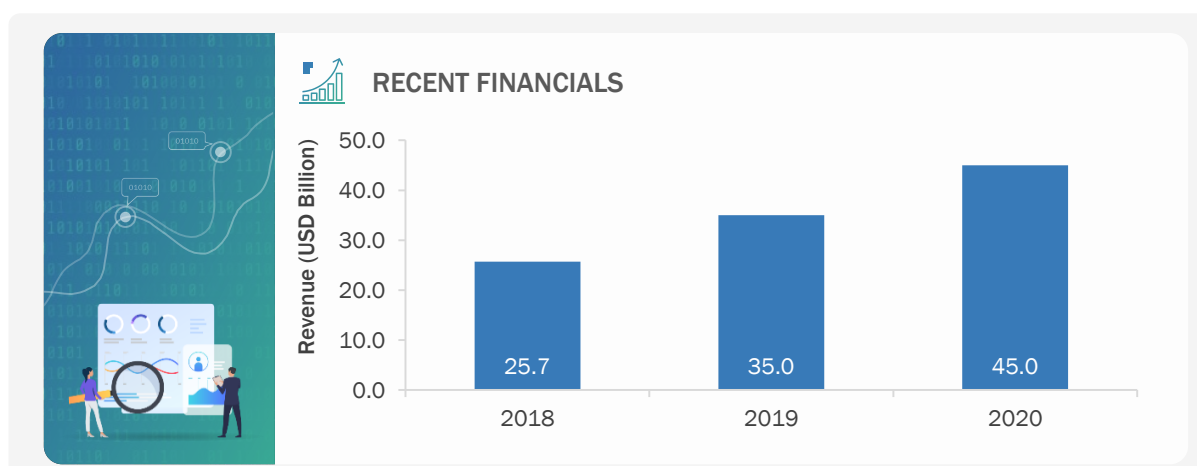
COMPANY AT A GLANCE - 2021	
Year Founded	2006
Country	US
City	Seattle, WA
No. of Employees	99,000+
Overall Annual Revenue	USD 45 billion
R&D Expenditure	USD 12 billion
Ownership	Subsidiary (Amazon)

Source: Company Website, Annual Reports, and SEC Filings

AWS is a subsidiary of Amazon and primarily offers cloud computing services in the form of web services. It also offers a wide range of products and services to customers present in 190 countries. Its product portfolio comprises segments, such as compute, storage, database, migration, network and content delivery, developer tools, management tools, media services, ML, and analytics.

AWS provides over 175 fully featured services for computing; storage; databases; networking; analytics; robotics; ML; AI; IoT; mobile; security; hybrid; virtual and augmented reality; media; and application development, deployment, and management from 76 availability zones within 24 geographic regions. The solutions segment offers website and web apps, mobile services, backup, storage and archive, financial services, and digital media. The company caters to various industry verticals, including media and entertainment, automotive, education, BFSI, game tech, government, healthcare and life sciences, manufacturing, retail, telecommunications, oil and gas, and power utilities. It has its geographical presence in North America, Europe, APAC, MEA, and Latin America.

In the predictive analytics market, AWS offers Amazon SageMaker on AWS Cloud. It is a cloud ML platform that enables developers to deploy ML models on embedded systems and edge devices. The platform removes heavy lifting from each step of the ML process to make it easier to develop high-quality models.

FIGURE 56 AWS: COMPANY SNAPSHOT

Note 1: The company's fiscal year ends on December 31.

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

12.2.8.2 Solutions offered

TABLE 228 AWS: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTIONS	DESCRIPTION	APPLICATION
Solution	Amazon SageMaker on AWS Cloud	It enables a data lake environment for building, training, and deploying ML models using Amazon SageMaker on AWS Cloud. It uses AWS services, such as Amazon Simple Storage Service (Amazon S3), Amazon Application Programming Interface (API) Gateway, AWS Lambda, Amazon Kinesis Data Streams, and Amazon Kinesis Data Firehose. It helps business users make predictive and prescriptive models for business value.	Predictive modeling

Source: Company Website

12.2.8.3 Recent developments

TABLE 229 AWS: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2020-May	AWS	Solution	Amazon Kendra	AWS introduced the general availability of Amazon Kendra, a highly accurate and easy-to-use enterprise search service powered by ML. With just a few clicks, Amazon Kendra would use ML to enable organizations to index all their internal data sources, make data searchable, and allow users to get precise answers to natural language queries.
2020-Mar	AWS	Service	AWS Managed Services	AWS Managed Services expanded its big data capabilities. The new big data capabilities would provide support for various AWS big data tools, including Amazon SageMaker, AWS Lake Formation, and Amazon CloudSearch.
2019-Dec	AWS	Solution	Amazon	AWS announced new analytics capabilities for

	Redshift	Amazon Redshift. (Advanced Query Accelerator (AQUA) for Amazon Redshift would provide a new innovative hardware-accelerated cache that would deliver up to 10× better query performance than other cloud data warehouse providers.
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Source: Press Releases

TABLE 230 **AWS: DEALS**

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2020-May	Collaboration	AWS	WHO	AWS collaborated with WHO to provide advanced cloud technologies and technical expertise to track the virus, understand the outbreak, and contain its spread. AWS launched AWS COVID-19 data lake as a centralized repository to provide up-to-date and curated information related to the spread and characteristics of the virus and its associated illnesses so that experts could access and analyze the latest data in their battle against the disease.
2019-Dec	Collaboration	AWS	Novartis	AWS collaborated with Novartis to accelerate the digital transformation of its business operations. This collaboration enables Novartis to reimagine its core pharmaceutical manufacturing, supply chain, and delivery operations using AWS' portfolio of cloud services.
2019-Nov	Strategic Partnership	AWS	Salesforce	AWS and Salesforce extended their global strategic partnership. With the expansion of the partnership, AWS and Salesforce addressed the growing technology skill gaps by enabling users to enhance their skills for the future.

Source: Press Releases

TABLE 231 **AWS: OTHERS**

YEAR-MONTH	COMPANY NAME	DEVELOPMENT TYPE	DESCRIPTION
2020-May	AWS	Initiative	AWS launched AWS Diagnostic Development Initiative, a program to support customers working to bring more accurate diagnostic solutions in the market for COVID-19. AWS' commitment to spend USD 20 million would accelerate the program and help its customers harness the cloud to tackle COVID-19 challenges.
2020-Apr	AWS	Business expansion	AWS expanded its business in Italy. With this expansion, AWS spanned 76 Availability Zones within 24 geographic regions around the world and announced plans for nine more Availability Zones and three more AWS Regions in Indonesia, Japan, and Spain.

Source: Press Releases

12.2.9 HPE

12.2.9.1 Business and financial overview

TABLE 232 HPE: BUSINESS OVERVIEW

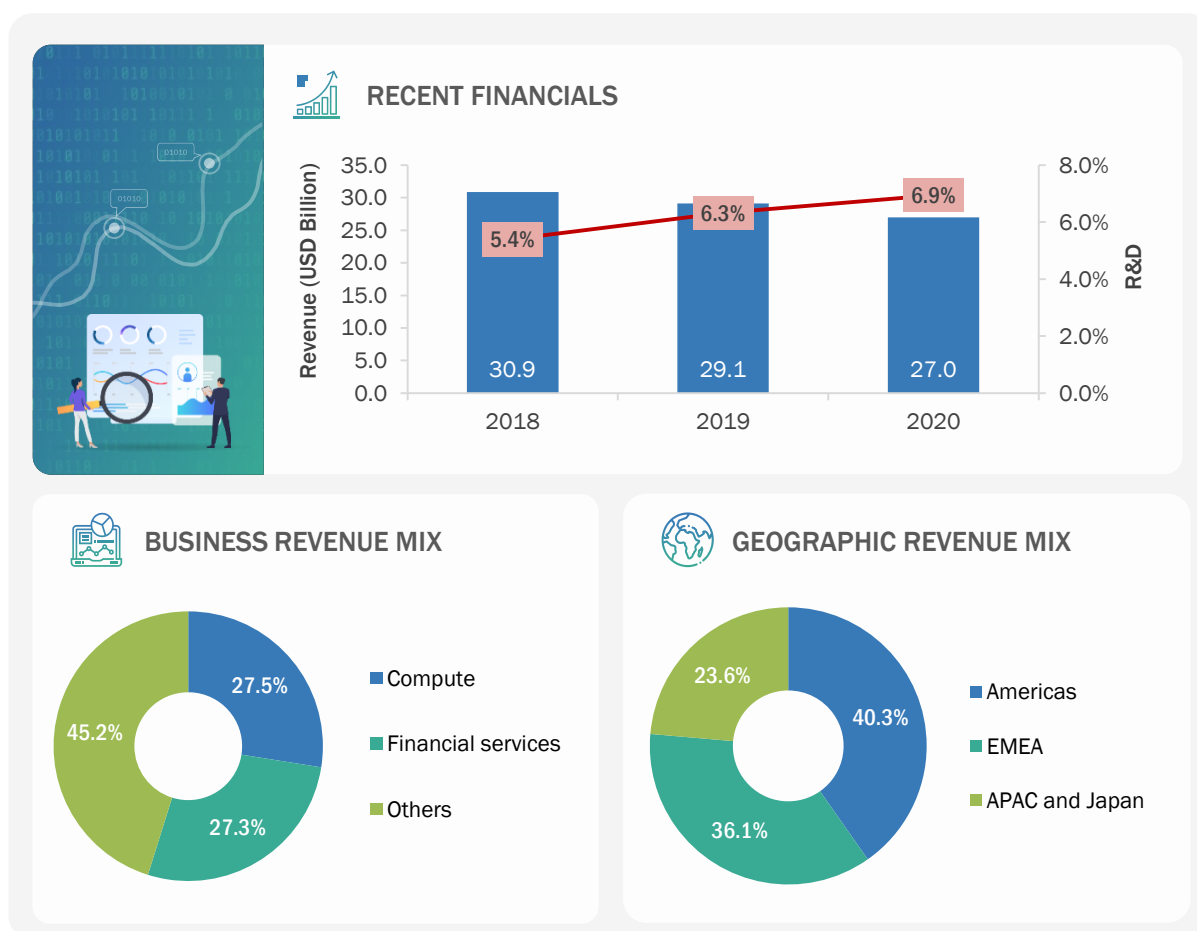
COMPANY AT A GLANCE - 2021	
Year Founded	2015
Country	US
City	Houston, TX
No. of Employees	59,400
Overall Annual Revenue	USD 27 billion
R&D Expenditure	USD 1.9 billion
Ownership	Public (NYSE: HPE)

Source: Company Website, Annual Reports, and SEC Filings

HPE provides IT solutions to businesses and public sector enterprises. In November 2015, HPE split from Hewlett Packard (HP) to form an independent company. It now operates through four segments: hybrid IT, intelligent edge, financial services, and corporate investments.

HPE delivers unique, open, and intelligent technology solutions with a consistent experience across all cloud and edge platforms to help customers develop new business models, engage in new ways, and increase operational performance. The company has invested in IoT-related projects to boost its revenue growth. In August 2019, HPE acquired MapR Technologies to strengthen its portfolio of products to drive AI and analytics applications. The organization offers advanced products for software-defined technologies that include Software-Defined Networking (SDN), Support Data Collector (SDC), and Software-Defined Storage (SDS) solutions. It has a strong presence in North America, Europe, MEA, APAC, and Latin America.

The company provides a complete flash storage portfolio, from entry-level to high-end. In the predictive analytics market, HPE offers HPE InfoSight that utilizes cloud-based predictive analytics to anticipate and prevent problems before they impact business.

FIGURE 57 HPE: COMPANY SNAPSHOT

Note 1: The company's fiscal year ends on October 31.

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

12.2.9.2 Solutions offered

TABLE 233 HPE: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTIONS	DESCRIPTION	APPLICATION
Solution	HPE InfoSight	The solution uses cloud-based ML for optimizing virtual performance by diagnosing the root cause and recommending the right remediation through app- and resource-centric modeling. It analyzes millions of sensors and predicts and prevents problems across the entire infrastructure stack every second. It predicts and automatically resolves 86% of customer issues proactively and helps deliver 99.9999% availability.	- Predictive modeling - Performance monitoring

Source: Company Website

12.2.9.3 Recent developments

TABLE 234 HPE: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2020-Jul	HPE	Solution	HPE InfoSight	HPE announced the enhancement of HPE InfoSight for Servers to expand its portfolio. HPE InfoSight for Servers would now include HPE Edgeline and HPE Moonshot. The latest enhancements would provide insights that simplify and automate infrastructure operations to maximize server uptime.
2020-Jun	HPE	Solution	HPE Ezmeral	This solution enabled organizations to increase agility and efficiency, unlock insights, and deliver business innovation.
2018-Nov	HPE	Solution	HPE InfoSight	HPE announced the extension of HPE InfoSight's predictive analytics and recommendation capabilities to HPE ProLiant servers, HPE Synergy compute modules, and HPE Apollo systems. By bringing predictive analytics to monitoring servers, InfoSight provides insights into the failure of selected parts. This enabled IT managers to make necessary maintenance requests for replacements to reduce downtime.

Source: Press Releases

TABLE 235 HPE: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2019-Nov	Strategic Partnership	HPE	National Renewable Energy Laboratory (NREL)	HPE partnered with NREL to develop AI and ML technologies for efficient data center operation. The partnership would bring together existing research efforts from both parties toward collaborative, innovative solutions that could have a major industry impact.
August-2019	Acquisition	HPE	MapR	HPE acquired the business assets of MapR. This acquisition accelerated the intelligent data platform capabilities of HPE. It enabled HPE to offer a complete portfolio of products to drive AI and analytics applications.

Source: Press Releases

TABLE 236 **HPE: OTHERS**

YEAR-MONTH	COMPANY NAME	DEVELOPMENT TYPE	DESCRIPTION
2020-May	HPE	Initiative	HPE deployed four COVID-19 test labs and four outpatient department centers across India to help contain the COVID-19 pandemic. HPE's COVID-19 test labs were self-contained, IoT-enabled, and telemedicine-ready facilities designed to enable quick and clean testing of suspected individuals. These facilities would operate on the basis of the specifications provided by the Ministry of Science & Technology.
2020-Apr	HPE	Initiative	HPE introduced an initiative in Italy to support front-line workers facing the global pandemic. HPE installed network infrastructure onboard named Grandi Navi Veloci (GNV) Splendid, a large passenger ferry owned by MSC Group. The ferry was transformed into a floating hospital to battle the global COVID-19 pandemic.

Source: Press Releases

12.2.10 TERADATA

12.2.10.1 Business and financial overview

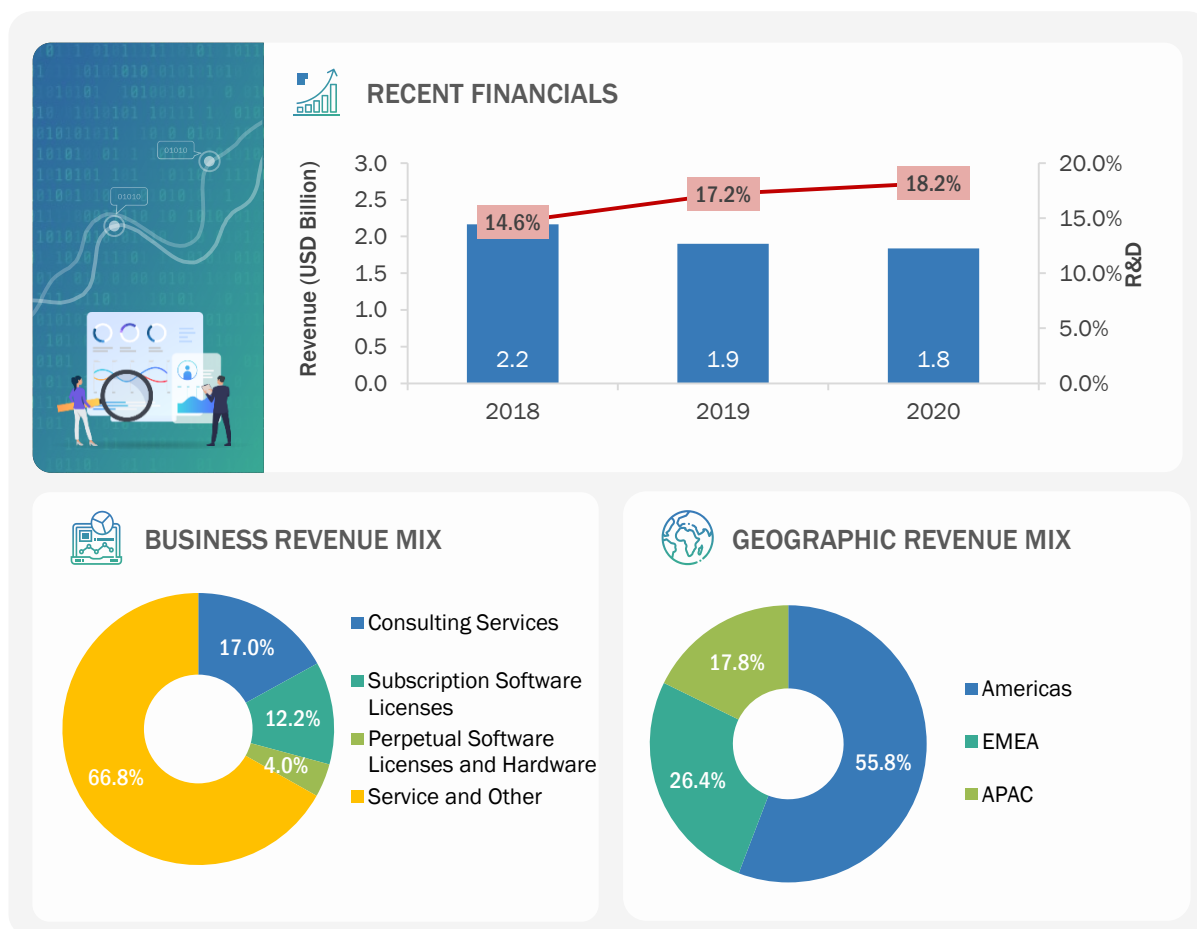
TABLE 237 TERADATA: BUSINESS OVERVIEW

COMPANY AT A GLANCE - 2021	
Year Founded	1979
Country	US
City	San Diego, CA
No. of Employees	8500+
Overall Annual Revenue	USD 1.8 billion
R&D Expenditure	USD 0.3 billion
Ownership	Public (NYSE: TDC)

Source: Company Website, Annual Reports, and SEC Filings

Teradata is a data intelligence company catering to more than 75 countries. It also caters to several industries, including communications, media and entertainment, financial services, life sciences, healthcare, retail, energy, travel and transportation, and manufacturing. Its broad customer base comprises O2 Czech Republic, BMW, Siemens Healthineers, Vodafone, American Red Cross, and US Bank. It has partnered with various companies, including Accenture, Alation, AlphaZetta, Alteryx, General Dynamics Information technology, Erwin, HCL Technologies, and Informatica, to deliver end-to-end analytics solutions. Its vast product portfolio consists of various categories, including Software, Cloud, Ecosystem Management, Hardware, and Applications.

The company offers the Teradata Vantage solution in the predictive analytics market. It is a business intelligence platform powered by the cloud.

FIGURE 58 TERADATA: COMPANY SNAPSHOT

Note 1: The company's fiscal year ends on December 31.

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

12.2.10.2 Solutions offered

TABLE 238 TERADATA: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTIONS	DESCRIPTION	APPLICATION
Solution	Teradata Vantage	It is a hybrid cloud solution with the ability to drive powerful analytics. It helps businesses solve the problem of fast-growing data volumes and enables discovering real-time business intelligence at scale. Teradata Vantage offers descriptive, predictive, and prescriptive analytics with integrated data that helps implement complex analytics and algorithms through a simple, intuitive SQL language compatible with preferred business analytics tools. It also includes a Vantage Analyst that enables business analysts to perform ML and advanced analytics.	- Fraud detection - Risk management - Predictive modeling - Customer analytics

Source: Company Website

12.2.10.3 Recent developments

TABLE 239 TERADATA: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2020-Oct	Teradata	Solution	Teradata Vantage	Teradata Vantage was made available on Google Cloud, allowing deployment flexibility across all three major public clouds. Using Google Cloud resources, this software-as-a-service offering would provide access to Vantage, a data and analytics platform that integrates analytics, data lakes, and data warehouses.
October-2019	Teradata	Solution	Teradata Vantage	Teradata announced the extension of the Teradata Vantage platform. Teradata extended its Teradata Vantage platform with Vantage Customer Experience (CX) and Vantage Analyst. Vantage CX delivers real-time personalized experiences to customers. Vantage Analyst enables business analysts to perform ML and advanced analytics.

Source: Press Releases

TABLE 240 TERADATA: DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2021-Sep	Strategic partnership	Teradata	Tesco PLC	Teradata stated that Tesco PLC is expanding its cooperation with Teradata by incorporating Vantage into its data platform. Tesco would use Vantage to eliminate silos and query all their data in real-time, regardless of whether in the cloud, multiple clouds, on-premises, or any mix thereof, to gain a comprehensive view of their business.
June-2020	Strategic partnership	Teradata	dotData	Teradata partnered with dotData. This partnership would leverage the robust enterprise data management and analytics capabilities of Teradata's Vantage platform and dotData's autoML 2.0 platform to create a powerful end-to-end data science solution.
October-2019	Strategic partnership	Teradata	Google Cloud	Teradata partnered with Google Cloud to expand its public cloud offerings. This partnership also added the feature of pay-as-you-go pricing for Teradata Vantage's users.

Source: Press Releases

12.2.11 ALTERYX

12.2.11.1 Business and financial overview

TABLE 241 ALTERYX: BUSINESS OVERVIEW

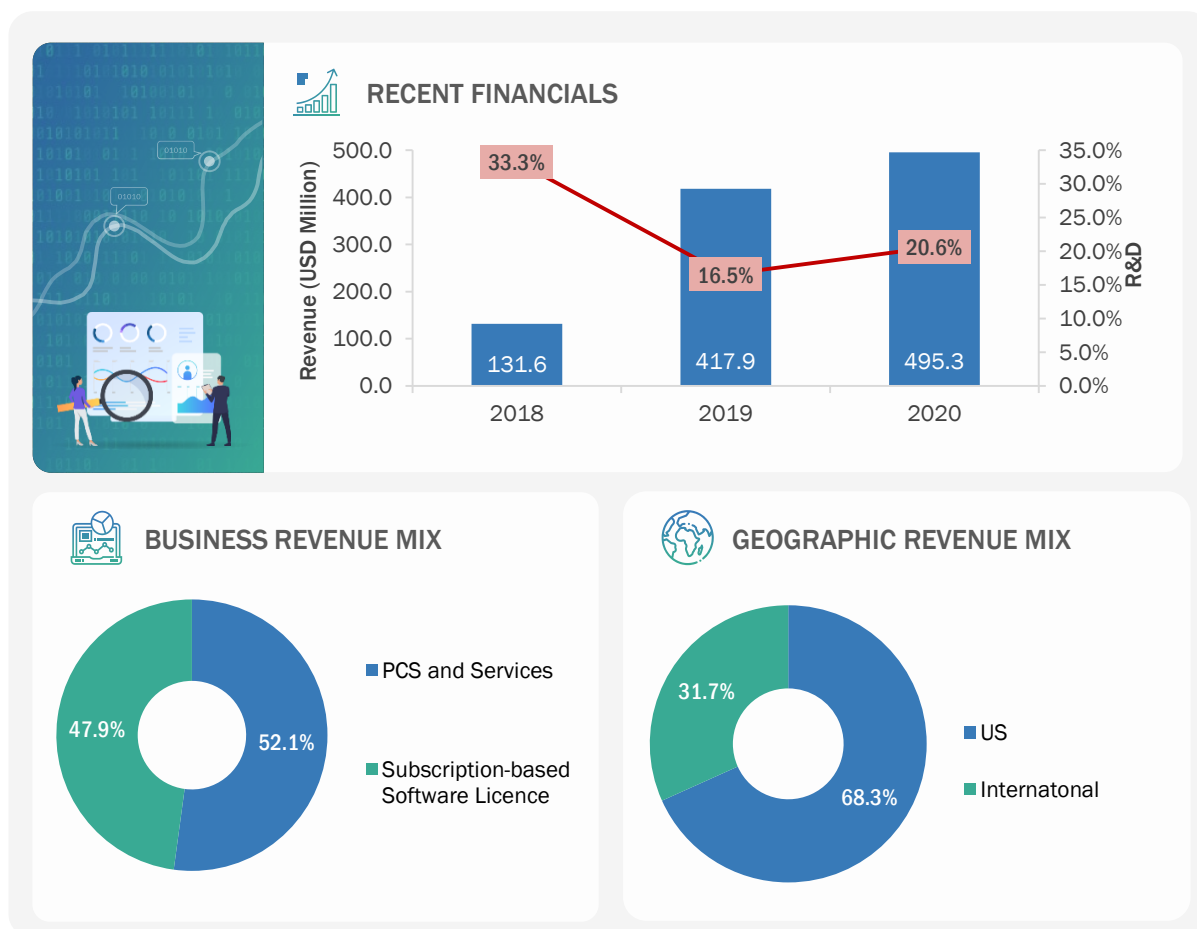
COMPANY AT A GLANCE - 2021	
Year Founded	2010
Country	US
City	Irvine, CA
No. of Employees	1,300+
Overall Annual Revenue	USD 495.3 million
R&D Expenditure	USD 102 million
Ownership	Public (NYSE: AYY)

Source: Company Website, Annual Reports, and SEC Filings

Alteryx is one of the leaders in data blending and advanced analytics software. It caters to over 200,000 users, and its customers include Kaiser, Experian, McDonald's, and Ford. It also caters to various growing organizations, such as Vertex, Rosenblatt Securities, and ConsumerOrbit. Alteryx has entered into partnerships and collaborations with top companies, such as Impala, Amazon RedShift, Teradata, and Spark. Since 2010, the company has been expanding its statistical and analytics capabilities, increasing its big data and SaaS connectivity, and delivering a cloud platform for analytics applications called Alteryx Analytics Gallery. Alteryx provides solutions and services to various industry verticals, including BFSI, healthcare, retail, transportation and logistics, oil and gas, real estate, communication, energy and utilities, education, manufacturing, media and entertainment, travel and hospitality, and the public sector. It has a geographical presence in North America, Europe, APAC, MEA, and Latin America. In May 2020, Alteryx launched a major new public upskilling initiative called Advancing Data and Analytics Potential Together (ADAPT), a free analytics training curriculum and certification program for those whose employment was impacted by COVID-19.

In the predictive analytics market, Alteryx offers the Alteryx Predictive Analytics solution, Alteryx Analytic Process Automation (APA) Platform, and Alteryx Intelligence Suite. Alteryx also offers predictive analytics solutions under the advanced analytics solutions category. It enables combining data access, preparation, and modeling and the sharing of analytics results on a single platform.

Alteryx Analytic Process Automation (APA) Platform delivers the end-to-end automation of analytics, ML, and data science processes, enabling the agility needed to accelerate digital transformation.

FIGURE 59 ALTERYX: COMPANY SNAPSHOT

Note 1: The company's fiscal year ends on December 31.

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

12.2.11.2 Solutions offered

TABLE 242 ALTERYX: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTIONS	DESCRIPTION	APPLICATION
Solution	Alteryx Predictive Analytics	Alteryx makes predictive analytics and applying ML more accessible and agile. Alteryx Designer allows applying different kinds of regression analysis, neural networks, decision trees, and time series modeling by offering complete predictive analytics capabilities. It also enables deep visual integration using industry-standard R statistics and predictive language.	- Time series modeling - Predictive modeling
	Alteryx APA Platform	- Alteryx APA Platform helps automate processes, embed intelligent decision-making, and empower people to deliver faster, better business outcomes. Four products are offered through this platform: - Alteryx Designer: It enables combining each data source with hundreds of ready-to-use automation building blocks and go through all the routine data prep to spend time discovering actionable insights through built-in predictive, prescriptive and location analytics.	- Fraud detection - Profitability management - Governance, risk and compliance management - Risk management

- Alteryx Server: It accelerates time to analytical insight and empowers analysts and business users across organizations to make informed, data-driven decisions.
- Alteryx Connect: It combines the power of data cataloging with human insight and empowers quickly finding, managing, understanding, and collaborating on the information that resides in the organization.
- Alteryx Promote: It helps analytic teams deploy more predictive models faster by eliminating the need to recode models for real-time application consumption.

Solution	Alteryx Intelligence Suite	It helps quickly build ML-powered predictive models without having to code or perform complex statistics. It offers a step-by-step process to create properly trained algorithms ready to deploy and build models without a data science degree.	Predictive modeling
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Source: Company Website

12.2.11.3 Recent developments

TABLE 243 ALTERYX: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2020-Jun	Alteryx	Solution	Analytics Hub and Intelligence Suite	Alteryx introduced new product innovations by launching Alteryx Analytics Hub to extend the power of Alteryx APA Platform and Alteryx Intelligence Suite, offering predictive modeling that will serve as an add-on for Alteryx Designer.
2020-May	Alteryx	Solution	Analytic Process Automation (APA) platform	Alteryx launched the Analytic Process Automation (APA) platform. APA unifies analytics, data science, and business process automation in an end-to-end platform. It enabled high-impact business outcomes and the rapid upskilling of employees across the organization.
2018-Aug	Alteryx	Solution	Visualytics	Alteryx announced a new version of the Alteryx Analytics Platform. Alteryx announced Visualytics that provides real-time, interactive visualizations across the Alteryx platform. Visualytics helped maximize the effectiveness of business-critical models by giving instant visual validation to data scientists on the health of their analysis.

Source: Press Releases

TABLE 244 **ALTERYX: DEALS**

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2020-Aug	Strategic partnership	Alteryx	UiPath	Alteryx partnered with UiPath. This partnership would accelerate digital transformation with hyperautomation. It would enable organizations to simplify end-to-end automation for faster, transformative business outcomes.
2019-Oct	Acquisition	Alteryx	Feature Labs	Alteryx acquired Feature Labs to advance the use of ML technology by bridging the gap between data science and ML.
2019-Apr	Acquisition	Alteryx	ClearStory Data	Alteryx acquired ClearStory Data to drive innovations in the field of data science and analytics.

Source: Press Releases

TABLE 245 **ALTERYX: OTHERS**

YEAR-MONTH	COMPANY NAME	DEVELOPMENT TYPE	DESCRIPTION
2019-Apr	Alteryx	Business expansion	Alteryx expanded in APAC. This expansion would amplify analytics in Japan to support the increasing customer base in the country.

Source: Press Releases

12.2.12 FICO

12.2.12.1 Business and financial overview

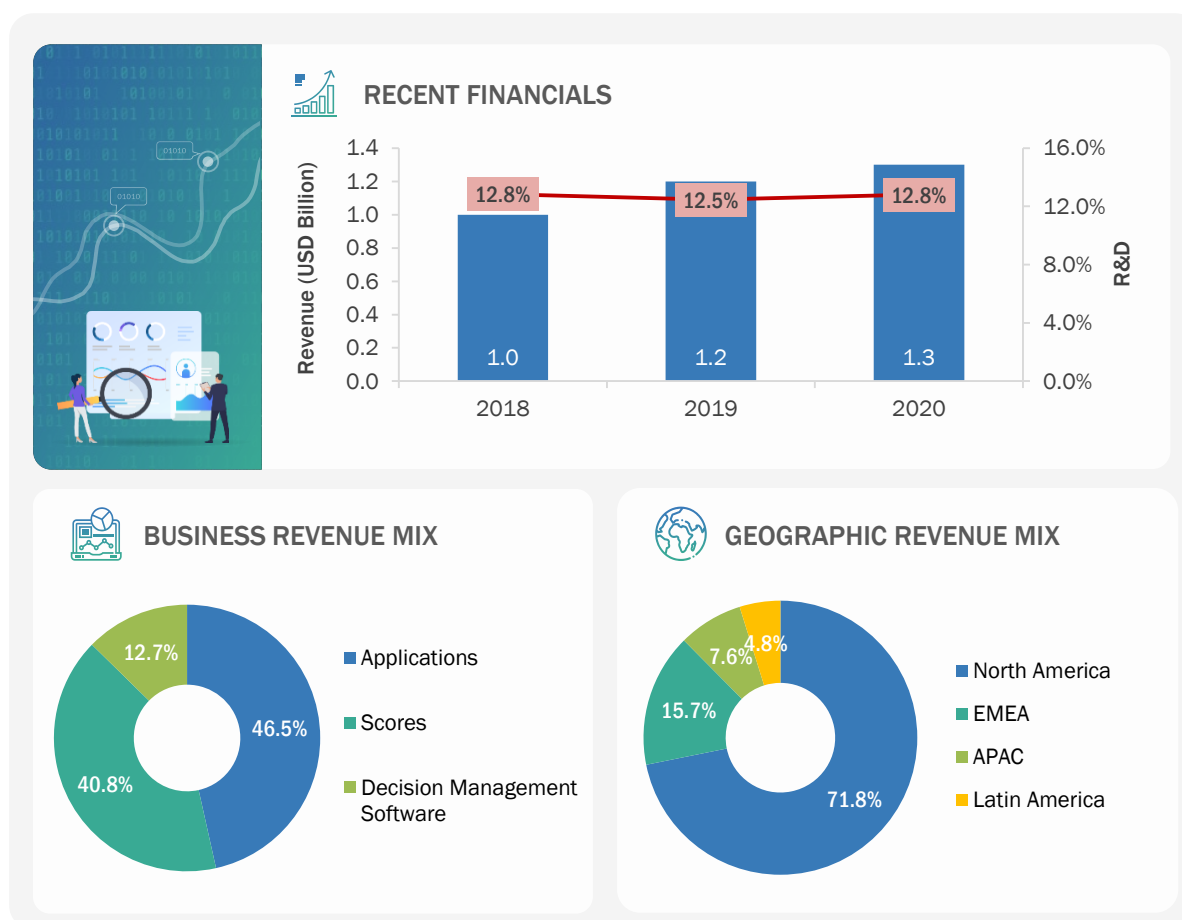
TABLE 246 FICO: BUSINESS OVERVIEW

COMPANY AT A GLANCE - 2021	
Year Founded	1956
Country	US
City	San Jose, CA
No. of Employees	4,000+
Overall Annual Revenue	USD 1.3 billion
R&D Expenditure	USD 0.2 billion
Ownership	Public (NYSE: FICO)

Source: Company Website, Annual Reports, and SEC Filings

Fair, Isaac and Company (FICO) is a leading analytics software company. The company's regional offices are in San Rafael, San Diego, and New York City in the US. It is spread across London and Birmingham in the UK; Moscow, Russia; Bangalore, India; Beijing, China; and Singapore. The company operates in more than 90 countries, helping client organizations make better decisions and attaining a higher level of growth, profits, and customer satisfaction. It offers various products, including FICO Debt Manager Solution, FICO Falcon Fraud Manager, FICO Engagement Analyzer, FICO Blaze Advisor Decision Rules Management Systems, FICO Score, FICO Decision Management Suite, FICO Score Open Access, and FICO Xpress Optimization Suite. FICO caters to various industries, including financial services, health care, insurance, automotive, public sector, retail, pharmaceuticals, telecommunications, travel and hospitality, media and entertainment, and high tech and utilities. However, it majorly focuses on banking, finance, insurance, and healthcare.

In the predictive analytics market, FICO offers FICO Predictive Analytics and FICO Analytics Workbench. These solutions help in decision-making using predictive analytics and enable businesses to mine data and predict future occurrences.

FIGURE 60 FICO: COMPANY SNAPSHOT


Note 1: The company's fiscal year ends on December 31.

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

12.2.12.2 Solutions offered

TABLE 247 FICO: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTIONS	DESCRIPTION	APPLICATION
Solution	FICO Predictive Analytics	FICO Predictive Analytics helps develop the best predictive models for business needs. The predictive models developed by FICO can be used on their own or as an integral part of FICO's decision management solutions. Some of the predictive modeling techniques and approaches used by FICO are FICO Scorecard Technology, Neural Networks, Time-to-Event Models, Action Effect/Uplift Models, and Economic Impact Models.	- Predictive modeling - Product and service development strategies - Financial and risk analytics
Solution	FICO Analytics Workbench	It is an integrated suite of state-of-the-art advanced analytics capabilities that enable companies to improve business decisions across the customer life cycle. It offers decision-making capabilities using tools such as AI and ML. The platform is built on the leading FICO Decision Management Platform to allow new predictive models and strategies to be deployed easily into production.	- Customer segmentation and clustering - Customer behavior analysis - Monitoring customer loyalty and satisfaction

12.2.12.3 Recent developments

TABLE 248 FICO: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2019-Sep	FICO	Solution	FICO Falcon X and FICO Financial Crimes Studio	FICO launched FICO Falcon X and FICO Financial Crimes Studio. The AI portfolio launched by FICO would help fight next-generation fraud and financial crimes by enabling data scientists to fuse open-source ML libraries with proven FICO ML techniques for the most accurate detection of criminal activities at any scale.
2018-Sep	FICO	Solution	FICO Analytics Workbench	FICO released a cloud-based Analytics Workbench with xAI Toolkit to enable data scientists to build transparent and predictive ML models. The solution helped directly embed predictive and prescriptive models into real-time business operations for creating faster and more impactful business outcomes.

Source: Press Releases

TABLE 249 FICO: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2020-Jun	Collaboration	FICO	FIS	FIS Teams collaborated with FICO. The two organizations collaborated to build an advanced anti-money laundering solution for the US financial institutions. Following this collaboration, FIS AML Compliance Manager was integrated with the FICO Falcon X technology to help North American financial institutions keep ahead of the increasingly sophisticated money launderers and other financial criminals
2019-May	Strategic Partnership	FICO	Equifax	Equifax and FICO announced a strategic partnership. This partnership would deliver Data Decisions Cloud and increase velocity to deploy predictive models. The new Data Decisions Cloud was an end-to-end data and analytics suite that addresses risk marketing and frauds taking place in financial institutions to enable financial institutions to meet the needs of consumers faster and more precisely than ever before.

Source: Press Releases

12.2.13 ALTAIR

12.2.13.1 Business and financial overview

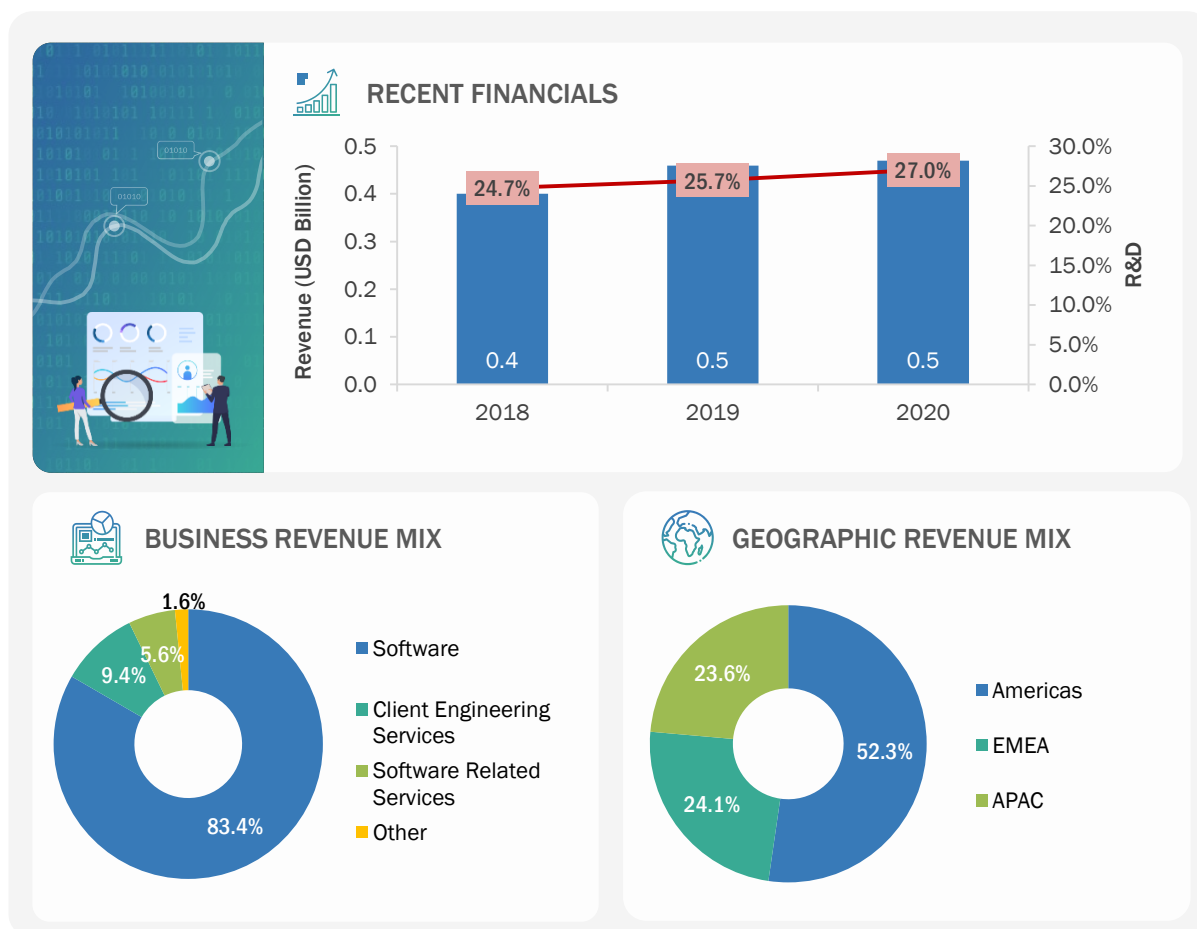
TABLE 250 ALTAIR: BUSINESS OVERVIEW

COMPANY AT A GLANCE - 2021	
Year Founded	1985
Country	US
City	Troy, MI
No. of Employees	3,300+
Overall Annual Revenue	USD 470 million
R&D Expenditure	USD 127 million
Ownership	Public (NYSE: ALTR)

Source: Company Website, Annual Reports, and SEC Filings

Altair is a global technology company and has a presence in 25 countries with 85 offices. The company provides solutions for product development, High-Performance Computing (HPC), and data analytics. It caters to various industries, including aerospace, architecture, engineering, construction, automotive and transportation, electronics, financial services, life sciences, and heavy industry.

The organization offers Altair Knowledge Works, Altair SmartSight, and Altair One in the predictive analytics market. Altair Knowledge Works is a data automation and ML platform comprising Altair Monarch, Altair Knowledge Hub, Altair Knowledge Studio, and Altair Panopticon. Altair SmartSight is a BI platform that can help users connect to data sources and create dashboards.

FIGURE 61 ALTAIR: COMPANY SNAPSHOT

Note 1: The company's fiscal year ends on December 31.

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

12.2.13.2 Solutions offered

TABLE 251 ALTAIR: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTIONS	DESCRIPTION	APPLICATION
Solution	Altair Knowledge Works	It is an ML and predictive analytics solution that rapidly visualizes data and generates explainable results without a single line of code. It brings transparency and automation to ML with features such as AutoML and Explainable AI without restricting how models are configured and tuned, giving control over the model building.	Predictive modeling
Solution	Altair SmartSight	Altair SmartSight is a self-service BI platform that helps unlock data insights. It is deployable on both on-premises and in the cloud. It is offered as a part of Altair SmartWorks Suite, an open-architecture solution enabling advanced edge-to-cloud IoT applications and augmented data analytics powered by ML to drive innovation. Altair SmartSight allows advanced data sets to be easily visualized using auto ML and multiple predictive models, including best fit or bring-your-own models.	- IoT - Data analytics

Solution	Altair One	Altair One offers dynamic, collaborative access to simulation and data analytics technologies with scalable HPC and cloud resources on a single platform.
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Source: Company Website

12.2.13.3 Recent developments

TABLE 252 ALTAIR: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2020-Jul	Altair	Solution	Altair Knowledge Studio	Altair Knowledge Studio was an ML and predictive analytics solution. The new features released by the company included automated Python code generation that supported all predictive modeling, direct data export to Altair Monarch, Altair's industry-leading data preparation tool, and support for R code 4.0 and higher.
2019-Aug	Altair	Solution	Altair GPU solutions	Altair launched Altair GPU solutions to provide NVIDIA graphics processing unit solutions to EMEA clients.

Source: Press Releases

TABLE 253 ALTAIR: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2021-Aug	Acquisition	Altair	S-FRAME	Altair announced the acquisition of S-FRAME Software, a structural analysis software platform used by engineers worldwide to evaluate a structure's ability to endure external loads (such as wind, water, and snow) and meet design code requirements.
2018-Dec	Acquisition	Altair	Datawatch	Altair announced the acquisition of Datawatch. Altair acquired Datawatch that provides data intelligence to boost business analytics.

Source: Press Releases

12.2.14 DOMO

12.2.14.1 Business and financial overview

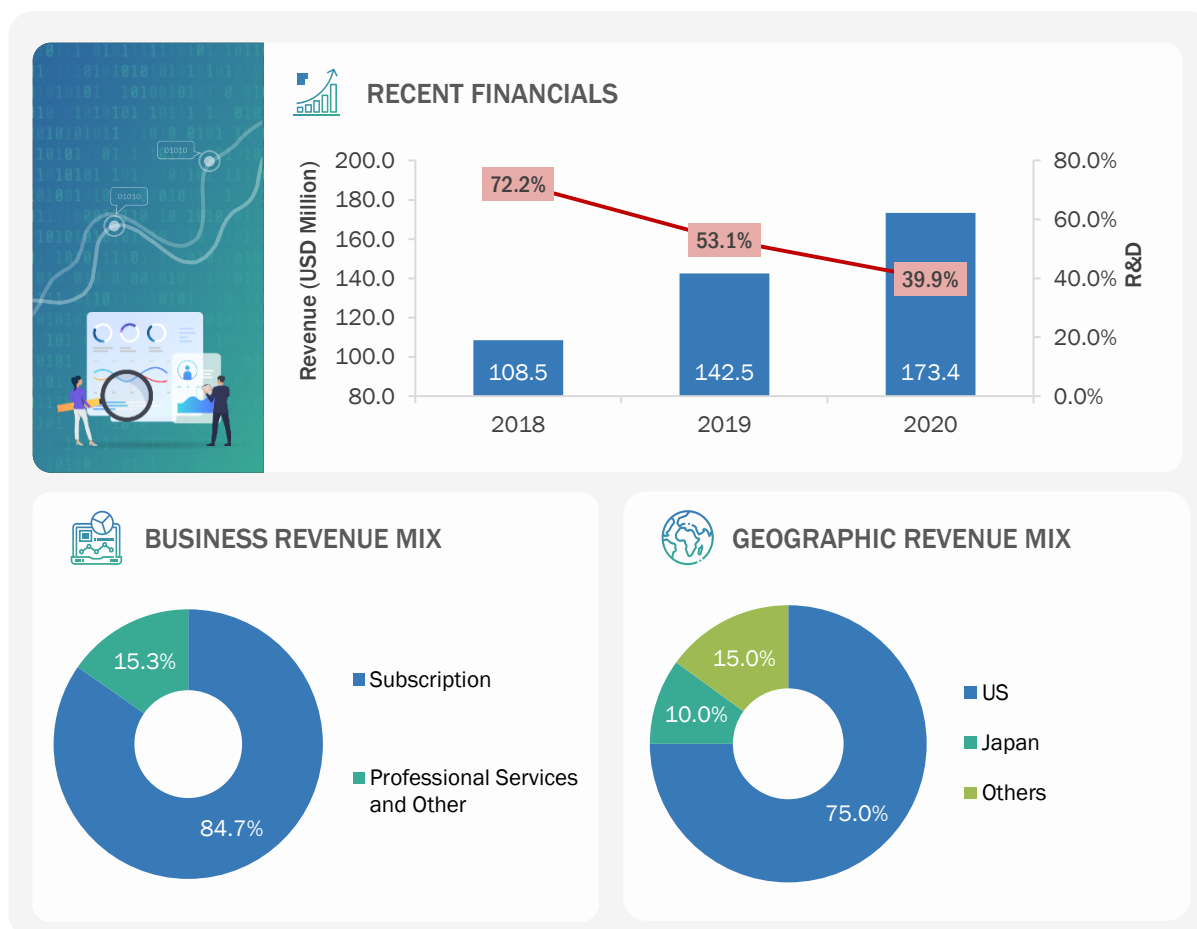
TABLE 254 DOMO: BUSINESS OVERVIEW

COMPANY AT A GLANCE - 2021	
Year Founded	2010
Country	US
City	American Fork, UT
No. of Employees	824
Overall Annual Revenue	USD 173.4 million
R&D Expenditure	USD 69 million
Ownership	Public (NYSE: DOMO)

Source: Company Website, Annual Reports, and SEC Filings

Domo is the fully mobile, cloud-based OS that unifies every business component and delivers it on all phones. It is a niche player. Domo went public in 2018 to gain access to equity markets and become more transparent about its financial performance. The company brings together people, data, and systems into one place for a digitally connected business. It provides both micro- and macro-level visibility and analysis, from cash balances and top-selling product lines to forecasted revenue by region and marketing RoI by channel, among others. It offers real-time dashboards with data pertaining to multiple business areas and performance metrics (such as KPIs and RoI). Domo also offers interactive visualization tools and instant access to company-wide data via a centralized dashboard.

The company is a cloud-based operating system and software, platform provider. It enables CEOs to digitally connect to frontline employees for managing the people, data, and systems in organizations. The platform facilitates companies to manage and optimize supply chains and access real-time data and insights from smartphones to manage businesses by leveraging AI, ML, correlations, alerts, and indices. The company operates its business through two segments: subscription and professional services & other. It caters to various industries, including technology, education, hospitality and transportation, manufacturing, media and entertainment, insurance, retail and eCommerce, financial markets, and healthcare.

FIGURE 62 DOMO: COMPANY SNAPSHOT

Note 1: The company's fiscal year ends on December 31.

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

12.2.14.2 Solutions offered

TABLE 255 DOMO: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTIONS	DESCRIPTION	APPLICATION
Solution	Domo Platform	Domo Platform helps to provide insights faster with automated data discovery, including deep learning and descriptive stats. It offers various data transformation tools for predictive modeling that accelerate discovery and decision-making. The platform offers personalized transformation options, gives alerts in real-time, and offers deep data science to prepare data for predictive modeling.	- Predictive modeling - Financial, risk, and customer analytics

Source: Company Website

12.2.14.3 Recent developments

TABLE 256 DOMO: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2020-Aug	Domo	Solution	COVID-19 Global Tracker	Domo enhanced its COVID-19 Global Tracker with Google and Apple Mobility Trends. This latest update helped decision-makers with anonymized data that can help them understand the movement of people in specific communities across categories that range from grocery and pharmacy to retail and recreation.
2020-May	Domo	Solution	COVID-19 global tracker	Domo updated its COVID-19 global tracker with data from AWS Data Exchange, FRED, and Thumbtack to deliver new insights into how different parts of the economy are responding to the current status of the pandemic.
2020-Apr	Domo	Solution	COVID-19 Crisis Command Center	Domo launched the COVID-19 Crisis Command Center for the State of Iowa, US. It is a robust data-driven solution designed to help Iowa initiate a coordinated, swift response to the COVID-19 pandemic.

Source: Press Releases

TABLE 257 DOMO: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2020	Strategic Partnership	Domo	Snowflake	Domo expanded its partnership with Snowflake to help customers build their business in the cloud. Domo joined Snowflake Partner Connect Program and released new federated query capabilities, including a native setup experience, exclusively for Snowflake.
2019-Oct	Strategic Partnership	Domo	Snowflake	Domo partnered with Snowflake to help customers leverage the speed and scale of the cloud to drive more business value from data. Through this partnership, customers would be able to use Domo's Integration Platform-as-a-Service (iPaaS) capabilities to manage their entire data ecosystem and easily bring additional data sources from their organization into Snowflake.

Source: Press Releases

12.2.15 CLUDERA

12.2.15.1 Business and financial overview

TABLE 258 CLUDERA: BUSINESS OVERVIEW

COMPANY AT A GLANCE - 2021	
Year Founded	2008
Country	US
City	Palo alto, CA
No. of Employees	3000
Overall Annual Revenue	USD 794.2 million
R&D Expenditure	USD 263.6 million
Ownership	Public (NYSE: CLDR)

Source: Company Website, Annual Reports, and SEC Filings

Cloudera delivers enterprise data cloud, from Edge to AI, comprising various offerings, including Cloudera DataFlow (CDF), Cloudera Data Warehouse, and Cloudera ML, helping businesses analyze data for getting actionable data insights. In January 2019, Cloudera and Hortonworks merged to deliver an enterprise data cloud platform to enterprises. The open-source enterprise data cloud platform would provide enterprises with the flexibility of running data on both hybrid and multi-cloud environments. The company offers professional, operational, and ML services to clients for improving business performance. It caters to various industries, including financial services, manufacturing, insurance, telecommunication, retail, technology, healthcare, energy and utilities, public sector, and education. Cloudera provides its offerings to a broad customer base comprising Magenta Telekom, Micron, Bombay Stock Exchange, Komatsu Mining Group, Airtel, Intel, and Yes Bank. It has a presence in South America, North America, EMEA, and APAC. The company offers various support services, including professional services, operational services, and ML advisory services.

In the predictive analytics market, the company offers Cloudera Data Science Workbench, Cloudera Enterprise Data Hub, Hortonworks Data Platform (HDP), Cloudera Distributed Hadoop (CDH), Cloudera Data Platform (CDP) Data Center, and Cloudera DataFlow (Ambari).

12.2.15.2 Solutions offered

TABLE 259 CLOUDERA: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTIONS	DESCRIPTION	APPLICATION
Solution	CDP	It comprises CDP Public Cloud Services and CDP Data Center. Its public cloud services include Data Hub, Data Warehouse, and ML. The platform helps manage data and provides security and governance policies demanded by IT and data leaders.	▪ Fraud detection ▪ Governance, risk, and compliance management
Solution	Cloudera Data Science Workbench	It accelerates ML from research to production with a secure, self-service enterprise data science platform built for enterprises. Cloudera Data Science Workbench lets data scientists manage their own analytics pipelines, including built-in scheduling, monitoring, and email alerting.	Performance monitoring
Solution	Cloudera Enterprise Data Hub	It is a data-driven platform that helps transform data into actionable insights. From data processing to data warehousing, Cloudera Enterprise Data Hub provides an integrated suite of analytic engines. It gives flexibility for data, applications, and business.	Predictive modeling
Solution	HDP	It provides an open-source framework for distributed storage and processing large datasets to gain insights from structured and unstructured data. It helps keep data secure in the cloud as well as on-premises. HDP enables agile application deployment, ML and deep learning workloads, real-time data warehousing, and security and governance.	▪ Governance, risk, and compliance management ▪ Performance monitoring
Solution	Ambari	It was formerly known as Hortonworks DataFlow. It is a scalable, real-time streaming analytics platform that ingests, curates, and analyzes data for key insights and immediate actionable intelligence.	Predictive modeling
Solution	CDP Data Center	It is an on-premises platform that provides enterprise data management and analytics built for hybrid cloud and optimized for bare metal. It delivers high-performance Structured Query Language (SQL) analytics, real-time stream processing and management, granular attribute-based access control, and dynamic column filtering and row masking.	Predictive modeling

Source: Company Website

12.2.15.3 Recent developments

TABLE 260 CLOUDERA: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2019-Nov	Cloudera	Solution	CDP on Microsoft Azure.	Cloudera launched CDP on Microsoft Azure. Cloudera's data platform was designed to be an integrated data platform that delivers self-service analytics powered by Microsoft Azure.
2019-Sep	Cloudera	Solution	Cloudera Data Warehouse.	Cloudera launched Cloudera Data Warehouse, a cloud-native service for Cloudera Data Platform (CDP).

Source: Press Releases

TABLE 261 CLOUDERA: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2020-Apr	Collaboration	Cloudera	RedHat OpenShift	Cloudera collaborated with RedHat OpenShift. Cloudera chose RedHat OpenShift as the preferred container solution for CDP Private Cloud. With this collaboration, CDP Private Cloud would be able to deliver powerful, self-service analytics and enterprise-grade performance with granular security and governance policies demanded by the leading IT companies.
2019-Sep	Acquisition	Cloudera	Arcadia Data	Cloudera acquired Arcadia Data to speed up data analytics' time-to-insight.

Source: Press Releases

TABLE 262 CLOUDERA: OTHERS

YEAR-MONTH	COMPANY NAME	DEVELOPMENT TYPE	DESCRIPTION
2018-Mar	Cloudera	Business expansion	Cloudera announced the expansion of the Budapest, Hungary headquarters. Cloudera expanded the R&D team of Budapest to increase the Hungarian employee base supported by Hungarian Investment Promotion Agency (HIPA).

Source: Press Releases

12.2.16 BOARD INTERNATIONAL

12.2.16.1 Business overview

TABLE 263 BOARD INTERNATIONAL: BUSINESS OVERVIEW

COMPANY AT A GLANCE - 2021	
Year Founded	1994
Country	Switzerland
City	Chiasso
Ownership	Private

Source: Company Website

Board International delivers a single, integrated system that provides BI, analytics, and financial planning capabilities. The company aims at providing an end-to-end decision-making platform. It offers Board Cloud, a SaaS version of the Board all-in-one platform, which allows analyzing, plan, simulate, and forecast easier than ever before. Combining cloud BI tools with cloud planning software, Board International's unified cloud-based analytics services make it easy to implement SaaS reporting and performance management solutions within organizations, providing true BI as a service. It functions across departments, such as finance, supply chain, HR, and sales, and caters to various industries, including retail, manufacturing, financial services, professional services, automotive, and Fast-Moving Consumer Goods (FMCG). Global enterprises such as Coca-Cola, Ricoh, KPMG, Puma, Siemens, and ZF Group have deployed the end-to-end decision-making applications of Board International. The company has its global presence in Europe, Latin America, North America, APAC, and MEA. In April 2020, Board International made access to its eLearning platform complimentary for the remainder of 2020. The move would help customers and partners to upskill themselves during the COVID-19 quarantine period and beyond.

12.2.16.2 Solutions offered

TABLE 264 BOARD INTERNATIONAL: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTION	DESCRIPTION	APPLICATION
Solution	Board BEAM	Board BEAM offers a revolutionary solution that incorporates all the power of predictive analytics into daily business operations and the decision-making process, making it easier for anyone to quickly gain powerful business insights and take action. It also offers a set of statistical functions, from the traditional min/max, average, and standard deviation to algorithms specifically designed for business analysis such as frequency, recency, dormancy, and nascency.	- Product and service development strategies - Predictive modeling
Solution	Board Platform	It is an all-in-one decision-making platform offering Business Intelligence, Performance Management, and Predictive Analytics. It unifies any data source and functionality required for decision-making into a single logical view.	Performance management and predictive analytics

Source: Company Website

12.2.16.3 Recent developments

TABLE 265 BOARD INTERNATIONAL: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2020-Jun	Board International	Solution	Reporting solution	Board International implemented a new reporting solution at Clinique La Prairie, a pioneer health and well-being clinic located in Montreux, Switzerland. Board International implemented a new financial reporting and management solution at Clinique La Prairie using its industry-leading decision-making platform.
2020-Apr	Board International	Solution	Board Marketplace	Board International launched Board Marketplace, an online hub offering a range of pre-built analysis, planning, and forecasting solutions. It featured a wide range of ready-made solutions built specifically for the Board Platform by trusted partners.
2020-Mar	Board International	Solution	Board 11.2	Board International released the latest version of its leading decision-making platform, Board 11.2. The most significant enhancement in the new version is the DeepLocker function, which enabled users to lock data values in the cell along any hierarchy and across any dimension.

Source: Press Releases

TABLE 266 BOARD INTERNATIONAL: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2020-Sep	Strategic partnership	Board International	Delaware	Board International signed a strategic partnership with a Singapore-based global IT consultancy firm, Delaware. The strategic partnership would enhance both parties' capabilities to drive digital transformation in customers' business-wide decision-making processes.
2020-Sep	Collaboration	Board International	Thrasio	Thrasio selected Board International for demand planning, forecasting, and reporting. Thrasio needed an application that could aid in long-range demand planning, forecasting, and reporting. The company wanted to generate segmented reports to break down its local and international business and segment business units in the US. Thrasio's team selected Board International because of its unified approach, which would provide Thrasio with a flexible solution that would easily adapt to changing business requirements.

2020-Aug	Strategic partnership	Board International	AttributeX	Board International and AttributeX entered into a strategic partnership to combine their Business Intelligence and Enterprise Performance Management expertise to enhance customers' strategic decision-making.
2020-Jul	Partnership	Board International	Mavericka	Board International partnered with Mavericka to focus on realizing business planning, analytics, and optimization projects. With a large customer base across the globe, Board International would be able to support Mavericka's go-to-market strategy with profound knowledge in the needs and requirements of CPG, retail, distribution, manufacturing, and high-tech industry segments.

Source: Press Releases

12.2.17 TIBCO SOFTWARE

12.2.17.1 Business overview

TABLE 267 TIBCO SOFTWARE: BUSINESS OVERVIEW

COMPANY AT A GLANCE - 2021	
Year Founded	1997
Country	US
City	Palo alto, CA
Ownership	Private

Source: Company Website

TIBCO Software is a US-based company that has been strengthening its position as a BI and analytics solution provider since its establishment. The company enhances digital business for its clients through its TIBCO Connected Intelligence Cloud platform that connects any application or data source; intelligently unifies data for greater access, trust, and control; and predicts outcomes in real-time and at scale. It captures data in real-time and augments the intelligence of businesses through analytical insights, ranging from APIs and systems to devices and people. Additionally, it provides products for business optimization, integration and core infrastructure, and process automation and collaboration. TIBCO Software offers consulting, education, and support services.

With API-led microservices, serverless functions, and integration flows, the company helps businesses of any size and users of any technical level to interconnect resources to accelerate digital transformation. In March 2019, TIBCO Software acquired SnappyData to complement the TIBCO Connected Intelligence Cloud platform with a unified analytics data fabric that enhances analytics, data science, streaming, and data management for various use cases requiring speed, volume, and agility. The company has developed an emergency task force to quickly address changing circumstances, such as monitoring compliance with applicable internal policies, to deal with COVID-19. It has launched a public website through which it offers analytics to track the spread and impact of the COVID-19 pandemic using the TIBCO Spotfire analytics software.

12.2.17.2 Solutions offered

TABLE 268 TIBCO SOFTWARE: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTION	DESCRIPTION	APPLICATION
Solution	TIBCO Spotfire	TIBCO Spotfire makes advanced predictive analytics easy, consumable, and accessible for everyone using user interfaces. It provides single-click data science with statistical and ML methods to predict outcomes in real-time, helping all users grow in efficiency, skills, and smarts. It enables predicting the behavior of businesses and customers and elevates analytics knowledge throughout the company. In November 2018, TIBCO Software announced the release of Spotfire X, a new analytics solution, which fuses multiple AI-powered interaction models that adapt to user preferences and styles.	- Customer segmentation and clustering - Customer behavior analysis - Monitoring customer loyalty and satisfaction - Predictive modeling

Solution	TIBCO Data Science	TIBCO Data Science helps organizations innovate and solve complex problems faster to ensure predictive findings quickly turn into optimal outcomes. It enables creating innovative pipelines with the latest deep learning, predictive, prescriptive, AI, and analytical techniques, which can run on any edge device, gateway, or microcontroller.	Predictive modeling
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Source: Company Website

12.2.17.3 Recent developments

TABLE 269 TIBCO SOFTWARE: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2021-May	TIBCO Software	Solution	TIBCO Data Virtualization and TIBCO EBX	TIBCO Software announced many innovative advancements to its Unify data management portfolio, including the TIBCO Data Virtualization and TIBCO EBX products. These solutions significantly strengthened an organization's data fabric, allowing customers to leverage the immense potential of data.
2021-May	TIBCO Software	Solution	TIBCO Hyperconverged Analytics	TIBCO Software announced many breakthroughs and features for its Predict product portfolio, dramatically improving time-to-insight and delivering smart, scalable analytics across the organization. TIBCO Spotfire, TIBCO Streaming, and TIBCO Data Science enhancements would be available in a single platform. This TIBCO Hyperconverged Analytics experience would boost company value by shortening the time between business events and analytics, insights, and action.

Source: Press Releases

TABLE 270 TIBCO SOFTWARE: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2021-Jan	Acquisition	TIBCO Software	ibi	TIBCO Software announced the successful completion of its acquisition of Information Builders, Inc. (ibi). ibi's data management and analytics capabilities would be added to the superior TIBCO Connected Intelligence platform as a result of the acquisition. Following the completion of the acquisition, TIBCO software would concentrate on business alignment and resource mapping to ensure the continued health and support of its increased client base and global partner network. TIBCO's innovative products would provide ibi customers and partners with modern solutions to address important business concerns.
2020-Nov	Strategic partnership	TIBCO Software	Mercedes-AMG Petronas Formula One Team	TIBCO Software announced the release of their TIBCO Cost Visualizer Tool, as well as a continued partnership with the Mercedes-AMG Petronas Formula One Team to provide real-time visual analytics and data science capabilities. The seven-time FIA Formula One World Constructors' Champions' team of designers, engineers, and budget planners could use the TIBCO Cost Visualizer Tool to better understand and anticipate the costs of building the Mercedes-AMG F1 W11 EQ Performance vehicle.
2020-Sep	Strategic Partnership	TIBCO software	Dartmouth College	Dartmouth College would benefit from TIBCO's Corporate Charitable Giving Program, which would provide data analytics tools and support. Dartmouth would use TIBCO GatherSmart, TIBCO Data Virtualization, and TIBCO Spotfire to collect, monitor, and communicate various data sets linked to COVID-19 to build a real-time perspective of its pandemic response. Dartmouth joined the list of top organizations using TIBCO GatherSmart to establish a safe environment for those returning to their places of work and learning, including TIBCO.

Source: Press Releases

12.2.18 HITACHI VANTARA

Founded	2017
Headquarters	Santa Clara, CA, US
Ownership	Private
Business Overview	Hitachi Vantara is the subsidiary of Hitachi Ltd, a Japanese manufacturing firm that combines technology, intellectual property, and industry knowledge to deliver data-managing solutions, helping enterprises improve customer experience, develop new revenue streams, and lower business costs. In June 2015, Hitachi acquired Pentaho, a leading data integration, visualization, and analytics company, which later in 2017 became a part of Hitachi Vantara. In March 2020, Hitachi Vantara acquired Waterline Data, Inc., a provider of intelligent data cataloging solutions. After the acquisition, the company introduced Lumada Data Catalog that incorporated the data cataloging technology of Waterline Data into its Lumada portfolio. The company provides a multi-cloud environment, which can be of private, public, and hybrid cloud, and an edge-to-cloud data infrastructure with the AI technology that provides the availability of data across data centers. It also offers consulting services. It focuses on catering to global Fortune 1000 companies with data management, infrastructure, content and analytics products, and industrial IoT solutions.
Solution Offered	Smart Data Center is a pre-engineered data center analytics solution. It utilizes AI to optimize data center performance for maximizing customer experience and lowering risks, and IT costs. Its predictive analytics feature helps highlight the areas of concern, and ML provides proactive insights. It also contains cooling optimization, which reduces energy costs.
Geographic Presence	North America, Europe, APAC, MEA, and Latin America

Source: Company Website and Press Releases

12.2.19 HAPPIEST MINDS

Founded	2011
Headquarters	Bangalore, India
Ownership	Public (NSE: HAPPSTMNDS)
Business Overview	Happiest Minds is an IT company that applies agile methodologies to enable digital transformation for enterprises and technology providers. It works in various segments, including digital business services, product engineering services, agile infra and security services, IoT, and analytics. It caters to various industries, including automotive, banking, consumer packaged goods, eCommerce, education publishing, manufacturing, retail, and travel and hospitality. In May 2020, the company contributed USD 45 million toward COVID-19 relief, including USD 1.7 million meals toward Akshaya Patra Foundation.
Solutions Offered	Happiest Minds' data science consulting and business analytics solutions leverage the power of predictive analytics for deriving real-time insights and reducing customer churn. These solutions help clients solve the toughest data challenges, predict demand for products and services to improve customer satisfaction and guide business strategies based on knowledge and foresight. Next Gen Enterprise Operations Analytics helps understand enterprise data at a glance to identify hidden opportunities, enable better decision-making, and increase efficiency across all departments. It provides demand analytics, asset analytics, security and risk analytics, sales analytics, HR analytics, and finance analytics.
Geographic Presence	North America, Europe, APAC, and MEA

Source: Company Website

12.2.20 DATAIKU

Founded	2013
Headquarters	New York, NY, US
Ownership	Private
Business Overview	Dataiku was formed to help companies in using data science, ML, and AI for continuous innovations. It works by enabling organizations to embed data systems and competencies at each stage. The company has started an initiative, 'AI for Good', to work with non-profit organizations. It has launched an academic program that provides free software licenses to academicians, researchers, and students. Dataiku's partners are analytics consultants, Original Equipment Manufacturer (OEM) partners, service partners, technology partners, and value-added resellers such as Micropole, InformationsFabrik, SVA, Ondata, Integra, Microsoft, and 4C. The valuable customers of Dataiku are GE, Sephora, Unilever, BNP Paribas, Ubisoft, Capgemini, and L'Oréal, among others. In December 2019, CapitalG partnered with Dataiku to directly acquire its shares from Serena Capital. Post the acquisition, Dataiku's valuation exceeded USD 1 billion. Dataiku offers Dataiku Data Science Studio (DSS) in the predictive analytics market. The solution collaborates with data analysts, engineers, and scientists on a single platform.
Solution Offered	Dataiku DSS is a data science software platform that provides visual interfaces at any step of the predictive dataflow prototyping process, from wrangling to analysis and modeling. It helps bundle the entire workflow as a single deployable package for real-time predictions with REST API and enables monitoring live data in-production with dashboard and data validation policies. The platform helps create powerful predictive analytics solutions using the latest data analysis and ML technologies.
Geographic Presence	North America, Europe, and APAC

Source: Company Website

12.2.21 RAPIDMINER

Founded	2007
Headquarters	Boston, MA, US
Ownership	Private
Business Overview	RapidMiner, formerly known as Yet Another Learning Environment (YALE), is an open data science platform for analytics. In 2013, it rebranded itself to RapidMiner. RapidMiner unites the life cycle of data science, from the preparation of data to ML and the deployment of predictive models that offer an end-to-end collaboration platform. It works on all the stages of Automated Data Science, from exploring data to modeling data, using RapidMiner Auto Model integrated with Turbo Prep and Model Ops. It offers different products in the field of Automated Data Science and managed as well as on-demand AI cloud. It also offers various platforms, including Studio, Server, Real-time Scoring, and Radoop. The customers of RapidMiner are BMW, CISCO, ezCater, Hewlett Packard Enterprise, Sanofi, and Transport for London.
Solutions Offered	▪ RapidMiner Platform is built for analytics teams. It unifies the entire data science life cycle, from data prep to ML and predictive model operations. It comprises various products, including Studio, Go, Notebooks, and AI Hub. Studio provides visual workflows for predictive analytics. ▪ RapidMiner Turbo Prep helps prepare the data for predictive modeling. Predictive models are prepared using RapidMiner Studio and RapidMiner Auto Model. RapidMiner Auto Model uses automated ML and best practices to build predictive models in five mouse clicks. ▪ RapidMiner AI Cloud enables employees across the enterprise to rapidly create and operate AI solutions and drive business impact. It provides high processing power that leverages the predictive power of big data. ▪ RapidMiner Radoop enables building and running predictive models in Hadoop using the RapidMiner Studio visual workflow designer without having to code in Spark.
Geographic Presence	North America and Europe

Source: Company Website

12.2.22 QLIK

Founded	1993
Headquarters	King of Prussia, PA, US
Ownership	Private
Business Overview	Qlik offers a platform with an end-to-end solution to get value out of data. It enables open-ended, curiosity-driven exploration for transformative changes. It caters to various industries, including healthcare, financial services, retail, manufacturing, high tech, public sector, energy and utilities, communications, life sciences, and consumer products, with more than 50,000 customers across the globe, including Bridgestone, Cisco, Lloyd's, Samsung, Deloitte, and Paccar. Qlik has over 1,700 partners across the globe. These partners include solution providers, consulting and service partners, various companies, managed service providers, technology partners, and OEM partners. In May 2019, Qlik acquired Attunity Ltd. to combine the data integration solutions of Attunity with the data management solutions of Qlik Data Catalyst. In the predictive analytics market, Qlik offers Qlik Sense. In September 2020, Tangent Works partnered with Qlik to directly bring their unique InstantML modeling solution into the Qlik Platform, giving business users the power of lightning-fast, incredibly accurate, and up-to-the-minute real-time predictive forecasting.
Solutions Offered	Qlik Sense is a BI and analytics platform that enables users to tackle every BI use case, from interactive dashboards and KPI reports to embedded, mobile, and conversational analytics. Qlik Sense combines data governance with AI-assisted data visualization and data exploration, advanced analytics, and intuitive, self-service data preparation capabilities to enable an organization to make impactful discoveries from data. It provides access to automated ML modeling and on-demand predictions directly from within Qlik Sense to generate predictive insights whenever and wherever needed. ▪ QlikView is a data analytics solution that provides all BI capabilities, dashboards, KPIs, balanced scorecards, and predictive analysis for enterprises to gain better data insights to make strategic business decisions. ▪ Qlik Data Catalyst builds a secure, enterprise-scale catalog that helps businesses understand data preparation and gain insights for business decisions. ▪ Qlik Analytics Platform helps build, embed, and integrate visual analytics for any device or app on a single platform.
Geographic Presence	North America, EMEA, and APAC

Source: Company Website

12.3 STARTUPS/SMES

12.3.1 BIOFOURMIS

Founded	2015
Headquarters	Boston, MA, US
Ownership	Private
Business Overview	Biofourmis is the fastest-moving digital health company. Its mission is to improve real-world outcomes by predicting and preventing serious medical events with AI-based physiology analytics. The company comprises engineers, clinicians, and scientists from around the globe, operating at the intersection of data science, biotechnology, and software technology. Biofourmis has collaborated with some of the world's most reputed healthcare organizations, research hospitals, and biopharma companies. It focuses on improving patient outcomes, advancing clinical research, reducing healthcare costs, and accelerating drug research and development. Its collaborators include Novartis, AstraZeneca, Mayo Clinic, Massachusetts general hospital, and SingHealth. Its investors are Sequoia, MassMutual, Openspace Ventures, and Aviva. The digital therapeutics portfolio of Biofourmis consists of products that cover multiple therapeutic areas, including heart failure, acute coronary syndrome, Chronic Obstructive Pulmonary Disease (COPD), oncology, and chronic pain. In September 2020, Biofourmis closed USD 100 million series C funding led by SoftBank Vision Fund 2, with participation from returning investors Openspace Ventures, MassMutual Ventures, Sequoia Capital, and EDBI. The company has raised a total of USD 145 million.
Solutions Offered	▪ Biofourmis Biovitals: It is a field-tested solution that enables pattern recognition to help care teams remotely monitor the physiological signs of patients with complex chronic and acute conditions. It also enables the comparison of the information with population-level data. The solution brings disease, data, and ML together to change the face of chronic care management. Biofourmis Biovitals predicts and detects serious medical events through advanced analytics and provides personalized feedback for patients through a patient-facing app. ▪ Biovitals HF: It is a prescription software that enables the early detection of heart failure exacerbation and augments guideline-directed use of heart failure medications. ▪ RhythmAnalytics: It is a cloud-based software that enables the automated detection of over 15 types of cardiac arrhythmias. ▪ Biovitals Sentinel: It is a solution that enables the continuous remote monitoring of critical biovitals and patient-reported symptoms using Everion, an innovative new multi-sensor armband wearable. It uses advanced AI/ML algorithms to process collected data, detect physiological changes, and predict events in various disease conditions, facilitating early intervention and dramatically improving the outcomes of many traditional therapeutics. ▪ Gaido: It is a prescription software that continuously monitors and detects the early signs of deterioration in oncology patients and pre- and post-treatments and enables preventing avoidable readmissions.
Geographic Presence	North America, Europe, and APAC

Source: Company Website

12.3.2 IN-MED PROGNOSTICS

Founded	2016
Headquarters	Hurstville, Australia
Ownership	Private
Business Overview	In-Med Prognostics brings affordable, accessible, reliable, and data-driven brain health prognostics and diagnostics tools to India and other emerging markets. Leveraging analytics and deep learning methodologies, the company develops a biomarker algorithm for various neurological disorders starting with pediatric epilepsy and dementia. The company offers segmentation services and comparative volumetric analysis reports to clinical end users and research support for clinical imaging-based projects to pharma and life science customers. The aim of In-Med Prognostics is to build an Indian databank for neurological disorders consisting of complete clinical data, including radiology. In January 2019, In-Med Prognostics received USD 71 million for 18 months from Biotechnology Industry Research Assistance Council (BIRAC).
Solution Offered	NEUROShield: It integrates MRI images, clinical reports, and psychological assessments to produce automated, quantified reports for supporting physicians' clinical impressions of various neurological disorders.
Geographic Presence	APAC

Source: Company Website

12.3.3 AITO.AI

Founded	2018
Headquarters	Helsinki, Finland
Ownership	Private
Business Overview	Aito.ai lets software developers run ML on relational datasets using predictive queries. It is the next generation of ML. It offers real-time learning through incrementally added data, immediately used for predictions without model re-training and deployment. Aito.ai replaces the current ML tools that have a steep learning curve and generate single-purpose models. In December 2019, Aito.ai announced that it raised USD 1.8 million from a B2B software investor, henQ, and launched the public version of its Predictive Database. The Predictive Database combines ML and databases into a single platform.
Solution Offered	Aito.ai: It is a predictive database for developers who value quick time-to-market. It is the fastest ML platform for Robotic Process Automation (RPA) developers.
Geographic Presence	North America and Europe

Source: Company Website

12.3.4 SYMEND

Founded	2016
Headquarters	Calgary, Alberta, Canada
Ownership	Private
Business Overview	Symend is scientifically transforming the debt recovery industry by combining behavioral science with advanced analytics to treat individuals with empathy and dignity when they need it most. Its platform helps service providers and financial institutions develop positive, individualized treatment programs that engage customers and build loyalty. In May 2020, Symend announced that it raised USD 52 million from investors led by a global technology venture firm, Inovia Capital. Other investors included Ignition Partners, Impression Ventures, BDC Capital's Women in Technology Fund, Mistral Venture Partners, and angel investor Markus Frind.
Solution Offered	Symend's digital engagement platform provides a simple, personalized experience that enables users to take desired actions while improving Net Promoter Score (NPS) and retaining KPIs.
Geographic Presence	North America and Europe

Source: Company Website

12.3.5 ONWARD HEALTH

Founded	2017
Headquarters	Hyderabad, Telangana, India
Ownership	Private
Business Overview	Onward Health provides oncology analytics tools that help Oncologists to look at the information of a large volume of patients and come to accurate clinical diagnoses. Cancer is a challenging disease; therefore, choosing the right treatment protocol is the significantly important step that oncologists seek maximum help with. Onward Health is focused on building a platform that would help in the clinical process. Its clinical collaborations with cancer institutes are focused on novel approaches, challenges faced in cancer prognostics, and important biomarkers.
Solution Offered	Onward Assist: It is a predictive analytics platform from Onward Health. It is a cancer prognostics platform built with clinicians and healthcare providers to assist in better diagnosis and prognosis.
Geographic Presence	APAC

Source: Company Website

12.3.6 UNIONCRATE

Founded	2016
Headquarters	New York, NY, US
Ownership	Private
Business Overview	Unioncrate is an AI supply chain platform that frees CPG brands from inaccurate predictions, manual processes, and wasted capital. The company has been solving CPG supply chain challenges since 2016. Its investors include Susa Ventures, Bain Capital Ventures, Laconia Capital, Founders Fund, and other notable angel investors. Its client base includes spindrift, COTY, and The Fishin' Company. In October 2019, Unioncrate raised USD 9 million through series A funding led by Gradient Ventures.
Solution Offered	Demand Planning AI: It helps in accurate forecasts. It also helps combine consumer behavior data with historical shipments to predict sales and inventory with up to 85.3% accuracy.
Geographic Presence	North America and Europe

Source: Company Website

12.3.7 CYBERLABS

Founded	2017
Headquarters	Rio de Janeiro, Brazil
Ownership	Partnership
Business Overview	CyberLabs is a laboratory that focuses on technology development in AI and computer vision. Its purpose is to attend to real market necessities with innovative technologies. It brings digital transformation to companies by extending the power of data and optimizing business decisions using AI.
Solution Offered	CyberLabs: It offers an image interpretation platform that counts people to know what is happening or what will happen each moment. It issues alerts and provides instant updates on crowded places. The solution is helping in containing COVID-19 in real-time by prioritizing cameras to identify more people.
Geographic Presence	Latin America

Source: Company Website

12.3.8 ACTIFY DATA LABS

Founded	2017
Headquarters	Bengaluru, Karnataka, India
Ownership	Private
Business Overview	Actify Data Labs is an augmented intelligence company that delivers rapid business impact using data science and engineering capabilities. It is a wholly-owned subsidiary of True North. True North is a USD 2.8 billion private equity fund with a 20-year history of investing in and transforming mid-sized companies. It offers various services, including risk management, marketing and customer analytics, and AI/ML enablement. In February 2018, Actify Data Labs won USD 10 million in funding.
Solution Offered	ADAPTify: It accelerates the complete data-to-decisions life cycle and democratizes AI by empowering data scientists. ADAPTify addresses all four stages of the data life cycle: data consolidation, visualization, ML model development, and model implementation.
Geographic Presence	APAC

Source: Company Website

12.3.9 AMLGO LABS

Founded	2017
Headquarters	Bengaluru, Karnataka, India
Ownership	Private
Business Overview	Amlgo Labs is an advanced analytics and ML consulting company. The company focuses on enhancing businesses and providing insights to organizations for making intelligent data-driven decisions across verticals. It has expertise in fraud risk analytics, regulatory and compliance analytics, customer analytics, Anti-money Laundering (AML), sales and marketing analytics, social media analytics, web analytics, and supply chain analytics. The company uses ML algorithms and data science tools that combine business understanding with quantitative analysis to support robust, reliable decision-making. Amlgo Labs offers various services, which include advanced analytics and ML, risk modeling and regulatory reporting, and data visualization and decisions. It specializes in traditional ML and deep learning, ML frameworks, AWS, Azure and Google Cloud, time series analysis, predictive analytics, big data, and natural language processing.
Solutions Offered	The company provides advanced analytics and ML solutions that collect data from various downstream systems of its clients, and then descriptive analytics is performed to present the data in the form of interactive charts or reports so that clients can easily understand it. Using ML algorithms, it performs predictive analytics to deliver business insights or forecast the future scenario.
Geographic Presence	APAC

Source: Company Website

12.3.10 VERIMOS

Founded	2018
Headquarters	Austin, TX, US
Ownership	Private
Business Overview	Verimos is purpose-built to support the well-being of the non-desk workforce. In July 2020, Verimos announced its first tranche of funding. The company closed its first funding round of USD 750 million. The participants were top executives, founders, and company operators.
Solution Offered	Verimos saves money by providing health access and healthcare to the non-desk workforce for a small fraction of the price of insurance-based products.
Geographic Presence	North America

Source: Company Website

13 APPENDIX

13.1 DISCUSSION GUIDE

Discovering a market is all about identifying opportunities in unrecognized and unexploited gaps that are worth pursuing. In most cases, these markets are not apparent or conventional. Discovering a product, unlike markets, is quite different and significantly depends on its value and usability. We want to understand and discuss how your company invests in discovering smart markets and smarter products to stay ahead of the curve.

Currently, we are conducting a global study on the predictive analytics market, with a focus on current trends and technology adoption. The report will provide a comprehensive market and forecast analysis of the overall market and enabling sub-markets. The report also features a detailed analysis of opportunities, market drivers, restraints, roadmaps, competitive landscape, and profiles of key industry players.

Q. 1. According to you, how would the COVID-19 pandemic impact the predictive analytics market?

SR. NO	POSITIVE IMPACT	NO IMPACT	NEGATIVE IMPACT
1			

Comments: _____

Q. 2. Please brief us about the solutions and services that your company offers in the predictive analytics market?

Solutions:

i) _____

ii) _____

Services:

i) _____

ii) _____

Q. 3. What does the predictive analytics market look like currently (2021), and where is it heading, i.e., how will it look after five years (2026)?

PREDICTIVE ANALYTICS	2021	AFTER 5 YEARS
(USD BILLION)		

Q. 4. What, according to you, are the approximate percentage contributions of solutions to the overall revenue generated by the predictive analytics market in 2021, and how do you think these segments will grow over the next five years (2026)?

SOLUTION	2021	2026
Financial Analytics		
Risk Analytics		
Marketing Analytics		
Sales Analytics		
Customer Analytics		
Web and Social Media Analytics		
Supply Chain Analytics		
Network Analytics		
Other solutions*		
Total		

*Other solutions include HR analytics and legal analytics.

- Q. 5.** What, according to you, are the approximate percentage contributions of the deployment modes to the overall revenue generated by the predictive analytics market in 2021, and how do you think these segments will grow over the next five years (2026)?

DEPLOYMENT MODE	2021	2026
On-premises		
Cloud		
Total		

- Q. 6.** What, according to you, are the approximate percentage contributions of the organization sizes to the overall revenue generated by the predictive analytics market in 2021, and how do you think these segments will grow over the next five years (2026)?

ORGANIZATION SIZE	2021	2026
SMEs		
Large Enterprises		
Total		

- Q. 7.** What, according to you, are the approximate percentage contributions of verticals to the overall revenue generated by the predictive analytics market in 2021, and how do you think these verticals will grow over the next five years (2026)?

VERTICAL	2020	2025
BFSI		
Healthcare and Life Sciences		
Retail and eCommerce		
Government and Defense		
Transportation and Logistics		
Energy and Utilities		
Telecommunications and IT		
Manufacturing		
Other verticals*		
Total		

*Other verticals include media and entertainment, travel and hospitality, and education.

Q. 8. Which region, according to you, contributes the highest (5) and the lowest (1) revenue currently (2021) and in the future (2026) to the predictive analytics market?

- Please rank on a scale of 1–5, with 5 being the highest.

S. NO.	REGION	2021 (SCALE 1–5)	2026 (SCALE 1–5)
1	North America		
2	Europe		
3	APAC		
4	MEA		
5	Latin America		

Q. 9. Please name your five main competitors.

Please assign the rank (scores from 1 to 5) relatively in the column named ‘Market Impact’ to each of these competitors, such that no two competitors have the same score. Kindly repeat the same for the columns ‘Innovation’ and ‘Solution/Product Feature.’

S. NO.	COMPETITORS	MARKET IMPACT (RANKING OF 1 TO 5, 1 BEING THE HIGHEST)	INNOVATION (RANKING OF 1 TO 5, 1 BEING THE HIGHEST)	SOLUTION/PRODUCT FEATURE (RANKING OF 1 TO 5, 1 BEING THE HIGHEST)
1				
2				
3				
4				
5				

Q. 10. What would you like to see covered additionally in the report, apart from the technology and market segmentation in the report structure attached?

Comments: _____

Q. 11. Please list some of the new technologies/innovations, along with their advantages, in the predictive analytics market, which are being introduced by vendors.

Comments: _____

Q. 12. What do you think about the predictive analytics market? Please give your opinion in two or three sentences about the growth of the predictive analytics market, considering the next five years (excluding company or product name).

Comments: _____

13.2 KNOWLEDGE STORE: MARKETSandMARKETS' SUBSCRIPTION PORTAL

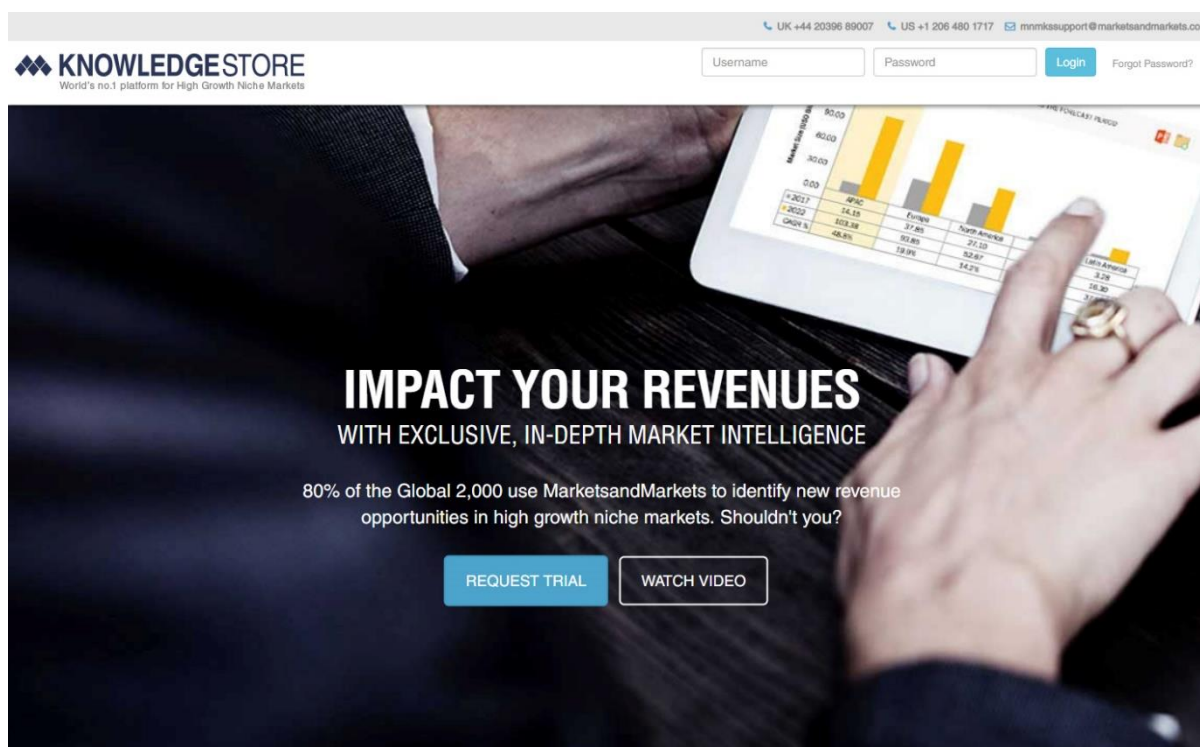
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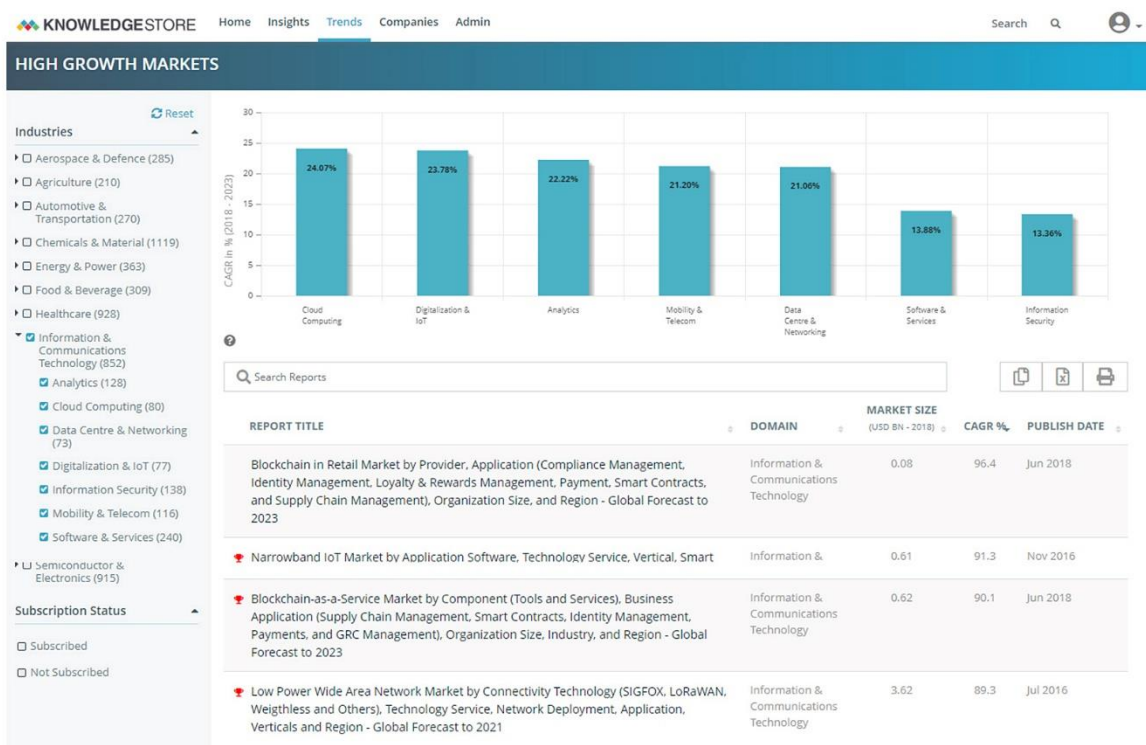
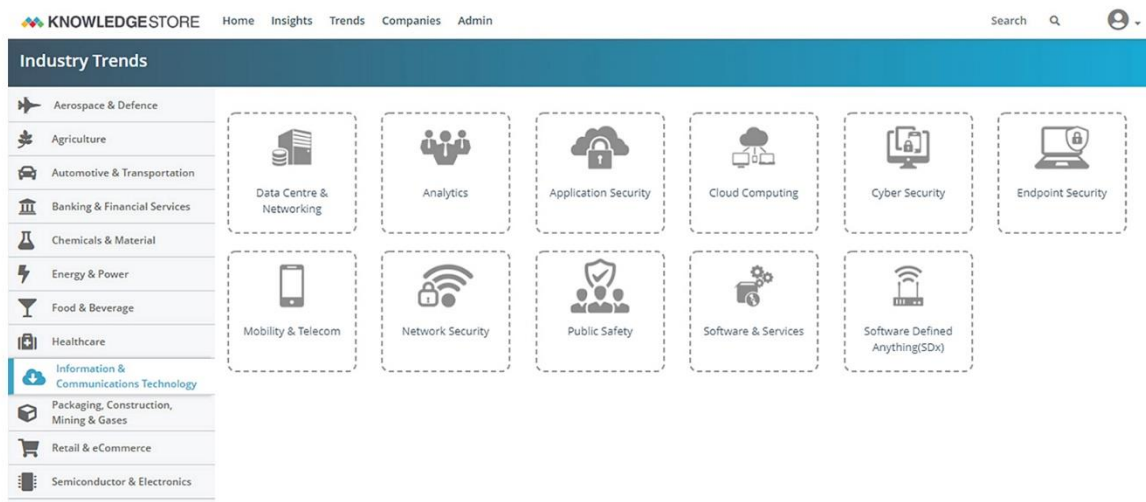
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	APAC	Europe	North America	Latin America
Q1 2021	48.1%	37.4%	21.1%	3.2%
Q2 2021	48.1%	37.4%	21.1%	3.2%
Q3 2021	48.1%	37.4%	21.1%	3.2%
Q4 2021	48.1%	37.4%	21.1%	3.2%

MARKETSANDMARKETS KNOWLEDGESTORE: INFORMATION and COMMUNICATION TECHNOLOGY SNAPSHOT



13.3 AVAILABLE CUSTOMIZATIONS

With the given market data, MarketsandMarkets offers customizations as per the company's specific needs. The following customization options are available for the report:

PRODUCT ANALYSIS

- Product Matrix, which gives a detailed comparison of the product portfolios of each company

GEOGRAPHIC ANALYSIS

- Further breakup of the North American predictive analytics market
- Further breakup of the European predictive analytics market
- Further breakup of the Asia Pacific predictive analytics market
- Further breakup of the Latin American predictive analytics market
- Further breakup of the MEA predictive analytics market

COMPANY INFORMATION

- Detailed analysis and profiling of additional market players (up to 5)

13.4 RELATED REPORTS

SR. NO	REPORT TITLE	PUBLISHED DATE
1	CUSTOMER JOURNEY ANALYTICS MARKET- GLOBAL FORECAST TO 2026 By Component, Organization Size, Deployment Mode, Data Source, Application (Campaign Management & Product Management), Vertical (BFSI, Retail & eCommerce, & Telecommunications and IT), and Region https://www.marketsandmarkets.com/Market-Reports/customer-journey-analytics-market-119398916.html	March 2021
2	BIG DATA MARKET- GLOBAL FORECAST TO 2025 By Component, Deployment Mode, Organization Size, Business Function (Operations, Finance, and Marketing and Sales), Industry Vertical (BFSI, Manufacturing, and Healthcare and Life Sciences), and Region https://www.marketsandmarkets.com/Market-Reports/big-data-market-1068.html	March 2020

13.5 AUTHOR DETAILS

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With an experience of over 14 years, he has been associated with ICT research for more than 10 years now. Prior to joining MarketsandMarkets, he worked with firms like Frost and Sullivan, AMI-Partners, TNS Market Research and global IT major, IBM, where he worked for more than 5 years, and his last role saw him as Market Segment Manager. He holds rich experience in the technology sector, working on several custom and syndicated research assignments with large Multinational Organizations.

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The role encompasses handling syndicated and custom research projects in the ICT domain. Having more than 14 years of experience with 10+ years in research, Sameer joined MarketsandMarkets from Forrester Research, where he was working as a Consultant, serving the Business Technology (BT) Professionals role. The role saw him working on varied consulting/custom-research projects and communicating directly with clients, internal resources (i.e., sales, research), and third-party vendors to address queries and manage expectations to create quality deliverables. Sameer has singlehandedly driven projects that include deliverables in the shape of data modeling, trend analysis, end-user requirements, competitive profiling, etc.

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